

4-6-6

1. Let h be a function defined for all $x \neq 0$ such that $h(4) = -3$ and the derivative of h is given by $h'(x) = \frac{x^2 - 2}{x}$ for all $x \neq 0$.
- (a) Find all values of x for which the graph of h has a horizontal tangent, and determine whether h has a local maximum, a local minimum, or neither at each of these values. Justify your answers.
- (b) On what intervals, if any, is the graph of h concave up? Justify your answer.
- (c) Write an equation for the line tangent to the graph of h at $x = 4$.
- (d) Does the line tangent to the graph of h at $x = 4$ lie above or below the graph of h for $x > 4$? Why?

Part A

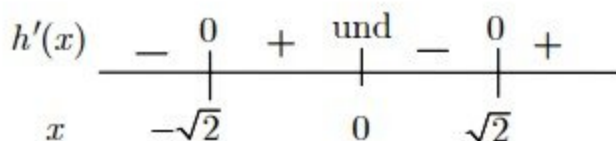
1 point is earned for: $x = \pm\sqrt{2}$

1 point is earned for the analysis

2 points are earned for the conclusions

<-1> not dealing with discontinuity at 0

$$h'(x) = 0 \text{ at } x = \pm\sqrt{2}$$



Local minima at $x = -\sqrt{2}$ and at $x = \sqrt{2}$



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for: $x = \pm\sqrt{2}$

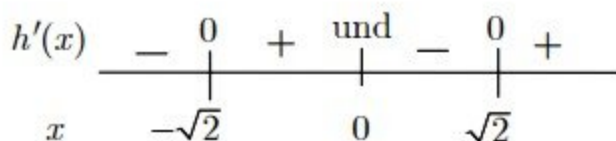
1 point is earned for the analysis

2 points are earned for the conclusions

4-6-6

<-1> not dealing with discontinuity at 0

$$h'(x) = 0 \text{ at } x = \pm\sqrt{2}$$



Local minima at $x = -\sqrt{2}$ and at $x = \sqrt{2}$

Part B

1 point is earned for: $h''(x)$

1 point is earned for: $h''(x) > 0$

1 point is earned for the answer

$h''(x) = 1 + \frac{2}{x^2} > 0$ for all $x \neq 0$. Therefore, the graph of h is concave up for all $x \neq 0$.



0	1	2	3
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The student response earns all of the following points:

1 point is earned for: $h''(x)$

1 point is earned for: $h''(x) > 0$

1 point is earned for the answer

$h''(x) = 1 + \frac{2}{x^2} > 0$ for all $x \neq 0$. Therefore, the graph of h is concave up for all $x \neq 0$.

Part C

1 point is earned for the tangent line equation

4-6-6

$$h'(4) = \frac{16-2}{4} = \frac{7}{2}$$

$$y + 3 = \frac{7}{2}(x - 4)$$



0

1

The student response earns all of the following points:

1 point is earned for the tangent line equation

$$h'(4) = \frac{16-2}{4} = \frac{7}{2}$$

$$y + 3 = \frac{7}{2}(x - 4)$$

Part D

1 point is earned for the answer with reason

The tangent line is below the graph because the graph of h is concave up for $x > 4$.



0

1

The student response earns all of the following points:

1 point is earned for the answer with reason

The tangent line is below the graph because the graph of h is concave up for $x > 4$.

4-6-6

2. Let h be a function defined for all $x \neq 0$ such that $h(4) = -3$ and the derivative of h is given by $h'(x) = \frac{x^2 - 2}{x}$ for all $x \neq 0$.

Does the line tangent to the graph of h at $x = 4$ lie above or below the graph of h for $x > 4$? Why?

Part D

1 point is earned for the answer with reason

The tangent line is below the graph because the graph of h is concave up for $x > 4$.



0	1
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The student response earns all of the following points:

1 point is earned for the answer with reason

The tangent line is below the graph because the graph of h is concave up for $x > 4$.

4-6-6

3. Let f be the function given by $f(x) = \frac{\ln x}{x}$ for all $x > 0$. The derivative of f is given by $f'(x) = \frac{1 - \ln x}{x^2}$.
- (a) Write an equation for the line tangent to the graph of f at $x = e^2$.
- (b) Find the x -coordinate of the critical point of f . Determine whether this point is a relative minimum, a relative maximum, or neither for the function f . Justify your answer.
- (c) The graph of the function f has exactly one point of inflection. Find the x -coordinate of this point.
- (d) Find $\lim_{x \rightarrow 0^+} f(x)$.

Part A

The response can earn up to 2 points:

1 point is earned for $f(e^2)$ and $f'(e^2)$

$$f(e^2) = \frac{\ln e^2}{e^2} = \frac{2}{e^2}, \quad f'(e^2) = \frac{1 - \ln e^2}{(e^2)^2} = -\frac{1}{e^4}$$

1 point is earned for correct answer

An equation for the tangent line is $y = \frac{2}{e^2} - \frac{1}{e^4}(x - e^2)$.



0	1	2
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The response earns all of the following points:

1 point is earned for $f(e^2)$ and $f'(e^2)$

$$f(e^2) = \frac{\ln e^2}{e^2} = \frac{2}{e^2}, \quad f'(e^2) = \frac{1 - \ln e^2}{(e^2)^2} = -\frac{1}{e^4}$$

1 point is earned for correct answer

An equation for the tangent line is $y = \frac{2}{e^2} - \frac{1}{e^4}(x - e^2)$.

Part B

The response can earn up to 3 points:

1 point is earned for $x = e$

4-6-6

1 point is earned for the relative maximum

1 point is earned for the justification

$f'(x) = 0$ when $x = e$. The function f has a relative maximum at $x = e$ because $f'(x)$ changes from positive to negative at $x = e$.



0	1	2	3
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The response earns all of the following points:

1 point is earned for $x = e$

1 point is earned for the relative maximum

1 point is earned for the justification

$f'(x) = 0$ when $x = e$. The function f has a relative maximum at $x = e$ because $f'(x)$ changes from positive to negative at $x = e$.

Part C

The response can earn up to 3 points:

2 points are earned for $f''(x)$

$$f''(x) = \frac{-\frac{1}{x}x^2 - (1 - \ln x)2x}{x^4} = \frac{-3 + 2 \ln x}{x^3} \text{ for all } x > 0$$

$$f''(x) = 0 \text{ when } -3 + 2 \ln x = 0$$

$$x = e^{3/2}$$

1 point is earned for the correct answer

The graph of f has a point of inflection at $x = e^{3/2}$ because $f''(x)$ changes sign at $x = e^{3/2}$.



0	1	2	3
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The response earns all of the following points:

2 points are earned for $f''(x)$

4-6-6

$$f''(x) = \frac{-\frac{1}{x}x^2 - (1 - \ln x)2x}{x^4} = \frac{-3 + 2 \ln x}{x^3} \text{ for all } x > 0$$

$$f''(x) = 0 \text{ when } -3 + 2 \ln x = 0$$

$$x = e^{3/2}$$

1 point is earned for the correct answer

The graph of f has a point of inflection at $x = e^{3/2}$ because $f''(x)$ changes sign at $x = e^{3/2}$.

Part D

The response can earn up to 1 point:

1 point is earned for the correct answer

$$\lim_{x \rightarrow 0^+} \frac{\ln x}{x} = -\infty \text{ or Does Not Exist}$$



0	1
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The response earns all of the following point:

1 point is earned for the correct answer

$$\lim_{x \rightarrow 0^+} \frac{\ln x}{x} = -\infty \text{ or Does Not Exist}$$

4-6-6

4.

x	-1.5	-1.0	-0.5	0	0.5	1.0	1.5
$f(x)$	-1	-4	-6	-7	-6	-4	-1
$f'(x)$	-7	-5	-3	0	3	5	7

Let f be a function that is differentiable for all real numbers. The table above gives the values of f and its derivative f' for selected points x in the closed interval $-1.5 \leq x \leq 1.5$. The second derivative of f has the property that $f''(x) > 0$ for $-1.5 \leq x \leq 1.5$.

(a) Evaluate $\int_0^{1.5} (3f'(x) + 4)dx$. Show the work that leads to your answer.

(b) Write an equation of the line tangent to the graph of f at the point where $x = 1$. Use this line to approximate the value of $f(1.2)$. Is this approximation greater than or less than the actual value of $f(1.2)$? Give a reason for your answer.

(c) Find a positive real number r having the property that there must exist a value c with $0 < c < 0.5$ and $f'(c) = r$. Give a reason for your answer.

(d) Let g be the function given by $g(x) = \begin{cases} 2x^2 - x - 7 & \text{for } x < 0 \\ 2x^2 + x - 7 & \text{for } x \geq 0. \end{cases}$

The graph of g passes through each of the points $(x, f(x))$ given in the table above. Is it possible that f and g are the same function? Give a reason for your answer.

Part A

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned} \int_0^{1.5} (3f'(x) + 4)dx &= 3 \int_0^{1.5} f'(x)dx + \int_0^{1.5} 4dx \\ &= 3f(x) + 4x \Big|_0^{1.5} = 3(-1 - (-7)) + 4(1.5) = 24 \end{aligned}$$



0	1	2
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The student response earns all of the following points:

1 point is earned for the antiderivative

1 point is earned for the answer

4-6-6

$$\begin{aligned}\int_0^{1.5} (3f'(x) + 4) dx &= 3 \int_0^{1.5} f'(x) dx + \int_0^{1.5} 4 dx \\ &= 3f(x) + 4x \Big|_0^{1.5} = 3(-1 - (-7)) + 4(1.5) = 24\end{aligned}$$

Part B

1 point is earned for the tangent line

1 point is earned for computing y on tangent line at $x = 1.2$

1 point is earned for the answer with reason

$$\begin{aligned}y &= 5(x - 1) - 4 \\ f(1.2) &\approx 5(0.2) - 4 = -3\end{aligned}$$

The approximation is less than $f(1.2)$ because the graph of f is concave up on the interval

$$1 < x < 1.2.$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the tangent line

1 point is earned for computing y on tangent line at $x = 1.2$

1 point is earned for the answer with reason

$$\begin{aligned}y &= 5(x - 1) - 4 \\ f(1.2) &\approx 5(0.2) - 4 = -3\end{aligned}$$

The approximation is less than $f(1.2)$ because the graph of f is concave up on the interval

$$1 < x < 1.2.$$

Part C

1 point is earned for the reference to MVT for f' (or differentiability of f)

4-6-6

1 point is earned for the value of r for interval $0 \leq x \leq 0.5$

By the Mean Value Theorem there is a c with $0 < c < 0.5$ such that

$$f''(c) = \frac{f'(0.5) - f'(0)}{0.5 - 0} = \frac{3 - 0}{0.5} = 6 = r$$



0	1	2
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The student response earns all of the following points:

1 point is earned for the reference to MVT for f' (or differentiability of f)

1 point is earned for the value of r for interval $0 \leq x \leq 0.5$

By the Mean Value Theorem there is a c with $0 < c < 0.5$ such that

$$f''(c) = \frac{f'(0.5) - f'(0)}{0.5 - 0} = \frac{3 - 0}{0.5} = 6 = r$$

Part D

1 point is earned for the answers "no" with reference to g' or g''

1 point is earned for the correct reason

$$\lim_{x \rightarrow 0^-} g'(x) = \lim_{x \rightarrow 0^-} (4x - 1) = -1$$

$$\lim_{x \rightarrow 0^+} g'(x) = \lim_{x \rightarrow 0^+} (4x + 1) = +1$$

Thus g' is not continuous at $x = 0$, but f' is continuous at $x = 0$, so $f \neq g$.

OR

$g''(x) = 4$ for all $x \neq 0$, but it was shown in part (c) that $f''(c) = 6$ for some $c \neq 0$, so $f \neq g$.

4-6-6



0	1	2
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The student response earns all of the following points:

1 point is earned for the answers "no" with reference to g' or g''

1 point is earned for the correct reason

$$\lim_{x \rightarrow 0^-} g'(x) = \lim_{x \rightarrow 0^-} (4x - 1) = -1$$

$$\lim_{x \rightarrow 0^+} g'(x) = \lim_{x \rightarrow 0^+} (4x + 1) = +1$$

Thus g' is not continuous at $x = 0$, but f' is continuous at $x = 0$, so $f \neq g$.

OR

$g''(x) = 4$ for all $x \neq 0$, but it was shown in part (c) that $f''(c) = 6$ for some $c \neq 0$, so $f \neq g$.

4-6-6

5.

x	-1.5	-1.0	-0.5	0	0.5	1.0	1.5
$f(x)$	-1	-4	-6	-7	-6	-4	-1
$f'(x)$	-7	-5	-3	0	3	5	7

Let f be a function that is differentiable for all real numbers. The table above gives the values of f and its derivative f' for selected points x in the closed interval $-1.5 \leq x \leq 1.5$. The second derivative of f has the property that $f''(x) > 0$ for $-1.5 \leq x \leq 1.5$.

Write an equation of the line tangent to the graph of f at the point where $x = 1$. Use this line to approximate the value of $f(1.2)$. Is this approximation greater than or less than the actual value of $f(1.2)$? Give a reason for your answer.

Part B

1 point is earned for the tangent line

1 point is earned for computes y on tangent line at $x = 1.2$

1 point is earned for the answer with reason

$$y = 5(x - 1) - 4$$

$$f(1.2) \approx 5(0.2) - 4 = -3$$

The approximation is less than $f(1.2)$ because the graph of f is concave up on the interval $1 < x < 1.2$.



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the tangent line

1 point is earned for computes y on tangent line at $x = 1.2$

1 point is earned for the answer with reason

$$y = 5(x - 1) - 4$$

$$f(1.2) \approx 5(0.2) - 4 = -3$$

The approximation is less than $f(1.2)$ because the graph of f is concave up on the interval $1 < x < 1.2$.

4-6-6

6. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let f be a twice-differentiable function with $f(0) = 4$. The derivative of f is given by $f'(x) = \sin(x^2 - 2x + 1)$ for $-2 \leq x \leq 2$.

- Find all values of x in the interval $-2 < x < 2$ at which f has a critical point. Classify each as the location of a relative minimum, a relative maximum, or neither. Justify your answers.
- Use the line tangent to the graph of f at $x = 0$ to approximate $f(0.25)$.
- On the interval $0 \leq x \leq 0.25$, $f'(x) > 0$ and $f''(x) < 0$. Is the approximation found in part (b) an overestimate or an underestimate for $f(0.25)$? Give a reason for your answer.
- Using the Mean Value Theorem, explain why the average rate of change of f over the interval $-2 \leq x \leq 2$ cannot equal 1.25.

Part A

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

A maximum of 1 out of 3 points is earned for reference $f'(x) = 0$ to and the three critical points without identification and/or justification, and no incorrect critical points are included.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

4-6-6

- ☐ relative maximum at $x = -1.507$ with justification
- ☐ relative minimum at $x = -0.772$ with justification
- ☐ neither at $x = 1$ with justification

Solution:

$$f'(x) = 0 \Rightarrow x = -1.507 \text{ (or } -1.506), x = -0.772, x = 1$$

f has a relative maximum at $x = -1.507$ (or -1.506) because f' changes from positive to negative there.

f has a relative minimum at $x = -0.772$ because f' changes from negative to positive there.

f has neither a relative minimum nor a relative maximum at $x = 1$ because f' does not change sign there.

Part B

A response of $4 + (0.841471)(0.25)$ or $4 + (\sin(1))(0.25)$ earns both points.

The second point requires substitution of values without a simplified answer. Trigonometric function values do not need to be evaluated.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ form of tangent line approximation
- ☐ answer

Solution:

$$f'(0) = \sin(1) = 0.841471$$

$$f(0.25) \approx f(0) + f'(0)(0.25) = 4.210$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

4-6-6



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ answer
- ☐ reason

Solution:

Because $f''(x) < 0$ for $0 \leq x \leq 0.25$, the graph of f is concave down for $0 \leq x \leq 0.25$.

Therefore, the tangent line approximation from part (b) is an overestimate for $f(0.25)$.

Part D

One point is earned for using MVT to conclude that there should be a number c for $-2 < c < 2$ such that $f'(c)$ equals the average rate of change of f over the interval $-2 \leq x \leq 2$. Two points are earned for showing that this is not possible.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes a correct explanation.

Solution:

f' is differentiable for $-2 \leq x \leq 2$.

$\Rightarrow f$ is continuous for $-2 \leq x \leq 2$.

Because the Mean Value Theorem can be applied to f on the interval $-2 \leq x \leq 2$, there should be a number c for $-2 < c < 2$ such that $f'(c)$ equals the average rate of change of f over the interval $-2 \leq x \leq 2$.

However, since $-1 \leq f'(x) \leq 1$, there can be no number c such that $f'(c) = 1.25$. Therefore, the average rate of change cannot equal 1.25.

4-6-6

7. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let f be a twice-differentiable function with $f(1.5) = 3$. The derivative of f is given by $f'(x) = (7x - 19) \sin(x^2 - 4x + 4)$ for $1 \leq x \leq 4$.

(a) Find all values of x in the interval $1 < x < 4$ at which f has a critical point. Classify each as the location of a relative minimum, a relative maximum, or neither. Justify your answers.

(b) Use the line tangent to the graph of f at $x = 1.5$ to approximate $f(1.8)$.

On the interval $1.5 \leq x \leq 1.8$, $f'(x) < 0$ and $f''(x) > 0$. Is the approximation found in part (b) an overestimate or an underestimate for $f(1.8)$? Give a reason for your answer.

Using the Mean Value Theorem, explain why the average rate of change of f over the interval $1 \leq x \leq 4$ cannot equal 6.5.

Part A

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

A maximum of 1 out of 3 points is earned for reference to $f'(x) = 0$ and the three critical points without identification and/or justification, and no incorrect critical points are included.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

4-6-6

- ☐ neither at $x = 2$ with justification
- ☐ relative minimum at $x = 2.714$ with justification
- ☐ relative maximum at $x = 3.772$ with justification

Solution:

$$f'(x) = 0 \Rightarrow x = 2, x = 2.714, x = 3.772$$

f has neither a relative minimum nor a relative maximum at $x = 2$ because f' does not change sign there.

f has a relative minimum at $x = 2.714$ because f' changes from negative to positive there.

f has a relative maximum at $x = 3.772$ because f' changes from positive to negative there.

Part B

A response of $3 + (-2.102934)(0.3)$ or $3 + (-8.5 \sin(0.25))(0.3)$ earns both points.

The second point requires substitution of values without a simplified answer. Trigonometric function values do not need to be evaluated.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ form of tangent line approximation
- ☐ answer

Solution:

$$f'(1.5) = -8.5 \sin(0.25) = -2.102934$$

$$f(1.8) \approx f(1.5) + f'(1.5)(0.3) = 2.369$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

4-6-6



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ answer
- ☐ reason

Solution:

Because $f''(x) > 0$ for $1.5 \leq x \leq 1.8$, the graph of f is concave up for $1.5 \leq x \leq 1.8$.

Therefore, the tangent line approximation from part (b) is an underestimate for $f(1.8)$.

Part D

The first point is earned for using MVT to conclude that there should be a number c for $1 < c < 4$ such that $f'(c)$ equals the average rate of change of f over the interval $1 \leq x \leq 4$. The second point is earned for showing that this is not possible.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ uses Mean Value Theorem to determine there should be a c such that $f'(c)$ equals the average rate of change of f
- ☐ explanation using Mean Value Theorem

Solution:

f' is differentiable for $1 \leq x \leq 4$. $\Rightarrow f$ is continuous for $1 \leq x \leq 4$.

Because the Mean Value Theorem can be applied to f on the interval $1 \leq x \leq 4$, there should be a number c for $1 < c < 4$ such that $f'(c)$ equals the average rate of change of f over the interval $1 \leq x \leq 4$.

However, because the equation $f'(x) = 6.5$ has no solution on the interval $1 \leq x \leq 4$, there can be no number c such that $f'(c) = 6.5$. Therefore, the average rate of change cannot equal 6.5.

4-6-6

8.  The function g is defined for $x > 0$ with $g(1) = 2$, $g'(x) = \sin(x + 1/x)$, and $g''(x) = (1 - 1/x^2)\cos(x + 1/x)$.
- (a) Find all values of x in the interval $0.12 \leq x \leq 1$ at which the graph of g has a horizontal tangent line.
- (b) On what subintervals of $(0.12, 1)$, if any, is the graph of g concave down? Justify your answer.
- (c) Write an equation for the line tangent to the graph of g at $x = 0.3$.
- (d) Does the line tangent to the graph of g at $x = 0.3$ lie above or below the graph of g for $0.3 < x < 1$? Why?

Part A

1 point is earned for sets $g'(x) = 0$

1 point is earned for the answer

The graph of g has a horizontal tangent line when $g'(x) = 0$. This occurs at $x = 0.163$ and $x = 0.359$.



0	1	2
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The student response earns all of the following points:

1 point is earned for sets $g'(x) = 0$

1 point is earned for the answer

The graph of g has a horizontal tangent line when $g'(x) = 0$. This occurs at $x = 0.163$ and $x = 0.359$.

Part B

1 point is earned for the answer

1 point is earned for the justification

$g''(x) = 0$ at $x = 0.129458$ and $x = 0.222734$

The graph of g is concave down on $(0.1295, 0.2227)$ because $g''(x) < 0$ on this interval.



0	1	2
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The student response earns all of the following points:

1 point is earned for the answer

4-6-6

1 point is earned for the justification

$$g''(x) = 0 \text{ at } x = 0.129458 \text{ and } x = 0.222734$$

The graph of g is concave down on $(0.1295, 0.2227)$ because $g''(x) < 0$ on this interval.

Part C

1 point is earned for $g'(0.3)$

1 point is earned for the integral expression

1 point is earned for $g(0.3)$

1 point is earned for the equation

$$g'(0.3) = -0.472161$$

$$g(0.3) = 2 + \int_1^{0.3} g'(x) dx = 1.546007$$

An equation for the line tangent to the graph of g is $y = 1.546 - 0.472(x - 0.3)$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $g'(0.3)$

1 point is earned for the integral expression

1 point is earned for $g(0.3)$

1 point is earned for the equation

$$g'(0.3) = -0.472161$$

$$g(0.3) = 2 + \int_1^{0.3} g'(x) dx = 1.546007$$

An equation for the line tangent to the graph of g is $y = 1.546 - 0.472(x - 0.3)$.

Part D

4-6-6

1 point is earned for answer with reason

$$g''(x) > 0 \text{ for } 0.3 < x < 1$$

Therefore the line tangent to the graph of g at $x = 0.3$ lies below the graph of g for $0.3 < x < 1$.



0	1
---	---

The student response earns all of the following points:

1 point is earned for answer with reason

$$g''(x) > 0 \text{ for } 0.3 < x < 1$$

Therefore the line tangent to the graph of g at $x = 0.3$ lies below the graph of g for $0.3 < x < 1$.

9. The function g is defined for $x > 0$ with $g(1)=2$, $g'(x)=\sin(x+1/x)$, and $g''(x)=(1-1/x^2)\cos(x+1/x)$.

Does the line tangent to the graph of g at $x=0.3$ lie above or below the graph of g for $0.3 < x < 1$? Why?

Part D

1 point is earned for answer with reason

$$g''(x) > 0 \text{ for } 0.3 < x < 1 \text{ Therefore the line tangent to the graph of } g \text{ at } x=0.3 \text{ lies below the graph of } g \text{ for } 0.3 < x < 1.$$



0	1
---	---

The student response earns one of the following points:

1 point is earned for answer with reason

$$g''(x) > 0 \text{ for } 0.3 < x < 1$$

Therefore the line tangent to the graph of g at $x=0.3$ lies below the graph of g for $0.3 < x < 1$.

4-6-6

10. Consider the curve defined by $-8x^2 + 5xy + y^3 = -149$.

(a) Find $\frac{dy}{dx}$.

(b) Write an equation for the line tangent to the curve at the point $(4, -1)$.

(c) There is a number k so that the point $(4.2, k)$ is on the curve. Using the tangent line found in part (b), approximate the value of k .

(d) Write an equation that can be solved to find the actual value of k so that the point $(4.2, k)$ is on the curve.

(e) Solve the equation found in part (d) for the value of k .

Part A

2 points are earned for the correct implicit differentiation

<-1> each error

$$-16x + 5y + 5x \frac{dy}{dx} + 3y^2 \frac{dy}{dx} = 0$$

1 point is earned for correctly solving for $\frac{dy}{dx}$

Must handle at least two $\frac{dy}{dx}$ terms

$$\frac{dy}{dx} = \frac{16x - 5y}{5x + 3y^2}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the correct implicit differentiation

<-1> each error

$$-16x + 5y + 5x \frac{dy}{dx} + 3y^2 \frac{dy}{dx} = 0$$

1 point is earned for correctly solving for $\frac{dy}{dx}$

4-6-6

Must handle at least two $\frac{dy}{dx}$ terms

$$\frac{dy}{dx} = \frac{16x - 5y}{5x + 3y^2}$$

Part B

1 point is earned for correctly evaluating $\frac{dy}{dx}$ at $(4, -1)$

$$\left. \frac{dy}{dx} \right|_{(4,-1)} = \frac{64+5}{20+3} = 3$$

1 point is earned for the correct equation of tangent line

$$y + 1 = 3(x - 4)$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for correctly evaluating $\frac{dy}{dx}$ at $(4, -1)$

$$\left. \frac{dy}{dx} \right|_{(4,-1)} = \frac{64+5}{20+3} = 3$$

1 point is earned for the correct equation of tangent line

$$y + 1 = 3(x - 4)$$

Part C

1 point is earned for correctly using $x = 4.2$ in tangent line equation

$$y + 1 = 3(4.2 - 4)$$

1 point is earned for the correct solution

$$y = -1$$

$$y = -0.4$$

$$k \approx -0.4$$

4-6-6



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for correctly using $x = 4.2$ in tangent line equation

$$y + 1 = 3(4.2 - 4)$$

1 point is earned for the correct solution

$k = -1$ if " $k =$ "

$$y = -0.4$$

$$k \approx -0.4$$

Part D

1 point is earned for correctly using $x = 4.2$ in equation of curve

$$-8(4.2)^2 + 5(4.2)k + k^3 = -149$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for correctly using $x = 4.2$ in equation of curve

$$-8(4.2)^2 + 5(4.2)k + k^3 = -149$$

Part E

1 point is earned for the correct answer

$$k = -0.373$$



0	1
---	---

4-6-6

The student response earns all of the following points:

1 point is earned for the correct answer

$$k = -0.373$$

11. Consider the curve defined by $-8x^2 + 5xy + y^3 = -149$.

There is a number k so that the point $(4.2, k)$ is on the curve. Using the tangent line found in part (b), approximate the value of k .

Part C

1 point is earned for correctly using $x = 4.2$ in tangent line equation

$$y + 1 = 3(4.2 - 4)$$

1 point is earned for the correct solution

$$< -1 > \text{ if } k =$$

$$y = -0.4$$

$$k \approx -0.4$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for correctly using $x = 4.2$ in tangent line equation

$$y + 1 = 3(4.2 - 4)$$

1 point is earned for the correct solution

$$< -1 > \text{ if } k =$$

$$y = -0.4$$

$$k \approx -0.4$$

4-6-6

12.

x	2	3	5	8	13
$f(x)$	1	4	-2	3	6

Let f be a function that is twice differentiable for all real numbers. The table above gives values of f for selected points in the closed interval $2 \leq x \leq 13$.

Suppose $f(5)=3$ and $f''(x)<0$ for all x in the closed interval $5 \leq x \leq 8$. Use the line tangent to the graph of f at $x=5$ to show that $f(7) \leq 4$. Use the secant line for the graph of f on $5 \leq x \leq 8$ to show that $f(7) \geq 4/3$.

Part D

1 point is earned for the tangent line

1 point is earned for showing $f(7) \leq 4$

1 point is earned for secant line

1 point is earned if shows $f(7) \geq \frac{4}{3}$

An equation for the tangent line is $y = -2 + 3(x - 5)$.

Since $f''(x) < 0$ for all x in the interval $5 \leq x \leq 8$, the line tangent to the graph of $y = f(x)$ at $x = 5$ lies above the graph for all x in the interval $5 < x \leq 8$.

Therefore, $f(7) \leq -2 + 3 \cdot 2 = 4$.

An equation for the secant line is $y = -2 + \frac{5}{3}(x - 5)$.

Since $f''(x) < 0$ for all x in the interval $5 \leq x \leq 8$, the secant line connecting $(5, f(5))$ and $(8, f(8))$ lies below the graph of $y = f(x)$ for all x in the interval $5 < x < 8$.

Therefore, $f(7) \geq -2 + \frac{5}{3} \cdot 2 = \frac{4}{3}$.



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

1 point is earned for the tangent line

1 point is earned for showing $f(7) \leq 4$

1 point is earned for secant line

4-6-6

1 point is earned if shows $f(7) \geq \frac{4}{3}$

An equation for the tangent line is $y = -2 + 3(x - 5)$.

Since $f'(x) < 0$ for all x in the interval $5 \leq x \leq 8$, the line tangent to the graph of $y = f(x)$ at $x = 5$ lies above the graph for all x in the interval $5 < x \leq 8$.

Therefore, $f(7) \leq -2 + 3 \cdot 2 = 4$.

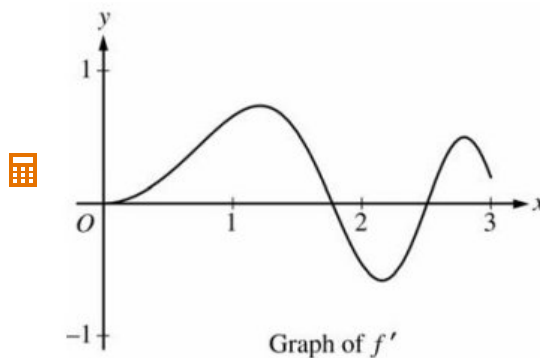
An equation for the secant line is $y = -2 + \frac{5}{3}(x - 5)$.

Since $f'(x) < 0$ for all x in the interval $5 \leq x \leq 8$, the secant line connecting $(5, f(5))$ and $(8, f(8))$ lies below the graph of $y = f(x)$ for all x in the interval $5 < x < 8$.

Therefore, $f(7) \geq -2 + \frac{5}{3} \cdot 2 = \frac{4}{3}$.

4-6-6

13.



Let f be the function defined for $x \geq 0$ with $f(0) = 5$ and f' , the first derivative of f , given by $f'(x) = e^{(-x/4)} \sin(x^2)$. The graph of $y = f'(x)$ is shown above.

(a) Use the graph of f' to determine whether the graph of f is concave up, concave down, or neither on the interval $1.7 < x < 1.9$. Explain your reasoning.

(b) On the interval $0 \leq x \leq 3$, find the value of x at which f has an absolute maximum. Justify your answer.

(c) Write an equation for the line tangent to the graph of f at $x = 2$.

Part A

1 point is earned for the answer

1 point is earned for the reason

On the interval $1.7 < x < 1.9$, f' is decreasing and thus f is concave down on this interval.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the reason

On the interval $1.7 < x < 1.9$, f' is decreasing and thus f is concave down on this interval.

Part B

1 point is earned if identified $\sqrt{\pi}$ and 3 as candidates **or** indicates that the graph of f increases, decreases, then increases

1 point is earned if justified $f(\sqrt{\pi}) > f(3)$

4-6-6

1 point is earned for answer

$$f'(x) = 0 \text{ when } x = 0, \sqrt{\pi}, \sqrt{2\pi}, \sqrt{3\pi}, \dots$$

On $[0, 3]$, f' changes from positive to negative only at $\sqrt{\pi}$. The absolute maximum must occur at $x = \sqrt{\pi}$ or at an endpoint.

$$f(0) = 5$$

$$f(\sqrt{\pi}) = f(0) + \int_0^{\sqrt{\pi}} f'(x) dx = 5.67911$$

$$f(3) = f(0) + \int_0^3 f'(x) dx = 5.57893$$

This shows that f has an absolute maximum at $x = \sqrt{\pi}$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if identified $\sqrt{\pi}$ and 3 as candidates **or** indicates that the graph of f increases, decreases, then increases

1 point is earned if justified $f(\sqrt{\pi}) > f(3)$

1 point is earned for answer

$$f'(x) = 0 \text{ when } x = 0, \sqrt{\pi}, \sqrt{2\pi}, \sqrt{3\pi}, \dots$$

On $[0, 3]$, f' changes from positive to negative only at $\sqrt{\pi}$. The absolute maximum must occur at $x = \sqrt{\pi}$ or at an endpoint.

$$f(0) = 5$$

4-6-6

$$f(\sqrt{\pi}) = f(0) + \int_0^{\sqrt{\pi}} f'(x) dx = 5.67911$$

$$f(3) = f(0) + \int_0^3 f'(x) dx = 5.57893$$

This shows that f has an absolute maximum at $x = \sqrt{\pi}$.

Part C

1 point is earned for $f(2)$ expression with integral

1 point is earned for $f(2)$ expression including $f(0)$ term

1 point is earned for $f'(2)$

1 point is earned for equation

$$f(2) = f(0) + \int_0^2 f'(x) dx = 5.62342$$

$$f'(2) = e^{-0.5} \sin(4) = -0.45902$$

$$y - 5.623 = (-0.459)(x - 2)$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $f(2)$ expression with integral

1 point is earned for $f(2)$ expression including $f(0)$ term

1 point is earned for $f'(2)$

1 point is earned for equation

$$f(2) = f(0) + \int_0^2 f'(x) dx = 5.62342$$

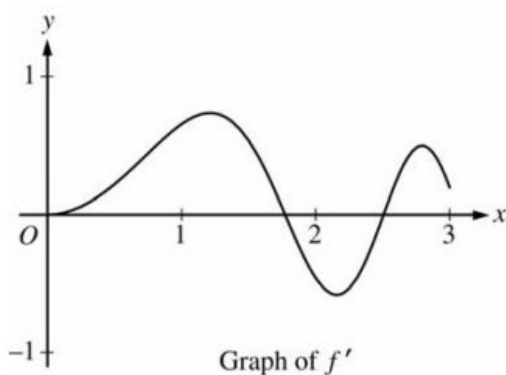
$$f'(2) = e^{-0.5} \sin(4) = -0.45902$$

$$y - 5.623 = (-0.459)(x - 2)$$

4-6-6

4-6-6

14.



Let f be the function defined for $x \geq 0$ with $f(0)=5$ and f' , the first derivative of f , given by $f'(x) = e^{(-x/4)} \sin(x^2)$. The graph of $y = f'(x)$ is shown above.



Write an equation for the line tangent to the graph of f at $x=2$.

Part C

1 point is earned for $f(2)$ expression with integral

1 point is earned for $f(2)$ expression including $f(0)$ term

1 point is earned for $f'(2)$

1 point is earned for equation

$$f(2) = f(0) + \int_0^2 f'(x) dx = 5.62342$$

$$f'(2) = e^{-0.5} \sin(4) = -0.45902$$

$$y - 5.623 = (-0.459)(x - 2)$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $f(2)$ expression with integral

1 point is earned for $f(2)$ expression including $f(0)$ term

1 point is earned for $f'(2)$

1 point is earned for equation

4-6-6

$$f(2)=f(0)+\int_0^2 f'(x)dx = 5.62342$$

$$f'(2)=e-0.5\sin(4) = -0.45902$$

$$y-5.623 = (-0.459)(x-2)$$

15.

x	1	1.1	1.2	1.3	1.4
$f'(x)$	8	10	12	13	14.5

The function f is twice differentiable for $x > 0$ with $f(1) = 15$ and $f''(1) = 20$. Values of f' , the derivative of f , are given for selected values of x in the table above.

Write an equation for the line tangent to the graph of f at $x = 1$. Use this line to approximate $f(1.4)$.

Part A

One point is earned for the tangent line

One point is earned for the approximation

$$f(1) = 15, f'(1) = 8$$

An equation for the tangent line is

$$y = 15 + 8(x - 1)$$

$$f(1.4) \approx 15 + 8(1.4 - 1) = 18.2$$



0	1	2
---	---	---

One point is earned for the tangent line

One point is earned for the approximation

$$f(1) = 15, f'(1) = 8$$

An equation for the tangent line is

$$y = 15 + 8(x - 1)$$

$$f(1.4) \approx 15 + 8(1.4 - 1) = 18.2$$

4-6-6

16. A differentiable function f has the property that $f(5) = 3$ and $f'(5) = 4$. What is the estimate for $f(4.8)$ using the local linear approximation for f at $x = 5$?

(A) 2.2



(B) 2.8

(C) 3.4

(D) 3.8

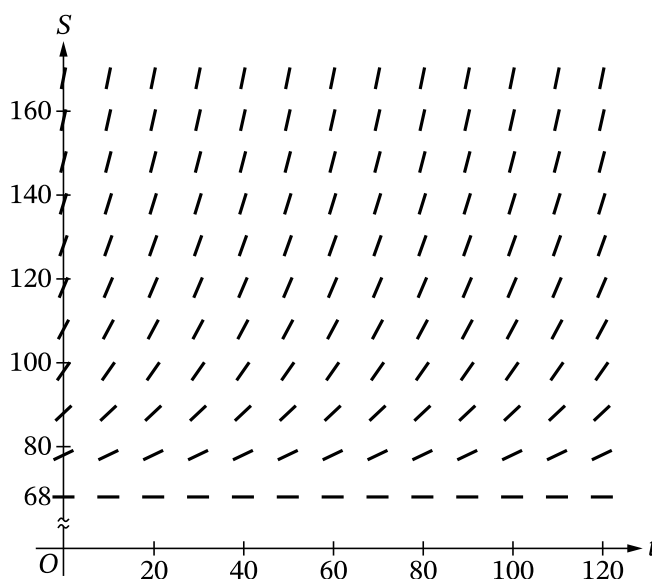
(E) 4.6

4-6-6

17. For $0 \leq t \leq 120$, the temperature of a bowl of soup at time t is modeled by the function S that satisfies the differential equation $\frac{dS}{dt} = -\frac{1}{20}(S - 68)$, where $S(t)$ is measured in degrees Fahrenheit ($^{\circ}\text{F}$) and t is measured in minutes. The temperature of the bowl of soup is always greater than 68°F and is 148°F at time $t = 0$.

Part A

Explain why the following could not be a slope field for the differential equation $\frac{dS}{dt} = -\frac{1}{20}(S - 68)$.

**Part B**

Use the line tangent to the graph of S at $t = 0$ to approximate the temperature of the soup at time $t = 5$ minutes.

Part C

According to the model, is the approximation found in part B an overestimate or an underestimate of the temperature of the soup at time $t = 5$ minutes? Give a reason for your answer.

Part D

Using separation of variables, find the particular solution $y = S(t)$ to the differential equation $\frac{dS}{dt} = -\frac{1}{20}(S - 68)$ with initial condition $S(0) = 148$.

Part A Point 1

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

4-6-6

- **P1** can be earned for a response of “because there are no negative slopes” or “because the slopes are positive.”
- **P1** can be earned by using a local argument at any point (t, S) with $0 \leq t \leq 120$ and $S > 68$, rather than a global argument.

Model Solution:

The slopes of the line segments in a slope field for the given differential equation should all be negative because for $S > 68$, $\frac{dS}{dt} < 0$. However, the slopes of the line segments in the slope field given are all nonnegative.



0	1
---	---

The response accurately includes the criteria below.

- Explanation

Part B Point 2

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response can earn **P2** with $\frac{dS}{dt} \Big|_{(0, 148)} = -4$, “slope is -4 ,” or equivalent.
- A response that presents a linear approximation with a slope of -4 also earns **P2**. For example, a response of $148 - 4(5)$ earns **P2**.
- A response that declares the slope equal to any nonzero value $k \neq -4$ does not earn **P2**.

Model Solution:

$$\frac{dS}{dt} \Big|_{(0, 148)} = -\frac{1}{20}(148 - 68) = -4$$



0	1
---	---

The response accurately includes the criteria below.

- Slope of tangent line

Part B Point 3

Select a point value to view scoring criteria, solutions, and examples or to score the response.

4-6-6

Scoring Notes:

- A response that declares the slope equal to any nonzero value $k \neq -4$:
 - Earns **P3** if there is evidence that the given differential equation was used to calculate k and that k was used correctly in a tangent line approximation of the form $148 + k(5)$. Subsequent errors in simplification will not be considered in scoring for **P3**.
 - Does not earn **P3** if k is simply declared to be a slope without a connection to the differential equation.
- **P3** cannot be earned for an approximation at any value of t other than 5.
- A response does not have to present the tangent line equation but must clearly demonstrate its use at $t = 5$ in finding the requested approximation in order to earn **P3**.
- A response of $148 - 4(5)$ earns **P3**. Subsequent errors in simplification will not be considered in scoring for **P3**.

Model Solution:

$$S(5) \approx 148 - 4(5) = 128$$

The temperature of the soup at $t = 5$ minutes is approximately 128°F .



0	1
---	---

The response accurately includes the criteria below.

- Tangent line approximation

Part C Point 4

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P4**, a response must include one of the following: $\frac{d^2S}{dt^2}$, $-\frac{1}{20}\frac{dS}{dt}$, $\frac{1}{400}(S - 68)$, “the second derivative,” or equivalent.
 - “ S' is increasing (or decreasing)” or a reference to concavity without a connection to the second derivative is not sufficient to earn **P4**.

Model Solution:

Because $S(t) > 68$, $\frac{d^2S}{dt^2} = -\frac{1}{20}\frac{dS}{dt} = \left(-\frac{1}{20}\right)\left(-\frac{1}{20}\right)(S - 68) = \frac{1}{400}(S - 68) > 0$.

4-6-6



0	1
---	---

The response accurately includes the criteria below.

- Considers $\frac{d^2S}{dt^2}$

Part C Point 5

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To be eligible to earn **P5**, a response must have earned **P4**.
- Responses do not need to say “concave up” to earn **P5**. A response of “underestimate because the second derivative is positive” is sufficient to earn **P5**.

Model Solution:

Because the second derivative is always positive, the tangent line approximation is an underestimate of the temperature predicted by the model.



0	1
---	---

The response accurately includes the criteria below.

- Answer with reason

Part D Point 6

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response with no separation of variables does not earn **P6**.
- A response earns **P6** if the separation of variables results in an equation of the form $A \frac{dS}{S-68} = B dt$, where A and B are nonzero constants.
- Special case: If the response begins with a separation of the form $A \frac{dS}{S+68} = B dt$, where A and B are correct coefficients, it earns **P6**.

Model Solution:

4-6-6

$$\frac{dS}{S-68} = -\frac{1}{20}dt$$



0	1
---	---

The response accurately includes the criteria below.

- Separate variables

Part D Point 7

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response with no separation of variables does not earn **P7**.
- A response is eligible to earn **P7** if the separation of variables resulted in an equation of the form $A \frac{dS}{S-68} = B dt$, where A and B are nonzero constants. A response that presents consistent antiderivatives of the form $A \ln(S - 68)$ (with either parentheses or absolute value) and Bt earns **P7**.
- Special case: If the response began with a separation of the form $A \frac{dS}{S+68} = B dt$, where A and B are correct coefficients, it earns **P7** with two consistent antiderivatives.
- A response with no constant of integration is eligible to earn **P7**.

Model Solution:

$$\ln|S - 68| = -\frac{t}{20} + C$$



0	1
---	---

The response accurately includes the criteria below.

- Finds antiderivatives

Part D Point 8

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response with no separation of variables does not earn **P8**.
- A response with no constant of integration does not earn **P8**.

4-6-6

- A response is eligible for **P8** if it has earned P6 and has one of the two antiderivatives correct OR if it has earned **P7**.
- An eligible response earns **P8** by correctly including the constant of integration in an equation and substituting 0 for t and 148 for S .

Model Solution:

$$\ln |148 - 68| = -\frac{0}{20} + C \Rightarrow \ln 80 = C$$

$$\ln (S - 68) = -\frac{t}{20} + \ln 80$$



0	1
---	---

The response accurately includes the criteria below.

- Constant of integration and uses initial condition

Part D Point 9

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response is eligible for **P9** only if it has earned **P6**, **P7**, and **P8**.
- A response earns **P9** only for an answer of $S(t) = 68 + 80e^{-t/20}$ or equivalent.

Model Solution:

$$S - 68 = e^{-t/20 + \ln 80} = 80e^{-t/20}$$

$$S(t) = 68 + 80e^{-t/20}$$



0	1
---	---

The response accurately includes the criteria below.

- Solves for $S(t)$

4-6-6

18. The following are related to this scenario:

Let f be a function with $f(2) = -8$ such that for all points (x, y) on the graph of f , the slope given by $\frac{3x^2}{y}$.

Write an equation of the line tangent to the graph of f at the point where $x = 2$ and use it to approximate $f(1.8)$.

Part A

The response can earn up to 3 points:

1 point: For correct slope

$$\text{Slope } \frac{(3)(4)}{-8} = -\frac{3}{2}$$

1 point: For the tangent line equation

$$\text{An equation for the tangent line is } y = -\frac{3}{2}(x - 2) - 8$$

1 point: For the correct approximation

$$f(1.8) \approx -\frac{3}{2}(1.8 - 2) - 8 = -7.7$$



0	1	2	3
---	---	---	---

The response earns all three of the following points:

1 point: For correct slope

$$\text{Slope } \frac{(3)(4)}{-8} = -\frac{3}{2}$$

1 point: For the tangent line equation

$$\text{An equation for the tangent line is } y = -\frac{3}{2}(x - 2) - 8$$

1 point: For the correct approximation

$$f(1.8) \approx -\frac{3}{2}(1.8 - 2) - 8 = -7.7$$

4-6-6

19. Let f be a function with $f(1) = 4$ such that for all points (x, y) on the graph of f the slope is given by $3x^2 + 1/2y$.

Write an equation for the line tangent to the graph of f at $x = 1$ and use it to approximate $f(1.2)$.

Part B

1 point is earned for the equation of tangent line

1 point is earned if uses equation to approximate $f(1.2)$

$$y - 4 = \frac{1}{2}(x - 1)$$

$$f(1.2) - 4 \approx \frac{1}{2}(1.2 - 1)$$

$$f(1.2) \approx 0.1 + 4 = 4.1$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for the equation of tangent line

1 point is earned if uses equation to approximate $f(1.2)$

$$y - 4 = \frac{1}{2}(x - 1)$$

$$f(1.2) - 4 \approx \frac{1}{2}(1.2 - 1)$$

$$f(1.2) \approx 0.1 + 4 = 4.1$$

4-6-6

20.

t (days)	0	10	22	30
$W'(t)$ (GL per day)	0.6	0.7	1.0	0.5

The twice-differentiable function W models the volume of water in a reservoir at time t , where $W(t)$ is measured in gigaleters (GL) and t is measured in days. The table above gives values of $W'(t)$ sampled at various times during the time interval $0 \leq t \leq 30$ days. At time $t=30$, the reservoir contains 125 gigaleters of water.

Use the tangent line approximation to W at time $t = 30$ to predict the volume of water $W(t)$, in gigaleters, in the reservoir at time $t = 32$. Show the computations that lead to your answer.

Part A

One point is earned for the answer

An equation of the tangent line is $y = 0.5(t - 30) + 125$.

$$W(32) \approx 0.5(32 - 30) + 125 = 126$$



0	1
---	---

The student earns all of the following points:

One point is earned for the answer

An equation of the tangent line is $y = 0.5(t - 30) + 125$.

$$W(32) \approx 0.5(32 - 30) + 125 = 126$$

4-6-6

21.

t (days)	0	10	22	30
$W'(t)$ (GL per day)	0.6	0.7	1.0	0.5

The twice-differentiable function W models the volume of water in a reservoir at time t , where $W(t)$ is measured in (GL) and t is measured in days. The table above gives values of $W'(t)$ sampled at various times during the time interval $0 \leq t \leq 30$ days. At time $t = 30$, the reservoir contains 125 gigaleters of water.

Use the tangent line approximation to W at time $t = 30$ to predict the volume of water $W(t)$, in gigaleters, in the reservoir at time $t = 32$. Show the computations that lead to your answer.

Part A

The response can earn up to 1 point:

1 point: For the correct answer

An equation of the tangent line is $y = 0.5(t - 30) + 125$.

$$W(32) \approx 0.5(32-30) + 125 = 126$$



0	1
---	---

The response earns the following point:

1 point: For the correct answer

An equation of the tangent line is $y = 0.5(t - 30) + 125$.

$$W(32) \approx 0.5(32-30) + 125 = 126$$

22. The function f is twice differentiable with $f(2) = 1$, $f'(2) = 4$, and $f''(2) = 3$. What is the value of the approximation of $f(1.9)$ using the line tangent to the graph of f at $x = 2$?

(A) 0.4

(B) 0.6

(C) 0.7

(D) 1.3

(E) 1.4



4-6-6

23. For the function f , $f'(x) = 2x + 1$ and $f(1) = 4$. What is the approximation for $f(1.2)$ found by using the line tangent to the graph of f at $x = 1$?
- (A) 0.6
(B) 3.4
(C) 4.2
(D) 4.6 ✓
(E) 4.64
24. Let f be the function defined by $f(x) = \sqrt[3]{x}$. What is the approximation for $f(10)$ found by using the line tangent to the graph of f at the point $(8, 2)$?
- (A) $11/6$
(B) $25/12$
(C) $13/6$ ✓
(D) $7/3$
25. Let f be the function given by $f(x) = 2 \cos x + 1$. What is the approximation for $f(1.5)$ found by using the line tangent to the graph of f at $x = \pi/2$?
- (A) -2
(B) 1
(C) $\pi - 2$ ✓
(D) $4 - \pi$

4-6-6

26. Let f be the function defined by $f(x) = (1 + \tan x)^{3/2}$ for $-\pi/4 < x < \pi/2$.

Using the equation found in part (a), approximate $f(0.02)$.

Part B

1 point is earned for evaluating their equation from part (a) at $x = 0.02$

$$f(0.02) \approx \frac{3}{2}(0.02) + 1$$



0	1
---	---

The student response earns one of the following points:

1 point is earned for evaluating their equation from part (a) at $x = 0.02$

$$f(0.02) \approx \frac{3}{2}(0.02) + 1$$

4-6-6

27. Solutions to the differential equation $dy/dx = xy^3$ also satisfy $d^2y/dx^2 = y^3(1 + 3x^2y^2)$. Let $y = f(x)$ be a particular solution to the differential equation $dy/dx = xy^3$ with $f(1) = 2$.

Use the tangent line equation from part (a) to approximate $f(1.1)$. Given that $f(x) > 0$ for $1 < x < 1.1$, is the approximation for $f(1.1)$ greater than or less than $f(1.1)$? Explain your reasoning.

Part B

1 point is earned for approximation

1 point is earned for conclusion with explanation

$$f(1.1) \approx 2.8$$

Since $y = f(x) > 0$ on the interval $1 \leq x < 1.1$,

$$\frac{d^2y}{dx^2} = y^3(1 + 3x^2y^2) > 0 \text{ on this interval.}$$

Therefore on the interval $1 < x < 1.1$, the line tangent to the graph of $y = f(x)$ at $x = 1$ lies below the curve and the approximation 2.8 is less than $f(1.1)$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for approximation

1 point is earned for conclusion with explanation

$$f(1.1) \approx 2.8$$

Since $y = f(x) > 0$ on the interval $1 \leq x < 1.1$,

$$\frac{d^2y}{dx^2} = y^3(1 + 3x^2y^2) > 0 \text{ on this interval.}$$

Therefore on the interval $1 < x < 1.1$, the line tangent to the graph of $y = f(x)$ at $x = 1$ lies below the curve and the approximation 2.8 is less than $f(1.1)$

4-6-6

28. At the beginning of 2010, a landfill contained 1400 tons of solid waste. The increasing function W models the total amount of solid waste stored at the landfill. Planners estimate that W will satisfy the differential equation $\frac{dW}{dt} = \frac{1}{25}(W - 300)$ for the next 20 years. W is measured in tons, and t is measured in years from the start of 2010.

Find d^2W/dt^2 in terms of W . Use d^2W/dt^2 to determine whether your answer in part (a) is an underestimate or an overestimate of the amount of solid waste that the landfill contains at time $t = 1/4$.

Part B

1 point is earned for $\frac{d^2W}{dt^2}$

1 point is earned for the answer with reason

$$\frac{d^2W}{dt^2} = \frac{1}{25} \frac{dW}{dt} = \frac{1}{625}(W - 300) \text{ and } W \geq 1400$$

Therefore $\frac{d^2W}{dt^2} > 0$ on the interval $0 \leq t < \frac{1}{4}$.

The answer in part(a) is an underestimate.



0	1	2
---	---	---

The student response earns none of the following points:

1 point is earned for $\frac{d^2W}{dt^2}$

1 point is earned for the answer with reason

$$\frac{d^2W}{dt^2} = \frac{1}{25} \frac{dW}{dt} = \frac{1}{625}(W - 300) \text{ and } W \geq 1400$$

Therefore $\frac{d^2W}{dt^2} > 0$ on the interval $0 \leq t < \frac{1}{4}$.

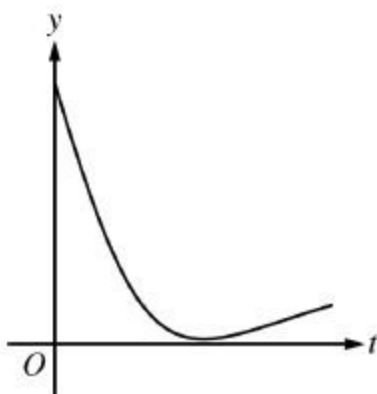
The answer in part(a) is an underestimate.

4-6-6

29. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

During a chemical reaction, the function $y = f(t)$ models the amount of a substance present, in grams, at time t seconds. At the start of the reaction ($t = 0$), there are 10 grams of the substance present. The function $y = f(t)$ satisfies the differential equation $\frac{dy}{dt} = -0.02y^2$.

- (a) Use the line tangent to the graph of $y = f(t)$ at $t = 0$ to approximate the amount of the substance remaining at time $t = 2$ seconds.
- (b) Using the given differential equation, determine whether the graph of f could resemble the following graph. Give a reason for your answer.



- (c) Find an expression for $y = f(t)$ by solving the differential equation $\frac{dy}{dt} = -0.02y^2$ with the initial condition $f(0) = 10$.
- (d) Determine whether the amount of the substance is changing at an increasing or a decreasing rate. Explain your reasoning.

Part A**1 point is earned for: $y'(0)$** **1 point is earned for: approximation**

$$y'(0) = -0.02(10^2) = -2$$

An equation for the line tangent to the graph of $y = f(t)$ at $t = 0$ is $y = 10 - 2t$.

$$y(2) \approx 10 - 2(2) = 6 \text{ grams}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: $y'(0)$

4-6-6**1 point is earned for:** approximation

$$y'(0) = -0.02(10^2) = -2$$

An equation for the line tangent to the graph of $y = f(t)$ at $t = 0$ is $y = 10 - 2t$.

$$y(2) \approx 10 - 2(2) = 6 \text{ grams}$$

Part B**1 point is earned for:** answer with reason

$$\frac{dy}{dt} = -0.02y^2 \leq 0, \text{ so the graph of } f \text{ is nonincreasing.}$$

The graph of f cannot resemble the given graph because the given graph is increasing on a portion of its domain.



0	1
---	---

The student response earns all of the following points:

1 point is earned for: answer with reason

$$\frac{dy}{dt} = -0.02y^2 \leq 0, \text{ so the graph of } f \text{ is nonincreasing.}$$

The graph of f cannot resemble the given graph because the given graph is increasing on a portion of its domain.

Part C**1 point is earned for:** separation of variables**1 point is earned for:** antiderivatives**1 point is earned for:** constant of integration and uses initial condition**1 point is earned for:** answer

$$\int \left(-\frac{1}{y^2} \right) dy = \int 0.02 dt$$

$$\frac{1}{y} = 0.02t + C$$

$$\frac{1}{10} = 0.02(0) + C \Rightarrow C = 0.1$$

$$\frac{1}{y} = 0.02t + 0.1 \Rightarrow y = \frac{1}{0.02t + 0.1} = \frac{50}{t + 5}$$

4-6-6

Note: this solution is valid for $t > -5$.

Note: max 2/4 [1-1-0-0] if no constant of integration

Note: 0/4 if no separation of variables



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for: separation of variables

1 point is earned for: antiderivatives

1 point is earned for: constant of integration and uses initial condition

1 point is earned for: answer

$$\int \left(-\frac{1}{y^2} \right) dy = \int 0.02 dt$$

$$\frac{1}{y} = 0.02t + C$$

$$\frac{1}{10} = 0.02(0) + C \Rightarrow C = 0.1$$

$$\frac{1}{y} = 0.02t + 0.1 \Rightarrow y = \frac{1}{0.02t+0.1} = \frac{50}{t+5}$$

Note: this solution is valid for $t > -5$.

Note: max 2/4 [1-1-0-0] if no constant of integration

Note: 0/4 if no separation of variables

Part D

1 point is earned for: $\frac{d^2y}{dt^2}$

1 point is earned for: answer with reason

$$\begin{aligned} \frac{d^2y}{dt^2} &= -0.04y \frac{dy}{dt} \\ &= -0.04y(-0.02y^2) \\ &= 0.0008y^3 \end{aligned}$$

Because $y > 0$, $0.0008y^3 > 0$.

The amount of substance is changing at an increasing rate.

4-6-6

— OR —

From part (c), $f(t) = \frac{50}{t+5}$, and from context, $t \geq 0$.

$$f'(t) = \frac{-50}{(t+5)^2} \text{ and } f''(t) = \frac{100}{(t+5)^3} > 0 \text{ for } t \geq 0.$$

The amount of substance is changing at an increasing rate.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: $\frac{d^2y}{dt^2}$

1 point is earned for: answer with reason

$$\begin{aligned} \frac{d^2y}{dt^2} &= -0.04y \frac{dy}{dt} \\ &= -0.04y(-0.02y^2) \\ &= 0.0008y^3 \end{aligned}$$

Because $y > 0$, $0.0008y^3 > 0$.

The amount of substance is changing at an increasing rate.

— OR —

From part (c), $f(t) = \frac{50}{t+5}$, and from context, $t \geq 0$.

$$f'(t) = \frac{-50}{(t+5)^2} \text{ and } f''(t) = \frac{100}{(t+5)^3} > 0 \text{ for } t \geq 0.$$

The amount of substance is changing at an increasing rate.

4-6-6

30.

t (minutes)	0	2	5	7	11	12
$r'(t)$ (feet per minute)	5.7	4.0	2.0	1.2	0.6	0.5

The volume of a spherical hot air balloon expands as the air inside the balloon is heated. The radius of the balloon, in feet, is modeled by a twice-differentiable function r of time t , where t is measured in minutes. For $0 < t < 12$, the graph of r is concave down. The table above gives selected values of the rate of change, $r'(t)$, of radius of the balloon over the time interval $0 \leq t \leq 12$. The radius of the balloon is 30 feet when $t = 5$.

(Note: The volume of a sphere of radius r is given by $V = \frac{4}{3}\pi r^3$.)

Estimate the radius of the balloon when $t = 5.4$ using the tangent line approximation at $t = 5$. Is your estimate greater than or less than the true value? Give a reason for your answer.

Part A

1 point is earned for the estimate

1 point is earned for the conclusion with reason

$$r(5.4) \approx r(5) + r'(5)\Delta t = 30 + 2(0.4) = 30.8 \text{ ft}$$

Since the graph of r is concave down on the interval $5 < t < 5.4$, this estimate is greater than $r(5.4)$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the estimate

1 point is earned for the conclusion with reason

$$r(5.4) \approx r(5) + r'(5)\Delta t = 30 + 2(0.4) = 30.8 \text{ ft}$$

Since the graph of r is concave down on the interval $5 < t < 5.4$, this estimate is greater than $r(5.4)$.

4-6-6

31. Let f be a twice-differentiable function such that $f''(x) < 0$ for all x . The graph of $y = S(x)$ is the secant line passing through the points $(3, f(3))$ and $(5, f(5))$. The graph of $y = T(x)$ is the line tangent to the graph of f at $x = 4$.

Which of the following is true?

- (A) $f(4.2) < S(4.2) < T(4.2)$
(B) $f(4.2) < T(4.2) < S(4.2)$
(C) $S(4.2) < f(4.2) < T(4.2)$ ✓
(D) $T(4.2) < f(4.2) < S(4.2)$

32. Let f be a differentiable function such that $f(3)=2$ and $f'(3)=5$. If the tangent line to the graph of f at $x=3$ is used to find an approximation to a zero of f , that approximation is

- (A) 0.4
(B) 0.5
(C) 2.6 ✓
(D) 3.4
(E) 5.5

33. Let f be a differentiable function such that $f(2) = 4$ and $f'(2) = -1/2$. What is the approximation for $f(2.1)$ found by using the line tangent to the graph of f at $x = 2$?

- (A) 2.95
(B) 3.95 ✓
(C) 4.05
(D) 4.1

Answer B

This option is correct. An equation of the tangent line at $x = a$ is $y = f(a) + f'(a)(x - a)$. In this question $a = 2$ and $f'(a) = -1/2$. The value of y when $x = 2.1$ would be an approximation to $f(2.1)$.

$$f(2.1) \approx f(2) - \frac{1}{2}(2.1 - 2) = 4 - \frac{1}{2}(0.1) = 4 - 0.05 = 3.95$$

4-6-6

34. Let f be a function with $f(2) = -8$ such that for all points (x, y) on the graph of f , the slope given by $\frac{3x^2}{y}$.
- (a) Write an equation of the line tangent to the graph of f at the point where $x = 2$ and use it to approximate $f(1.8)$.
- (b) Find an expression for $y = f(x)$ by solving the differential equation $\frac{dy}{dx} = \frac{3x^2}{y}$ with initial condition $f(2) = -8$.

Part A

The response can earn up to 3 points:

1 point earned for correct slope

$$\text{Slope} = \frac{(3)(4)}{-8} = -\frac{3}{2}$$

1 point earned for the tangent line equation

$$\text{An equation for the tangent line is } y = -\frac{3}{2}(x - 2) - 8.$$

1 point earned for the correct approximation

$$f(1.8) \approx -\frac{3}{2}(1.8 - 2) - 8 = -7.7$$



0	1	2	3
---	---	---	---

The response earns all of the following points:

1 point earned for correct slope

$$\text{Slope} = \frac{(3)(4)}{-8} = -\frac{3}{2}$$

1 point earned for the tangent line equation

$$\text{An equation for the tangent line is } y = -\frac{3}{2}(x - 2) - 8.$$

1 point earned for the correct approximation

$$f(1.8) \approx -\frac{3}{2}(1.8 - 2) - 8 = -7.7$$

4-6-6

Part B

The response can earn up to 6 points:

1 point earned for separation of variables

2 point earned for the antiderivatives

1 point earned for the constant of integration

1 point earned for use of the initial condition

1 point earned for the solution of y

Note: max [1-2-0-0-0] if no constant of integration

Note: 0/6 if no separation of variables

$$\int y \, dy = \int 3x^2 \, dx$$

$$\frac{1}{2}y^2 = x^3 + C$$

$$\frac{1}{2}(-8)^2 = 2^3 + C \Rightarrow C = 24$$

$$y^2 = 2(x^3 + 24) = 2x^3 + 48$$

$$y = -\sqrt{2x^3 + 48}$$

Note: This solution is valid for $x > -\sqrt[3]{24}$.



0	1	2	3	4	5	6
---	---	---	---	---	---	---

The response earns all of the following points:

1 point earned for separation of variables

2 point earned for the antiderivatives

1 point earned for the constant of integration

1 point earned for use of the initial condition

1 point earned for the solution of y

Note: max [1-2-0-0-0] if no constant of integration

Note: 0/6 if no separation of variables

4-6-6

$$\int y \, dy = \int 3x^2 \, dx$$

$$\frac{1}{2}y^2 = x^3 + C$$

$$\frac{1}{2}(-8)^2 = 2^3 + C \Rightarrow C = 24$$

$$y^2 = 2(x^3 + 24) = 2x^3 + 48$$

$$y = -\sqrt{2x^3 + 48}$$

Note: This solution is valid for $x > -\sqrt[3]{24}$.

4-6-6

35. Solutions to the differential equation $\frac{dy}{dx} = xy^3$ also satisfy $\frac{d^2y}{dx^2} = y^3(1 + 3x^2y^2)$. Let $y = f(x)$ be a particular solution to the differential equation $\frac{dy}{dx} = xy^3$ with $f(1) = 2$.

- (a) Write an equation for the line tangent to the graph of $y = f(x)$ at $x = 1$.
- (b) Use the tangent line equation from part (a) to approximate $f(1.1)$. Given that $f(x) > 0$ for $1 < x < 1.1$, is the approximation for $f(1.1)$ greater than or less than $f(1.1)$? Explain your reasoning.
- (c) Find the particular solution $y = f(x)$ with initial condition $f(1) = 2$.

Part A

1 point is earned for $f'(1)$

1 point is earned for the answer

$$f'(1) = \left. \frac{dy}{dx} \right|_{(1,2)} = 8$$

An equation of the tangent line is $y = 2 + 8(x - 1)$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $f'(1)$

1 point is earned for the answer

$$f'(1) = \left. \frac{dy}{dx} \right|_{(1,2)} = 8$$

An equation of the tangent line is $y = 2 + 8(x - 1)$.

Part B

1 point is earned for approximation

1 point is earned for conclusion with explanation

$$f(1.1) \approx 2.8$$

Since $y = f(x) > 0$ on the interval $1 \leq x < 1.1$,

4-6-6

$$\frac{d^2y}{dx^2} = y^3 (1 + 3x^2y^2) > 0 \text{ on this interval.}$$

Therefore on the interval $1 < x < 1.1$, the line tangent to the graph of $y = f(x)$ at $x = 1$ lies below the curve and the approximation 2.8 is less than $f(1.1)$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for approximation

1 point is earned for conclusion with explanation

$$f(1.1) \approx 2.8$$

Since $y = f(x) > 0$ on the interval $1 \leq x < 1.1$,

$$\frac{d^2y}{dx^2} = y^3 (1 + 3x^2y^2) > 0 \text{ on this interval.}$$

Therefore on the interval $1 < x < 1.1$, the line tangent to the graph of $y = f(x)$ at $x = 1$ lies below the curve and the approximation 2.8 is less than $f(1.1)$

Part C

1 point is earned for separation of variables

1 point is earned for antiderivatives

1 point is earned for constant of integration

1 point is earned for using initial condition

1 point is earned for solving for y

Note: max 2/5 [1-1-0-0-0] if no constant of integration

Note: 0/5 if no separation of variables

4-6-6

$$\frac{dy}{dx} = xy^3$$

$$\int \frac{1}{y^3} dy = \int x dx$$

$$-\frac{1}{2y^2} = \frac{x^2}{2} + C$$

$$-\frac{1}{2 \cdot 2^2} = \frac{1^2}{2} + C \Rightarrow C = -\frac{5}{8}$$

$$y^2 = \frac{1}{\frac{5}{4} - x^2}$$

$$f(x) = \frac{2}{\sqrt{5-4x^2}}, \frac{-\sqrt{5}}{2} < x < \frac{\sqrt{5}}{2}$$



0	1	2	3	4	5
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The student response earns all of the following points:

1 point is earned for separation of variables

1 point is earned for antiderivatives

1 point is earned for constant of integration

1 point is earned for using initial condition

1 point is earned for solving for y

Note: max 2/5 [1-1-0-0-0] if no constant of integration

Note: 0/5 if no separation of variables

$$\frac{dy}{dx} = xy^3$$

$$\int \frac{1}{y^3} dy = \int x dx$$

$$-\frac{1}{2y^2} = \frac{x^2}{2} + C$$

$$-\frac{1}{2 \cdot 2^2} = \frac{1^2}{2} + C \Rightarrow C = -\frac{5}{8}$$

$$y^2 = \frac{1}{\frac{5}{4} - x^2}$$

$$f(x) = \frac{2}{\sqrt{5-4x^2}}, \frac{-\sqrt{5}}{2} < x < \frac{\sqrt{5}}{2}$$

4-6-6

36. Let f be a function with $f(1) = 4$ such that for all points (x, y) on the graph of f the slope is given by $3x^2 + 1 / 2y$.
- Find the slope of the graph of f at the point where $x = 1$.
 - Write an equation for the line tangent to the graph of f at $x = 1$ and use it to approximate $f(1.2)$.
 - Find $f(x)$ by solving the separable differential equation $dy/dx = 3x^2 + 1 / 2y$ with the initial condition $f(1) = 4$.
 - Use your solution from part (c) to find $f(1.2)$.

Part A

1 point is earned for the answer

$$\frac{dy}{dx} = \frac{3x^2 + 1}{2y}$$

$$\left. \frac{dy}{dx} \right|_{\substack{x=1 \\ y=4}} = \frac{3 + 1}{2 \cdot 4} = \frac{4}{8} = \frac{1}{2}$$



0

1

The student response earns all of the following points:

1 point is earned for the answer

$$\frac{dy}{dx} = \frac{3x^2 + 1}{2y}$$

$$\left. \frac{dy}{dx} \right|_{\substack{x=1 \\ y=4}} = \frac{3 + 1}{2 \cdot 4} = \frac{4}{8} = \frac{1}{2}$$

Part B

1 point is earned for the equation of tangent line

1 point is earned if uses equation to approximate $f(1.2)$

$$y - 4 = \frac{1}{2}(x - 1)$$

4-6-6

$$f(1.2) - 4 \approx \frac{1}{2}(1.2 - 1)$$

$$f(1.2) \approx 0.1 + 4 = 4.1$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the equation of tangent line

1 point is earned if uses equation to approximate $f(1.2)$

$$y - 4 = \frac{1}{2}(x - 1)$$

$$f(1.2) - 4 \approx \frac{1}{2}(1.2 - 1)$$

$$f(1.2) \approx 0.1 + 4 = 4.1$$

Part C

1 point is earned if separates variables

1 point is earned for antiderivative of dy term

1 point is earned for antiderivative of dx term

1 point is earned if uses $y = 4$ when $x = 1$ to pick one function out of a family of functions

1 point is earned if solves for y

$$2y \, dy = (3x^2 + 1) \, dx$$

$$\int 2y \, dy = \int (3x^2 + 1) \, dx$$

$$y^2 = x^3 + x + C$$

$$4^2 = 1 + 1 + C$$

$$14 = C$$

4-6-6

$$y^2 = x^3 + x + 14$$

$y = \sqrt{x^3 + x + 14}$ is branch with point (1, 4)

$$f(x) = \sqrt{x^3 + x + 14}$$



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned if separates variables

1 point is earned for antiderivative of dy term

1 point is earned for antiderivative of dx term

1 point is earned if uses $y = 4$ when $x = 1$ to pick one function out of a family of functions

1 point is earned if solves for y

$$2y \, dy = (3x^2 + 1) \, dx$$

$$\int 2y \, dy = \int (3x^2 + 1) \, dx$$

$$y^2 = x^3 + x + C$$

$$4^2 = 1 + 1 + C$$

$$14 = C$$

$$y^2 = x^3 + x + 14$$

$y = \sqrt{x^3 + x + 14}$ is branch with point (1, 4)

$$f(x) = \sqrt{x^3 + x + 14}$$

Part D

4-6-6

1 point is earned if answer, from student's solution to the given differential equation in (c)

$$f(1.2) = \sqrt{1.2^3 + 1.2 + 14} \approx 4.114$$



0	1
---	---

The student response earns all of the following points:

1 point is earned if answer, from student's solution to the given differential equation in (c)

$$f(1.2) = \sqrt{1.2^3 + 1.2 + 14} \approx 4.114$$

4-6-6

37. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

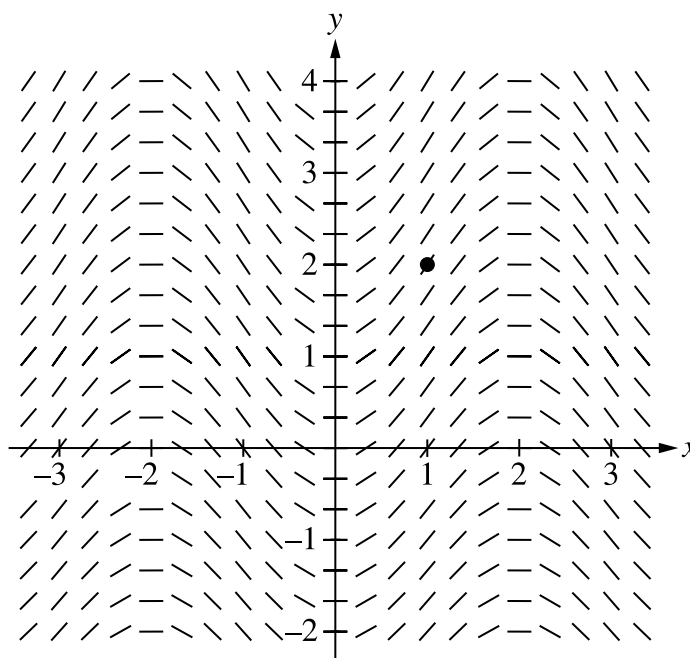
Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the differential equation $\frac{dy}{dx} = \frac{1}{2} \sin\left(\frac{\pi}{2}x\right) \sqrt{y+7}$. Let $y = f(x)$ be the particular solution to the differential equation with the initial condition $f(1) = 2$. The function f is defined for all real numbers.

- (a) A portion of the slope field for the differential equation is given. Sketch the solution curve through the point $(1, 2)$.



- (b) Write an equation for the line tangent to the solution curve in part (a) at the point $(1, 2)$. Use the equation to approximate $f(0.8)$.
- (c) It is known that $f''(x) > 0$ for $-1 \leq x \leq 1$. Is the approximation found in part (b) an overestimate or an underestimate for $f(0.8)$? Give a reason for your answer.
- (d) Use separation of variables to find $y = f(x)$, the particular solution to the differential equation $\frac{dy}{dx} = \frac{1}{2} \sin\left(\frac{\pi}{2}x\right) \sqrt{y+7}$ with the initial condition $f(1) = 2$.

Part A

4-6-6

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The solution curve must pass through the point $(1, 2)$, extend reasonably close to the left and right edges of the square and have no obvious conflicts with the given slope lines.

Only portions of the solution curve within the given slope field are considered.

The solution curve must indicate $f'(x) > 0$ for all points on the curve.

All local maximum/minimum points on the solution curve must occur at horizontal line segments in the slope field.



0

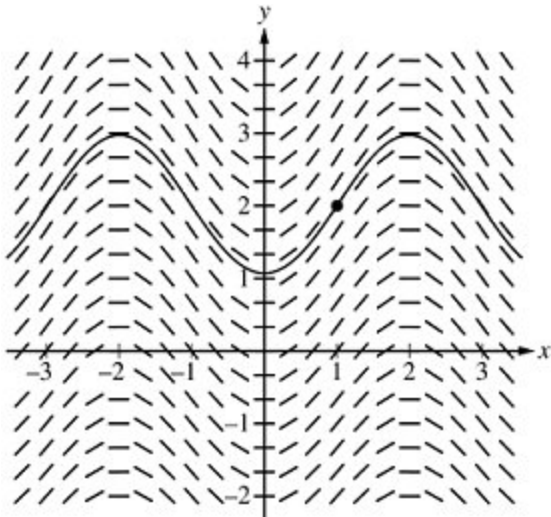
1

The student response accurately includes the criteria below.

☐

Solution curve

Solution:



Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The tangent line equation can be presented in any equivalent form.

An incorrect tangent line equation with a slope of $\frac{3}{2}$ is eligible to earn the second point for a consistent answer.

A response of only $2 + \frac{3}{2}(0.8 - 1)$ earns the second point but not the first.

4-6-6



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Tangent line equation
- ☐ Approximation

Solution:

$$\left. \frac{dy}{dx} \right|_{(x,y)=(1,2)} = \frac{1}{2} \cdot 3 \cdot \sin\left(\frac{\pi}{2}\right) = \frac{3}{2}$$

An equation for the tangent line is $y = 2 + \frac{3}{2}(x - 1)$.

$$f(0.8) \approx 2 + \frac{3}{2}(0.8 - 1) = 1.7$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The reason must include $f''(x) > 0$, $f'(x)$ is increasing, or $f(x)$ is concave up.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer with reason

Solution:

Because $f''(x) > 0$, f is concave up on $-1 \leq x \leq 1$, the tangent line lies below the graph of $y = f(x)$ at $x = 0.8$, and the approximation for $f(0.8)$ is an underestimate.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with no separation of variables earns 0 out of 5 points.

4-6-6

A response with no constant of integration can earn at most the first 3 points.

A response is eligible for the fourth point only if it has earned the first point and at least 1 of the 2 antiderivative points.

· Special case: The incorrect separation of $\sqrt{y+7} \, dy = \frac{1}{2} \sin\left(\frac{\pi}{2}x\right) \, dx$ does not earn the first point, is only eligible for the antiderivative point for $-\frac{1}{\pi} \cos\left(\frac{\pi}{2}x\right)$, and is eligible for the fourth point.

An eligible response earns the fourth point by correctly including the constant of integration in an equation and substituting 1 for x and 2 for y .

A response is eligible for the fifth point only if it has earned the first 4 points.



0	1	2	3	4	5
---	---	---	---	---	---

The student response accurately includes **all five** of the criteria below.

- ☐ Separation of variables
- ☐ One correct antiderivative
- ☐ The other correct antiderivative
- ☐ Constant of integration and uses initial condition
- ☐ Solves for y

Solution:

$$\int \frac{dy}{\sqrt{y+7}} = \int \frac{1}{2} \sin\left(\frac{\pi}{2}x\right) dx$$

$$2\sqrt{y+7} = -\frac{1}{\pi} \cos\left(\frac{\pi}{2}x\right) + C$$

$$f(1) = 2 \Rightarrow 2\sqrt{2+7} = -\frac{1}{\pi} \cos\left(\frac{\pi}{2} \cdot 1\right) + C$$

$$\Rightarrow 6 = -\frac{1}{\pi} \cos\left(\frac{\pi}{2}\right) + C \Rightarrow C = 6$$

$$\sqrt{y+7} = 3 - \frac{1}{2\pi} \cos\left(\frac{\pi}{2}x\right)$$

$$y = \left(3 - \frac{1}{2\pi} \cos\left(\frac{\pi}{2}x\right)\right)^2 - 7$$

4-6-6

38. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the differential equation $\frac{dy}{dx} = 10 - 2y$. Let $y = f(x)$ be the particular solution to the differential equation with the initial condition $f(0) = 2$.

- Write an equation for the line tangent to the graph of $y = f(x)$ at $x = 0$. Use the tangent line to approximate $f(0.5)$.
- Find the value of $\frac{d^2y}{dx^2}$ at the point $(0, 2)$. Is the graph of $y = f(x)$ concave up or concave down at the point $(0, 2)$? Give a reason for your answer.
- Find $y = f(x)$, the particular solution to the differential equation with the initial condition $f(0) = 2$.
- For the particular solution $y = f(x)$ found in part (c), find $\lim_{x \rightarrow \infty} f(x)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ tangent line equation
- ☐ approximation

Solution:

$$\left. \frac{dy}{dx} \right|_{(x,y)=(0,2)} = 10 - 2 \cdot 2 = 6$$

An equation for the line tangent to the graph of $y = f(x)$ at $x = 0$ is $y = 6x + 2$.

$$f(0.5) \approx 6 \cdot 0.5 + 2 = 5$$

4-6-6

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\frac{d^2y}{dx^2} \Big|_{(x,y)=(0,2)}$
- ☐ concave down with reason

Solution:

$$\frac{d^2y}{dx^2} = -2 \frac{dy}{dx} = -2(10 - 2y) = 4y - 20$$

$$\frac{d^2y}{dx^2} \Big|_{(x,y)=(0,2)} = 4 \cdot 2 - 20 = -12$$

Because $\frac{d^2y}{dx^2} \Big|_{(x,y)=(0,2)} < 0$ and $\frac{d^2y}{dx^2}$ is continuous, the graph of $y = f(x)$ is concave down at the point $(0, 2)$.

Part C

Zero out of 4 points earned if no separation of variables.

At most 2 out of 4 points earned [1-1-0-0] if no constant of integration.

The fourth point requires an expression for y .

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ separation of variables
- ☐ antiderivative
- ☐ constant of integration and uses initial condition

4-6-6

☐ solves for y **Solution:**

$$\frac{1}{10-2y} dy = dx$$

$$\int \frac{1}{10-2y} dy = \int 1 dx$$

$$-\frac{1}{2} \ln |10-2y| = x + C$$

Since the solution curve includes the point $(0, 2)$,

$$-\frac{1}{2} \ln (10-2y) = x + C$$

$$-\frac{1}{2} \ln (10-2 \cdot 2) = 0 + C \Rightarrow C = -\frac{1}{2} \ln 6$$

$$-\frac{1}{2} \ln (10-2y) = x - \frac{1}{2} \ln 6$$

$$\ln (10-2y) = -2x + \ln 6$$

$$10-2y = e^{-2x+\ln 6} = 6e^{-2x}$$

$$y = 5 - 3e^{-2x}$$

Part D

No supporting work is required to earn the point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct value.

Solution:

$$\lim_{x \rightarrow \infty} (5 - 3e^{-2x}) = 5 - 3 \cdot 0 = 5$$

4-6-6

39. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

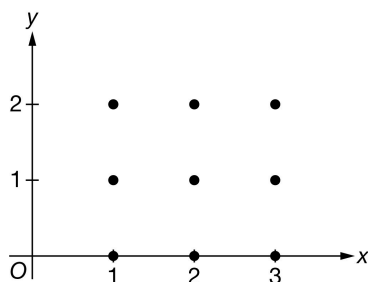
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Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the differential equation $\frac{dy}{dx} = \frac{y-1}{x}$.

(a) On the axes provided, sketch a slope field for the given differential equation at the nine points indicated.



(b) Let $y = f(x)$ be the particular solution to the given differential equation with the initial condition $f(3) = 2$. Write an equation for the line tangent to the graph of $y = f(x)$ at $x = 3$. Use the equation to approximate the value of $f(3.3)$.

(c) Find the particular solution $y = f(x)$ to the given differential equation with the initial condition $f(3) = 2$.

Part A

The first point is earned for having all 3 correct segments of zero slope.

The second point is earned for having all 6 correct segments for the other slopes.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

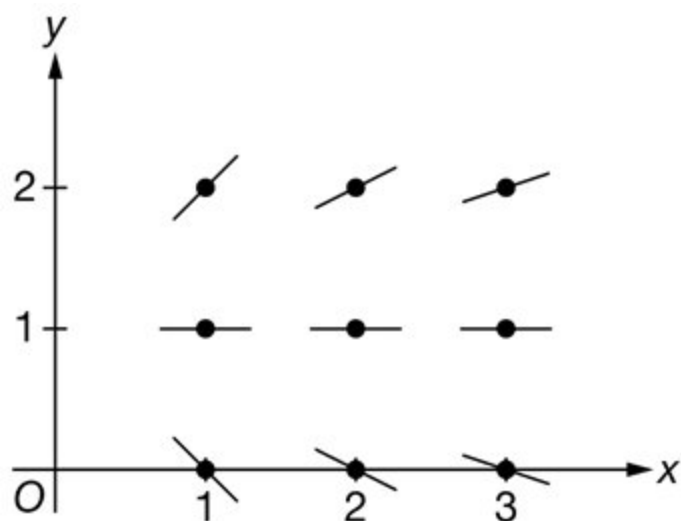
The student response accurately includes both of the criteria below.

zero slopes

4-6-6

☐☐

other slopes

Solution:**Part B**

The tangent line equation is not required to be in slope-intercept form.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

☐

tangent line equation

☐

approximation

Solution:

$$\left. \frac{dy}{dx} \right|_{(x,y)=(3,2)} = \frac{1}{3}$$

An equation for the line tangent to the graph of $y = f(x)$ at $x = 3$ is $y = 2 + \frac{1}{3}(x - 3)$.

$$f(3.3) \approx 2 + \frac{1}{3}(3.3 - 3) = 2.1$$

Part C

4-6-6

Zero out of 5 points earned if no separation of variables.

At most 2 out of 5 points earned [1-1-0-0-0] if no constant of integration.

If response has only one correct antiderivative, the second point is not earned. The response is eligible for the third, fourth, and fifth points if the antiderivative error is carried through the rest of the problem consistently and correctly.

The fifth point requires an expression for y . The domain is included with the solution; this is not a requirement to earn the fifth point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3	4	5
---	---	---	---	---	---

The student response accurately includes all five of the criteria below.

- ☐ separation of variables
- ☐ both antiderivatives
- ☐ constant of integration
- ☐ uses initial condition
- ☐ solves for y

Solution:

$$\frac{1}{y-1} dy = \frac{1}{x} dx$$

$$\int \frac{1}{y-1} dy = \int \frac{1}{x} dx$$

$$\ln |y-1| = \ln |x| + C$$

Since the solution curve includes the point (3, 2),

$$\ln(y-1) = \ln x + C$$

$$\ln 1 = \ln 3 + C \Rightarrow C = -\ln 3$$

$$\ln(y-1) = \ln x - \ln 3 = \ln\left(\frac{x}{3}\right)$$

$$y-1 = \frac{x}{3} \Rightarrow y = 1 + \frac{x}{3}$$

Note: This solution is valid for $x > 0$.

4-6-6

40. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

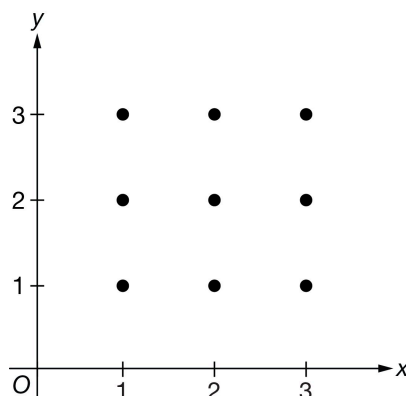
Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the differential equation $\frac{dy}{dx} = \frac{2-y}{x}$.

(a) On the axes provided, sketch a slope field for the given differential equation at the nine points indicated.



(b) Let $y = f(x)$ be the particular solution to the given differential equation with the initial condition $f(3) = 1$. Write an equation for the line tangent to the graph of $y = f(x)$ at $x = 3$. Use the equation to approximate the value of $f(3.5)$.

(c) Find the particular solution $y = f(x)$ to the given differential equation with the initial condition $f(3) = 1$.

Part A

The first point is earned for having all 3 correct segments of zero slope.

The second point is earned for having all 6 correct segments for the other slopes.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



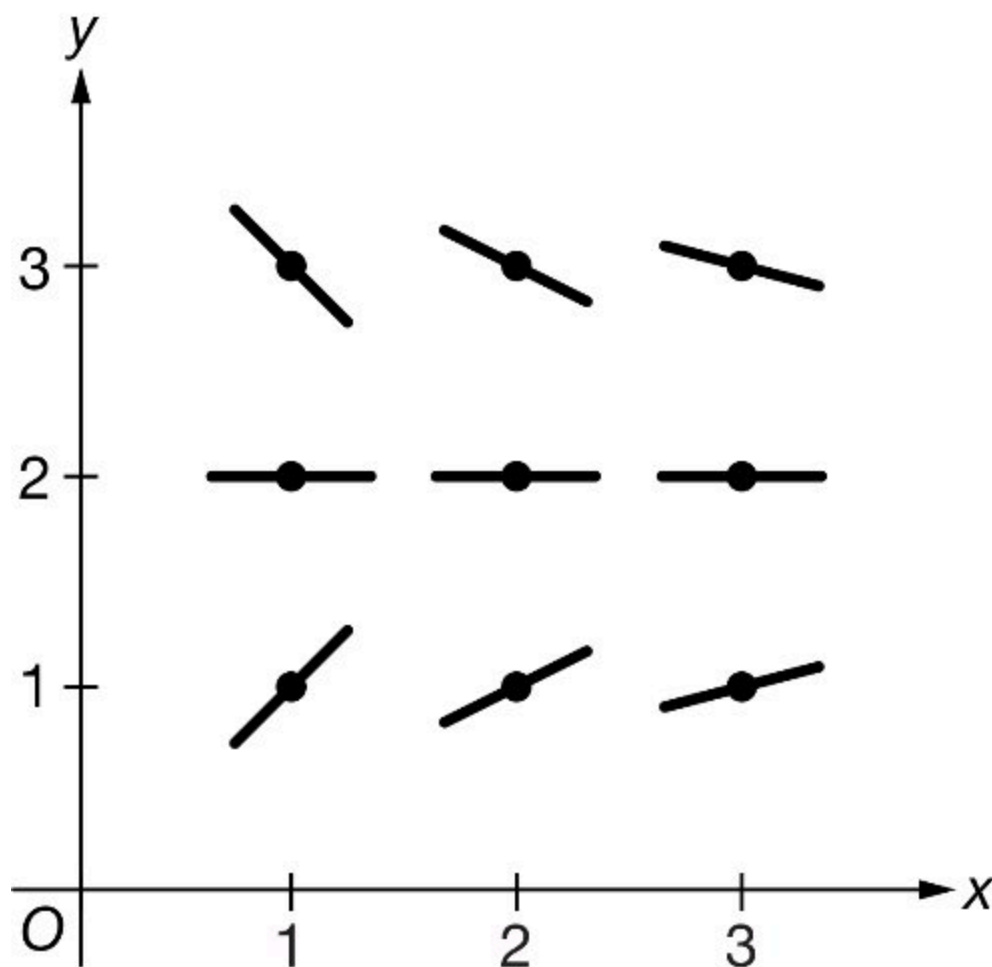
0	1	2
---	---	---

4-6-6

The student response accurately includes both of the criteria below.

- ☐ zero slopes
- ☐ other slopes

Solution:



Part B

The tangent line equation is not required to be in slope-intercept form.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

4-6-6

- ☐ tangent line equation
- ☐ approximation

Solution:

$$\left. \frac{dy}{dx} \right|_{(x,y)=(3,1)} = \frac{1}{3}$$

An equation for the line tangent to the graph of $y = f(x)$ at $x = 3$ is $y = \frac{1}{3}(x - 3) + 1 = \frac{1}{3}x$.

$$f(3.5) \approx \frac{1}{3}(3.5) = \frac{7}{6}$$

Part C

Zero out of 5 points earned if no separation of variables.

At most 2 out of 5 points earned [1-1-0-0-0] if no constant of integration.

The fifth point requires an expression for y . The domain is included with the solution; this is not a requirement to earn the fifth point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3	4	5
---	---	---	---	---	---

The student response accurately includes all five of the criteria below.

- ☐ separation of variables
- ☐ both antiderivatives
- ☐ constant of integration
- ☐ uses initial condition
- ☐ solves for y

Solution:

$$\frac{1}{2-y} dy = \frac{1}{x} dx$$

$$\int \frac{1}{2-y} dy = \int \frac{1}{x} dx$$

$$-\ln|2-y| = \ln|x| + C$$

Since the solution curve includes the point $(3, 1)$,

4-6-6

$$-\ln(2 - y) = \ln x + C$$

$$-\ln 1 = \ln 3 + C \Rightarrow C = -\ln 3$$

$$\ln(2 - y) = -\ln x + \ln 3 = \ln\left(\frac{3}{x}\right)$$

$$2 - y = \frac{3}{x} \Rightarrow y = 2 - \frac{3}{x}$$

1Note: This solution is valid for $x > 0$.

4-6-6

41.

t (minutes)	0	2	5	7	11	12
$r'(t)$ (feet per minute)	5.7	4.0	2.0	1.2	0.6	0.5

The volume of a spherical hot air balloon expands as the air inside the balloon is heated. The radius of the balloon, in feet, is modeled by a twice-differentiable function r of time t , where t is measured in minutes. For $0 < t < 12$, the graph of r is concave down. The table above gives selected values of the rate of change, $r'(t)$, of the radius of the balloon over the time interval $0 \leq t \leq 12$. The radius of the balloon is 30 feet when $t = 5$.

(Note: The volume of a sphere of radius r is given by $V = \frac{4}{3}\pi r^3$.)

(a) Estimate the radius of the balloon when $t = 5.4$ using the tangent line approximation at $t = 5$. Is your estimate greater than or less than the true value? Give a reason for your answer.

(b) Find the rate of change of the volume of the balloon with respect to time when $t = 5$. Indicate units of measure.

(c) Use a right Riemann sum with the five subintervals indicated by the data in the table to approximate $\int_0^{12} r'(t) dt$. Using correct units, explain the meaning of $\int_0^{12} r'(t) dt$ in terms of the radius of the balloon.

(d) Is your approximation in part (c) greater than or less than $\int_0^{12} r'(t) dt$? Give a reason for your answer.

Part A

1 point is earned for the estimate

1 point is earned for the conclusion with reason

$$r(5.4) \approx r(5) + r'(5)\Delta t = 30 + 2(0.4) = 30.8 \text{ ft}$$

Since the graph of r is concave down on the interval $5 < t < 5.4$, this estimate is greater than $r(5.4)$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the estimate

1 point is earned for the conclusion with reason

4-6-6

$$r(5.4) \approx r(5) + r'(5)\Delta t = 30 + 2(0.4) = 30.8 \text{ ft}$$

Since the graph of r is concave down on the interval $5 < t < 5.4$, this estimate is greater than $r(5.4)$.

Part B

2 points are earned for : $\frac{dV}{dt}$

1 point is earned for the answer

$$\frac{dV}{dt} = 3\left(\frac{4}{3}\right)\pi r^2 \frac{dr}{dt}$$

$$\frac{dV}{dt}\bigg|_{t=5} = 4\pi(30)^2 2 = 7200\pi \text{ ft}^3/\text{min}$$

1 point is earned for units in (b) and (c)

Units of ft^3/min in part (b) and ft in part (c)



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for : $\frac{dV}{dt}$

1 point is earned for the answer

$$\frac{dV}{dt} = 3\left(\frac{4}{3}\right)\pi r^2 \frac{dr}{dt}$$

$$\frac{dV}{dt}\bigg|_{t=5} = 4\pi(30)^2 2 = 7200\pi \text{ ft}^3/\text{min}$$

1 point is earned for units in (b) and (c)

Units of ft^3/min in part (b) and ft in part (c)

Part C

4-6-6

1 point is earned for the approximation

1 point is earned for the explanation

$$\int_0^{12} r'(t) dt \approx 2(4.0) + 3(2.0) + 2(1.2) + 4(0.6) + 1(0.5)$$

$$= 19.3 \text{ ft}$$

$$\int_0^{12} r'(t) dt \text{ is the change in the radius, in feet, from}$$

$t = 0$ to $t = 12$ minutes.

1 point is earned for units in (b) and (c)

Units of ft^3/min in part (b) and ft in part (c)



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the approximation

1 point is earned for the explanation

$$\int_0^{12} r'(t) dt \approx 2(4.0) + 3(2.0) + 2(1.2) + 4(0.6) + 1(0.5)$$

$$= 19.3 \text{ ft}$$

$$\int_0^{12} r'(t) dt \text{ is the change in the radius, in feet, from}$$

$t = 0$ to $t = 12$ minutes.

1 point is earned for units in (b) and (c)

Units of ft^3/min in part (b) and ft in part (c)

4-6-6

Part D

1 point is earned for the conclusion with reason

Since r is concave down, r' is decreasing on $0 < t < 12$.

Therefore, this approximation, 19.3 ft, is less than

$$\int_0^{12} r'(t) dt.$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the conclusion with reason

Since r is concave down, r' is decreasing on $0 < t < 12$.

Therefore, this approximation, 19.3 ft, is less than

$$\int_0^{12} r'(t) dt.$$

4-6-6

42. At the beginning of 2010, a landfill contained 1400 tons of solid waste. The increasing function W models the total amount of solid waste stored at the landfill. Planners estimate that W will satisfy the differential equation $\frac{dW}{dt} = \frac{1}{25}(W - 300)$ for the next 20 years. W is measured in tons, and t is measured in years from the start of 2010.
- (a) Use the line tangent to the graph of W at $t = 0$ to approximate the amount of solid waste that the landfill contains at the end of the first 3 months of 2010 (time $t = \frac{1}{4}$).
- (b) Find d^2W/dt^2 in terms of W . Use d^2W/dt^2 to determine whether your answer in part (a) is an underestimate or an overestimate of the amount of solid waste that the landfill contains at time $t = 1/4$.
- (c) Find the particular solution $W = W(t)$ to the differential equation $\frac{dW}{dt} = \frac{1}{25}(W - 300)$ with initial condition $W(0) = 1400$.

Part A

1 point is earned for $\frac{dW}{dt}$ at $t = 0$

1 point is earned for the answer

$$\left. \frac{dW}{dt} \right|_{t=0} = \frac{1}{25}(W(0) - 300) = \frac{1}{25}(1400 - 300) = 44$$

The tangent line is $y = 1400 + 44t$

$$W\left(\frac{1}{4}\right) \approx 1400 + 44\left(\frac{1}{4}\right) = 1411 \text{ tons}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $\frac{dW}{dt}$ at $t = 0$

1 point is earned for the answer

4-6-6

$$\left. \frac{dW}{dt} \right|_{t=0} = \frac{1}{25}(W(0) - 300) = \frac{1}{25}(1400 - 300) = 44$$

The tangent line is $y = 1400 + 44t$

$$W\left(\frac{1}{4}\right) \approx 1400 + 44\left(\frac{1}{4}\right) = 1411 \text{ tons}$$

Part B

1 point is earned for $\frac{d^2W}{dt^2}$

1 point is earned for the answer with reason

$$\frac{d^2W}{dt^2} = \frac{1}{25} \frac{dW}{dt} = \frac{1}{625}(W - 300) \text{ and } W \geq 1400$$

Therefore $\frac{d^2W}{dt^2} > 0$ on the interval $0 \leq t \leq \frac{1}{4}$.

The answer in part(a) is an underestimate.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $\frac{d^2W}{dt^2}$

1 point is earned for the answer with reason

$$\frac{d^2W}{dt^2} = \frac{1}{25} \frac{dW}{dt} = \frac{1}{625}(W - 300) \text{ and } W \geq 1400$$

4-6-6

Therefore $\frac{d^2W}{dt^2} > 0$ on the interval $0 \leq t \leq \frac{1}{4}$.

The answer in part(a) is an underestimate.

Part C

1 point is earned for separation of variables

1 point is earned for antiderivatives

1 point is earned for constant of integration

1 point is earned for using initial condition

1 point is earned for solving for W

$$\frac{dW}{dt} = \frac{1}{25}(W - 300)$$

$$\int \frac{1}{W - 300} dW = \int \frac{1}{25} dt$$

$$\ln |W - 300| = \frac{1}{25}t + C$$

$$\ln(1400 - 300) = \frac{1}{25}(0) + C \Rightarrow \ln(1100) = C$$

$$W - 300 = 1100e^{\frac{1}{25}t}$$

$$W(t) = 300 + 1100e^{\frac{1}{25}t}, 0 \leq t \leq 20$$



0	1	2	3	4	5
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The student response earns all of the following points:

1 point is earned for separation of variables

1 point is earned for antiderivatives

1 point is earned for constant of integration

1 point is earned for using initial condition

1 point is earned for solving for W

4-6-6

$$\frac{dW}{dt} = \frac{1}{25}(W - 300)$$

$$\int \frac{1}{W - 300} dW = \int \frac{1}{25} dt$$

$$\ln |W - 300| = \frac{1}{25}t + C$$

$$\ln(1400 - 300) = \frac{1}{25}(0) + C \Rightarrow \ln(1100) = C$$

$$W - 300 = 1100e^{\frac{1}{25}t}$$

$$W(t) = 300 + 1100e^{\frac{1}{25}t}, \quad 0 \leq t \leq 20$$

4-6-6

43. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$W(t) = \begin{cases} \frac{17}{2} - \frac{3}{2} \cos\left(\frac{\pi t}{6}\right) & \text{for } 0 \leq t \leq 6 \\ 10 - \frac{1}{5}(t-6)^2 & \text{for } 6 < t \leq 10 \end{cases}$$

The depth of a river at a certain point is modeled by the function W defined above, where $W(t)$ is measured in feet and time t is measured in hours.

- (a) Find $W'(8)$. Using correct units, explain the meaning of $W'(8)$ in the context of the problem.
- (b) The graph of W is concave down for $3 \leq t \leq 3.5$. Use the line tangent to the graph of W at $t = 3$ to show that $W(3.5) \leq 9$.
- (c) Find $\lim_{t \rightarrow 2} \frac{W(t) - t^3 + \frac{1}{4}}{t-2}$.

Part A

The first 2 points may be earned with the presence of a correct numerical answer.

The second point may be earned with a maximum of one sign error in $W'(t)$.

The third point requires a time reference, rate, and units. The response is eligible for the third point if the second point is earned AND provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $W'(t)$
- ☐ $W'(8)$

4-6-6

☐ interpretation with units

Solution:

For $6 < t < 10$, $W'(t) = -\frac{2}{5}(t - 6)$.

$$W'(8) = -\frac{2}{5}(8 - 6) = -\frac{4}{5}$$

At time $t = 8$ hours, the depth of the river is changing at a rate of $-\frac{4}{5}$ foot per hour.

Part B

The second point may be earned with a maximum of one error in $W'(t)$.

For the second and third points, trigonometric function values do not need to be evaluated.

The third point may be earned if the response is consistent with previous results.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ $W'(t)$
- ☐ $W'(3)$
- ☐ approximation
- ☐ explanation

Solution:

$$W(3) = \frac{17}{2} - \frac{3}{2}\cos\left(\frac{3\pi}{6}\right) = \frac{17}{2}$$

For $0 < t < 6$,

$$W'(t) = \frac{3}{2}\sin\left(\frac{\pi t}{6}\right) \cdot \frac{\pi}{6} = \frac{\pi}{4}\sin\left(\frac{\pi t}{6}\right).$$

$$W'(3) = \frac{\pi}{4}\sin\left(\frac{3\pi}{6}\right) = \frac{\pi}{4}$$

$$W(3.5) \approx \frac{17}{2} + \frac{\pi}{4}(3.5 - 3) = \frac{17}{2} + \frac{\pi}{8}$$

4-6-6

Because the graph of W is concave down for $3 \leq t \leq 3.5$, approximation of $W(3.5)$ is an overestimate of the actual value of $W(3.5)$.

Therefore, $W(3.5) < \frac{17}{2} + \frac{\pi}{8} < \frac{17}{2} + \frac{1}{2} = 9$.

Part C

The first point requires showing that an indeterminate form $\frac{0}{0}$ is present in order to apply L'Hospital's Rule. Any error in mathematical communication (e.g., writing " $= \frac{0}{0}$ " or not using limit notation) impacts the first point.

The second point requires substitution of function values. A simplified answer is not required; trigonometric function values do not need to be evaluated. Any differentiation or computation error impacts the second point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ applies L'Hospital's Rule
- ☐ answer

Solution:

$$\lim_{t \rightarrow 2} \left(W(t) - t^3 + \frac{1}{4} \right) = W(2) - 2^3 + \frac{1}{4}$$

$$= \frac{17}{2} - \frac{3}{2} \cos\left(\frac{2\pi}{6}\right) - 8 + \frac{1}{4} = 0$$

$$\lim_{t \rightarrow 2} (t - 2) = 0$$

By L'Hospital's Rule,

$$\lim_{t \rightarrow 2} \frac{W(t) - t^3 + \frac{1}{4}}{t - 2} = \lim_{t \rightarrow 2} \frac{W'(t) - 3t^2}{1} = W'(2) - 3 \cdot 2^2$$

$$= \frac{\pi}{4} \sin\left(\frac{2\pi}{6}\right) - 12 = \frac{\pi\sqrt{3}}{8} - 12$$

4-6-6

44. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$W(t) = \begin{cases} \frac{32}{5} + \frac{2}{5}\cos\left(\frac{\pi t}{4}\right) & \text{for } 0 \leq t \leq 4 \\ 6 + \frac{1}{8}(t-4)^2 & \text{for } 4 < t \leq 9 \end{cases}$$

The depth of water in a zoo aquarium at a certain point is modeled by the function W defined above, where $W(t)$ is measured in feet and time t is measured in hours.

- (a) Find $W'(7)$. Using correct units, explain the meaning of $W'(7)$ in the context of the problem.
- (b) The graph of W is concave up for $2 \leq t \leq 2.5$. Use the line tangent to the graph of W at $t = 2$ to show that $W(2.5) \geq 6$.
- (c) Find $\lim_{t \rightarrow 2} \frac{W(t) - t^2 - \frac{12}{5}}{t-2}$.

Part A

The first 2 points may be earned with the presence of a correct numerical answer.

The second point may be earned with a maximum of one sign error in $W'(t)$.

The third point requires a time reference, rate, and units. The response is eligible for the third point if the second point is earned AND provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $W'(t)$
- ☐ $W'(7)$

4-6-6

☐ interpretation with unit

Solution:

For $4 < t < 9$, $W'(t) = \frac{1}{4}(t - 4)$.

$$W'(7) = \frac{1}{4}(7 - 4) = \frac{3}{4}$$

At time $t = 7$ hours, the depth of the water in the zoo aquarium is changing at a rate of $\frac{3}{4}$ foot per hour.

Part B

The second point may be earned with a maximum of one error in $W'(t)$.

For the second and third points, trigonometric function values do not need to be evaluated.

The third point may be earned if the response is consistent with previous results. Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ $W'(t)$
- ☐ $W'(2)$
- ☐ approximation
- ☐ explanation

Solution:

$$W(2) = \frac{32}{5} + \frac{2}{5}\cos\left(\frac{2\pi}{4}\right) = \frac{32}{5}$$

For $0 < t < 4$,

$$W'(t) = -\frac{2}{5}\sin\left(\frac{\pi t}{4}\right) \cdot \frac{\pi}{4} = -\frac{\pi}{10}\sin\left(\frac{\pi t}{4}\right).$$

$$W'(2) = -\frac{\pi}{10}\sin\left(\frac{2\pi}{4}\right) = -\frac{\pi}{10}$$

$$W(2.5) \approx \frac{32}{5} - \frac{\pi}{10}(2.5 - 2) = \frac{32}{5} - \frac{\pi}{20}$$

4-6-6

Because the graph of W is concave up for $2 \leq t \leq 2.5$, the tangent line approximation of $W(2.5)$ is an underestimate of the actual value of $W(2.5)$. Therefore, $W(2.5) > \frac{32}{5} - \frac{\pi}{20} > 6$.

Part C

The first point requires showing that an indeterminate form $\frac{0}{0}$ is present in order to apply L'Hospital's Rule. Any error in mathematical communication (e.g., writing " $\frac{0}{0}$ " or not using limit notation) impacts the first point.

The second point requires substitution of function values. A simplified answer is not required; trigonometric function values do not need to be evaluated. Any differentiation or computation error impacts the second point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ applies L'Hospital's Rule
- ☐ answer

Solution:

$$\begin{aligned}\lim_{t \rightarrow 2} \left(W(t) - t^2 - \frac{12}{5} \right) &= W(2) - 2^2 - \frac{12}{5} \\ &= \frac{32}{5} + \frac{2}{5} \cos\left(\frac{2\pi}{4}\right) - 4 - \frac{12}{5} = 0\end{aligned}$$

$$\lim_{t \rightarrow 2} (t - 2) = 0$$

By L'Hospital's Rule,

$$\begin{aligned}\lim_{t \rightarrow 2} \frac{W(t) - t^2 - \frac{12}{5}}{t - 2} &= \lim_{t \rightarrow 2} \frac{W'(t) - 2t}{1} = W'(2) - 2 \cdot 2 \\ &= -\frac{\pi}{10} \sin\left(\frac{2\pi}{4}\right) - 4 = -\frac{\pi}{10} - 4\end{aligned}$$

4-6-6

45. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	5	15	30	35
$A(t)$ (gallons)	10	18	25	16	8

The number of gallons of olive oil in a tank at time t is given by the twice-differentiable function A , where t is measured in hours and $0 \leq t \leq 35$. Values of $A(t)$ at selected times t are given in the table above.

- (a) Use the data in the table to estimate the rate at which the number of gallons of olive oil in the tank is changing at time $t = 10$ hours. Show the computations that lead to your answer. Indicate units of measure.
- (b) For $0 \leq t \leq 30$, is there a time t at which $A'(t) = \frac{1}{5}$? Justify your answer.
- (c) The number of gallons of olive oil in the tank at time t is also modeled by the function G defined by $G(t) = 5t - \frac{2}{3}(t+9)^{\frac{3}{2}} + 28$, where t is measured in hours and $0 \leq t \leq 35$. Based on the model, at what time t , for $0 \leq t \leq 35$, is the number of gallons of olive oil in the tank an absolute maximum? Justify your answer.
- (d) For the function G defined in part (c), the locally linear approximation near $t = 7$ is used to approximate $G(8)$. Is this approximation an overestimate or an underestimate for the value of $G(8)$? Give a reason for your answer.

Part A

Substitution of function values is required. The answer does not need to be simplified.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes the criteria below.

4-6-6

- ☐ approximation with units

Solution:

$$A'(10) \approx \frac{A(15) - A(5)}{15 - 5} = \frac{25 - 18}{15 - 5} = \frac{7}{5} \text{ gallons per hour}$$

Part B

The first point requires identification of the difference quotient. The substitution of function values and conclusion impacts the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\frac{A(30) - A(0)}{30 - 0}$
- ☐ justification using Mean Value Theorem

Solution:

A is twice differentiable. $\Rightarrow A$ is differentiable. $\Rightarrow A$ is continuous.

$$\frac{A(30) - A(0)}{30 - 0} = \frac{16 - 10}{30 - 0} = \frac{6}{30} = \frac{1}{5}$$

Therefore, by the Mean Value Theorem, there is a time t , for $0 \leq t \leq 30$, at which $A'(t) = \frac{1}{5}$.

Part C

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

The response is eligible for additional points based on consistent answers using an incorrect $G'(t)$ with a maximum of one computational error.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3	4
---	---	---	---	---

4-6-6

The student response accurately includes all four of the criteria below.

- ☐ $G'(t)$
- ☐ sets $G'(t) = 0$
- ☐ identifies $t = 16$ as a candidate
- ☐ answer with justification

Solution:

$$G'(t) = 5 - (t + 9)^{\frac{1}{2}} = 0 \Rightarrow (t + 9)^{\frac{1}{2}} = 5$$

$$\Rightarrow t + 9 = 25 \Rightarrow t = 16$$

Because $G'(t)$ changes from positive to negative at $t = 16$, G has a relative maximum at $t = 16$.

Since $t = 16$ is the only critical point of G on the interval $0 \leq t \leq 35$ that is the location of a relative maximum, it is also the location of the absolute maximum of G on the interval $0 \leq t \leq 35$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $G''(t)$
- ☐ overestimate with reason

Solution:

$$G''(t) = -\frac{1}{2}(t + 9)^{-\frac{1}{2}}$$

$G''(t) < 0$ for $7 \leq t \leq 8$, and so the graph of G is concave down for $7 \leq t \leq 8$. Therefore, the approximation of $G(8)$ using the locally linear approximation of G near $t = 7$ is an overestimate.

4-6-6

46. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	1	5	8	10
$G(t)$ (thousands of gallons)	22	18.5	19.5	22	24

A gasoline storage tank is filled by a pipeline from a refinery. At the same time, gasoline flows from the tank into trucks that will make deliveries to gasoline stations. The amount of gasoline in the storage tank at time t is given by the twice-differentiable function G , where t is measured in hours and $0 \leq t \leq 10$. Values of $G(t)$, in thousands of gallons, at selected times t are given in the table above. It is known that $G''(t) > 0$ for $5 \leq t < 10$.

- Use the data in the table to estimate the rate of change of the amount of gasoline in the storage tank at time $t = 3$ hours. Show the computations that lead to your answer. Indicate units of measure.
- For $0 < t < 10$, is there a time t at which $G(t)$ is increasing at a rate of 0.2 thousand gallons per hour? Justify your answer.
- It is known that $G'(5) = 0.5$. Use the locally linear approximation for G at time $t = 5$ to approximate the amount of gasoline in the storage tank at time $t = 7$. Is this approximation an overestimate or an underestimate for the actual amount of gasoline in the storage tank at time $t = 7$? Give a reason for your answer.
- The rate at which gasoline flows out of the storage tank into trucks at time t can be modeled by the function R defined by $R(t) = \frac{100t}{t^2+4}$, where t is measured in hours, and $R(t)$ is measured in thousands of gallons. Based on the model, at what time t , for $0 \leq t \leq 10$, is the rate at which gasoline flows out of the storage tank an absolute maximum? Justify your answer.

Part A

Substitution of function values is required. The answer does not need to be simplified.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct approximation with units.

4-6-6

Solution:

$$G'(3) \approx \frac{G(5) - G(1)}{5 - 1} = \frac{19.5 - 18.5}{5 - 1} = \frac{1}{4} = 0.25 \text{ thousand gallons per hour (which is equivalent to 250 gallons per hour)}$$

Part B

The first point requires identification of the difference quotient. The substitution of function values and conclusion impacts the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\frac{G(10) - G(0)}{10 - 0}$
- ☐ justification using Mean Value Theorem

Solution:

G is twice differentiable. $\Rightarrow G$ is differentiable. $\Rightarrow G$ is continuous.

$$\frac{G(10) - G(0)}{10 - 0} = \frac{24 - 22}{10 - 0} = \frac{2}{10} = 0.2 \text{ thousand gallons per hour (which is equivalent to 200 gallons per hour)}$$

Therefore, by the Mean Value Theorem, there is a time t , for $0 < t < 10$, at which $G'(t) = 0.2$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ approximation of $G(7)$
- ☐ underestimate with reason

4-6-6

Solution:

$$G(7) \approx G(5) + G'(5)(7 - 5) = 19.5 + (0.5)(2) = 20.5 \text{ thousand gallons}$$

$G''(t) > 0$ for $5 \leq t \leq 7$, and so the graph of G is concave up for $5 \leq t \leq 7$. Therefore, the approximation for $G(7)$ using the locally linear approximation for G at time $t = 5$ is an underestimate for the actual amount.

Part D

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

The response is eligible for additional points based on consistent answers using an incorrect $R'(t)$ with a maximum of one computational error.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ $R'(t)$
- ☐ sets $R'(t) = 0$
- ☐ identifies $t = 2$ as a candidate
- ☐ answer with justification

Solution:

$$R'(t) = \frac{100(t^2+4) - 100t(2t)}{(t^2+4)^2} = \frac{400 - 100t^2}{(t^2+4)^2} = \frac{100(4-t^2)}{(t^2+4)^2} = 0$$

$$\Rightarrow 4 - t^2 = 0 \Rightarrow t = 2$$

Because $R'(t)$ changes from positive to negative at $t = 2$, R has a relative maximum at $t = 2$. R is increasing on $0 \leq t \leq 2$ and R is decreasing on $2 \leq t \leq 10$. Since $t = 2$ is the only critical point of R on the interval $0 \leq t \leq 10$ that is the location of a relative maximum, it is also the location of the absolute maximum of R on the interval $0 \leq t \leq 10$.

4-6-6

47. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	3	8	12	14
$S'(t)$ (centimeters per hour)	2.3	2.1	1.9	1.7	1.6

The depth of snow in a field is given by the twice-differentiable function S for $0 \leq t \leq 14$, where $S(t)$ is measured in centimeters and time t is measured in hours. Values of $S'(t)$, the derivative of S , at selected values of time t are shown in the table above. It is known that the graph of S is concave down for $0 \leq t \leq 14$.

- Use the data in the table to approximate $S''(10)$. Show the computations that lead to your answer. Using correct units, explain the meaning of $S''(10)$ in the context of the problem.
- Is there a time t , for $0 \leq t \leq 14$, at which the depth of snow is changing at a rate of 2 centimeters per hour? Justify your answer.
- At time $t = 8$, the depth of snow is 45 centimeters. Use the line tangent to the graph of S at $t = 8$ to approximate the depth of snow at time $t = 10$. Is the approximation an underestimate or an overestimate of the actual depth of snow at time $t = 10$? Justify your answer.
- The depth of snow, in centimeters, is also modeled by the twice-differentiable function D for $0 \leq t \leq 14$, where $D(t) = 120 - 92e^{-\frac{t}{40}}$ and time t is measured in hours. Use the model to find $D'(10)$. Using correct units, explain the meaning of $D'(10)$ in the context of the problem.

Part A

The first point requires a correct substitution of values. A simplified answer is not required.

The second point requires a time reference, rate, and units. The point may be earned with an incorrect approximation based on one computational error and provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

4-6-6

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ approximation
- ☐ interpretation with units

Solution:

$$S''(10) \approx \frac{S'(12) - S'(8)}{12 - 8} = \frac{1.7 - 1.9}{4} = -0.05$$

At time $t = 10$ hours, the rate of change of the depth of snow is changing at a rate of -0.05 centimeter per hour per hour.

Part B

The second point may be earned if response contains conditions of IVT without naming the theorem.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ S' is continuous
- ☐ justification using Intermediate Value Theorem

Solution:

S is twice differentiable. $\Rightarrow S'$ is differentiable and thus continuous.

$$S'(8) = 1.9 < 2 < 2.1 = S'(3)$$

By the Intermediate Value Theorem, there is a value of t , for $0 \leq t \leq 14$, such that $S'(t) = 2$.

Part C

The second point may be earned if using an incorrect value for slope, provided the value is labeled as $S'(8)$.

4-6-6

The second point does not require a simplified answer. Substitution of function values is required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ equation for tangent line
- ☐ approximation
- ☐ overestimate with justification using concavity

Solution:

$$S'(8) = 1.9 \text{ from table}$$

An equation for the tangent line is $y = 45 + 1.9(t - 8)$.

$$S(10) \approx 45 + S'(8)(10 - 8) = 48.8 \text{ centimeters}$$

This approximation is an overestimate because the graph of S is concave down for $8 \leq t \leq 10$.

Part D

The second point requires a time reference, rate, and units. The point may be earned with an incorrect approximation based on one computational error and provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $D'(10)$
- ☐ interpretation with units

Solution:

$$D'(10) = 1.791$$

4-6-6

At time $t = 10$ hours, the depth of snow is changing by 1.791 centimeters per hour.

4-6-6

48. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	3	8	12
$H(t)$ (degrees Celsius)	90	80	67	58

As a bowl of soup cools, the temperature of the soup is given by the twice-differentiable function H for $0 \leq t \leq 12$, where $H(t)$ is measured in degrees Celsius ($^{\circ}\text{C}$) and time t is measured in minutes. Values of $H(t)$ at selected values of time t are shown in the table above.

- Use the data in the table to approximate $H'(5)$. Show the computations that lead to your answer. Using correct units, explain the meaning of $H'(5)$ in the context of the problem.
- Is there a time t , for $0 \leq t \leq 12$, at which the temperature of the soup is 60°C ? Justify your answer.
- The temperature of the soup is also modeled by the twice-differentiable function C for $0 \leq t \leq 12$, where $C(t)$ is measured in degrees Celsius and time t is measured in minutes. It is known that $C(3) = 80$ and the derivative of C is given by $C'(t) = -3.6e^{-0.05t}$. Write an equation for the line tangent to the graph of C at $t = 3$ and use it to approximate $C(5)$.
- Based on the model given in part (c), at what rate, in degrees Celsius per minute per minute, is the rate of change of the temperature of the soup changing at time $t = 3$ minutes?

Part A

The first point requires a correct substitution of values. A simplified answer is not required.

The second point requires a time reference, rate, and units. The point may be earned with an incorrect approximation based on one computational error and provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

4-6-6

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ approximation
- ☐ interpretation with units

Solution:

$$H'(5) \approx \frac{H(8) - H(3)}{8 - 3} = \frac{67 - 80}{5} = -2.6$$

At time $t = 5$ minutes, the temperature of the soup is changing at a rate of -2.6 degrees Celsius per minute.

Part B

The second point may be earned if response contains conditions of IVT without naming the theorem.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ H is continuous
- ☐ justification using Intermediate Value Theorem

Solution:

H is twice differentiable. $\Rightarrow H$ is continuous.

$$H(12) = 58 < 60 < 90 = H(0)$$

By the Intermediate Value Theorem, there is a value of t , for $0 \leq t \leq 12$, such that $H(t) = 60$.

Part C

The second point may be earned if using an incorrect value for slope, provided the value is labeled as $C'(3)$.

The third point does not require a simplified answer. Substitution of function values is required.

4-6-6

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ slope
- ☐ equation for tangent line
- ☐ approximation

Solution:

$$C'(3) = -3.098549$$

An equation for the tangent line is $y = 80 - 3.098549(t - 3)$.

$$C(5) \approx 80 - C'(3)(5 - 3) = 73.802903$$

$$C(5) = 73.803 \text{ (or } 73.802\text{)}$$

Part D

The second point requires a time reference, rate, and units. The point may be earned with an incorrect approximation based on one computational error and provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $C''(3)$
- ☐ interpretation with units

Solution:

$$C''(3) = 0.154927$$

At time $t = 3$ minutes, the rate of change of the temperature is changing by 0.155 (or 0.154) degree Celsius per minute per minute.

4-6-6

49. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

A book in a popular book series goes on sale at time $t = 0$. The total number of books sold by time t hours is modeled by the function B that satisfies the differential equation $\frac{dB}{dt} = \frac{5}{\sqrt{t+4}} + \frac{1}{9}(t-6)^2$ with $B(12) = 36$, where B is measured in hundreds of books and $0 \leq t \leq 24$.

- Write an equation for the line tangent to the graph of B at $t = 12$. Use this equation to approximate the total number of books sold by time $t = 10$.
- Find $B''(5)$. Using correct units, interpret the meaning of $B''(5)$ in the context of the problem.
- Find an expression for $B(t)$. Based on this model, what is the total number of books sold by time $t = 21$?

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ tangent line equation
- ☐ approximation

Solution:

$$B(12) = 36$$

$$B'(12) = \frac{5}{\sqrt{12+4}} + \frac{1}{9}(12-6)^2 = \frac{5}{4} + 4 = \frac{21}{4}$$

An equation for the line tangent to the graph of B at $t = 12$ is $y = 36 + \frac{21}{4}(t - 12)$.

4-6-6

$$B(10) \approx 36 + \frac{21}{4}(10 - 12) = 36 - \frac{21}{2} = \frac{51}{2} = 25.5$$

The number of books sold by time $t = 10$ is approximately 2550.

Part B

The third point requires (1) rate of $-\frac{17}{54}$, (2) units of hundred books per hour per hour, and (3) reference to $t = 5$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $B''(t)$
- ☐ $B''(5)$
- ☐ interpretation

Solution:

$$B''(t) = 5\left(-\frac{1}{2}\right)(t+4)^{-\frac{3}{2}} + \frac{1}{9} \cdot 2(t-6) = -\frac{5}{2(t+4)^{\frac{3}{2}}} + \frac{2}{9}(t-6)$$

$$B''(5) = -\frac{5}{2(5+4)^{\frac{3}{2}}} + \frac{2}{9}(5-6) = -\frac{5}{54} - \frac{2}{9} = -\frac{17}{54}$$

At time $t = 5$, the rate at which books are selling is changing at a rate of $-\frac{17}{54}$ hundred books per hour per hour.

Part C

At most 1 out of 4 points earned [1-0-0-0] if no constant of integration.

The fourth point requires both an expression for $B(t)$ and value for $B(21)$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

4-6-6

- ☐ antiderivative
- ☐ constant of integration
- ☐ uses initial conditions
- ☐ $B(t)$ and $B(21)$

Solution:

$$\frac{dB}{dt} = \frac{5}{\sqrt{t+4}} + \frac{1}{9}(t-6)^2$$

$$B(t) = 5 \cdot 2\sqrt{t+4} + \frac{1}{9} \cdot \frac{1}{3}(t-6)^3 + C = 10\sqrt{t+4} + \frac{1}{27}(t-6)^3 + C$$

$$B(12) = 10\sqrt{12+4} + \frac{1}{27}(12-6)^3 + C = 48 + C = 36 \Rightarrow C = -12$$

$$B(t) = 10\sqrt{t+4} + \frac{1}{27}(t-6)^3 - 12$$

$$B(21) = 10\sqrt{21+4} + \frac{1}{27}(21-6)^3 - 12 = 50 + 125 - 12 = 163$$

The number of books sold by time $t = 21$ is 16,300.

4-6-6

50. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

A large vat is initially filled with a saltwater solution. A solution with a higher concentration of salt flows into the vat, and solution flows out of the vat at the same rate. The number of pounds of salt in the vat at time t minutes is modeled by the function A that satisfies the differential equation $\frac{dA}{dt} = 6 - 0.02A$. At time $t = 10$ minutes, the vat contains 50 pounds of salt.

(a) Write an equation for the line tangent to the graph of A at $t = 10$. Use the tangent line to approximate the number of pounds of salt in the vat at time $t = 12$ minutes.

(b) Show that $A(t) = 300 - 250e^{0.2-0.02t}$ satisfies the differential equation $\frac{dA}{dt} = 6 - 0.02A$ with initial condition $A(10) = 50$.

(c) The flow of solution into the vat is stopped, and the solution is drained. The depth of solution in the vat is modeled by the function h that satisfies the differential equation $\frac{dh}{dt} = -k\sqrt{h}$, where $h(t)$ is measured in meters, t is the number of minutes since draining began, and k is a constant. If the depth of the solution is 16 meters at time $t = 0$ minutes and 4 meters at time $t = 30$ minutes, what is $h(t)$ in terms of t ?

Part A

For the second point, it is incorrect to state $A(12) = 60$ rather than $A(12) \approx 60$.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ tangent line equation
- ☐ approximation

Solution:

4-6-6

$$A(10) = 50$$

$$\left. \frac{dA}{dt} \right|_{t=10} = 6 - 0.02 \cdot 50 = 5$$

An equation for the line tangent to the graph of A at $t = 10$ is $y = 50 + 5(t - 10)$.

At time $t = 12$ minutes, the vat contains approximately $50 + 5(12 - 10) = 60$ pounds of salt. $A(12) \approx 60$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ verification of initial condition
- ☐ $\frac{dA}{dt} = 5e^{0.2-0.02t}$
- ☐ verification that $\frac{dA}{dt} = 6 - 0.02A$

Solution:

$$A(10) = 300 - 250e^{0.2-0.02 \cdot 10} = 300 - 250e^0 = 50$$

$$\begin{aligned} \frac{dA}{dt} &= \frac{d}{dt}(300 - 250e^{0.2-0.02t}) \\ &= -250e^{0.2-0.02t}(-0.02) = 5e^{0.2-0.02t} \end{aligned}$$

$$\begin{aligned} 6 - 0.02A &= 6 - 0.02(300 - 250e^{0.2-0.02t}) \\ &= 6 - 6 + 5e^{0.2-0.02t} = 5e^{0.2-0.02t} \end{aligned}$$

Part C

Zero out of 4 points earned if no separation of variables.

At most 2 out of 4 points earned [1-1-0-0] if no constant of integration.

Both antiderivatives must be correct to earn the second point.

The fourth point requires an expression for $h(t)$. The domain of $h(t)$ is included with the solution; this is not a requirement to earn the fourth point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response

4-6-6



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ separation of variables
- ☐ antiderivatives
- ☐ constant of integration and uses initial conditions
- ☐ $h(t)$

Solution:

$$\frac{dh}{dt} = -k\sqrt{h}$$

$$\frac{dh}{\sqrt{h}} = -k dt$$

$$\int \frac{1}{\sqrt{h}} dh = - \int k dt$$

$$2\sqrt{h} = -kt + C$$

$$2\sqrt{16} = 0 + C \Rightarrow C = 8$$

$$2\sqrt{4} = -30k + 8 \Rightarrow k = \frac{2}{15}$$

$$2\sqrt{h} = -\frac{2}{15}t + 8$$

$$h(t) = \left(-\frac{t}{15} + 4\right)^2$$

Note: This is valid for $0 \leq t \leq 60$.

4-6-6

51. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

On a certain day at a diner, the total number of food orders taken by time t is modeled by the function F that satisfies the differential equation $\frac{dF}{dt} = \frac{48}{\sqrt{3t+1}}$ with $F(1) = 32$, where t is the number of hours since the diner opened and $0 \leq t \leq 8$.

- Write an equation for the line tangent to the graph of F at $t = 1$. Use this equation to approximate the total number of food orders taken by time $t = 2$.
- Find $F''(5)$. Using correct units, interpret the meaning of $F''(5)$ in the context of the problem.
- Find an expression for $F(t)$. Based on this model, what is the total number of food orders taken by time $t = 8$?

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ tangent line equation
- ☐ approximation

Solution:

$$F(1) = 32$$

$$F'(1) = \frac{48}{\sqrt{3 \cdot 1 + 1}} = 24$$

An equation for the line tangent to the graph of F at $t = 1$ is $y = 32 + 24(t - 1)$.

By time $t = 2$, the total number of food orders taken is approximately $32 + 24(2 - 1) = 56$.

4-6-6

Part B

The third point requires (1) rate of $-\frac{9}{8}$, (2) units of orders per hour per hour, and (3) reference to $t = 5$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $F''(t)$
- ☐ $F''(5)$
- ☐ interpretation

Solution:

$$F''(t) = 48\left(-\frac{1}{2}\right)(3t+1)^{-\frac{3}{2}} \cdot 3 = -72(3t+1)^{-\frac{3}{2}}$$

$$F''(5) = -72(3 \cdot 5 + 1)^{-\frac{3}{2}} = -72(16)^{-\frac{3}{2}} = -\frac{72}{64} = -\frac{9}{8}$$

At time $t = 5$, the rate at which food orders are taken is changing at a rate of $-\frac{9}{8}$ orders per hour per hour.

Part C

At most 1 out of 4 points earned [1-0-0-0] if no constant of integration.

The fourth point requires both an expression for $F(t)$ and value for $F(8)$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ antiderivative
- ☐ constant of integration
- ☐ uses initial conditions
- $F(t)$ and $F(8)$

4-6-6**Solution:**

$$\frac{dF}{dt} = \frac{48}{\sqrt{3t+1}}$$

$$F(t) = 48 \cdot 2\sqrt{3t+1} \cdot \frac{1}{3} + C = 32\sqrt{3t+1} + C$$

$$F(1) = 32\sqrt{3 \cdot 1 + 1} + C = 64 + C = 32 \Rightarrow C = -32$$

$$F(t) = 32\sqrt{3t+1} - 32$$

The number of food orders taken by time $t = 8$ is $32\sqrt{3 \cdot 8 + 1} - 32 = 128$.

4-6-6

52. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

The population of a virus in a host can be modeled by the function A that satisfies the differential equation $\frac{dA}{dt} = t - \frac{1}{5}A$, where A is measured in millions of virus cells and t is measured in days for $5 \leq t < 10$. At time $t = 5$ days, there are 10 million cells of the virus in the host.

- (a) Write an equation for the line tangent to the graph of A at $t = 5$. Use the tangent line to approximate the number of virus cells in the host, in millions, at time $t = 7$ days.
- (b) Show that $A(t) = -25 + 5t + 10e^{-\frac{t}{5}+1}$ satisfies the differential equation $\frac{dA}{dt} = t - \frac{1}{5}A$ with initial condition $A(5) = 10$.
- (c) The host receives an antiviral medication. The amount of medication in the host is modeled by the function M that satisfies the differential equation $\frac{dM}{dt} = -\frac{1}{2}\left(\frac{M}{t+k}\right)$, where M is measured in milligrams, t is measured in days since the host received the medication, and k is a positive constant. If the amount of medication in the host is 30 milligrams at time $t = 0$ days and 15 milligrams at time $t = 3$ days, what is $M(t)$ in terms of t ?

Part A

For the second point, it is incorrect to state $A(7) = 16$ rather than $A(7) \approx 16$.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ tangent line equation
- ☐ approximation

Solution:

$$A(5) = 10$$

4-6-6

$$\left. \frac{dA}{dt} \right|_{t=5} = 5 - \frac{1}{5} \cdot 10 = 3$$

An equation for the line tangent to the graph of A at $t = 5$ is $y = 10 + 3(t - 5)$.

At time $t = 7$ days, there are approximately $10 + 3(7 - 5) = 16$ million virus cells in the host. $A(7) \approx 16$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ verification of initial condition
- ☐ $\frac{dA}{dt} = 5 - 2e^{-\frac{t}{5}+1}$
- ☐ verification that $\frac{dA}{dt} = t - \frac{1}{5}A$

Solution:

$$A(5) = -25 + 5 \cdot 5 + 10e^{-1+1} = 10$$

$$\frac{dA}{dt} = \frac{d}{dt} \left(-25 + 5t + 10e^{-\frac{t}{5}+1} \right) = 5 + 10e^{-\frac{t}{5}+1} \left(-\frac{1}{5} \right) = 5 - 2e^{-\frac{t}{5}+1}$$

$$t - \frac{1}{5}A = t - \frac{1}{5} \left(-25 + 5t + 10e^{-\frac{t}{5}+1} \right) = t + 5 - t - 2e^{-\frac{t}{5}+1} = 5 - 2e^{-\frac{t}{5}+1}$$

Part C

Zero out of 4 points earned if no separation of variables.

At most 2 out of 4 points earned [1-1-0-0] if no constant of integration.

Both antiderivatives must be correct to earn the second point.

The fourth point requires an expression for $M(t)$. The domain of $M(t)$ is included with the solution; this is not a requirement to earn the fourth point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

4-6-6



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ separation of variables
- ☐ antiderivatives
- ☐ constant of integration and uses initial conditions
- ☐ $M(t)$

Solution:

$$\frac{dM}{dt} = -\frac{1}{2}\left(\frac{M}{t+k}\right)$$

$$\frac{dM}{M} = -\frac{1}{2}\left(\frac{dt}{t+k}\right)$$

$$\int \frac{1}{M} dM = -\frac{1}{2} \int \left(\frac{1}{t+k}\right) dt$$

$$\ln |M| = -\frac{1}{2} \ln |t+k| + C$$

From the context, M and t are nonnegative. It is given that $k > 0$.

$$\ln M = -\frac{1}{2} \ln(t+k) + C$$

$$\ln M = \ln \frac{1}{\sqrt{t+k}} + C$$

$$M = \frac{D}{\sqrt{t+k}}$$

$$30 = \frac{D}{\sqrt{k}} \Rightarrow D = 30\sqrt{k}$$

$$15 = \frac{D}{\sqrt{3+k}} \Rightarrow D = 15\sqrt{3+k}$$

$$30\sqrt{k} = 15\sqrt{3+k} \Rightarrow k = 1 \text{ and } D = 30$$

$$M(t) = \frac{30}{\sqrt{t+1}}$$

Note: This is valid for $t \geq 0$.

4-6-6

53. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

For time $0 \leq t \leq 8$, people arrive at a venue for an outdoor concert at a rate modeled by the function A defined by $A(t) = 0.3 \sin(1.9t) + 0.3 \cos(0.6t) + 1.3$. For time $0 \leq t \leq 1$, no one leaves the venue, and for time $1 \leq t \leq 8$, people leave the venue at a rate modeled by the function L defined by $L(t) = 0.2 \cos(1.9t) + 0.2 \sin t + 0.8$. Both $A(t)$ and $L(t)$ are measured in hundreds of people per hour, and t is measured in hours. The number of people at the venue, in hundreds, at time t hours is given by $P(t)$.

- (a) At time $t = 2$, there are 6 hundred people at the venue. Write an equation for the locally linear approximation of P at $t = 2$, and use it to approximate the number of people at the venue at time $t = 2.5$.
- (b) Find $P''(5)$. Using correct units, interpret the meaning of $P''(5)$ in the context of the problem.
- (c) Is there a time t , for $1 < t < 8$, at which the rate of change of the number of people at the venue changes from negative to positive? Give a reason for your answer.
- (d) A rectangular “standing-only” section at the venue changes size as t increases in order to manage the flow of people. Let x represent the length, in feet, of the section, and let y represent the width, in feet, of the section. The length of the section is increasing at a rate of 6 feet per hour, and the width of the section is decreasing at a rate of 3 feet per hour. What is the rate of change of the area of the section with respect to time when $x = 16$ and $y = 10$? Indicate units of measure.

Part A

The second point may be earned if response contains an incorrect value for slope based on one computational error.

The third point **does not** require a simplified answer. Substitution of function values is required. An answer of 6.201 (or 6.200) alone does not earn the point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-6-6



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ slope
- ☐ equation for tangent line
- ☐ approximation

Solution:

$$P(2) = 6$$

$$P'(2) = A(2) - L(2) = 0.401484$$

$$y = P(2) + P'(2)(t - 2) = 6 + 0.401484(t - 2)$$

$$P(2.5) \approx P(2) + P'(2)(2.5 - 2) = 6.201 \text{ (or 6.200) hundred people (} \approx 620 \text{ OR } \approx 621 \text{ people)}$$

Part B

The second point requires a time reference, rate, and units. The point may be earned with an incorrect approximation based on one computational error and provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $P''(5)$
- ☐ interpretation with units

Solution:

$$P''(5) = A'(5) - L'(5) = -0.679$$

At time $t = 5$ hours, the rate of change of the number of people at the stage is changing at a rate of -0.679 hundred people (≈ -68 people) per hour per hour.

4-6-6

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct answer with reason.

Solution:

$$P'(t) = A(t) - L(t) = 0 \text{ for } 1 < t < 8.$$

$$P'(t) = 0 \Rightarrow t = 5.811 \text{ (or 5.810), } t = 6.548 \text{ (or 6.547)}$$

Yes, there is such a time t , because $P'(t) < 0$ for $5.811 < t < 6.548$ and $P'(t) > 0$ for $6.548 < t < 8$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\frac{dA}{dt} = x \frac{dy}{dt} + y \frac{dx}{dt}$
- ☐ $\frac{dx}{dt} = 6$ and $\frac{dy}{dt} = -3$
- ☐ answer with units

Solution:

Let x represent the length of the section, in feet, and let y represent the width of the section, in feet.

The area of the rectangular section is $A = xy$.

$$\frac{dA}{dt} = x \frac{dy}{dt} + y \frac{dx}{dt}$$

$$\text{When } x = 16 \text{ and } y = 10, \frac{dA}{dt} = x \frac{dy}{dt} + y \frac{dx}{dt} = (16)(-3) + (10)(6) = 12.$$

Therefore, the rate of change of the area of the section with respect to time is 12 square feet per hour.

4-6-6

54. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

For time $0 \leq t \leq 10$, water is flowing into a small tub at a rate given by the function F defined by $F(t) = \arctan\left(\frac{\pi}{2} - \frac{t}{10}\right)$. For time $5 \leq t \leq 10$, water is leaking from the tub at a rate given by the function L defined by $L(t) = 0.03(20t - t^2 - 75)$. Both $F(t)$ and $L(t)$ are measured in cubic feet per minute, and t is measured in minutes. The volume of water in the tub, in cubic feet, at time t minutes is given by $W(t)$.

- (a) At time $t = 3$, there are 2.5 cubic feet of water in the tub. Write an equation for the locally linear approximation of W at $t = 3$, and use it to approximate the volume of water in the tub at time $t = 3.5$.
- (b) Find $W''(8)$. Using correct units, interpret the meaning of $W''(8)$ in the context of the problem.
- (c) Is there a time t , for $5 < t < 10$, at which the rate of change of the volume of water in the tub changes from positive to negative? Give a reason for your answer.
- (d) The tub is in the shape of a rectangular box that is 0.5 foot wide, 4 feet long, and 3 feet deep. What is the rate of change of the depth of the water in the tub at time $t = 6$?

Part A

The second point may be earned if response contains an incorrect value for slope based on one computational error.

The third point does not require a simplified answer. Substitution of function values is required. Units are not required.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-6-6



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ slope
- ☐ equation for tangent line
- ☐ approximation

Solution:

$$W(3) = 2.5$$

$$W'(3) = F(3) = 0.904089$$

$$y = W(3) + W'(3)(t - 3) = 2.5 + 0.904(t - 3)$$

$$W(3.5) \approx W(3) + W'(3)(3.5 - 3) = 2.952 \text{ cubic feet}$$

Part B

The second point requires a time reference, rate, and units. The point may be earned with an incorrect approximation based on one computational error and provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $W''(8)$
- ☐ interpretation with units

Solution:

$$W''(8) = F'(8) - L'(8) = -0.183 \text{ (or } -0.182)$$

At time $t = 8$ minutes, the rate at which the rate of change of the volume of water is changing by -0.183 (or -0.182) cubic feet per minute per minute.

4-6-6

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct answer with reason.

Solution:

$$W'(t) = F(t) - L(t) = 0 \text{ for } 5 < t < 10.$$

$$W'(t) = 0 \Rightarrow t = 8.149287$$

Yes, there is such a time t , because $W'(t) > 0$ for $5 < t < 8.149$ and $W'(t) < 0$ for $8.149 < t < 10$.

Part D

The first point is earned with $\frac{dV}{dt} = 2\frac{dh}{dt}$.

The third point may be earned with a maximum of one error in the evaluation of $\frac{dV}{dt}$.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ relationship between $\frac{dV}{dt}$ and $\frac{dh}{dt}$
- ☐ $\frac{dV}{dt} \big|_{t=6}$
- ☐ answer

The volume of water in the tub is $V = (0.5)(4)h = 2h$, where h is the depth of water, in feet.

$$\frac{dV}{dt} = 2\frac{dh}{dt}$$

$$\text{At time } t = 6, \frac{dV}{dt} \big|_{t=6} = W'(t) = F(6) - L(6) = 0.500581.$$

4-6-6

Therefore, $\left. \frac{dh}{dt} \right|_{t=6} = \frac{1}{2} \frac{dV}{dt} = 0.250$

4-6-6

55. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	2	4	6	8	10	12
$R(t)$ (vehicles per hour)	2935	3653	3442	3010	3604	1986	2201

On a certain weekday, the rate at which vehicles cross a bridge is modeled by the differentiable function R for $0 \leq t \leq 12$, where $R(t)$ is measured in vehicles per hour and t is the number of hours since 7:00 A.M. ($t = 0$). Values of $R(t)$ for selected values of t are given in the table above.

Use the data in the table to approximate $R'(5)$. Show the computations that lead to your answer. Using correct units, explain the meaning of $R'(5)$ in the context of the problem.

Use a midpoint sum with three subintervals of equal length indicated by the data in the table to approximate the value of $\int_0^{12} R(t) \, dt$. Indicate units of measure.

On a certain weekend day, the rate at which vehicles cross the bridge is modeled by the function H defined by $H(t) = -t^3 - 3t^2 + 288t + 1300$ for $0 \leq t \leq 17$, where $H(t)$ is measured in vehicles per hour and t is the number of hours since 7:00 A.M. ($t = 0$). According to this model, what is the average number of vehicles crossing the bridge per hour on the weekend day for $0 \leq t \leq 12$?

For $12 < t < 17$, $L(t)$, the local linear approximation to the function H given in part (c) at $t = 12$, is a better model for the rate at which vehicles cross the bridge on the weekend day. Use $L(t)$ to find the time t , for $12 < t < 17$, at which the rate of vehicles crossing the bridge is 2000 vehicles per hour. Show the work that leads to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-6-6

The first point does not require a simplified answer. Substitution of function values is required.

The second point requires a time reference, a rate, and units. The point may be earned with an incorrect approximation based on one computational error and provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ approximation using values from table
- ☐ interpretation with units

Solution:

$$R'(5) \approx \frac{R(6) - R(4)}{6 - 4} = \frac{3010 - 3442}{2} = -216$$

At time $t = 5$ hours (12 p.m.), the rate at which vehicles cross the bridge is decreasing at a rate of approximately 216 vehicles per hour per hour.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

- The first point does not require a simplified answer. Substitution of function values is required. A single substitution error for one and only one of $R(2)$, $R(6)$, or $R(10)$ will still earn this point.

The second point requires earning the first point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ midpoint sum setup
- ☐ approximation using values from table with units

Solution:

4-6-6

$$\begin{aligned}\int_0^{12} R(t) \, dt &\approx 4(R(2) + R(6) + R(10)) \\ &= 4(3653 + 3010 + 1986) \\ &= 34,596 \text{ vehicles}\end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct integrand and limits of integration. The constant $\frac{1}{12}$ is not needed to earn this point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ definite integral
- ☐ answer

Solution:

$$\frac{1}{12-0} \int_0^{12} H(t) \, dt = 2452$$

On the weekend day for $0 \leq t \leq 12$, the average number of vehicles crossing the bridge is 2452 vehicles per hour.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ slope
- ☐ $L(t) = 2000$
- ☐ answer

4-6-6**Solution:**

$$L(t) = H(12) + H'(12)(t - 12)$$

$$H(12) = 2596$$

$$H'(12) = -216$$

$$L(t) = 2000 \Rightarrow t = 14.759259$$

On the weekend day, the rate at which vehicles cross the bridge is 2000 vehicles per hour at time $t = 14.759$ hours.

4-6-6

56. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	2	4	5	8	10	14
$h'(t)$ (feet per hour)	0.3	0.6	0.7	1.4	2.3	7.7

The height of water in a storage tank during a 15-hour period is given by a twice-differentiable function h , where $h(t)$ is measured in feet and t is measured in hours since midnight for $0 \leq t \leq 15$. The graph of h is concave up on the interval $0 \leq t \leq 15$. Selected values of the derivative of h , $h'(t)$, are given in the table above. At time $t = 2$, the height of the water is 7 feet.

- (a) Use the tangent line approximation for h at time $t = 2$ to estimate $h(2.8)$, the height of the water at time $t = 2.8$. Is the approximation an overestimate or an underestimate for $h(2.8)$? Give a reason for your answer.
- (b) Use a right Riemann sum with the five subintervals indicated by the data in the table to approximate $\int_2^{14} h'(t) \, dt$. Indicate units of measure.
- (c) Is the approximation in part (b) an overestimate or underestimate for $\int_2^{14} h'(t) \, dt$? Give a reason for your answer.
- (d) The sum $\sum_{k=1}^n h'\left(\frac{15}{n}(k-1)\right) \frac{15}{n}$ is a left Riemann sum with n subintervals of equal length. The limit of this sum as n goes to infinity can be interpreted as a definite integral. Express the limit as a definite integral.

Part A

For the first point, the approximation must show the form of a correct tangent line equation and evaluation of that equation at $t = 2.8$.

Numerical answers do not need to be simplified, and units are not required to earn the approximation point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-6-6



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ approximation
- ☐ underestimate with reason

Solution:

$$h(2.8) \approx h(2) + h'(2)(2.8 - 2) = 7 + 0.3(0.8) = 7.24 \text{ feet}$$

Since the graph of h is concave up on $2 \leq t \leq 2.8$, the tangent line lies under the graph of h on this interval. Therefore, the approximation is an underestimate for $h(2.8)$.

Part B

To earn the first point, the response must present the form of a right Riemann sum.

To earn the second point, values for h' must be substituted into the sum. Therefore, $2 \cdot 0.6 + 1 \cdot 0.7 + 3 \cdot 1.4 + 2 \cdot 2.3 + 4 \cdot 7.7$ earns the first and second points.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ right Riemann sum
- ☐ approximation
- ☐ units

Solution:

$$\begin{aligned} \int_2^{14} h'(t) dt &\approx (4 - 2)h'(4) + (5 - 4)h'(5) + (8 - 5)h'(8) + (10 - 8)h'(10) + (14 - 10)h'(14) \\ &= 2 \cdot 0.6 + 1 \cdot 0.7 + 3 \cdot 1.4 + 2 \cdot 2.3 + 4 \cdot 7.7 = 41.5 \text{ feet} \end{aligned}$$

Part C

4-6-6

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ overestimate with reason that references the behavior of h'

Solution:

Since the graph of h is concave up on $0 \leq t \leq 15$, h' is increasing on $2 \leq t \leq 14$. Therefore, the right Riemann sum approximation is an overestimate for $\int_2^{14} h'(t) dt$.

Part D

The first point may be earned as presented in the solution or by using that information to determine that the interval of integration is of length 15. A response that produces the correct definite integral earns all 3 points.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\Delta t = \frac{15}{n}$
- ☐ limits of integration
- ☐ integrand

Solution:

For this Riemann sum, $\Delta t = \frac{15}{n}$.

Thus, the integral is over an interval of length 15. This interval can be taken to be $0 \leq t \leq 15$.

The left endpoints of the subintervals used in the Riemann sum would be of the form $t_k = \frac{15}{n}(k-1)$ for k from 1 to n .

4-6-6

Therefore, $\lim_{n \rightarrow \infty} \left(\sum_{k=1}^n h' \left(\frac{15}{n}(k-1) \right) \frac{15}{n} \right) = \int_0^{15} h'(t) \, dt.$

Note: The limit of the Riemann sum can be written as any definite integral of the form $\int_a^{a+15} h'(t-a) \, dt.$

4-6-6

57. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (weeks)	2	4	5	8	10
$L'(t)$ (feet per week)	1.0	0.8	0.7	0.4	0.2

The length of a vine during a 12-hour period is given by a twice-differentiable function L , where $L(t)$ is measured in feet and t is measured in weeks for $0 \leq t \leq 12$. The graph of L is concave down on the interval $0 \leq t \leq 12$. Selected values of the derivative of L , $L'(t)$, are given in the table above. At time $t = 4$, the length of the vine is 5 feet.

(a) Use the tangent line approximation for L at time $t = 4$ to estimate $L(4.3)$, the length of the vine at time $t = 4.3$. Is the approximation an overestimate or an underestimate for $L(4.3)$? Give a reason for your answer.

(b) Use a left Riemann sum with the four subintervals indicated by the data in the table to approximate $\int_2^{10} L'(t) \, dt$. Indicate units of measure.

(c) Is the approximation in part (b) an overestimate or an underestimate for $\int_2^{10} L'(t) \, dt$? Give a reason for your answer.

(d) The sum $\sum_{k=1}^n L'\left(\frac{10k}{n}\right) \frac{10}{n}$ is a right Riemann sum with n subintervals of equal length. The limit of this sum as n goes to infinity can be interpreted as a definite integral. Express the limit as a definite integral.

Part A

For the first point, the approximation must show the form of a correct tangent line equation and evaluation of that equation at $t = 4.3$. Numerical answers do not need to be simplified, and units are not required to earn the approximation point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-6-6



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ approximation
- ☐ underestimate with reason

Solution:

$$L(4.3) \approx L(4) + L'(4)(4.3 - 4) = 5 + 0.8(0.3) = 5.24 \text{ feet}$$

Since the graph of L is concave down on $4 \leq t \leq 4.3$, the tangent line lies above the graph of L on this interval. Therefore, the approximation is an overestimate for $L(4.3)$.

Part B

To earn the first point, the response must present the form of a left Riemann sum. To earn the second point, values for L' must be substituted into the sum. Therefore, $2 \cdot 1.0 + 1 \cdot 0.8 + 3 \cdot 0.7 + 2 \cdot 0.4$ earns the first and second points. Numerical answers do not need to be simplified for the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ left Riemann sum
- ☐ approximation
- ☐ units

Solution:

$$\begin{aligned} \int_2^{10} L'(t) dt &\approx (4 - 2)L'(2) + (5 - 4)L'(4) + (8 - 5)L'(5) + (10 - 8)L'(8) \\ &= 2 \cdot 1.0 + 1 \cdot 0.8 + 3 \cdot 0.7 + 2 \cdot 0.4 = 5.7 \text{ feet} \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

4-6-6



0	1
---	---

The student response accurately includes overestimate with reason that references the behavior of L'

Solution:

Since the graph of L is concave down on $0 \leq t \leq 12$, L' is decreasing on $2 \leq t \leq 10$. Therefore, the left Riemann sum approximation is an overestimate for $\int_2^{10} L'(t) dt$.

Part D

The first point may be earned as presented in the solution or by using that information to determine that the interval of integration is of length 10. A response that produces the correct definite integral earns all 3 points.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\Delta t = \frac{10}{n}$
- ☐ limits of integration
- ☐ integrand

Solution:

For this Riemann sum, $\Delta t = \frac{10}{n}$. Thus, the integral is over an interval of length 10. This interval can be taken to be $0 \leq t \leq 10$.

The right endpoints of the subintervals used in the Riemann sum would be of the form $t_k = \frac{10k}{n}$ for k from 1 to n .

Therefore, $\lim_{n \rightarrow \infty} \left(\sum_{k=1}^n L' \left(\frac{10k}{n} \right) \frac{10}{n} \right) = \int_0^{10} L'(t) dt$.

Note: The limit of the Riemann sum can be written as any definite integral of the form $\int_a^{a+10} L'(t-a) dt$.

4-6-6

58. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	6	13	18
$M(t)$ (coins per day)	42	38	35	30

A class collected 1-cent coins to raise money for classroom supplies. Students in the class deposited the coins into a container where all the coins were kept until the end of the collection period. For time t days, $0 \leq t \leq 18$, the rate at which coins are being added to the container is modeled by a decreasing, twice-differentiable function M . Values of $M(t)$ at various times t are shown in the table above.

(a) Use a right Riemann sum with the three subintervals indicated by the table to approximate

$\int_0^{18} M(t) \, dt$. Show the computations that lead to your answer.

(b) Is the approximation from part (a) an overestimate or an underestimate for $\int_0^{18} M(t) \, dt$? Give a reason for your answer.

(c) Using correct units, approximate $M'(13)$ using the average rate of change of M over the interval $6 \leq t \leq 18$.

(d) Explain the meaning of $M'(13)$ in the context of the problem.

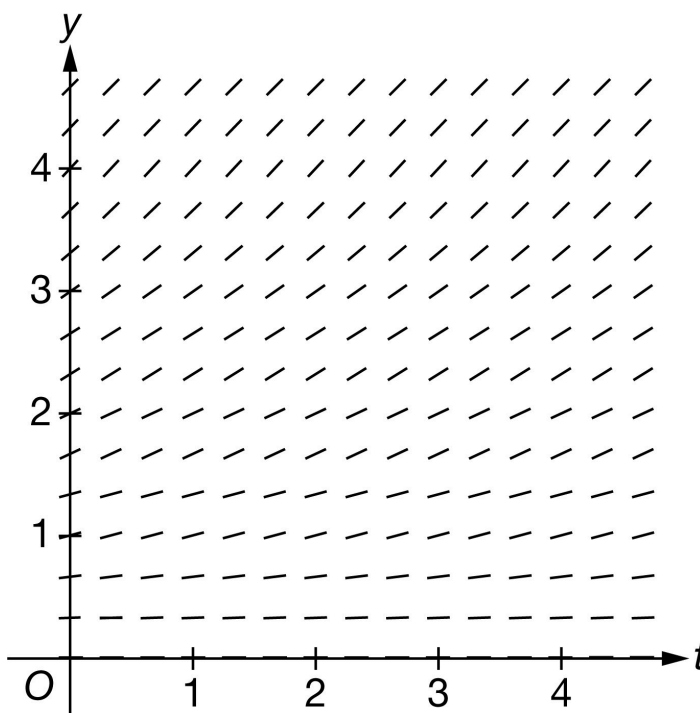
(e) It is known that $M'(6) = -0.5$. Write an equation for the line tangent to the graph of $y = M(t)$ at $t = 6$.

(f) Find the value of $\int_0^{18} M'\left(\frac{t}{3}\right) \, dt$. Show the computations that lead to your answer.

(g) A second class is also collecting 1-cent coins. For $0 \leq t \leq 18$, the total number of coins collected by the second class can be modeled by the decreasing and differentiable function G , where G is a function of t . Assume the weight of each coin is 3 grams. Write an expression for the rate of change, in grams per day, of the total weight of the coins for the second class at $t = 8$ days.

4-6-6

(h) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = G'(t)$ for the function G defined in part (g).

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a right Riemann sum. The setup must be a sum of three products.

A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example below).

Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns, like so.

38	6	228
35	7	245
30	5	+150
		623

or

4-6-6

$(6 - 0)(38)$	$= 228$
$(13 - 6)(35)$	$= 245$
$(18 - 13)(30)$	$= 150$
	623

Implicit responses such as those above are eligible for both points.

A response presenting a left Riemann sum (instead of right), will not earn any points for part (a).

The second point is for the correct numerical result.

In order to earn the second point, there must be supporting calculations.

The numerical values for $M(6)$, $M(13)$, and $M(18)$ must be pulled from the table

Any simplification shown must be correct.

Use the following as an illustration how to score both points. (In most cases, you see scores for which the expression represents the entirety of relevant work shown for the response.)

- A response of $38 \cdot (6 - 0) + 35 \cdot (13 - 6) + 30 \cdot (18 - 13)$ earns both points.
- A response of $6 \cdot M(6) + 7 \cdot M(13) + 5 \cdot M(18)$ earns the first point. Eligible for second point if more work is on the page.
- A response of $6 \cdot M(6) + 7 \cdot M(13) + 5 \cdot M(18) = 623$ earns both points.
- A response of $6 \cdot 38 + 7 \cdot 35 + 5 \cdot 30$ earns both points.
- A response of $6 \cdot 38 + 7 \cdot 35 + 6 \cdot 30$ contains one numerical error. Earns the first point. Not eligible for the second point.
- A response of $6 \cdot 38 + 7 \cdot 35 + 5 \cdot 30 = 641$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.
- A response of $228 + 245 + 150$ or $228 + 245 + 150 = 623$ does not earn the first point because we cannot see the sum of three products. However, this is the correct sum, so we are not seeing an unsupported answer. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points. $6 \cdot 38 + 7 \cdot 35 + 5 \cdot 30$



0	1	2
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4-6-6

The student response accurately includes **both** of the criteria below.

- ☐ Right Riemann sum
- ☐ Answer

Solution:

$$\int_0^{18} M(t) dt \approx M(6)(6-0) + M(13)(13-6) + M(18)(18-13)$$

$$= 38 \cdot 6 + 35 \cdot 7 + 30 \cdot 5 = 623$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an underestimate because M is decreasing.

The response does not need an explicit reference to part (a) or to a right Riemann sum.

A response that appeals to concavity as a reason will NOT earn this point.

The following examples earn the point:

- Underestimate because M is decreasing
- Underestimate because $M'(t) < 0$
- Underestimate because the graph of M is decreasing (we know they are talking about the given function M)
- Underestimate because the function is decreasing. (There is only one function in the problem at this point.)

The following examples do not earn the point:

- Underestimate because M is concave down/up
- Underestimate because the graph/curve is decreasing. Since no graph/curve is provided, we do not know what graph/curve they are talking about.
- Underestimate because the table [or table values] is/are decreasing. The behavior of the values shown in the table is not the same as the behavior of the function M . The table does not tell us what is happening to M between the given points.



0	1
---	---

4-6-6

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because M is a decreasing function, and this is a right Riemann sum, this is an underestimate.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

We need to see both a difference and a quotient. They don't have to be together.

The expression does not need to be simplified, but values for $M(18)$ and $M(6)$ must be pulled from the table.

Decimal approximation is not expected, but if presented, then the decimal value must be correct to three decimal places, either rounded or truncated. We do not read anything past the third decimal place.

- If the first three places match -0.666 or -0.667 then the point is earned.
- It does not matter whether any subsequent decimal places match the correct value or not. (For example, the value -0.6672 will earn the point because the first three decimal places match the correct rounded value.)
- Responses with fewer than three decimal places (-0.66 , -0.67 , -0.7 , etc.) will not earn the point

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

The second point is for correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols. For example:

- Coins per day per day or Coins/day/day
- Coins per day² or Coins/day²
- Coins per day squared

We do not read work shown in part (d) for this point. The units must appear in part (c).

The letter “C” is not a standard abbreviation for “coins.” It will not be interpreted as such.

The response will not be penalized for any attempt to verbally describe their answer after a correct answer has been presented.

4-6-6

Minimalist response: The following expression and units represent the minimum work that must be seen in a response in order to earn both points. $\frac{30-38}{12}$ coins/day²



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$M'(13) \approx \frac{M(18) - M(6)}{18 - 6} = \frac{30 - 38}{12} = -\frac{2}{3} \text{ coin per day per day}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

- Indicating a rate of a rate
- Including the context of coin collection
- Explicitly referring to the specified point in time ($t = 13$)

Incorrect or missing units will not be penalized as long as there is a clear connection to the context of the problem.

The words “increasing” and “decreasing” are interpreted as modifiers describing one of the specified rates.

If the numerical value from part (c) is used in this interpretation, then we expect that the choice of “increasing” or “decreasing” is consistent with that value.

The following examples use $-\frac{2}{3}$ as the imported value from part (c). If a different [incorrect] value is imported from part (c), then these statements should be consistent with that value. (Remember that we are reading $=$ and $\approx$ as the same thing.)

- “... increasing at a rate of $-\frac{2}{3}$ ” is OK [Units not required in part (d)]. (the point is not yet earned with this phrase alone)
- “... decreasing at a rate of $\frac{2}{3}$ ” is OK [Units not required in part (d)]. (the point is not yet earned with this phrase alone)

4-6-6

- “... decreasing at a rate of $-\frac{2}{3}$ ” is not OK and the point is not earned.

Rate of a rate

To earn this point, the response must explicitly discuss the rate of change of a rate of change. For example:

- The rate of change in the rate at which coins are changing
- The rate of change in coins is changing (or decreasing) at a rate of . . .
- The rate at which the rate of change of coins is changing

The words “increasing,” “decreasing,” or “changing” are not interpreted as rates.

Units are not interpreted as rates. (The phrase “coins per day” is not enough to indicate a rate.)

Slope of tangent line, velocity, and acceleration are not appropriate interpretations for this context.

“Rate of change of M ” is not enough without describing what M represents in this context.

Context

A key component of the context for this explanation is “coins.”

Explicit reference to $t = 13$

Acceptable reference to specific point in time include (but are not limited to):

- On day 13 (where day could be replaced with any other unit of time)
- At $t = 13$ (with or without units of time)
- After 13 days (where days could be replaced with any other unit of time).

Note: we are choosing to assume that “after” is being used in lieu of “at” unless we have additional evidence to the contrary.

If the response is explicitly indicating an interval of time (e.g., “from day 13 on” or “during the first 13 days”), this point will not be earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

4-6-6

$M'(13)$ is the rate at which the rate of change of the number of 1-cent coins in the container is changing, in coins per day per day, on day 13.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. Any of these equations can be written as is without any explanation or additional supporting work. If rewriting of equations is done (transforming to slope-intercept form, perhaps), it must be done correctly.

· $y = -0.5(t - 6) + 38$

· $y - 38 = -0.5(t - 6)$

· $y = -0.5t + 41$

The key calculus concept being assessed in this part of the problem is that the value of $M'(6)$ is the slope of the tangent line. In that spirit:

- The response will not be penalized for using x instead of t .
- We are accepting $M(t)$ in lieu of y in all forms of the tangent line equation.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = M'(6)(t - 6) + M(6)$$

$$y = -0.5(t - 6) + 38$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are three components of substitution that we need to see:

- Uses $u = \frac{t}{3}$, which may be done implicitly
- Uses the correct constant of 3

4-6-6

· Uses the correct limits for the substitution (0 and 6). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.

Either of the expressions $3M\left(\frac{t}{3}\right)\big|_0^{18}$ or $3M(u)\big|_0^6$ will earn the substitution point without any additional supporting work.

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point.

- Limits for t attached to an integral with u as the variable of integration (or vice versa)
- Improper equals signs, such as an indefinite integral equal to a definite integral
- Reversed limits of integration, for example: $\int_6^0 M'(u) du$

The second point is for the Fundamental Theorem of Calculus.

This point can be earned with any of the following, where k is a nonzero constant and $b = 6$ or $b = 18$. No explanation or additional supporting work is necessary.

- o $kM\left(\frac{t}{3}\right)\big|_0^b$
- o $k(M(b) - M(0))$

This point can be earned without earning the substitution point.

The third point is for the correct numerical value, which is -12 .

Values from the table must be used. $3(M(6) - M(0))$ is not enough to earn the answer point.

The expression $3(38 - 42)$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they need not be correct in order to earn this point.

The response cannot earn this point without also earning first two points.

Minimalist response: The expression $3(38 - 42)$ represents the minimum work that must be seen in a response in order to earn all three points.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

4-6-6

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = \frac{t}{3}$. Then $du = \frac{dt}{3}$.

When $t = 0$, $u = 0$.

When $t = 18$, $u = 6$.

$$\begin{aligned} \int_0^{18} M' \left(\frac{t}{3} \right) dt &= 3 \int_0^6 M' (u) du \\ &= 3 (M (6) - M (0)) \\ &= 3 (38 - 42) = -12 \end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression at $t = 8$ earns the point. It may be written as $3 \cdot G'(8)$ or $3 \cdot \frac{dG}{dt} \big|_{t=8}$.

The expression $3 \cdot G'(t)$ is not sufficient because it does not use $t = 8$.

The response must use G (not M or any other function name) in order to earn the point.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Expression

Solution:

An expression for the rate of change, in grams per day, of the total weight of the coins for the second class at $t = 8$ days is $3G'(8)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-6-6

This point is earned for a correct statement identifying the distinction between the differential equation

$$\frac{dy}{dt} = G'(t) \text{ and the given slope field.}$$

In order to earn the point, the response must state that the values of the differential equation are negative while the slopes in the given slope field are positive or 0 (nonnegative).

Some ways to say that the values of differential equation are negative:

- G is decreasing
- $G'(t) < 0$
- G has negative slopes

When discussing the slopes in the slope field, the response must explicitly refer to the signs of the slopes.

The following examples could earn the point:

- The slope field has nonnegative slopes.
- The slopes shown are positive or 0.
- The slope field contains positive slopes.
- “The graph” or “The field” could be used in lieu of “The slope field.”

The following examples will not earn the point:

- The slope field is increasing/positive.
- The slope field looks like it represents an increasing function.
- The slope field represents a quadratic [or exponential] function.
- “It” (rather than “the slope field”)

Statements like “ G is a decreasing function, so the signs of the slopes in the slope field should be negative” can earn this point because “should be” is indicating contrast with what is known.

The response will not be penalized for saying that the slopes in the slope field are “always” positive or “all” positive.



0	1
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The student response accurately includes the criteria below.

☐ Explanation

4-6-6**Solution:**

G is a decreasing function, so $\frac{dy}{dt} = G'(t) \leq 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are positive.

4-6-6

59. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

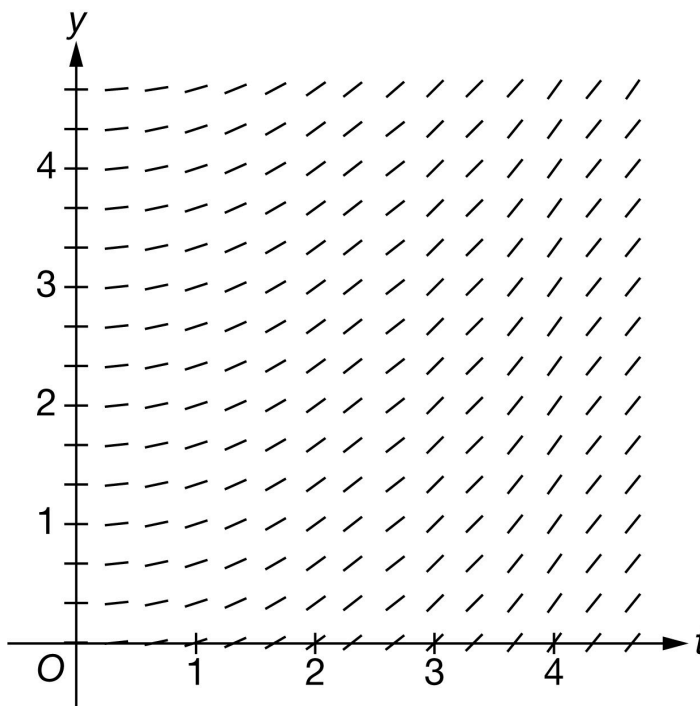
t (days)	0	6	18
$M(t)$ (coins per day)	40	54	46

A class collected 1-cent coins to raise money for classroom supplies. Students in the class deposited the coins into a container where all the coins were kept until the end of the collection period. For time t days, $0 \leq t \leq 18$, the rate at which coins are being added to the container, in coins per day, is modeled by a twice-differentiable function M . Values of $M(t)$ at various times t are shown in the table above.

- (a) Using correct units, approximate $M'(6)$ using the average rate of change of M over the interval $0 \leq t \leq 18$.
- (b) Explain the meaning of $M'(6)$ in the context of the problem.
- (c) The function $C(t)$ models the number of coins in the container at time t days, where $C'(t) = M(t)$. It is known that $C(6) = 300$. Write an equation for the line tangent to the graph of $y = C(t)$ at $t = 6$.
- (d) Find the value of $\int_0^6 M'(3t) \, dt$. Show the calculations that lead to your answer.
- (e) Use a trapezoidal sum with the two subintervals indicated by the table to approximate $\int_0^{18} M(t) \, dt$. Show the calculations that lead to your answer.
- (f) A second class is also collecting 1-cent coins. For $0 \leq t \leq 18$, the rate at which coins are being collected by the second class can be modeled by the differentiable function B , where B is a function of t . Assume the weight of each coin is 3 grams. Write an expression in terms of M and B for the rate of change, in grams per day, of the total weight of the coins collected by both classes at $t = 7$ days.

4-6-6

(g) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = 0.3y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

Answers without supporting work do not earn this point. Supporting work must be at least a difference and a quotient.

Response does not need to be simplified, but any simplification must be done correctly.

If a decimal is presented, it must be correct to three decimal places. We do not read anything past the third decimal place.

The second point is earned for the correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols.

The response will not be penalized for any verbal attempt to discuss the answer after the correct presentation of the approximation.

The minimum response needed to earn both points is: $\frac{46-40}{18}$ coins per day per day.

4-6-6



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$M'(6) \approx \frac{M(18) - M(0)}{18 - 0} = \frac{46 - 40}{18} = \frac{1}{3} \text{ coin per day per day}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This part is intended to be independent from part (a), so we do not want or need to see any reference to the value obtained in (a). If that value is referenced, conclusion must be consistent.

Three components of a correct answer:

- Indicating a rate of a rate (look for the word rate to be used correctly twice)
- Including the context of coin collection
- Referring to the specific point in time

Units will not be considered for this point. They do not need to be present, and if they are, they do not need to be correct.

Any incorrect statements make the response ineligible for the point. For example:

- $M'(6)$ is the average (or an approximation of, or an estimate of) rate of change of
- $M'(6)$ is the acceleration (or rate of change of velocity) of

“Increasing” or “decreasing” may be used in place of “changing” as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Beware of phrases such as “decreasing at a negative rate.”

- Example: $M'(6)$ is the rate of increase of the rate of change of the number of coins at 6 days. (earns the point)
- Example: $M'(6)$ means the number of coins per day is increasing by 6 coins/day/day. (does not earn the point)

Acceptable references to time include (but are not limited to):

- On day 6

4-6-6

- At $t = 6$ (with correct units of time, without correct units of time, or even with incorrect units of time)
- After $t = 6$ (with correct units of time, without correct units of time, or even with incorrect units of time)

If the response clearly indicates an interval of time (e.g. “from day 0 to day 6”) the response will not earn the point.

Examples that earn the point include:

- $M'(6)$ represents the rate at which the rate of change of the number of coins is changing on day 6.
- $M'(6)$ is the rate of change in the rate of change of the number of coins at 6 days.
- $M'(6)$ represents the rate at which the rate of change of the number of coins is increasing at $t = 6$.
- $M'(6)$ equals the answer from part (a) means the rate of change in the number of coins per day is increasing at a rate of (answer from part (a)) at $t = 6$.
- $M'(6)$ represents the rate at which the rate of change of the number of coins is decreasing on the sixth day.

Examples that do not earn the point include:

- $M'(6)$ represents the rate at which the number of coins is changing on day 6.
- $M'(6)$ is the rate of change in the number of coins per day at $t = 6$ days
- The rate of change in the number of coins per day is increasing at $t = 6$.
- The rate of change in the rate of change of the number of coins per day is increasing at $t = 6$.
- $M'(6) = -\frac{1}{3}$ means that the rate of change in the number of coins is decreasing at a rate of $-\frac{1}{3}$ at $t = 6$. (double negative).



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$M'(6)$ is the rate at which the rate of change of the number of 1-cent coins in the container is changing, in coins per day per day, on day 6.

Part C

4-6-6

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned with a correct tangent line equation in any form. Equivalent forms include (but are not limited to):

- $y = 54(t - 6) + 300$

- $y - 300 = 54(t - 6)$

- $y = 54t - 24$

The student may write x in place of t and still earn the point.

No work is required to earn the point.

We will accept $C(t)$ in lieu of y in the equation of the tangent line.



0	1
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The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = C'(6)(t - 6) + C(6)$$

$$y = 54(t - 6) + 300$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution.

Two things are required to award this point:

- Coefficient of $\frac{1}{3}$ correctly applied to an integral or an antiderivative
- Evidence that the correct limits of 0 to 18 will be or have been applied

Examples that will earn the first point (these are just a few):

- $\frac{1}{3} \int_0^{18} M'(u) \, du$

4-6-6

$$\cdot \frac{1}{3}M(u)\big|_0^{18} \text{ or } \frac{1}{3}M(3t)\big|_0^6$$

Responses that include misstatements such as using t -values for limits with u as the variable of integration in an intermediate step which is later corrected may still earn this point. Intermediate errors in bounds that are subsequently corrected will disqualify the student from earning the third point. If the bounds are never appropriately handled, the response cannot earn the first point. Scratch work that is not connected to the solution will not be read.

Examples:

$$\cdot \int_0^6 M'(3t) dt = \frac{1}{3} \int_0^6 M'(u) du = \frac{1}{3}M(3t)\big|_0^6 \text{ earns the first point, but will not earn the third point.}$$

$$\cdot \int_0^6 M'(3t) dt = \frac{1}{3} \int_0^6 M'(u) du = \frac{1}{3}[M(6) - M(0)] \text{ does not earn the first point.}$$

The second point is for the Fundamental Theorem of Calculus, demonstrating the knowledge that

$$\int_a^b M'(u) du = M(u)\big|_a^b = M(b) - M(a)$$

This point can be earned with either $kM(3t)\big|_a^b$, $kM(t)\big|_a^b$, $k(M(b) - M(a))$ where k is a nonzero constant [not necessarily correct], $a = 0$, and $b = 6$ or $b = 18$.

A response can earn this point without earning the first point.

$$\frac{1}{3}[M(18) - M(0)] \text{ earns the first two points.}$$

The third point is for the correct numerical value. Our answer only.

Values from the table must be used (i.e., $\frac{1}{3}[M(18) - M(0)]$ is not sufficient to earn the third point).

If the response includes any linkage error, (i.e., the use of equal signs to equate two expressions that are not equal), the third point will not be earned. If an indefinite integral is linked with a definite integral the response will not be penalized.

$$\cdot \int_0^6 M'(3t) dt = \frac{1}{3} \int M'(u) du = \frac{1}{3}M(3t)\big|_0^6 = \dots \text{ is eligible for the third point.}$$

Eligibility: Must have earned the first two points.

The minimum response needed to earn all three points is: $\frac{1}{3}(46 - 40)$.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

4-6-6

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 3t$. Then $du = 3 dt$.

When $t = 0$, $u = 0$.

When $t = 6$, $u = 18$.

$$\begin{aligned}\int_0^6 M'(3t) dt &= \frac{1}{3} \int_0^{18} M'(u) du \\ &= \frac{1}{3}(M(18) - M(0)) \\ &= \frac{1}{3}(46 - 40) = 2\end{aligned}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the form of a trapezoidal sum.

Point is earned for the setup of the trapezoidal sum. Numerical values do not need to be pulled from the table for this point, but Δt values must be presented. Responses that contain one incorrect numerical value in the trapezoidal sum setup are still eligible to earn the first point, but will not earn the second.

Response may present a left Riemann sum and a right Riemann sum, then average the two. Again, one incorrect numerical value will not disqualify the response from earning this first point.

Response may compute trapezoids as areas of rectangles and triangles.

Trapezoidal areas may be given in a graph. In order to earn this point, the response must clearly show the computations leading to the values presented in the graph, and must show that these areas are added to obtain the final approximation value.

A response using subtraction instead of addition will not earn the point (e.g., $\frac{1}{2}[6(M(0) - M(6)) + 12(M(6) - M(18))]$ earns no points).

Response utilizing “Trapezoidal Rule” will not earn this point, unless this represents only one incorrect numerical values.

$\cdot \frac{1}{2}(6)[M(0) + 2M(6) + M(18)]$ is equivalent to $\frac{1}{2}[6(M(0) + M(6)) + 6(M(6) + M(18))]$, thus has only one incorrect numerical value (the second interval width is 6 instead of 12), so can earn the point.

4-6-6

$\cdot \frac{1}{2}(9) [M(0) + 2M(6) + M(18)]$ has two incorrect numerical values (both interval widths are incorrect) so will not earn the point.

The second point is for the correct numerical result.

Our answer only. Response does not need to be simplified, but if it is, it must be simplified correctly.

The answer must be accompanied with supporting work. Supporting work may appear in a graph, with clearly labeled areas. It is possible to earn the second point without the first if the supporting work is not sufficient to earn the first point.

Pictures without clear area computations and a clear summation of quantities can earn at most the second point.

Special case: A positive constant multiple of a complete and correctly expressed trapezoidal sum (with the correct values pulled from the table) will earn only the first point:

$\cdot k \left[\left(\frac{40+54}{2} \right) \cdot 6 + \left(\frac{54+46}{2} \cdot 12 \right) \right]$ for $k > 0$ earns the first but not the second point.

The minimum response needed to earn both points: $\left(\frac{94}{2} \right) \cdot 6 + \left(\frac{100}{2} \right) \cdot 12$



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Form of trapezoidal sum
- ☐ Answer

Solution:

$$\begin{aligned} \int_0^{18} M(t) dt &\approx \left(\frac{M(0)+M(6)}{2} \right) (6-0) + \left(\frac{M(6)+M(18)}{2} \right) (18-6) \\ &= \left(\frac{40+54}{2} \right) \cdot 6 + \left(\frac{54+46}{2} \right) \cdot 12 = 882 \end{aligned}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for a correct constant multiple or correct sum.

Response must contain the constant 3 as a factor of the expression, and/or the correct sum $B(7) + M(7)$.

If the sum is incorrect, then the constant factor of 3 must be applied either to a sum that involves only $B(t)$ and $M(t)$, $B'(t)$ and $M'(t)$, or integrals of $B(t)$ and $M(t)$, or to one of the correct terms of the sum $B(7)$ and

4-6-6

$M(7)$, with no additional incorrect terms.

Examples that earn the point include:

· $3[M(t) + B(t)]$

· $3 \int [M(t) + B(t)] dt$

· $3[M' + B']$

· $3B(7)$

Examples that do not earn the point include:

· $3[M(t) + B'(t)]$

· $3 \int M(t) dt$

· $3M' + B'$

· $3B(7) + t$

Special case: $3M(7) + B(7)$ is an error in parentheses, so will earn the first point but is not eligible to earn the second point.

If the constant multiple is missing or incorrect, the sum must be correct. $k[B(7) + M(7)]$ for any nonzero constant k will earn the first point.

The second point is for the correct expression.

$3B(7) + 3M(7)$ (or an equivalent expression) earns both points.

Inclusion of units (correct or incorrect) will not affect the score.

A completely correct expression with one additional constant term added/subtracted would earn just the first point. For example, $3B(7) + 3M(7) + 3(2)$ earns only the first point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Constant multiple or sum
- ☐ Expression

Solution:

4-6-6

An expression for the rate of change, in grams per day, of the total weight of the coins collected by both classes at time $t = 7$ days is $3B(7) + 3M(7)$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for a correct statement identifying specific inconsistencies between the given differential equation and the given slope field. There are many such correct statements, such as:

- The response references a specific point on the y -axis (such as $(0, 4)$), and indicates that the given slope field shows a slope of zero, yet the differential equation would yield a positive slope.
- The response references a pattern of slopes, and explains why that pattern does not fit our differential equation (e.g. “the slope field shows slopes that change as t changes, but our differential equation depends only on y , so slopes should remain constant for a given value of y as t changes.”)

Identifying a mismatch between the differential equation and the slope field at one point is sufficient to earn the point.

Response is not penalized for referencing “ x ” instead of “ t ”.

If the response contains any incorrect statement, the response will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point, nor will responses that present information beyond what we are given. For example:

- Responses that specify that the slope field shows a particular nonzero slope at a particular point will not earn the point, as this cannot actually be determined visually (i.e., the response should not indicate that at $(1, 2)$ the slope shown on the slope field is 1)
- Responses that state a specific slope value on the slope field for a general value of x or y that are not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

A response may involve a solution to the differential equation, with an argument based on the resulting function. If done correctly, and presented with a correct argument, this will earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

4-6-6

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

4-6-6

60. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	5	10	15	20
$M(t)$ (number of 1-cent coins)	0	300	500	870	1100

A class collected 1-cent coins to raise money for classroom supplies. Students in the class deposited the coins into a container where all the coins were kept until the end of the collection period. For time t days, $0 \leq t \leq 20$, the total number of coins in the container is modeled by a twice-differentiable function M . Values of $M(t)$ at various times t are shown in the table above.

- (a) Use a midpoint Riemann sum with the two subintervals indicated by the table to approximate

$$\int_0^{20} M(t) \, dt.$$

Show the computations that lead to your answer.

- (b) Find the value of $\int_0^5 M'(3t) \, dt$. Show the computations that lead to your answer.

- (c) Using correct units, approximate $M'(15)$ using the average rate of change of M over the interval $10 \leq t \leq 20$.

- (d) Explain the meaning of $M'(15)$ in the context of the problem.

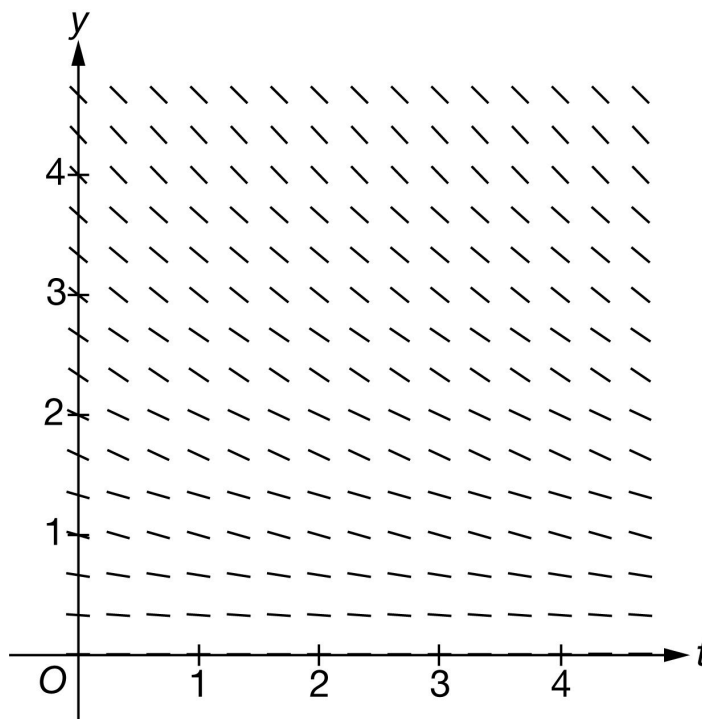
- (e) It is known that $M'(10) = 55$. Write an equation for the line tangent to the graph of $y = M(t)$ at $t = 10$.

- (f) It is known that $M''(t) > 0$ for $10 \leq t \leq 15$. If $M(14)$ is approximated using the equation from part (e), would the approximation be an overestimate or underestimate for the actual value of $M(14)$? Give a reason for your answer.

- (g) For $0 \leq t \leq 20$, the total number of coins collected can also be modeled by the differentiable function R , where R is a function of t . Assume the weight of each coin is 3.5 grams. Write an expression for the rate of change, in grams per day, of the total weight of the coins at $t = 11$ days.

4-6-6

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = 0.3y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Form of midpoint sum): Philosophy: The response should provide clear evidence adding the area of two rectangles whose heights are found using the midpoint of the interval.

Acceptable evidence:

$$\cdot 10 \cdot (M(5) + M(15))$$

$$\cdot 10 \cdot M(5) + 10 \cdot M(15)$$

$$\cdot (10 - 0) \cdot 300 + (20 - 10) \cdot 870$$

$$\cdot 10 \cdot (300 + 870) \text{ or } 10 \cdot 300 + 10 \cdot 870$$

To earn this point the response can have at most one mistake in the expression or equation. So, expressions of the form $\frac{10}{b} \cdot (M(5) + M(15))$, for any nonzero value of b would earn the point. Note: The value $\frac{10}{b}$ might be distributed across the sum.

Expressions of the form $\frac{a}{b} \cdot (M(5) + M(15))$ for $a \neq 10$ will not earn this point.

Once the first point is earned, it cannot be lost.

Second Point (Answer)

4-6-6

To earn the point, the answer must be correct. It does not need to be simplified, but if it is, the simplification must be correct.

The answer point cannot be earned unless the first (Form of midpoint sum) point was earned.

Minimal complete answers that earn both points:

$$\cdot 10 \cdot (300 + 870)$$

$$\cdot 10 \cdot 300 + 10 \cdot 870$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Form of midpoint sum

☐ Answer

Solution:

$$\begin{aligned} \int_0^{20} M(t) dt &\approx M(5)(10 - 0) + M(15)(20 - 10) \\ &= 300 \cdot 10 + 870 \cdot 10 = 11,700 \end{aligned}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Substitution) Philosophy: This point is earned when the response has all three components of the substitution done correctly.

Uses $u = 3t$, perhaps implicitly (any variable could be used for u)

Uses the correct constant of $\frac{1}{3}$

Uses the correct limits for their substitution.

· The limits might not be seen until the final evaluation, and at that point, response may have returned to the variable t .

· Limits for t attached to an integral involving u or vice versa will not earn this substitution point but could still earn the answer point, if the numerical answer is correct.

4-6-6

Second point: (Fundamental Theorem of Calculus) Philosophy: This point is earned when the FTC has been applied correctly to either $\int M'(3t) dt$ or $\int M'(u) du$.

The point is earned for any of the following:

- The equation $\int M'(3t) dt = M(3t)$ or $\int M'(u) du = M(u)$
- $\frac{1}{3}M(3t)|_0^5$ or $\frac{1}{3}M(u)|_0^{15}$
- $M(15) - M(0)$, $M(15)$, or $M(0)$

This point can be earned if the substitution point is not.

Note: $\int_a^b M'(u) du = M'(b) - M'(a)$ does not earn the point. The FTC point is for the response communicating that the antiderivative of M' is M .

Third Point (Answer)

Must be the correct numerical answer.

Must have earned the second point for the FTC.

Might have not earned the first point for substitution because of an incorrect use of limits, but will still earn the third point for the correct answer.

$M(15)$ and $M(0)$ must be evaluated; the expression $\frac{1}{3}(M(15) - M(0))$ by itself does not earn the third point.

The answer may be unsimplified, but if simplified it must be done correctly.

Scoring common responses:

The minimal work required to earn all three points is $\frac{1}{3}(870 - 0)$ or $\frac{1}{3} \cdot 870$

The following responses would earn the first two points and be eligible for the answer point if the response continues:

- $\frac{1}{3}M(3t)|_0^5$
- $\frac{1}{3}(M(15) - M(0))$

Responses of only the following would earn only the second point:

- $M(15) - M(0)$
- $M(5) - M(0)$

4-6-6

$$\cdot M(15)$$

$$\cdot \int_0^5 M'(u) \, du = M(5)$$

$$\cdot M(3t)|_0^5 \text{ or } M(u)|_0^{15}$$

The following responses would earn no points:

· 290 in the absence of one of the first two points.

$$\cdot \int_0^5 M'(3t) \, dt = M(5)$$



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 3t$. Then $du = 3 \, dt$.

When $t = 0$, $u = 0$.

When $t = 5$, $u = 15$.

$$\int_0^5 M'(3t) \, dt = \frac{1}{3} \int_0^{15} M'(u) \, du$$

$$= \frac{1}{3}(M(15) - M(0))$$

$$= \frac{1}{3}(870 - 0) = 290$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Approximation)

The response must be a difference and a quotient.

4-6-6

The answer does not need to be simplified, but if it is simplified it must be done correctly.

The minimal work that must be shown to earn this point is either $\frac{1100-500}{10}$ or $\frac{600}{20-10}$.

Second point (Units)

The units must be attached to a correct or incorrect value.

The response will not be penalized for any verbal attempt to discuss the answer after the correct approximation with units is presented.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Approximation

☐ Units

Solution:

$$M'(15) \approx \frac{M(20) - M(10)}{20 - 10} = \frac{1100 - 500}{10} = 60 \text{ coins per day}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The interpretation must include three components:

- indicate the rate of change
- the specific context of the number of coins in the container
- the specific point in time, $t = 15$

Responses do not need to include correct units or any units.

The word “increasing” may be used in place of the word “changing”.

The response is not earned for “the rate of coins deposited on day 15 “ as this does not indicate a change in the rate of deposit.

“the rate of change of M “ without specifying that M is the number of coins deposited, does not earn the point.

Acceptable ways to provide the specific point in time:

4-6-6

- On day 15 or after 15 days.
- At $t = 15$ (missing the unit of time)
- On week 15 (incorrect unit of time)

A response that clearly indicates an interval of time (e.g. “during the first 15 days” or “from day 15 on”) does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$M'(15)$ is the rate at which the number of 1-cent coins in the container is changing, in coins per day, on day 15.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The use of $M(t)$ in place of y earns the point.

The use of x in place of t earns the point.

Tangent lines do not need to be simplified, but any simplification must be done correctly.

Any correct form of the tangent line earns the point. For example:

- $y = 500 + 55(t - 10)$
- $y - 500 = 55(t - 10)$
- $y = 55t - 550 + 500$
- $y = 55t - 50$



0	1
---	---

The student response accurately includes the criteria below.

4-6-6

☐ Tangent line equation
Solution:

$$y = M'(10)(t - 10) + M(10)$$

$$y = 55(t - 10) + 500$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the point the response must include $M''(t) > 0$ and underestimate.

“ $M(t)$ is concave up” can be used instead of $M''(t) > 0$.

Information about $M'(t)$ by itself without reference to $M''(t) > 0$ is not sufficient and does not earn this point.

“Underestimate because it is concave up” is not sufficient by itself because of the use of “it”. The response must clearly, directly or indirectly, indicate to what “it” is referring.

If a response makes the correct conclusion and states, $M''(t) > 0$, and then goes on and makes an incorrect statement or a statement that cannot be verified will not earn this point.

For example: “ $M'(t)$ is concave up, underestimate” does not earn this point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason
Solution:

Because $M''(t) > 0$, the tangent line approximation would produce an underestimate.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression of only $3.5R'(11)$ earns this point.

Only expression $3.5M'(11)$ also earns this point.

No justifications or explanations are required.

4-6-6

Any incorrect statement, for example $R'(11) = 3.5M'(11)$, does not earn this point.



0	1
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The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in grams per day, of the total weight of the coins at time $t = 11$ days is $3.5R'(11)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Response clearly references what part of the differential equation is different from the slope field and the response needs only to clearly and explicitly reference one of them.

The response does not need to explicitly reference both the differential equation and the given slope field.

The distinction can be made with explicit reference to one of the two things and an implicit reference to the other

For example:

· “ $\frac{dy}{dt}$ is positive, not negative”. We will infer that negative is referring to the slopes shown in the slope field.

· “This would be correct for a negative derivative, but $\frac{dy}{dt}$ is positive.” We will read “This” as “the slope field” in this sentence.

· “All of the slopes in the slope field are negative, not positive.” We will infer that the values of $\frac{dy}{dt}$ at points in the first quadrant are positive.

A response that lacks even an implicit reference to either the differential equation or the slope field will not earn the point. For example:

· “ $\frac{dy}{dt}$ is positive” lacks any reference to the slope field and so will not earn the point.

· “All the slopes in the graph are negative” lacks any reference to the differential equation, and so will not earn the point.

· Vague answers such as “The slope field has negative points” or “The field is decreasing” will not earn the point.

4-6-6



0

1

The student response accurately includes the criteria below.

☐ Explanation

Solution:

When $y > 0$, $\frac{dy}{dt} > 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are negative.

4-6-6

61. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	4	10	18
$M(t)$ (number of 1-cent coins)	0	200	540	900

A class collected 1-cent coins to raise money for classroom supplies. Students in the class deposited the coins into a container where all the coins were kept until the end of the collection period. For time t days, $0 \leq t \leq 18$, the total number of coins in the container is modeled by an increasing, twice-differentiable function M . Values of $M(t)$ at various times t are shown in the table above.

(a) Using correct units, approximate $M'(10)$ using the average rate of change of M over the interval $4 \leq t \leq 18$.

(b) Explain the meaning of $M'(10)$ in the context of the problem.

(c) It is known that $M'(4) = 48$. Write an equation for the line tangent to the graph of $y = M(t)$ at $t = 4$.

(d) Use a right Riemann sum with the three subintervals indicated by the table to approximate

$\int_0^{18} M(t) \, dt$. Show the computations that lead to your answer.

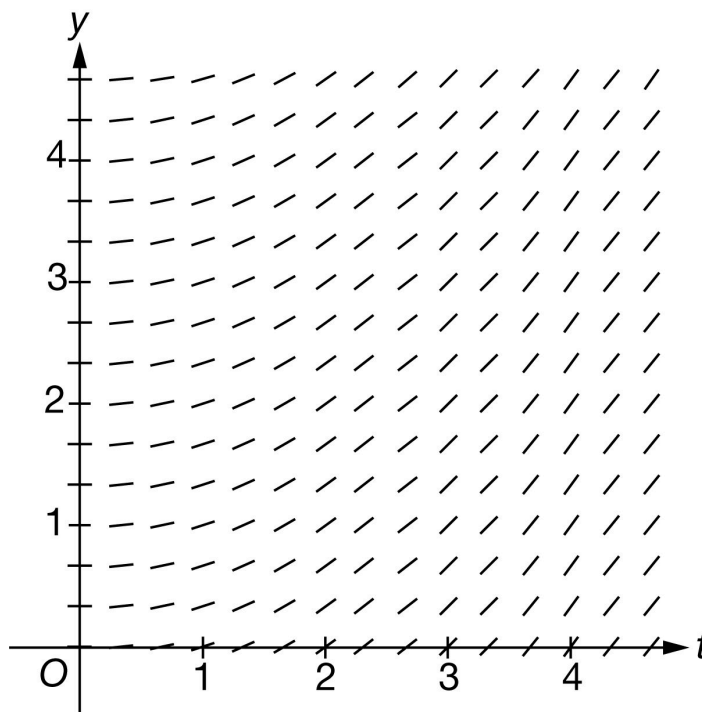
(e) Is the approximation from part (d) an overestimate or an underestimate for $\int_0^{18} M(t) \, dt$? Give a reason for your answer.

(f) Find the value of $\int_0^6 M'(3t) \, dt$. Show the computations that lead to your answer.

(g) For $0 \leq t \leq 18$, the total number of coins collected can also be modeled by the differentiable function N , where N is a function of t . Assume the weight of each coin is 2.5 grams. Write an expression for the rate of change, in grams per day, of the total weight of the coins at $t = 9$ days.

4-6-6

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = 0.4y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work. The supporting work must be at least a difference and a quotient. Correct values must be taken from the table to earn the point. The difference quotient does not need to be simplified.

Examples that earn the first point include:

$$\cdot \frac{900 - 200}{18 - 4}$$

$$\cdot \frac{M(18) - M(4)}{14} = \frac{700}{14}$$

$$\cdot \frac{M(18) - M(4)}{14} = 50$$

Examples that do not earn the first point include:

$$\cdot \frac{M(18) - M(4)}{14}$$

$$\cdot \frac{M(18) - M(4)}{18 - 4}$$

Minimal responses:

4-6-6

· $\frac{900-200}{14}$ or $\frac{700}{18-4}$ earns first point

· Neither $\frac{700}{14}$ nor 50, with no supporting work, will earn this point

The second point is for correct units

The correct units must be attached to a numerical value: either correct or incorrect.

The units may be written using words or a combination of words and symbols. For example:

- Coins per day
- Pennies per day
- Cents per day
- Coins/day

The response will not be penalized for any verbal attempt to discuss the answer after the correct answer has been presented.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$M'(10) \approx \frac{M(18)-M(4)}{18-4} = \frac{900-200}{14} = 50 \text{ coins per day}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

- Indicating a rate of change
- Including the context of coin collection
- Referring to the specific point in time, $t = 10$

4-6-6

Students are not penalized for incorrect units in this part. (Whether correct or incorrect, units have no effect on the score in this part.)

Responses may use “changing” or “increasing” as a modifier as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Example: “The number of coins is increasing at a rate of 50 coins per day...” is okay.

Acceptable references to a specific point in time include (but are not limited to):

- On day 10 and after 10 days
- At $t = 10$
- At $t = 10$ weeks or any incorrect unit of time

If the response clearly indicates an interval of time (e.g., “from day 10 on,” “during the first 10 days,” or “for $t = 0$ to $t = 10$ ”), the point is not earned.

Examples that earn the point include:

- $M'(10)$ is the rate of change of the number of coins per day at time $t = 10$
- The rate of change of number of coins is 50 coins per day on day 10
- $M'(10)$ is the rate of change in the number of coins at time $t = 10$ in coins per *month*
- 50 is the rate of increase in the number of coins per day on day 10

Examples that do not earn the point include:

- $M'(10)$ is the average rate of change of the total number of coins per day on day 10
- $M'(10)$ is the total number of coins per day per day on day 10
- The rate of change is 50 coins per day (Missing “at $t = 10$ ”)
- $M'(10)$ is the slope of the tangent line at $t = 10$
- Rate of change of M at time $t = 10$
- Coins per day increases on day 10



0	1
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The student response accurately includes the criteria below.

☐ Explanation

4-6-6

Solution:

$M'(10)$ is the rate at which the number of 1-cent coins in the container is changing, in coins per day, on day 10.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. The use of x in place of t is acceptable. Some examples include

· $y - 200 = 48(t - 4)$, $y - 200 = 48(x - 4)$

· $y = 48(t - 4) + 200$, $y = 48(x - 4) + 200$

· $y = 48t + 8$, $y = 48x + 8$

The key calculus concept being assessed in this part of the problem is that the value of $M'(4)$ is the slope of the tangent line. In that spirit, we are accepting both $y = \dots$ and $M(t) = \dots$, but not $M(4) = \dots$

· $M(t) = 48(t - 4) + 200$ earns the point

· However, $M(4) = 48(t - 4) + 200$ does not earn the point



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = M'(4)(t - 4) + M(4)$$

$$y = 48(t - 4) + 200$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a right Riemann sum. The setup must be a sum of three products.

· A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example below).

4-6-6

- Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns. Implicit responses are eligible for both points.

A response presenting a left Riemann sum (instead of right), will not earn any points for part (d).

The second point is for the correct numerical result.

- In order to earn the second point, there must be supporting calculations.
- The numerical values for $M(4)$, $M(10)$, and $M(18)$ must be pulled from the table.
- Any simplification shown must be correct.

Use the following as an illustration to score both points. (In most cases, the expression on the left side represents the entirety of relevant work shown for the response.)

- A response of $200 \cdot (4 - 0) + 540 \cdot (10 - 4) + 900 \cdot (18 - 10)$ earns both points.
- A response of $4M(4) + 6M(10) + 8M(18)$ earns the first point. Eligible for second point if more work is on the page.
- A response of $4M(4) + 6M(10) + 8M(18) = 11,240$ earns both points.
- A response of $4 \cdot 200 + 6 \cdot 540 + 8 \cdot 900$ earns both points.
- A response of $4 \cdot 200 + 10 \cdot 540 + 8 \cdot 900$ contains one numerical error and earns the first point, but is not eligible for the second point.
- A response of $4 \cdot 200 + 6 \cdot 540 + 8 \cdot 900 = 10,240$ earns the first point for left side of equation but does not earn second point due to incorrect simplification.
- A response of $800 + 3240 + 7200$ or $800 + 3240 + 7200 = 11,240$ does not earn the first point because we cannot see the sum of 3 products. However, this is the correct sum, so we see an answer with supporting work. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points.

$$4 \cdot 200 + 6 \cdot 540 + 8 \cdot 900$$



0	1	2
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The student response accurately includes **both** of the criteria below.

4-6-6

- ☐ Right Riemann sum
- ☐ Answer

Solution:

$$\int_0^{18} M(t) dt \approx M(4)(4-0) + M(10)(10-4) + M(18)(18-10)$$
$$= 200 \cdot 4 + 540 \cdot 6 + 900 \cdot 8 = 11,240$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned when the response states that the approximation is an overestimate because M is increasing.

The response does not need an explicit reference to the approximation in part (d), nor does there need to be an answer given in part (d).

The following statements are sufficient for earning this point:

- Overestimate because M is increasing
- Overestimate because the function is increasing

Note: At this point, there is only one function being discussed in this problem so we will assume “the function” is M .

When a response includes additional information about concavity, interpret that as a lack of understanding about the question, so this will not earn the point. When the response references a graph or the given table rather than the function M , the point is not earned.

Examples that earn the point:

- The right Riemann sum overestimates because the function is increasing.
- The sum is an overestimate because $M(t)$ is increasing.
- The sum is an overestimate because $M'(t) > 0$.

Examples that do not earn the point:

- The sum is an overestimate because $M'(t)$ is increasing. ($M'(t)$ increasing is not known.)
- The right Riemann sum overestimates because the graph is increasing.
- The right Riemann sum overestimates because the function is increasing and concave up.

4-6-6



0	1
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The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because M is an increasing function, and this is a right Riemann sum, the answer from part (d) is an overestimate.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are three components of substitution that need to be seen:

- Uses $u = 3t$, which may be done implicitly
- Uses the correct constant of $\frac{1}{3}$
- Uses the correct limits for the substitution (0 and 18). These limits may only be seen when evaluating the antiderivative. These don't need to be seen as limits on a definite integral.
- Either of the expressions below will earn the substitution point without any additional supporting work:

$$\frac{1}{3} M(3t) \Big|_0^6 \text{ or } \frac{1}{3} M(u) \Big|_0^{18}$$

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point.

- Limits for t attached to an integral with u as the variable of integration (or vice versa),
- Improper equals signs, such as an indefinite integral equal to a definite integral
- Reversed limits of integration, for example: $\int_{18}^0 M'(u) du$

The second point is for the Fundamental Theorem of Calculus. This point can be earned with any of the following, where k is a nonzero constant and $b = 6$ or $b = 18$. No explanation or additional supporting work is necessary.

$$\cdot kM(3t) \Big|_0^b$$

4-6-6

$$\cdot k(M(b) - M(0))$$

This point can be earned without earning the substitution point.

The third point is for the correct numerical value, which is 300.

Values from the table must be used. $\frac{1}{3}[M(18) - M(0)]$ is not enough to earn the answer point.

The expression $\frac{1}{3}(900 - 0)$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they need not be correct in order to earn this point.

The response cannot earn this point without also earning the FTC point.

Minimalist response: The expression $\frac{1}{3}(900)$ represents the minimum work that must be seen in a response in order to earn all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 3t$. Then $du = 3 dt$.

When $t = 0$, $u = 0$.

When $t = 6$, $u = 18$.

$$\begin{aligned} \int_0^6 M'(3t) dt &= \frac{1}{3} \int_0^{18} M'(u) du \\ &= \frac{1}{3}(M(18) - M(0)) \\ &= \frac{1}{3}(900 - 0) = 300 \end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-6-6

This point is earned for a correct expression at $t = 9$. No supporting work or justification is required.

The correct expression of $2.5M'(9)$ or $2.5N'(9)$ earns this point.

No justifications or explanations are required.

Any incorrect statement such as, $M'(9) = 2.5N'(9)$, does not earn this point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Expression

Solution:

An expression for the rate of change, in grams per day, of the total weight of the coins at $t = 9$ days is $2.5N'(9)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between two things: the differential equation, $\frac{dy}{dt} = 0.4y$, and the given slope field.

The response does not need to explicitly reference both the differential equation and the given slope field., but it must explicitly reference one and implicitly reference the other.

Acceptable responses include:

- Identifying an inconsistency between the differential equation and the slope field at one point, e.g., at the point $(4, 0)$, the slope of the segment should be 0, but it is positive.
- Referencing a pattern of slopes, and explaining why that pattern does not fit the given differential equation.

A response that lacks even an implicit reference to the one of two things being compared will not earn the point.

For example, merely stating either $\frac{dy}{dt}$ is positive, or that the slope field is positive discretely does not earn the point.

- “The values for y are greater than or equal to zero so $\frac{dy}{dt} \geq 0$ ” lacks any reference to the slope field and will not earn the point.
- “All the slopes in the slope field are positive” lacks any reference to the differential equation and will not earn the point.

4-6-6

Vague answers which have the potential to be incorrect will not earn the point. Some examples are:

- “The field is increasing”
- Responses that state a specific slope value on the slope field for a general value of x or y that is not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

A response may correctly solve or attempt to solve the differential equation and then provide a correct argument without reference to their solution. Such a response would earn the point.

A response might actually solve the differential equation and provide a correct argument on the resulting function.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

4-6-6

62. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

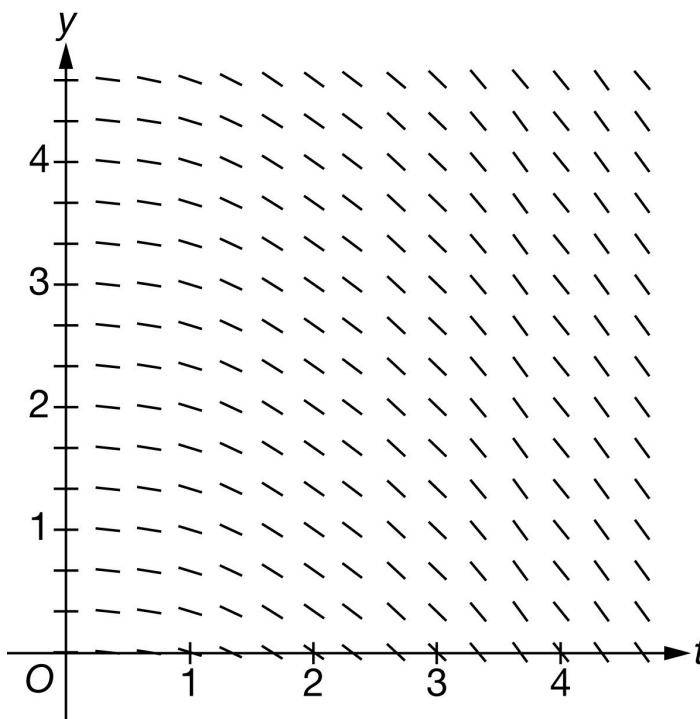
t (minutes)	0	10	35	40
$N(t)$ (number of 5-cent coins)	1000	810	200	0

A child has collected 5-cent coins and is preparing them to deposit in the bank by counting them into bags. The child begins with 1000 coins. For time t minutes, $0 \leq t \leq 40$, the total number of coins remaining to prepare is modeled by a decreasing, twice-differentiable function N . Values of $N(t)$ at various times t are shown in the table above.

- Using correct units, approximate $N'(35)$ using the average rate of change of N over the interval $10 \leq t \leq 40$.
- Explain the meaning of $N'(35)$ in the context of the problem.
- It is known that $N'(10) = -20$. Write an equation for the line tangent to the graph of $y = N(t)$ at $t = 10$.
- Use a right Riemann sum with the three subintervals indicated by the table to approximate $\int_0^{40} N(t) \, dt$. Show the computations that lead to your answer.
- Is the approximation from part (d) an overestimate or an underestimate for $\int_0^{40} N(t) \, dt$? Give a reason for your answer.
- Find the value of $\int_0^{10} N'(4t) \, dt$. Show the computations that lead to your answer.
- For $0 \leq t \leq 40$, the total number of 5-cent coins remaining can also be modeled by the differentiable function $\bar{F}$, where F is a function of t . Assume the weight of each coin is 5 grams. Write an expression for the rate of change, in grams per minute, of the total weight of the 5-cent coins remaining at $t = 11$ minutes.

4-6-6

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = -0.3y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work. The supporting work must be at least a difference and a quotient. Correct values must be taken from the table to earn the point. The difference quotient does not need to be simplified.

Examples that earn the first point include:

$$\cdot \frac{0-810}{40-10}$$

$$\cdot \frac{N(40)-N(10)}{30} = \frac{-810}{30}$$

$$\cdot \frac{N(40)-N(10)}{30} = -27$$

An example that does not earn the first point is $\frac{N(40)-N(10)}{30}$.

Minimal response that earns the first point: $\frac{0-810}{40-10}$ or $\frac{0-810}{30}$

· Neither $\frac{-810}{30}$ nor -27 , with no supporting work, will earn the point.

The second point is for correct units

4-6-6

The correct units must be attached to a numerical value: either correct or incorrect.

The units may be written using words or a combination of words and symbols. For example:

- Coins per minute
- Nickels per minute,
- 5-cents per minute
- Coins/minute

The response will not be penalized for any verbal attempt to discuss the answer after the correct answer has been presented.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$N'(35) \approx \frac{N(40) - N(10)}{40 - 10} = \frac{0 - 810}{30} = -27 \text{ coins per minute}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

- Indicating a *rate of change*
- Including the context of coin collection
- Referring to the specific point in time, $t = 35$

The response is not penalized for incorrect units in this part. (Whether correct or incorrect, units have no effect on the score in this part.)

Responses may use “changing” or “decreasing” as a modifier as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Example: “the number of coins is decreasing at a rate of 27 coins per day...” is okay, but “the number of coins is decreasing at a rate of -27 coins per day...” would not earn the point.

4-6-6

Acceptable references to a specific point in time include (but are not limited to):

- On minute 35 and after 35 minutes
- At $t = 35$
- At $t = 35$ hours or any incorrect unit of time

If the response clearly indicates an interval of time (e.g., “from minute 35 on,” “during the first 35 minutes,” or “for $t = 0$ to $t = 35$ ”), the point is not earned.

Examples that earn the point include:

- $N'(35)$ is the rate of change of the number of coins per minute at time $t = 35$
- The rate of change of number of coins is -27 coins per minute on minute 35
- $N'(35)$ is the rate of change in the number of coins at time $t = 35$ in coins per minute
- -27 is the rate of decrease in the number of coins per minute on minute 35

Examples that do not earn the point include:

- $N'(35)$ is the average rate of change of the total number of coins per minute on minute 35
- $N'(35)$ is the total number of coins per minute on minute 35
- The rate of change is -27 coins per minute (Missing “at $t = 35$ ”)
- $N'(35)$ is the slope of the tangent line at $t = 35$
- Rate of change of N at time $t = 35$
- Coins per minute decreases at minute 35



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$N'(35)$ is the rate at which the number of 5-cent coins remaining is changing, in coins per minute, at time $t = 35$ minutes.

Part C

4-6-6

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. The use of x in place of t is acceptable.

· $y = -20(t - 10) + 810, y = -20(x - 10) + 810$

· $y - 810 = -20(t - 10), y - 810 = -20(x - 10)$

· $y = -20t + 1010, y = -20x + 1010$

The key calculus concept being assessed in this part of the problem is that the value of $N'(10)$ is the slope of the tangent line. In that spirit, we are accepting $N'(t) = \dots$ in lieu of $y = \dots$, but we are not accepting $N'(10) = \dots$

Note that $N(t) = -20(t - 10) + 810$ earns the point, but $N(10) = -20(t - 10) + 810$ does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = N'(10)(t - 10) + N(10)$$

$$y = -20(t - 10) + 810$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a right Riemann sum. The setup must be a sum of three products.

· A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example in the table below).

· Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns. Implicit responses are eligible for both points.

A response presenting a left Riemann sum (instead of right), will not earn any points for part (d).

The second point is for the correct numerical result.

4-6-6

- In order to earn the second point, there must be supporting calculations.
- The numerical values for $N(10)$, $N(35)$, and $N(40)$ must be pulled from the table
- Any simplification shown must be correct.

Use the following as an illustration how to score both points. (In most cases, you see scores for which the expression on the left side represents the entirety of relevant work shown for the response.)

- A response of $810 \cdot (10 - 0) + 200 \cdot (35 - 10) + 0 \cdot (40 - 35)$ earns both points.
- A response of $10 \cdot N(10) + 25 \cdot N(35) + 5 \cdot N(40)$ earns the first point. Eligible for second point if more work is on the page.
- A response of $10 \cdot N(10) + 25 \cdot N(35) + 5 \cdot N(40) = 13,100$ earns both points.
- A response of $10 \cdot 810 + 25 \cdot 200 + 5 \cdot 0$ earns both points.
- A response of $10 \cdot 810 + 15 \cdot 200 + 5 \cdot 0$ contains one numerical error. Earns the first point. Not eligible for the second point.
- A response of $10 \cdot 810 + 25 \cdot 200 + 5 \cdot 0 = 12,100$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.
- A response of $8100 + 5000 + 0$ or $8100 + 5000 + 0 = 13,100$ does not earn the first point because we cannot see the sum of three products. However, this is the correct sum, so we see an answer with supporting work. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points.

$$10 \cdot 810 + 25 \cdot 200 + 5 \cdot 0$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Right Riemann sum
- ☐ Answer

Solution:

$$\int_0^{40} N(t) dt \approx N(10)(10 - 0) + N(35)(35 - 10) + N(40)(40 - 35)$$

4-6-6

$$= 810 \cdot 10 + 200 \cdot 25 + 0 \cdot 5 = 13,100$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an underestimate because N is decreasing.

The response does not need an explicit reference in part (d). The following statements are sufficient for earning this point:

- Underestimate because N is decreasing.
- Underestimate because the function is decreasing

Note: At this point, there is only one function being discussed in this problem, so we will assume “the function” is N .

When a response includes additional information about concavity, we interpret that as a lack of understanding about the question, so this will not earn the point. When the response references a graph or the given table rather than the function N , the point is not earned.

Examples that earn the point:

- The right Riemann sum underestimates because the function is decreasing.
- The sum is an underestimate because $N(t)$ is decreasing.
- The sum is an underestimate because $N'(t) < 0$.

Examples that do not earn the point:

- The sum is an underestimate because $N'(t)$ is decreasing. ($N'(t)$ decreasing is not known)
- The right Riemann sum underestimates because the graph is decreasing.
- The right Riemann sum underestimates because the function is decreasing and concave up.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because N is a decreasing function, and this is a right Riemann sum, the answer from part (d) is an underestimate.

4-6-6

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are 3 components of substitution that we need to see:

- Uses $u = 4t$, which may be done implicitly
- Uses the correct constant of $\frac{1}{4}$
- Uses the correct limits for the substitution (0 and 40). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.
- Either of the expressions below will earn the substitution point without any additional supporting work:

$$\frac{1}{4} N(4t) \Big|_0^{10} \text{ or } \frac{1}{4} N(u) \Big|_0^{40}$$

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point

- Limits for t attached to an integral with u as the variable of integration (or vice versa),
- Improper equals signs, such as an indefinite integral equal to a definite integral
- Reversed limits of integration, for example: $\int_{40}^0 N'(u) du$

The second point is for the Fundamental Theorem of Calculus. This point can be earned with any of the following, where k is a nonzero constant and $b = 10$ or $b = 40$. No explanation or additional supporting work is necessary.

$$\circ kN(4t) \Big|_0^b$$

$$\circ k(N(b) - N(0))$$

This point can be earned without earning the substitution point.

The third point is for the correct numerical value, which is -250 .

Values from the table must be used. $\frac{1}{4}[N(40) - N(0)]$ is not enough to earn the answer point.

The expression $\frac{1}{4}(0 - 1000)$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they need not be correct in order to earn this point.

4-6-6

The response cannot earn this point without also earning the FTC point.

Minimalist response: The expression $\frac{1}{4}(-1000)$ represents the minimum work that must be seen in a response in order to earn all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 4t$. Then $du = 4 dt$.

When $t = 0$, $u = 0$.

When $t = 10$, $u = 40$.

$$\int_0^{10} N'(4t) dt = \frac{1}{4} \int_0^{40} N'(u) du$$

$$= \frac{1}{4}(N(40) - N(0))$$

$$= \frac{1}{4}(0 - 1000) = -250$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct expression at $t = 11$. No supporting work or justification is required.

The correct expression of just $5F'(11)$ or $5N'(11)$ earns this point.

No justifications or explanations are required.

Any incorrect statement, such as $N'(11) = 5F'(11)$, does not earn this point.



0	1
---	---

4-6-6

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in grams per minute, of the total weight of the 5 -cent coins remaining at $t = 11$ minutes is $5F'(11)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between two things: the differential equation, $\frac{dy}{dt} = -0.3y$, and the given slope field.

The response does not need to explicitly reference both the differential equation and the given slope field., but it must explicitly reference one and implicitly reference the other.

Acceptable responses include

- Identifying an inconsistency between the differential equation and the slope field at one point , e.g., at the point $(4, 0)$, the slope of the segment should be 0, but it is negative.
- Referencing a pattern of slopes, and explaining why that pattern does not fit the given differential equation.

A response that lacks even an implicit reference to one of the two things being compared will not earn the point.

For example, merely stating either $\frac{dy}{dt}$ is negative, or that the slope field is negative does not earn the point.

- “The values for y are less than or equal to zero so $\frac{dy}{dt} \leq 0$ ” lacks any reference to the slope field and will not earn the point.
- “All the slopes in the slope field are negative” lacks any reference to the differential equation and will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point. Some examples are:

- “The field is decreasing”
- Responses that state a specific slope value on the slope field for a general value of x or y that is not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

A response may correctly solve or attempt to solve the differential equation and then provide a correct argument without reference to their solution. Such a response would earn the point.

4-6-6

A response might actually solve the differential equation and provide a correct argument on the resulting function.



0

1

The student response accurately includes the criteria below.

☐

Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

4-6-6

63. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	10	20	30	40
$N(t)$ (number of 5-cent coins remaining)	1200	1000	750	400	0

A child has collected 5-cent coins and is preparing them to deposit in the bank by counting them into bags. The child begins with 1200 coins. For time t minutes, $0 \leq t \leq 40$, the total number of coins remaining to prepare is modeled by a decreasing, twice-differentiable function N . Values of $N(t)$ at various times t are shown in the table above.

(a) Use a midpoint Riemann sum with the two subintervals indicated by the table to approximate

$$\int_0^{40} N(t) \, dt.$$

Show the computations that lead to your answer.

(b) Find the value of $\int_0^{10} N'(2t) \, dt$. Show the computations that lead to your answer.

(c) Using correct units, approximate $N'(20)$ using the average rate of change of N over the interval $10 \leq t \leq 30$.

(d) Explain the meaning of $N'(20)$ in the context of the problem.

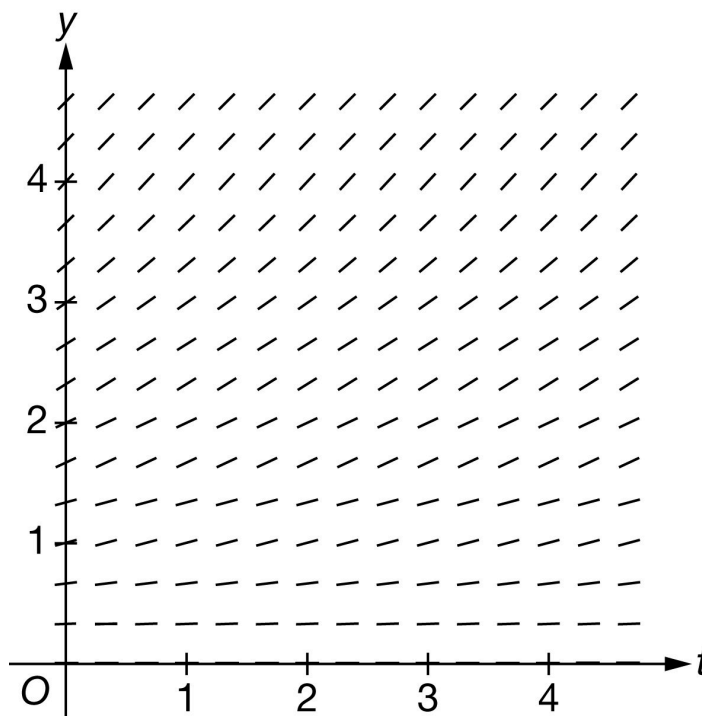
(e) It is known that $N'(10) = -20$. Write an equation for the line tangent to the graph of $y = N(t)$ at $t = 10$.

(f) It is known that $N''(t) < 0$ for $10 \leq t \leq 20$. If $N(17)$ is approximated using the equation from part (e), would the approximation be an overestimate or underestimate for the actual value of $N(17)$? Give a reason for your answer.

(g) For $0 \leq t \leq 40$, the total number of 5-cent coins remaining can also be modeled by the differentiable function G , where G is a function of t . Assume the weight of each coin is 6 grams. Write an expression for the rate of change, in grams per minute, of the total weight of the 5-cent coins remaining at $t = 37$ minutes.

4-6-6

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = -0.4y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Form of midpoint sum): Philosophy: The response should provide clear evidence adding the area of two rectangles whose heights are found using the midpoint of the interval.

Acceptable evidence:

$$\cdot 20 \cdot (N(10) + N(30)) \text{ or } 20 \cdot N(10) + 20 \cdot N(30)$$

$$\cdot (20 - 0) \cdot 1000 + (40 - 20) \cdot 400$$

$$\cdot 20 \cdot (1000 + 400) \text{ or } 20 \cdot 1000 + 20 \cdot 400$$

To earn this point the response can have at most one mistake in the expression or equation. So, expressions of the form $\frac{20}{b} \cdot (N(10) + N(30))$, for any nonzero value of b would earn the point. Note: The value $\frac{20}{b}$ might be distributed across the sum.

Expressions of the form $\frac{a}{b} \cdot (N(10) + N(30))$ for $a \neq 20$ will not earn this point.

Once the first point is earned, it cannot be lost.

Second Point (Answer)

4-6-6

To earn the point, the answer must be correct. It does not need to be simplified, but if it is, the simplification must be correct.

The answer point cannot be earned unless the first (midpoint sum) point was earned.

Minimal complete answers that earn both points:

$$\cdot 20 \cdot (1000 + 400)$$

$$\cdot 20 \cdot 1000 + 20 \cdot 400$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint sum
- ☐ Answer

Solution:

$$\begin{aligned} \int_0^{40} N(t) dt &\approx N(10)(20 - 0) + N(30)(40 - 20), \\ &= 1000 \cdot 20 + 400 \cdot 20 = 28,000 \end{aligned}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Substitution) Philosophy: This point is earned when the response has all three components of the substitution done correctly.

Uses $u = 2t$, perhaps implicitly (any variable could be used for u)

Uses the correct constant of $\frac{1}{2}$

Uses the correct limits for their substitution.

· The limits might not be seen until the final evaluation, and at that point, response may have returned to the variable t .

· Limits for t attached to an integral involving u or vice versa will not earn this substitution point but could still earn the answer point, if the numerical answer is correct.

4-6-6

Second point: (Fundamental Theorem of Calculus) Philosophy: This point is earned when the FTC has been applied correctly to either $\int N'(2t) dt$ or $\int N'(u) du$.

The point is earned for any of the following:

- The equation $\int N'(2t) dt = N(2t)$ or $\int N'(u) du = N(u)$
- $\frac{1}{2}N(2t)|_0^{10}$ or $\frac{1}{2}N(u)|_0^{20}$
- $N(20) - N(0)$, $N(20)$, or $N(0)$

This point can be earned if the substitution point is not.

Note: $\int_a^b N'(u) du = N(b) - N(a)$ does not earn the point. The FTC point is for the response communicating that the antiderivative of N' is N .

Third Point (Answer)

Must be the correct numerical answer.

Must have earned the second point for the FTC.

Might have not earned the first point for substitution because of an incorrect use of limits, but will still earn the third point for the correct answer.

$N(20)$ and $N(0)$ must be evaluated; the expression $\frac{1}{2}(N(20) - N(0))$ by itself does not earn the third point.

The answer may be unsimplified, but if simplified it must be done correctly.

Scoring common responses:

The minimal work required to earn all three points is $\frac{1}{2}(750 - 1200)$ or $\frac{1}{2}(750) - \frac{1}{2}(1200)$.

The following responses would earn the first two points and be eligible for the answer point if the response continues:

- $\frac{1}{2}N(2t)|_0^{10}$
- $\frac{1}{2}(N(20) - N(0))$

Responses of only the following would earn only the Second point:

- $N(20) - N(0)$
- $N(10) - N(0)$

4-6-6

$$\cdot \int_0^{10} N'(u) \, du = N(10) - N(0)$$

$$\cdot N(2t)|_0^{10} \text{ or } N(u)|_0^{20}$$

The following responses would earn no points:

$$\cdot -225 \text{ in the absence of one of the first two points.}$$

$$\cdot \int_0^{10} N'(2t) \, dt = N(10) - N(0)$$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 2t$. Then $du = 2 \, dt$.

When $t = 0$, $u = 0$.

When $t = 10$, $u = 20$.

$$\int_0^{10} N'(2t) \, dt = \frac{1}{2} \int_0^{20} N'(u) \, du$$

$$= \frac{1}{2}(N(20) - N(0))$$

$$= \frac{1}{2}(750 - 1200) = -225$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Approximation)

The response must be a difference and a quotient.

The answer does not need to be simplified, but if it is simplified it must be done correctly.

4-6-6

The minimal work that must be shown to earn this point is either $\frac{400-1000}{20}$ or $\frac{-600}{30-10}$.

Second point (Units)

The units must be attached to a value (either correct or incorrect).

The response will not be penalized for any verbal attempt to discuss the answer after the correct approximation with units is presented.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$N'(20) \approx \frac{N(30)-N(10)}{30-10} = \frac{400-1000}{20} = -30 \text{ coins per minute}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The interpretation must include three components:

- indicate the rate of change
- the specific context of the number of coins remaining
- the specific point in time, $t = 20$.

Responses do not need to include correct units or any units.

The word “decreasing” may be used in place of the word “changing”.

The response is not earned for:

- “the rate of coins remaining at minute 20 ” as this does not indicate a change in the rate of deposit.
- “decreasing at a rate of -30 coins per minute”

“the rate of change of N ” without specifying that N is the number of coins remaining, does not earn the point.

Acceptable ways to provide the specific point in time:

4-6-6

- At minute 20 or after 20 minutes.
- At $t = 20$ (missing the unit of time)
- At 20 seconds (incorrect unit of time)

A response that clearly indicates an interval of time (e.g. “during the first 20 minutes” or “from minute 20 on”) does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$N'(20)$ is the rate at which the number of 5-cent coins remaining is changing, in coins per minute, at time $t = 20$ minutes.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The use of $N(t)$ in place of y earns the point.

The use of x in place of t earns the point.

Tangent lines do not need to be simplified, but any simplification must be done correctly.

Any correct form of the tangent line earns the point. For example:

- $y = 1000 - 20(t - 10)$
- $y - 1000 = 20(t - 10)$
- $y = -20t + 200 + 1000$
- $y = -20t + 1200$



0	1
---	---

The student response accurately includes the criteria below.

4-6-6

☐ Tangent line equation
Solution:

$$y = N'(10)(t - 10) + N(10)$$

$$y = -20(t - 10) + 1000$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point the response must include $N''(t) < 0$ and overestimate.

“ $N(t)$ is concave down” can be used instead of $N''(t) < 0$.

$N'(t)$ by itself is not sufficient and does not earn this point

“Overestimate because it is concave down” is not sufficient by itself because of the use of "it". The response must clearly, directly or indirectly, indicate to what "it" is referring.

If a response makes the correct conclusion and states $N''(t) < 0$ and then goes on and makes an incorrect statement or a statement that cannot be verified will not earn this point.

For example: “ $N'(t)$ is concave down, overestimate” does not earn this point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason
Solution:

Because $N''(t) < 0$, the tangent line approximation would produce an overestimate.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression of only $6 \cdot G'(37)$ earns this point.

Only expression $6 \cdot N'(37)$ also earns this point.

No justifications or explanations are required.

4-6-6

Any incorrect statement such as, $G'(37) = 6 \cdot N'(37)$, does not earn this point.



0

1

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in grams per minute, of the total weight of the 5-cent coins remaining at $t = 37$ minutes is $6G'(37)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Response clearly references what part of the differential equation is different from the slope field and the response needs only to clearly and explicitly reference one of them.

The response does not need to explicitly reference both the differential equation and the given slope field.

The distinction can be made with explicit reference to one of the two things and an implicit reference to the other

For example:

· “ $\frac{dy}{dt}$ is negative, not positive”. We will infer that positive is referring to the slopes shown in the slope field.

· “This would be correct for a positive derivative, but $\frac{dy}{dt}$ is negative.” We will read “This” as “the slope field” in this sentence.

· “All of the slopes in the slope field are positive, not negative.” We will infer that the values of $\frac{dy}{dt}$ at points in the first quadrant are negative.

A response that lacks even an implicit reference to either the differential equation or the slope field will not earn the point. For example:

· “ $\frac{dy}{dt}$ is negative” lacks any reference to the slope field and so will not earn the point.

· “All the slopes in the graph are positive” lacks any reference to the differential equation, and so will not earn the point.

· Vague answers such as “The slope field has positive points” or “The field is increasing” will not earn the point.

4-6-6



0

1

The student response accurately includes the criteria below.

☐

Explanation

Solution:

When $y > 0$, $\frac{dy}{dt} < 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are positive.

4-6-6

64. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	10	35	40
$N(t)$ (coins per minute)	-8	-11	-13	-15

A child has collected 5-cent coins and is preparing them to deposit in the bank by counting them into bags. For time t minutes, $0 \leq t \leq 40$, the rate of the total number of coins remaining to prepare, in 5-cent coins per minute, is modeled by a decreasing, twice-differentiable function N . Values of $N(t)$ at various times t are shown in the table above.

(a) Use a left Riemann sum with the three subintervals indicated by the table to approximate

$\int_0^{40} N(t) \, dt$. Show the calculations that lead to your answer.

(b) Is your answer from part (a) an overestimate or an underestimate for $\int_0^{40} N(t) \, dt$? Give a reason for your answer.

(c) Using correct units, approximate $N'(10)$ using the average rate of change of N over the interval $0 \leq t \leq 35$.

(d) Explain the meaning of $N'(10)$ in the context of the problem.

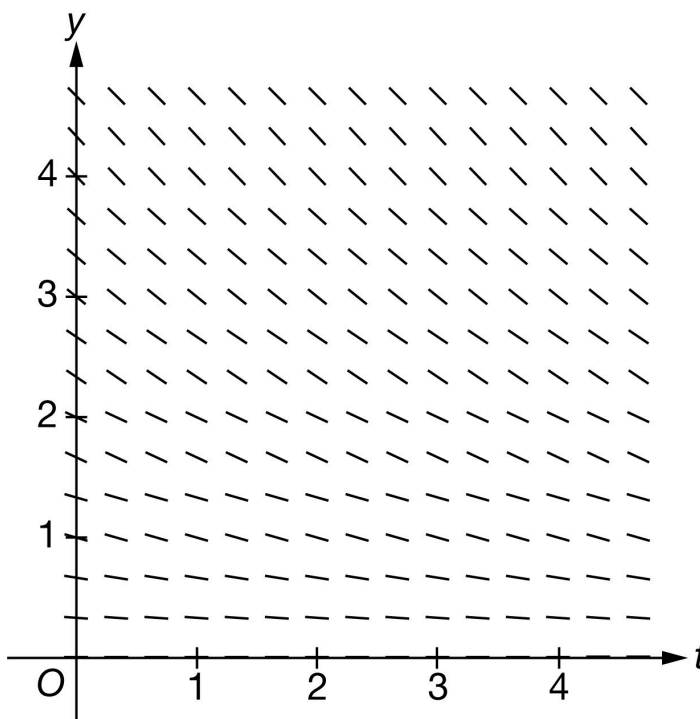
(e) It is known that $N'(35) = -0.2$. Write an equation for the line tangent to the graph of $y = N(t)$ at $t = 35$.

(f) Find the value of $\int_0^{40} N'\left(\frac{t}{4}\right) \, dt$. Show the calculations that lead to your answer.

(g) A second child is also preparing 5-cent coins for deposit. For $0 \leq t \leq 40$, the total number of 5-cent coins prepared by the second child for deposit can be modeled by the increasing, differentiable function F , where F is a function of t . Assume the weight of each coin is 5 grams. Write an expression for the rate of change, in grams per minute, of the total weight of the 5-cent coins remaining for the second child at $t = 23$ minutes.

4-6-6

(h) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = F'(t)$ for the function F defined in part (g).



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a left Riemann sum. The setup must be a sum of three products.

A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example below).

Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns, like so.

10	(-8)	-80
25	(-11)	-275
5	(-13)	+(-65)
		-420

or

4-6-6

$(10 - 0)(-8)$	$= -80$
$(35 - 10)(-11)$	$= -275$
$(40 - 35)(-13)$	$= -65$
	-420

Implicit responses such as those above are eligible for both points.

A response presenting a right Riemann sum (instead of left), will not earn any points for part (a).

The second point is for the correct numerical result.

In order to earn the second point, there must be supporting calculations.

The numerical values for $N(0)$, $N(10)$, and $N(35)$ must be pulled from the table.

Any simplification shown must be correct.

Use the following as an illustration of how to score both points. (In most cases, you see scores for which the expression on the left side represents the entirety of relevant work shown for the response.)

- A response of $(-8) \cdot (10 - 0) + (-11) \cdot (35 - 10) + (-13) \cdot (40 - 35)$ earns both points.
- A response of $10 \cdot N(0) + 25 \cdot N(10) + 5 \cdot N(35)$ earns the first point. Eligible for second point if more work is on the page.
- A response of $10 \cdot N(0) + 25 \cdot N(10) + 5 \cdot N(35) = -420$ earns both points.
- A response of $10 \cdot (-8) + 25 \cdot (-11) + 5 \cdot (-13)$ earns both points.
- A response of $10 \cdot (-8) + 35 \cdot (-11) + 5 \cdot (-13)$ contains one numerical error. Earns the first point. Not eligible for the second point.
- A response of $10 \cdot (-8) + 25 \cdot (-11) + 5 \cdot (-13) = -400$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.
- A response of $(-80) + (-275) + (-65)$ or $(-80) + (-275) + (-65) = -420$ does not earn the first point because we cannot see the sum of three products. However, this is the correct sum, so we are not seeing an unsupported answer. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points.

$$10 \cdot (-8) + 25 \cdot (-11) + 5 \cdot (-13)$$

4-6-6



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Left Riemann sum
- ☐ Answer

Solution:

$$\int_0^{40} N(t) dt \approx N(0)(10 - 0) + N(10)(35 - 10) + N(35)(40 - 35)$$

$$= (-8) \cdot 10 + (-11) \cdot 25 + (-13) \cdot 5 = -420$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an overestimate because N is decreasing.

The response does not need an explicit reference to part (a) or to a left Riemann sum.

A response that appeals to concavity as a reason will NOT earn this point.

The following table contains some examples of what WILL or will NOT earn the point.

The following examples earn the point:

- Underestimate because N is decreasing
- Underestimate because $N'(t) < 0$
- Underestimate because the graph of N is decreasing (we know they are talking about the given function N)
- Underestimate because the function is decreasing. (There is only one function in the problem at this point.)

The following examples do not earn the point:

- Underestimate because N is concave down/up
- Underestimate because the graph/curve is decreasing. Since no graph/curve is provided, we do not know what graph/curve they are talking about.
- Underestimate because the table [or table values] is/are decreasing. The behavior of the values shown in the table is not the same as the behavior of the function N . The table does not tell us what is happening to N between the

4-6-6

given points.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because N is a decreasing function, and this is a left Riemann sum, this is an overestimate.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

We need to see both a difference and a quotient. They don't have to be together.

The expression does not need to be simplified, but values for $N(35)$ and $N(0)$ must be pulled from the table.

Decimal approximation is not expected, but if presented, then the decimal value must be correct to three decimal places, either rounded or truncated. We do not read anything past the third decimal place.

- If the first three places match -0.142 or -0.143 then the point is earned.
- It does not matter whether any subsequent decimal places match the correct value or not. (For example, the value -0.1424 will earn the point because the first three decimal places match the correct value.)
- Responses with fewer than 3 decimal places (-0.14 , -0.1 , etc.) will not earn the point

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

The second point is for correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols. For example:

- Coins per minute per minute or Coins/min/min
- Coins per min^2 or Coins/ min^2
- Coins per minute squared

We do not read work shown in part (d) for this point. The units must appear in part (c).

4-6-6

The letter “ C ” is not a standard abbreviation for “Coins.” It will not be interpreted as such.

The response will not be penalized for any attempt to verbally describe their answer after a correct answer has been presented.

Minimalist response: The following expression and units represent the minimum work that must be seen in a response in order to earn both points. $\frac{-13+8}{35}$ coins/min² or $\frac{-13-(-8)}{35}$ coins/min²



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$N'(10) \approx \frac{N(35)-N(0)}{35-0} = \frac{-13-(-8)}{35} = -\frac{1}{7} \text{ coin per minute per minute}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

- Indicating a rate of a rate
- Including the context of coin preparation (coins remaining)
- Explicitly referring to the specified point in time ($t = 10$)

Incorrect or missing units will not be penalized as long as there is a clear connection to the context of the problem.

The words “increasing” and “decreasing” are interpreted as modifiers describing one of the specified rates.

If the numerical value from part (c) is used in this interpretation, then we expect that the choice of “increasing” or “decreasing” is consistent with that value.

The following examples use $\frac{-1}{7}$ as the imported value from part (c). If a different [incorrect] value is imported from part (c), then these statements should be consistent with that value. (Remember that we are reading $=$ and $\approx$ as the same thing.)

4-6-6

- “... increasing at a rate of $\frac{-1}{7}$ ” is OK [Units not required in part (d)] (the point is not earned with this phrase alone)
- “... decreasing at a rate of $\frac{-1}{7}$ ” is OK [Units not required in part (d)] (the point is not earned with this phrase alone)
- “... decreasing at a rate of $\frac{-1}{7}$ ” is not OK and the point is not earned.

Note: the context is complicated. $N(t)$ is describing the rate of change of the remaining unprepared coins. The values in the table indicate that the number of unprepared coins is decreasing more rapidly as time passes. Common sense might tell us that this equates to the idea that the rate at which coins are prepared is increasing as time passes. Explaining the meaning using “coins prepared” rather than “coins remaining” will not prevent the response from earning this point.

Rate of a rate

To earn this point, the response must explicitly discuss the rate of change of a rate of change. For example:

- The rate of change in the rate at which coins are changing
- The rate of change in coins is changing/increasing/decreasing at a rate of . . .
- The rate at which the rate of change of coins is changing

The words “increasing,” “decreasing,” or “changing” are not interpreted as rates.

Units are not interpreted as rates. (The phrase “coins per minute” is not enough to indicate a rate.)

Slope of tangent line, velocity, and acceleration are not appropriate interpretations for this context.

“Rate of change of N ” is not enough without describing what N represents in this context.

Context

A key component of the context for this explanation is “coins.”

Any misstatement about coins prepared vs. coins remaining will not prevent the response from earning this point.

Explicit reference to $t = 10$

Acceptable references to a specific point in time include (but are not limited to):

- At minute 10 (where minute could be replaced with any other unit of time)
- At $t = 10$ (with or without units of time)
- After 10 minutes (where minutes could be replaced with any other unit of time).

4-6-6

Note: we are choosing to assume that “after” is being used in lieu of “at” unless we have additional evidence to the contrary.

If the response is explicitly indicating an interval of time (e.g., “from minute 10 on” or “during the first 10 minutes”), this point will not be earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$N'(10)$ is the rate at which the rate of change in the number of 5-cent coins remaining is changing, in coins per minute per minute, at minute 10.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. Any of these equations can be written as is without any explanation or additional supporting work. If rewriting of equations is done (transforming to slope-intercept form, perhaps), it must be done correctly.

· $y = -0.2(t - 35) - 13$

· $y + 13 = -0.2(t - 35)$

· $y = -0.2t - 6$

The key calculus concept being assessed in this part of the problem is that the value of $N'(35)$ is the slope of the tangent line. In that spirit:

- The response will not be penalized for using x instead of t .
- We are accepting $N(t)$ in lieu of y in all forms of the tangent line equation.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

4-6-6

Solution:

$$y = N'(35)(t - 35) + N(35)$$

$$y = -0.2(t - 35) - 13$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are 3 components of substitution that we need to see:

- Uses $u = \frac{t}{4}$, which may be done implicitly
- Uses the correct constant of 4
- Uses the correct limits for the substitution (0 and 10). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.

Either of the expressions $4N\left(\frac{t}{4}\right)\bigg|_0^{40}$ or $4N(u)\bigg|_0^{10}$ will earn the substitution point without any additional supporting work.

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point.

- Limits for t attached to an integral with u as the variable of integration (or vice versa)
- Improper equals signs, such as an indefinite integral equal to a definite integral
- Reversed limits of integration, for example: $\int_{10}^0 N'(u) du$

The second point is for the Fundamental Theorem of Calculus.

This point can be earned with any of the following, where k is a nonzero constant and $b = 10$ or $b = 40$. No explanation or additional supporting work is necessary.

- $kN\left(\frac{t}{4}\right)\bigg|_0^b$
- $k(N(b) - N(0))$

This point can be earned without the substitution point.

The third point is for the correct numerical value, which is -12 .

Values from the table must be used. $4(N(10) - N(0))$ is not enough to earn the answer point.

4-6-6

The expression $4(-11 - (-8))$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they do not need to be correct to earn this point.

The response cannot earn this point without also earning first two points.

Minimalist response: The expression $4(-11 - (-8))$ represents the minimum work that must be seen in a response in order to earn all three points.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = \frac{t}{4}$. Then $du = \frac{dt}{4}$.

When $t = 0$, $u = 0$.

When $t = 40$, $u = 10$.

$$\int_0^{40} N' \left(\frac{t}{4} \right) dt = 4 \int_0^{10} N' (u) du$$

$$= 4(N(10) - N(0))$$

$$= 4(-11 - (-8)) = -12$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy for this part: This question was difficult due to the fact that the new function introduced here represents the total number of coins prepared by the second child, but the question asks about the total weight of the coins remaining.

The key calculus concept being assessed in this part of the problem is to understand that the rate of change in the total weight of the coins (either prepared or remaining) at time $t = 23$ could be written as an expression involving

4-6-6

a constant multiple of $F'(23)$. In that spirit, either of the following expressions will earn the point: $5F'(23)$ or $-5F'(23)$.

Notations such as $5 \cdot \frac{dF}{dt} \Big|_{t=23}$ may also be used and earn the point.

The expression must use the time $t = 23$, so $5F'(t)$ is not sufficient to earn the point.



0	1
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The student response accurately includes the criteria below.

☐ Expression

Solution:

Let $C(0)$ be the total number of coins available to be prepared at time $t = 0$. Let $R(t)$ be a model for the number of coins remaining at time t with $R(t) = C(0) - F(t)$. The rate of change of the number of coins remaining at time t , in grams per minute, is $R'(t) = -F'(t)$. Therefore, the rate of change of the total weight of the 5-cent coins remaining for the second child at $t = 23$ minutes, in grams per minute, is $-5F'(23)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between the differential equation

$$\frac{dy}{dt} = F'(t) \text{ and the given slope field.}$$

In order to earn the point, the response must:

State that the values of the differential equation are positive while the slopes in the given slope field are negative or 0 (non-positive).

Some ways to say that the values of differential equation are positive:

- F is increasing
- $F'(t) > 0$
- F has positive slopes

When discussing the slopes in the slope field, the response must explicitly refer to the signs of the slopes.

The following examples could earn the point:

- The slope field has nonpositive slopes.

4-6-6

- The slopes shown are negative or 0.
- The slope field contains negative slopes.
- “The graph” or “The field” could be used in lieu of “The slope field.”

The following examples will not earn the point:

- The slope field is decreasing/negative.
- The slope field looks like it represents a decreasing function.
- The slope field represents a quadratic [or exponential] function.
- “It” (rather than “the slope field”)

Statements like “ F is an increasing function, so the signs of the slopes in the slope field should be positive” can earn this point because “should be” is indicating contrast with what is known.

The response will not be penalized for saying that the slopes in the slope field are “always” negative or “all” negative.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

F is an increasing function, so $\frac{dy}{dt} = F'(t) \geq 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are negative.

4-6-6

65. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

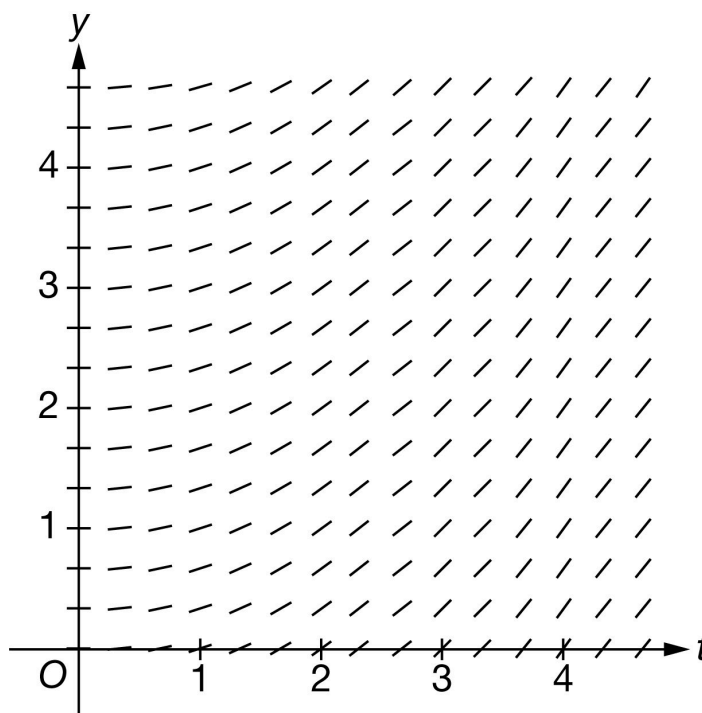
t (minutes)	0	10	40
$N(t)$ (coins per minute)	-10	-16	-8

A child has collected 5-cent coins and is preparing them to deposit in the bank by putting them into bags. For time t minutes, $0 \leq t \leq 40$, the rate of change in the number of coins remaining to prepare, in coins per minute, is modeled by a twice-differentiable function N . Values of $N(t)$ at various times t are shown in the table above.

- (a) Using correct units, approximate $N'(10)$ using the average rate of change of N over the interval $0 \leq t \leq 40$.
- (b) Explain the meaning of $N'(10)$ in the context of the problem.
- (c) The function $C(t)$ models the number of coins remaining to be put into bags at time t minutes, where $C'(t) = N(t)$. It is known that $C(10) = 400$. Write an equation for the line tangent to the graph of $y = C(t)$ at $t = 10$.
- (d) Find the value of $\int_0^{10} N'(4t) \, dt$. Show the calculations that lead to your answer.
- (e) Use a trapezoidal sum with the two subintervals indicated by the table to approximate $\int_0^{40} N(t) \, dt$. Show the calculations that lead to your answer.
- (f) A second child is preparing another deposit of 5-cent coins. For $0 \leq t \leq 40$, the rate of change in the number of coins remaining for the second child to prepare, in coins per minute, can be modeled by the differentiable function F , where F is a function of t . Assume the weight of each coin is 5 grams. Write an expression in terms of N and F for the rate of change, in grams per minute, in the combined weight of the 5-cent coins remaining for the two children to prepare at $t = 11$ minutes.

4-6-6

(g) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = 0.4y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

Answers without supporting work do not earn this point. Supporting work must be at least a difference and a quotient.

Response does not need to be simplified, but any simplification must be done correctly.

If a decimal is presented, it must be correct to 3 decimal places. We do not read anything past the third decimal place.

The second point is earned for the correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols.

The response will not be penalized for any verbal attempt to discuss the answer after the correct presentation of the approximation.

The minimum response needed to earn both points is: $\frac{-8+10}{40}$ coins per minute per minute

4-6-6



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$N'(10) \approx \frac{N(40) - N(0)}{40 - 0} = \frac{-8 - (-10)}{40} = \frac{1}{20} \text{ coin per minute per minute}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This part is intended to be independent from part (a), so we do not want or need to see any reference to the value obtained in (a). If that value is referenced, conclusion must be consistent.

Three components of a correct answer:

- Indicating a rate of a rate (look for the word rate to be used correctly twice)
- Including the context of coin collection
- Referring to the specific point in time

Units will not be considered for this point. They do not need to be present, and if they are, they do not need to be correct.

Any incorrect statements make the response ineligible for the point. For example:

- $N'(10)$ is the average (or an approximation of, or an estimate of) rate of change of
- $N'(10)$ is the acceleration (or rate of change of velocity) of

“Increasing” or “decreasing” may be used in place of “changing” as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Beware of phrases such as “decreasing at a negative rate.”

· Example: $N'(10)$ is the rate of increase of the rate of change of the number of coins at 10 minutes. (earns the point)

· Example: $N'(10)$ means the number of coins per minute is increasing by $\frac{1}{20}$ coins/min/min (does not earn the point)

4-6-6

Acceptable references to time include (but are not limited to):

- At the tenth minute
- At $t = 10$ (with correct units of time, without correct units of time, or even with incorrect units of time)
- After $t = 10$ (with correct units of time, without correct units of time, or even with incorrect units of time)

If the response clearly indicates an interval of time (e.g. “over the first ten minutes”) the response will not earn the point.

Examples that earn the point include:

- $N'(10)$ represents the rate at which the rate of change of the number of coins is changing at minute 10.
- $N'(10)$ is the rate of change in the rate of change of the number of coins at 10 minutes.
- $N'(10)$ represents the rate at which the rate of change of the number of coins is increasing at $t = 10$.
- $N'(10)$ equals (answer from part (a)) means the rate of change in the number of coins per minute is increasing at a rate of (answer from part (a)) at $t = 10$.
- $N'(10)$ represents the rate at which the rate of change of the number of coins is decreasing after 10 10 minutes.

Examples that do not earn the point include:

- $N'(10)$ represents the rate at which the number of coins is changing at minute 10.
- $N'(10)$ is the rate of change in the number of coins per minute at 10 minutes.
- The rate of change in the number of coins per minute is increasing at $t = 10$.
- The rate of change in the rate of change of the number of coins per minute is increasing at $t = 10$.
- $N'(10) = -\frac{1}{20}$ means that the rate of change in the number of coins is decreasing at a rate of $-\frac{1}{20}$ after 10 minutes. (double negative)



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$N'(10)$ is the rate at which the rate of change of the number of 5-cent coins remaining is changing, in coins per minute per minute, at time $t = 10$ minutes.

4-6-6

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned with a correct tangent line equation in any form. Equivalent forms include (but are not limited to):

· $y = -16(t - 10) + 400$

· $y - 400 = -16(t - 10)$

· $y = -16t + 560$

The student may write x in place of t and still earn the point.

No work is required to earn the point.

We will accept $C(t)$ in lieu of y in the equation of the tangent line.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = C'(10)(t - 10) + C(10)$$

$$y = -16(t - 10) + 400$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution.

Two things are required to award this point:

· Coefficient of $\frac{1}{4}$ correctly applied to an integral or an antiderivative

· Evidence that the correct limits of 0 to 40 will be or have been applied

Examples that will earn the first point (these are just a few):

4-6-6

$$\cdot \frac{1}{4} \int_0^{40} N'(u) \, du$$

$$\cdot \frac{1}{4} N(u) \Big|_0^{40} \text{ or } \frac{1}{4} N(4t) \Big|_0^{10}$$

Responses that include misstatements such as using t -values for limits with u as the variable of integration in an intermediate step which is later corrected may still earn this point. Intermediate errors in bounds that are subsequently corrected will disqualify the student from earning the third point. If the bounds are never appropriately handled, the response cannot earn the first point. Scratch work that is not connected to the solution will not be read.

Examples:

$$\cdot \int_0^{10} N'(4t) \, dt = \frac{1}{4} \int_0^{10} N'(u) \, du = \frac{1}{4} N(4t) \Big|_0^{10} \text{ earns the first point, but will not earn the third point.}$$

$$\cdot \int_0^{10} N'(4t) \, dt = \frac{1}{4} \int_0^{10} N'(u) \, du = \frac{1}{4} [N(10) - N(0)] \text{ does not earn the first point.}$$

The second point is for the Fundamental Theorem of Calculus, demonstrating the knowledge that

$$\int_a^b N'(u) \, du = N(u) \Big|_a^b = N(b) - N(a)$$

This point can be earned with either $kN(4t) \Big|_a^b$, $kN(t) \Big|_a^b$, or $k(N(b) - N(a))$ where k is a nonzero constant [not necessarily correct], $a = 0$, and $b = 10$ or $b = 40$.

A response can earn this point without earning the first point.

$$\frac{1}{4} [N(40) - N(0)] \text{ earns the first two points.}$$

The third point is for the correct numerical value. Our answer only.

Values from the table must be used (i.e., $\frac{1}{4} [N(40) - N(0)]$ is not sufficient to earn the third point).

If the response includes any linkage error, (i.e., the use of equal signs to equate two expressions that are not equal), the third point will not be earned. If an indefinite integral is linked with a definite integral the response will not be penalized.

$$\cdot \int_0^{10} N'(4t) \, dt = \frac{1}{4} \int N'(u) \, du = \frac{1}{4} N(4t) \Big|_0^{10} = \dots \text{ is eligible for the third point.}$$

Eligibility: Must have earned the first two points.

The minimum response needed to earn all three points is: $\frac{1}{4}(-8 + 10)$.

4-6-6



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 4t$. Then $du = 4 dt$.

When $t = 0$, $u = 0$.

When $t = 10$, $u = 40$.

$$\begin{aligned}
 \int_0^{10} N'(4t) dt &= \frac{1}{4} \int_0^{40} N'(u) du \\
 &= \frac{1}{4}(N(40) - N(0)) \\
 &= \frac{1}{4}(-8 - (-10)) = \frac{1}{2}
 \end{aligned}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a trapezoidal sum

Point is earned for the setup of the trapezoidal sum. Numerical values do not need to be pulled from the table for this point, but Δt values must be presented. Responses that contain one incorrect numerical value in the trapezoidal sum setup are still eligible to earn the first point, but will not earn the second.

Response may present a left Riemann sum and a right Riemann sum, then average the two. Again, one incorrect numerical value will not disqualify the response from earning this first point.

Response may compute trapezoids as areas of rectangles and triangles.

Trapezoidal areas may be given in a graph. In order to earn this point, the response must clearly show the computations leading to the values presented in the graph, and must show that these areas are added to obtain the final approximation value.

A response using subtraction instead of addition will not earn the point

4-6-6

(e.g. $\frac{1}{2}[10(N(0) - N(10)) + 30(N(10) - N(40))]$ earns no points.)

Response utilizing “Trapezoidal Rule” will not earn this point, unless this represents only one incorrect numerical value.

$\cdot \frac{1}{2}(10)[N(0) + 2N(10) + N(40)]$ is equivalent to $\frac{1}{2}[10(N(0) + N(10)) + 10(N(10) + N(40))]$
 $\frac{1}{2}[10(N(0) + N(10)) + 10(N(10) + N(40))]$, thus has only one incorrect numerical value (the second interval width is 10 instead of 30), so can earn the point.

$\cdot \frac{1}{2}(20)[N(0) + 2N(10) + N(40)]$ has two incorrect numerical values (both interval widths are incorrect) so will not earn the point.

The second point is for the correct numerical result.

Our answer only. Response does not need to be simplified, but if it is, it must be simplified correctly.

The answer must be accompanied with supporting work. Supporting work may appear in a graph, with clearly labeled areas. It is possible to earn the second point without earning the first point, if the supporting work is not sufficient to earn the first point.

Pictures without clear area computations and a clear summation of quantities can earn at most the second point.

Special case: A positive constant multiple of a complete and correctly expressed trapezoidal sum (with the correct values pulled from the table) will earn the first, but not the second point:

$\cdot k \left[\left(\frac{-10-16}{2} \right) \cdot 10 + \left(\frac{-16-8}{2} \cdot 30 \right) \right]$ for $k > 0$ earns only the first point.

The minimum response needed to earn both points: $\left(\frac{-26}{2} \right) \cdot 10 + \left(\frac{-24}{2} \right) \cdot 30$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of trapezoidal sum
- ☐ Answer

Solution:

$$\int_0^{40} N(t) dt \approx \left(\frac{N(0)+N(10)}{2} \right) (10-0) + \left(\frac{N(10)+N(40)}{2} \right) (40-10)$$

$$= \left(\frac{-10-16}{2} \right) \cdot 10 + \left(\frac{-16-8}{2} \right) \cdot 30 = -490$$

4-6-6

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for a correct constant multiple or correct sum.

Response must contain the constant 5 as a factor of the expression, and/or the correct sum $N(11) + F(11)$.

If the sum is incorrect, then the constant factor of 5 must be applied either to a sum that involves only $N(t)$ and $F(t)$, $N'(t)$ and $F'(t)$, or integrals of $N(t)$ and $F(t)$ or to one of the correct terms of the sum $N(11)$ or $F(11)$, with no additional incorrect terms.

Examples that earn the point include:

$$\cdot 5 [N (t) + F (t)]$$

$$\cdot 5 \int [N (t) + F (t)] dt$$

$$\cdot 5 [N' + F']$$

$$\cdot 5N (11)$$

Examples that do not earn the point include:

$$\cdot 5 [N (t) + F' (t)]$$

$$\cdot 5 \int N (t) dt$$

$$\cdot 5N' + F'$$

$$\cdot 5N (11) + t$$

Special case: $5N (11) + F (11)$ is an error in parentheses, so will earn the first point but is not eligible to earn the second point.

If the constant multiple is missing or incorrect, the sum must be correct. $k [N (11) + F (11)]$, for any nonzero constant k will earn the first point.

The second point is for the correct expression.

$5 [N (11) + F (11)]$ (or an equivalent expression) earns both points.

Inclusion of units (correct or incorrect) will not affect the score.

A completely correct expression with one additional constant term added/subtracted would earn the first point only. For example: $5 [N (11) + F (11)] + 5 (2)$ earns only the first point.

4-6-6



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Constant multiple or sum
- ☐ Expression

Solution:

The rate of change, in grams per minute, in the combined weight of the 5-cent coins remaining for the two children to prepare at $t = 11$ minutes is $5N(11) + 5F(11)$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for a correct statement identifying specific inconsistencies between the given differential equation and the given slope field. There are many such correct statements, such as:

- The response references a specific point on the y -axis (such as $(0, 4)$), and indicates that the given slope field shows a slope of zero, yet the differential equation would yield a positive slope.
- The response references a pattern of slopes, and explains why that pattern does not fit our differential equation (e.g. “the slope field shows slopes that change as t changes, but our differential equation depends only on y , so slopes should remain constant for a given value of y , as t changes.”)

Identifying a mismatch between the differential equation and the slope field at one point is sufficient to earn the point.

Response is not penalized for referencing “ x ” instead of “ t ”.

If the response contains any incorrect statement, the response will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point, nor will responses that present information beyond what we are given. For example:

- Responses that specify that the slope field shows a particular nonzero slope at a particular point will not earn the point, as this cannot actually be determined visually (i.e., the response should not indicate that at $(1, 2)$ the slope shown on the slope field is 1)
- Responses that state a specific slope value on the slope field for a general value of x or y that are not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

4-6-6

A response may involve a solution to the differential equation, with an argument based on the resulting function. If done correctly, and presented with a correct argument, this will earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

4-6-6

66. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	3	6	9	12
$S(t)$ (number of boxes of snacks remaining)	600	500	440	240	20

A student organization is selling boxes of snacks to raise money for a school trip. The student organization begins with 600 boxes of snacks and sells them over a twelve-day period. For time t days, $0 \leq t \leq 12$, the total number of boxes of snacks remaining to sell is modeled by a twice-differentiable function S . Values of $S(t)$ at various times t are shown in the table above.

(a) Use a midpoint Riemann sum with the two subintervals indicated by the table to approximate

$$\int_0^{12} S(t) \, dt.$$

Show the computations that lead to your answer.

(b) Find the value of $\int_0^3 S'(3t) \, dt$. Show the computations that lead to your answer.

(c) Using correct units, approximate $S'(9)$ using the average rate of change of S over the interval $6 \leq t \leq 12$.

(d) Explain the meaning of $S'(9)$ in the context of the problem.

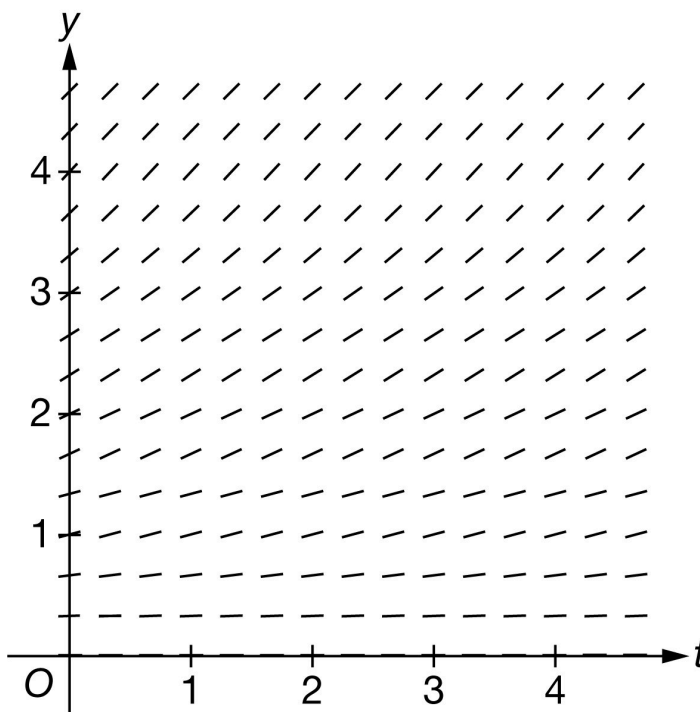
(e) It is known that $S'(6) = -40$. Write an equation for the line tangent to the graph of $y = S(t)$ at $t = 6$.

(f) It is known that $S''(t) < 0$ for $6 \leq t \leq 9$. If $S(8)$ is approximated using the equation from part (e), would the approximation be an overestimate or underestimate for the actual value of $S(8)$? Give a reason for your answer.

(g) For $0 \leq t \leq 12$, the total number of boxes of snacks remaining can be modeled by the differentiable function Q , where Q is a function of t . Assume the weight of each box is 8 ounces. Write an expression for the rate of change, in ounces per day, of the total weight of the boxes of snacks remaining at $t = 5$ days.

4-6-6

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = -0.25y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Form of midpoint sum): Philosophy: The response should provide clear evidence adding the area of two rectangles whose heights are found using the midpoint of the interval.

Acceptable evidence:

$$\cdot 6 \cdot (S(3) + S(9)) \text{ or } 6 \cdot S(3) + 6 \cdot S(9)$$

$$\cdot (6 - 0) \cdot 500 + (12 - 6) \cdot 240$$

$$\cdot 6 \cdot (500 + 240) \text{ or } 6 \cdot 500 + 6 \cdot 240$$

To earn this point the response can have at most one mistake in the expression or equation. So, expressions of the form $\frac{6}{b} \cdot (S(3) + S(9))$, for any nonzero value of b would earn the point. Note: The value $\frac{6}{b}$ might be distributed across the sum.

Expressions of the form $\frac{a}{b} \cdot (S(3) + S(9))$, for $a \neq 6$ will not earn this point.

Once the first point is earned, it cannot be lost.

Second Point (Answer)

To earn the point, the answer must be correct. It does not need to be simplified, but if it is, the simplification must be correct.

4-6-6

The answer point cannot be earned unless the first (midpoint sum) point was earned.

Minimal complete answers that earn both points:

$$\cdot 6 \cdot (500 + 240)$$

$$\cdot 6 \cdot 500 + 6 \cdot 240$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Form of midpoint sum

☐ Answer

Solution:

$$\int_0^{12} S(t) dt \approx S(3)(6 - 0) + S(9)(12 - 6)$$

$$= 500 \cdot 6 + 240 \cdot 6 = 4440$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Substitution) Philosophy: This point is earned when the response has all three components of the substitution done correctly.

Uses $u = 3t$, perhaps implicitly (any variable could be used for u)

Uses the correct constant of $\frac{1}{3}$

Uses the correct limits for their substitution.

· The limits might not be seen until the final evaluation, and at that point, response may have returned to the variable t .

· Limits for t attached to an integral involving u or vice versa will not earn this substitution point but could still earn the answer point, if the numerical answer is correct.

Second point: (Fundamental Theorem of Calculus) Philosophy: This point is earned when the FTC has been applied correctly to either $\int S'(3t) dt$ or $\int S'(u) du$.

The point is earned for any of the following:

4-6-6

· The equation $\int S'(3t) dt = S(3t)$ or $\int S'(u) du = S(u)$

· $\frac{1}{3}S(3t)\big|_0^3$ or $\frac{1}{3}S(u)\big|_0^9$

· $S(9) - S(0)$, $S(9)$, or $S(0)$

This point can be earned if the substitution point is not.

Note: $\int_a^b S'(u) du = S(b) - S(a)$ does not earn the point. The FTC point is for the response communicating that the antiderivative of S' is S .

Third Point (Answer)

Must be the correct numerical answer.

Must have earned the second point for the FTC.

Might have not earned the first point for substitution because of an incorrect use of limits but will still earn the third point for the correct answer.

$S(9)$ and $S(0)$ must be evaluated; the expression $\frac{1}{3}(S(9) - S(0))$ by itself does not earn the third point.

The answer may be unsimplified, but if simplified it must be done correctly.

Scoring common responses:

The minimal work required to earn all three points is $\frac{1}{3}(240 - 600)$.

The following responses would earn the first two points and be eligible for the answer point if the response continues:

· $\frac{1}{3}S(3t)\big|_0^3$

· $\frac{1}{3}(S(9) - S(0))$

Responses of only the following would earn only the second point:

· $S(9) - S(0)$

· $S(3) - S(0)$

· $\int_0^9 S'(u) du = S(9) - S(0)$

4-6-6

$$\cdot S(3t) \Big|_0^3 \text{ or } S(u) \Big|_0^9$$

The following response would earn no points: -120 in the absence of one of the first two points.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 3t$. Then $du = 3 \, dt$.

When $t = 0$, $u = 0$.

When $t = 3$, $u = 9$.

$$\begin{aligned} \int_0^3 S'(3t) \, dt &= \frac{1}{3} \int_0^9 S'(u) \, du \\ &= \frac{1}{3}(S(9) - S(0)) \\ &= \frac{1}{3}(240 - 600) = -120 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Approximation)

The response must be a difference and a quotient.

The answer does not need to be simplified, but if it is simplified it must be done correctly.

The minimal work that must be shown to earn this point is either $\frac{20-440}{6}$ or $\frac{-420}{12-6}$.

Second point (Units)

The units must be attached to a correct or incorrect value.

4-6-6

The response will not be penalized for any verbal attempt to discuss the answer after the correct approximation with units is presented.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Approximation

☐ Units

Solution:

$$S'(9) \approx \frac{S(12) - S(6)}{12 - 6} = \frac{20 - 440}{6} = -70 \text{ boxes per day}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The interpretation must include three components:

- indicate the rate of change
- the specific context of the number of boxes remaining
- the specific point in time, $t = 9$.

Responses do not need to include correct units or any units.

The word “decreasing” may be used in place of the word “changing”.

The response is not earned for:

- “the rate of boxes sold on day 9” as this does not indicate a change in the rate of boxes remaining.
- “decreasing at a rate of -70 boxes per day”

“the rate of change of S ” without specifying that S is the number of boxes remaining, does not earn the point.

Acceptable ways to provide the specific point in time:

- On day 9 or after 9 days.
- At $t = 9$ (missing the unit of time)
- On week 9 (incorrect unit of time)

4-6-6

A response that clearly indicates an interval of time (e.g. “during the first 9 days” or “from day 9 on”) does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$S'(9)$ is the rate at which the number of boxes of snacks remaining is changing, in boxes per day, on day 9.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The use of $S(t)$ in place of y earns the point.

The use of x in place of t earns the point.

Tangent lines do not need to be simplified, but any simplification must be done correctly.

Any correct form of the tangent line earns the point. For example:

· $y = 440 - 40(t - 6)$

· $y - 440 = -40(t - 6)$

· $y = -40t + 240 + 440$

· $y = -40t + 680$



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = S'(6)(t - 6) + S(6)$$

4-6-6

$$y = -40(t - 6) + 440$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the point the response must include $S''(t) < 0$ and overestimate.

“ $S(t)$ is concave down” can be used instead of $S''(t) < 0$.

Information about $S'(t)$ by itself without reference to $S''(t) < 0$ is not sufficient and does not earn this point.

“Overestimate because it is concave down” is not sufficient by itself because of the use of “it”. The response must clearly, directly or indirectly, indicate to what “it” is referring.

If a response makes the correct conclusion and states, $S''(t) < 0$, and then goes on and makes an incorrect statement or a statement that cannot be verified will not earn this point.

For example: “ $S'(t)$ is concave down, overestimate” does not earn this point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because $S''(t) < 0$, the tangent line approximation would produce an overestimate.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression of only $8 \cdot Q'(5)$ earns this point.

Only expression $8 \cdot S'(5)$ also earns this point.

No justifications or explanations are required.

Any incorrect statement such as, $Q'(5) = 8 \cdot S'(5)$, does not earn this point.



0	1
---	---

4-6-6

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in ounces per day, of the total weight of the boxes of snacks remaining at $t = 5$ days is $8Q'(5)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Response clearly references what part of the differential equation is different from the slope field and the response needs only to clearly and explicitly reference one of them.

The response does not need to explicitly reference both the differential equation and the given slope field.

The distinction can be made with explicit reference to one of the two things and an implicit reference to the other

For example:

· “ $\frac{dy}{dt}$ is negative, not positive”. We will infer that negative is referring to the slopes shown in the slope field.

· “This would be correct for a positive derivative, but $\frac{dy}{dt}$ is negative.” We will read “This” as “the slope field” in this sentence.

· “All of the slopes in the slope field are positive, not negative.” We will infer that the values of $\frac{dy}{dt}$ at points in the first quadrant are negative.

A response that lacks even an implicit reference to either the differential equation or the slope field will not earn the point. For example:

· “ $\frac{dy}{dt}$ is negative” lacks any reference to the slope field and so will not earn the point.

· “All the slopes in the graph are positive” lacks any reference to the differential equation, and so will not earn the point.

· Vague answers such as “The slope field has positive points” or “The field is increasing” will not earn the point.



0

1

The student response accurately includes the criteria below.

4-6-6☐

Explanation

Solution:

When $y > 0$, $\frac{dy}{dt} < 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are positive.

4-6-6

67. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	2	5	10
$S(t)$ (number of boxes of snacks remaining)	500	340	200	20

A student organization is selling boxes of snacks to raise money for a school trip. The student organization begins with 500 boxes of snacks and sells them over a ten-day period. For time t days, $0 \leq t \leq 10$, the total number of boxes of snacks remaining to sell is modeled by a decreasing, twice-differentiable function S . Values of $S(t)$ at various times t are shown in the table above.

(a) Using correct units, approximate $S'(5)$ using the average rate of change of S over the interval $2 \leq t \leq 10$.

(b) Explain the meaning of $S'(5)$ in the context of the problem.

(c) It is known that $S'(2) = -55$. Write an equation for the line tangent to the graph of $y = S(t)$ at $t = 2$.

(d) Use a left Riemann sum with the three subintervals indicated by the table to approximate $\int_0^{10} S(t) \, dt$. Show the computations that lead to your answer.

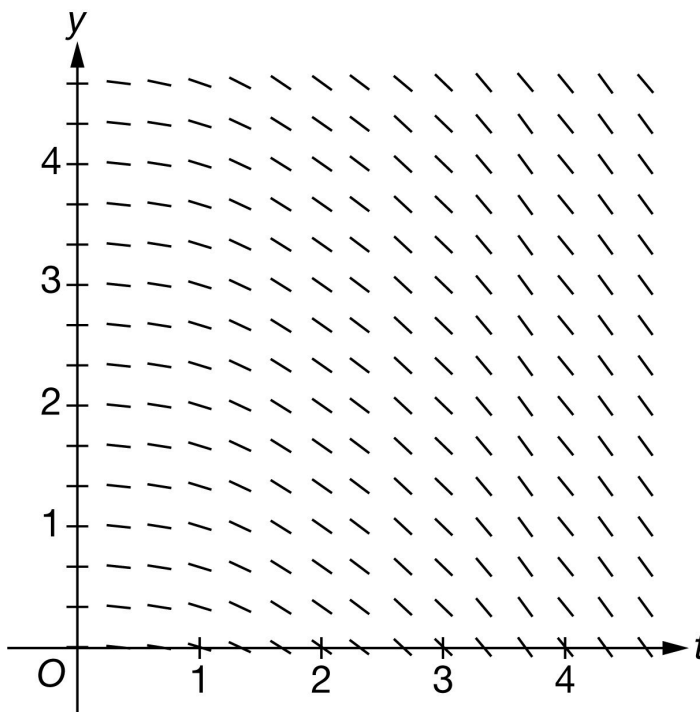
(e) Is the approximation from part (d) an overestimate or an underestimate for $\int_0^{10} S(t) \, dt$? Give a reason for your answer.

(f) Find the value of $\int_0^5 S'(2t) \, dt$. Show the computations that lead to your answer.

(g) For $0 \leq t \leq 10$, the total number of boxes of snacks remaining can also be modeled by the differentiable function M , where M is a function of t . Assume the weight of each box is 7 ounces. Write an expression for the rate of change, in ounces per day, of the total weight of the boxes of snacks remaining at $t = 8$ days.

4-6-6

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = -0.2y$.

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work. The supporting work must be at least a difference and a quotient. Correct values must be taken from the table to earn the point. The difference quotient does not need to be simplified.

Examples that earn the first point include:

$$\begin{aligned} & \cdot \frac{20-340}{10-2} \\ & \cdot \frac{S(10)-S(2)}{8} = \frac{-320}{8} \\ & \cdot \frac{S(10)-S(2)}{8} = -40 \end{aligned}$$

Examples that do not earn the first point include:

$$\begin{aligned} & \cdot \frac{S(10)-S(2)}{8} \\ & \cdot \frac{S(10)-S(2)}{10-2} \end{aligned}$$

Minimal response that earns the point: $\frac{20-340}{10-2}$ or $\frac{20-340}{8}$

4-6-6

Neither $\frac{-320}{8}$ nor -40 , with no supporting work, will earn this point

The second point is for correct units

The correct units must be attached to a numerical value: either correct or incorrect.

The units may be written using words or a combination of words and symbols. For example:

- Boxes per day
- Boxes of snacks per day
- Snacks per day
- Boxes/day

The response will not be penalized for any verbal attempt to discuss the answer after the correct answer has been presented.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$S'(5) \approx \frac{S(10) - S(2)}{10 - 2} = \frac{20 - 340}{8} = -40$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

- Indicating a rate of change
- Including the context of boxes of snack collection
- Referring to the specific point in time, $t = 5$

Students are not penalized for incorrect units as a modifier in this part. (Whether correct or incorrect, units have no effect on the score in this part.) Example: “The number of coins is decreasing at a rate of 40 coins per day.”

4-6-6

Responses may use “changing” or “decreasing” as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Example: “The number of boxes is decreasing at a rate of 40 boxes per day” is okay, but “The number of boxes is decreasing at a rate of -40 boxes per day” would not earn the point.

Acceptable references to a specific point in time include (but are not limited to):

- On day 5 and after 5 days
- At $t = 5$
- At $t = 5$ weeks or any incorrect unit of time

If the response clearly indicates an interval of time (e.g., “from day 5 on,” “during the first 5 days,” or “for $t = 0$ to $t = 5$ ”), the point is not earned.

Examples that earn the point include:

- $S'(5)$ is the rate of change of the number of boxes of snacks per day at time $t = 5$
- The rate of change of number of boxes of snacks per day is -40 boxes of snacks per day on day 5
- $S'(5)$ is the rate of change in the number of boxes of snacks at time $t = 5$ in boxes of snacks per *month*
- 40 is the rate of decrease in the number of boxes of snacks per day on day 5

Examples that do not earn the point include:

- $S'(5)$ is the average rate of change of the total number of boxes of snacks per day on day 5
- $S'(5)$ is the total number of boxes of snacks per day on day 5
- The rate of change is -40 boxes of snacks per day (Missing “at $t = 5$ ”)
- $S'(5)$ is the slope of the tangent line at $t = 5$
- Rate of change of S at time $t = 5$
- Increase in number of boxes of snacks at $t = 5$
- Boxes per day increases on day 5



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

4-6-6

Solution:

$S'(5)$ is the rate at which the number of boxes of snacks remaining is changing, in boxes per day, on day 5.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. The use of x in place of t is acceptable.

Some examples include

· $y - 340 = -55(t - 2), y - 340 = -55(x - 2)$

· $y = -55(t - 2) + 340, y = -55(x - 2) + 340$

· $y = -55t + 450, y = -55x + 450$

The key calculus concept being assessed in this part of the problem is that the value of $S'(2)$ is the slope of the tangent line. In that spirit, we are accepting $S(t) = \dots$ in lieu of $y = \dots$, but not $S(2) = \dots$



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = S'(2)(t - 2) + S(2)$$

$$y = -55(t - 2) + 340$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a left Riemann sum. The setup must be a sum of 3 products.

· A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example below).

· Responses with the numerical error described above are not eligible for the second point.

4-6-6

NOTE: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns. Implicit responses are eligible for both points.

A response presenting a right Riemann sum (instead of left), will not earn any points for part (a).

The second point is for the correct numerical result.

- In order to earn the second point, there must be supporting calculations.
- The numerical values for $S(2)$, $S(5)$, and $S(10)$ must be pulled from the table
- Any simplification shown must be correct.

Use the following as an illustration to score both points. (In most cases, you see scores for which the expression on the left side represents the entirety of relevant work shown for the response.)

- A response of $500 \cdot (2 - 0) + 340 \cdot (5 - 2) + 200 \cdot (10 - 5)$ earns both points.
- A response of $2 \cdot S(0) + 3 \cdot S(2) + 5 \cdot S(5)$ earns the first point. Eligible for second point if more work is on the page.
- A response of $2 \cdot S(0) + 3 \cdot S(2) + 5 \cdot S(5) = 3020$ earns both points.
- A response of $2 \cdot 500 + 3 \cdot 340 + 5 \cdot 200$ earns both points.
- A response of $2 \cdot 500 + 10 \cdot 340 + 5 \cdot 200$ contains one numerical error. Earns the first point. Not eligible for the second point.
- A response of $2 \cdot 500 + 3 \cdot 340 + 5 \cdot 200 = 3200$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.
- A response of $1000 + 1020 + 1000$ or $1000 + 1020 + 1000 = 3020$ does not earn the first point because we cannot see the sum of 3 products. However, this is the correct sum, so we see an answer with supporting work. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points.

$$2 \cdot 500 + 10 \cdot 340 + 5 \cdot 200$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Left Riemann sum

4-6-6

☐ Answer**Solution:**

$$\int_0^{10} S(t) dt \approx S(0)(2-0) + S(2)(5-2) + S(5)(10-5)$$

$$= 500 \cdot 2 + 340 \cdot 3 + 200 \cdot 5 = 3020$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an overestimate because S is decreasing.

The response does not need an explicit reference to the approximation in part (d).

The following statements are sufficient for earning this point:

- Overestimate because S is decreasing
- Overestimate because the function is decreasing

NOTE: At this point, there is only one function being discussed in this problem, so we will assume “the function” is S .

When a response includes additional information about concavity, we interpret that as a lack of understanding about the question, so this will not earn the point. When the response references a graph or the given table rather than the function S , the point is not earned.

Examples that earn the point:

- The left Riemann sum overestimates because the function is decreasing.
- The sum is an overestimate because $S(t)$ is decreasing.
- The sum is an overestimate because $S'(t) < 0$.

Examples that do not earn the point:

- The sum is an overestimate because $S'(t)$ is decreasing. ($S'(t)$ decreasing is not known)
- The left Riemann sum overestimates because the graph is decreasing.
- The left Riemann sum overestimates because the function is decreasing and concave up.



0

1

4-6-6

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because S is a decreasing function, and this is a left Riemann sum, the answer from part (d) is an overestimate.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are three components of substitution that need to be seen:

- Uses $u = 2t$, which may be done implicitly
- Uses the correct constant of $\frac{1}{2}$
- Uses the correct limits for the substitution (0 and 10). These limits may only be seen when evaluating the antiderivative. These don't need to see them as limits on a definite integral.
- Either of the expressions below will earn the substitution point without any additional supporting work:
 $\frac{1}{2} S(2t) \Big|_0^5$ or $\frac{1}{2} S(u) \Big|_0^{10}$

Special Handling: (Notation Errors)

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point.

- Limits for t attached to an integral with u as the variable of integration (or vice versa),
- Improper equals signs, such as an indefinite integral equal to a definite integral
- Reversed limits of integration, for example: $\int_{10}^0 S'(u) du$

The second point is for the Fundamental Theorem of Calculus. This point can be earned with any of the following, where k is a nonzero constant and $b = 5$ or $b = 10$. No explanation or additional supporting work is necessary.

- $k S(2t) \Big|_0^b$
- $k (S(b) - S(0))$
- This point can be earned without earning the substitution point.

4-6-6

The third point is for the correct numerical value, which is -240 .

Values from the table must be used. $\frac{1}{2}[S(10) - S(0)]$ is not enough to earn the answer point

The expression $\frac{1}{2}(20 - 500)$ close parenthesis is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they need not be correct in order to earn this point.

The response cannot earn this point without also earning the FTC point.

Minimalist response: The expression $\frac{1}{2}(-480)$ represents the minimum work that must be seen in a response in order to earn all three points.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 2t$. Then $du = 2 dt$.

When $t = 0$, $u = 0$.

When $t = 5$, $u = 10$.

$$\begin{aligned}\int_0^5 S'(2t) dt &= \frac{1}{2} \int_0^{10} S'(u) du \\ &= \frac{1}{2}(S(10) - S(0)) \\ &= \frac{1}{2}(20 - 500) = -240\end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct expression at $t = 8$. No supporting work or justification is required.

The correct expression of $7 \cdot S'(8)$ or $7 \cdot M'(8)$ earns this point.

4-6-6

No justifications or explanations are required.

Any incorrect statement such as, $S'(8) = 7 \cdot M'(8)$, does not earn this point.



0	1
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The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in ounces per day, of the total weight of the boxes of snacks remaining at $t = 8$ days is $7M'(8)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between two things: the differential equation, $\frac{dy}{dt} = -0.2y$, and the given slope field.

The response does not need to explicitly reference both the differential equation and the given slope field, but it must explicitly reference one and implicitly reference the other.

Acceptable responses include

- Identifying an inconsistency between the differential equation and the slope field at one point, e.g., at the point $(4, 0)$, the slope of the segment should be 0, but it is negative.
- Referencing a pattern of slopes, and explaining why that pattern does not fit the given differential equation.

A response that lacks even an implicit reference to the one of two things being compared will not earn the point.

For example, merely stating either $\frac{dy}{dt}$ is negative, or that the slope field is negative does not earn the point.

· “The values for y are less than or equal to zero so $\frac{dy}{dt} \leq 0$ ” lacks any reference to the slope field and will not earn the point.

· “All the slopes in the slope field are negative” lacks any reference to the differential equation and will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point. Some examples are:

- “The field is decreasing”

4-6-6

· Responses that state a specific slope value on the slope field for a general value of x or y that is not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

A response may correctly solve or attempt to solve the differential equation and then provide a correct argument without reference to their solution. Such a response would earn the point.

A response might actually solve the differential equation and provide a correct argument on the resulting function.



0	1
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The student response accurately includes the criteria below.

☐ Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

4-6-6

68. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	2	5	10
$S(t)$ (boxes per day)	-60	-40	-30	-5

A student organization is selling boxes of snacks to raise money for a school trip. For time t days, $0 \leq t \leq 10$, the rate of change in the number of boxes of snacks remaining to sell is modeled by an increasing, twice-differentiable function S . Values of $S(t)$ at various times t are shown in the table above.

(a) Use a right Riemann sum with the three subintervals indicated by the table to approximate

$\int_0^{10} S(t) \, dt$. Show the calculations that lead to your answer.

(b) Is your answer from part (a) an overestimate or an underestimate for $\int_0^{10} S(t) \, dt$? Give a reason for your answer.

(c) Using correct units, approximate $S'(2)$ using the average rate of change of S over the interval $0 \leq t \leq 5$.

(d) Explain the meaning of $S'(2)$ in the context of the problem.

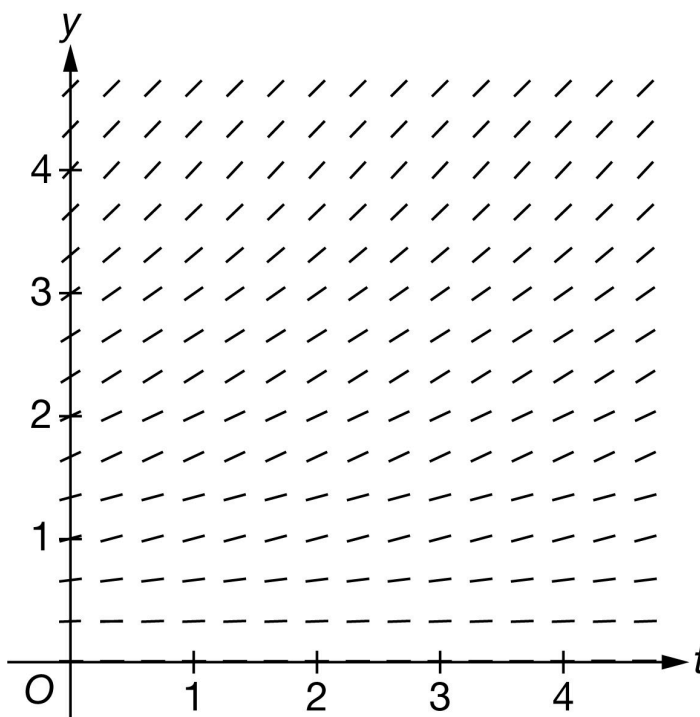
(e) It is known that $S'(5) = 6$. Write an equation for the line tangent to the graph of $y = S(t)$ at $t = 5$.

(f) Find the value of $\int_0^{10} S'\left(\frac{t}{2}\right) \, dt$. Show the calculations that lead to your answer.

(g) Another student organization is also selling boxes of snacks. For $0 \leq t \leq 10$, the total number of boxes of snacks remaining for the second student organization can be modeled by the decreasing and differentiable function M , where M is a function of t . Assume the weight of each box is 7.5 ounces. Write an expression for the rate of change, in ounces per day, of the total weight of the boxes of snacks remaining for the second student organization at $t = 6$ days.

4-6-6

(h) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = M'(t)$ for the function M defined in part (g).



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a right Riemann sum. The setup must be a sum of three products.

A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example in the table below).

Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns, like so.

-40	2	-80	or	$(2 - 0)(-40)$	$= -80$
-30	3	-90		$(5 - 2)(-30)$	$= -90$
-5	5	$+(-25)$		$(10 - 5)(-5)$	$= -25$
		-195			-195

Implicit responses such as those above are eligible for both points.

A response presenting a left Riemann sum (instead of right), will not earn any points for part (a).

The second point is for the correct numerical result.

4-6-6

In order to earn the second point, there must be supporting calculations.

The numerical values for $S(2)$, $S(5)$, and $S(10)$ must be pulled from the table

Any simplification shown must be correct.

Use the following as an illustration of how to score both points. (In most cases, you see scores for which the expression represents the entirety of relevant work shown for the response.)

- A response of $(-40) \cdot (2 - 0) + (-30) \cdot (5 - 2) + (-5) \cdot (10 - 5)$ earns both points.
- A response of $2 \cdot S(2) + 3 \cdot S(5) + 5 \cdot S(10)$ earns the first point. Eligible for second point if more work is on the page.
- A response of $2 \cdot S(2) + 3 \cdot S(5) + 5 \cdot S(10) = -195$ earns both points.
- A response of $2 \cdot (-40) + 3 \cdot (-30) + 5 \cdot (-5)$ earns both points.
- A response of $2 \cdot (-40) + 3 \cdot (-30) + 4 \cdot (-5)$ contains one numerical error. Earns the first point. Not eligible for the second point.
- A response of $2 \cdot (-40) + 3 \cdot (-30) + 5 \cdot (-5) = -190$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.
- A response of $(-80) + (-90) + (-25)$ or $(-80) + (-90) + (-25) = -195$ does not earn the first point because we cannot see the sum of three products. However, this is the correct sum, so we are not seeing an unsupported answer. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points.

$$2 \cdot (-40) + 3 \cdot (-30) + 5 \cdot (-5)$$



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Right Riemann sum
- ☐ Answer

Solution:

$$\int_0^{10} S(t) \, dt \approx S(2)(2 - 0) + S(5)(5 - 2) + S(10)(10 - 5)$$

4-6-6

$$= (-40) \cdot 2 + (-30) \cdot 3 + (-5) \cdot 5 = -195$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an overestimate because S is increasing.

The response does not need an explicit reference to part (a) or to a right Riemann sum.

A response that appeals to concavity as a reason will NOT earn this point.

The following table contains some examples of what WILL or will NOT earn the point.

The following examples earn the point:

- Overestimate because S is increasing
- Overestimate because $S'(t) > 0$
- Overestimate because the graph of S is increasing (we know they are talking about the given function S)
- Overestimate because the function is increasing. (There is only one function in the problem at this point.)

The following examples do not earn the point:

- Overestimate because S is concave down/up
- Overestimate because the graph/curve is increasing. Since no graph/curve is provided, we do not know what graph/curve they are talking about.
- Overestimate because the table [or table values] is/are increasing. The behavior of the values shown in the table is not the same as the behavior of the function S . The table does not tell us what is happening to S between the given points.



0	1
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The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because S is an increasing function, and this is a right Riemann sum, this is an overestimate.

Part C

4-6-6

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

We need to see both a difference and a quotient. They don't have to be together.

The expression does not need to be simplified, but values for $S(5)$ and $S(0)$ must be pulled from the table.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

The second point is for correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols. For example:

- Boxes per day per day or Boxes/day/day
- Boxes per day² or Boxes/day²
- Boxes per day squared

We do not read work shown in part (d) for this point. The units must appear in part (c).

The letter “ B ” is not a standard abbreviation for “boxes.” It will not be interpreted as such.

The response will not be penalized for any attempt to verbally describe their answer after a correct answer has been presented.

Minimalist response: The following expression and units represent the minimum work that must be seen in a response in order to earn both points.

$$\frac{-30+60}{5} \text{ Boxes/day}^2 \text{ or } \frac{-30-(-60)}{5} \text{ Boxes/day}^2$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$S'(2) \approx \frac{S(5)-S(0)}{5-0} = \frac{-30-(-60)}{5} = 6 \text{ boxes per day per day}$$

4-6-6

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

- Indicating a rate of a rate
- Including the context of boxes [remaining to be sold]
- Explicitly referring to the specified point in time $t = 2$

Incorrect or missing units will not be penalized as long as there is a clear connection to the context of the problem.

The words “increasing” and “decreasing” are interpreted as modifiers describing one of the specified rates.

If the numerical value from part (c) is used in this interpretation, then we expect that the choice of “increasing” or “decreasing” is consistent with that value.

The following examples use 6 as the imported value from part (c). If a different [incorrect] value is imported from part (c), then these statements should be consistent with that value. (Remember that we are reading $=$ and $\approx$ as the same thing.)

- “... increasing at a rate of 6 ” is OK [Units not required in part (d)]. (the point is not yet earned with this phrase alone)
- “... decreasing at a rate of 6 ” is OK [Units not required in part (d)]. (the point is not yet earned with this phrase alone)

Note: the context is complicated. $S(t)$ is describing the rate of change of the remaining inventory of boxes. The values in the table indicate that the inventory is decreasing more slowly as time passes. Common sense might tell us that this equates to the idea that the rate at which boxes are sold is decreasing. Explaining the meaning using “boxes sold” rather than “boxes remaining” will not prevent the response from earning this point.

Rate of a rate

To earn this point, the response must explicitly discuss the rate of change of a rate of change. For example:

- The rate of change in the rate at which boxes are changing
- The rate of change in boxes is changing (or decreasing) at a rate of ...
- The rate at which the rate of change of boxes is changing

The words “increasing,” “decreasing,” or “changing” are not interpreted as rates.

Units are not interpreted as rates. (The phrase “boxes per day” is not enough to indicate a rate.)

Slope of tangent line, velocity, and acceleration are not appropriate interpretations for this context.

4-6-6

“Rate of change of S ” is not enough without describing what S represents in this context.

Context

A key component of the context for this explanation is “boxes.”

Any misstatement about boxes sold vs. boxes remaining to be sold will not prevent the response from earning this point.

Explicit reference to $t = 2$

Acceptable references to a specific point in time include (but are not limited to):

- On day 2 (where day could be replaced with any other unit of time)
- At $t = 2$ (with or without units of time)
- After 2 days (where days could be replaced with any other unit of time).

Note: we are choosing to assume that “after” is being used in lieu of “at” unless we have additional evidence to the contrary.

If the response is explicitly indicating an interval of time (e.g., “from day 2 on” or “during the first 2 days”), this point will not be earned.



0	1
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The student response accurately includes the criteria below.

☐ Explanation

Solution:

$S'(2)$ is the rate at which the rate of change in the number of boxes of snacks remaining is changing, in boxes per day per day, on day 2.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. Any of these equations can be written as is without any explanation or additional supporting work. If rewriting of equations is done (transforming to slope-intercept form, perhaps), it must be done correctly.

$$y = 6(t - 5) - 30$$

4-6-6

$$y + 30 = 6(t - 5)$$

$$y = 6t - 60$$

The key calculus concept being assessed in this part of the problem is that the value of $S'(5)$ is the slope of the tangent line. In that spirit...

- The response will not be penalized for using x instead of t .

We are accepting $S(t)$ in lieu of y in all forms of the tangent line equation.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Tangent line equation

Solution:

$$y = S'(5)(t - 5) + S(5)$$

$$y = 6(t - 5) - 30$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are three components of substitution that we need to see:

- Uses $u = \frac{t}{2}$, which may be done implicitly
- Uses the correct constant of 2
- Uses the correct limits for the substitution (0 and 5). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.

Either of the expressions $2S\left(\frac{t}{2}\right)\bigg|_0^{10}$ or $2S(u)\bigg|_0^5$ will earn the substitution point without any additional supporting work.

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point.

- Limits for t attached to an integral with u as the variable of integration (or vice versa),

4-6-6

· Improper equals signs, such as an indefinite integral equal to a definite integral

· Reversed limits of integration, for example: $\int_5^0 S'(u) \, du$

The second point is for the Fundamental Theorem of Calculus.

This point can be earned with any of the following, where k is a nonzero constant and $b = 5$ or $b = 10$. No explanation or additional supporting work is necessary.

· $kS\left(\frac{t}{2}\right)\Big|_0^b$

· $k(S(b) - S(0))$

This point can be earned without earning the substitution point.

The third point is for the correct numerical value, which is 60.

Values from the table must be used. $2(S(5) - S(0))$ is not enough to earn the answer point.

The expression $2(-30 - (-60))$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown they do not need to be correct to earn this point.

The response cannot earn this point without also earning the first two points.

Minimalist response: The expression $2(-30 - (-60))$ represents the minimum work that must be seen in a response in order to earn all 3 points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = \frac{t}{2}$. Then $du = \frac{dt}{2}$.

When $t = 0$, $u = 0$.

When $t = 10$, $u = 5$.

4-6-6

$$\begin{aligned}\int_0^{10} S' \left(\frac{t}{2} \right) dt &= 2 \int_0^5 S' (u) du \\ &= 2 (S(5) - S(0)) \\ &= 2 (-30 - (-60)) = 60\end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression at $t = 6$ earns the point. It may be written as $(7.5) \cdot M'(6)$ or $(7.5) \cdot \frac{dM}{dt} \Big|_{t=6}$

The expression $(7.5) \cdot M'(t)$ is not sufficient because it does not use $t = 6$.

The response must use M (not S or any other function name) in order to earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in ounces per day, of the total weight of the boxes of snacks remaining for the second student organization at $t = 6$ days is $7.5M'(6)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between the differential equation

$$\frac{dy}{dt} = M'(t) \text{ and the given slope field.}$$

In order to earn the point, the response must state that the values of the differential equation are negative while the slopes in the given slope field are positive or 0 (nonnegative).

Some ways to say that the values of differential equation are negative:

- M is decreasing
- $M'(t) < 0$
- M has negative slopes

4-6-6

When discussing the slopes in the slope field, the response must explicitly refer to the signs of the slopes.

The following examples could earn the point:

- The slope field has nonnegative slopes.
- The slopes shown are positive or 0.
- The slope field contains positive slopes.
- “The graph” or “The field” could be used in lieu of “The slope field.”

The following examples will not earn the point:

- The slope field is increasing/positive.
- The slope field looks like it represents an increasing function.
- The slope field represents a quadratic [or exponential] function.
- “It” (rather than “the slope field”)

Statements like “ M is a decreasing function, so the signs of the slopes in the slope field should be negative” can earn this point because “should be” is indicating contrast with what is known.

The response will not be penalized for saying that the slopes in the slope field are “always” positive or “all” positive.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

M is a decreasing function, so $\frac{dy}{dt} = M'(t) \leq 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are positive.

4-6-6

69. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	5	9
$S(t)$ (boxes per day)	-70	-30	-16

A student organization is selling boxes of snacks to raise money for a school trip. For time t days, $0 \leq t \leq 9$, the rate of change in the number of boxes of snacks remaining to sell is modeled by an increasing, twice-differentiable function S . Values of $S(t)$ at various times t are shown in the table above.

(a) Using correct units, approximate $S'(5)$ using the average rate of change of S over the interval $0 \leq t \leq 9$.

(b) Explain the meaning of $S'(5)$ in the context of the problem.

(c) The function $B(t)$ models the number of boxes remaining to sell at time t days, where $B'(t) = S(t)$. It is known that $B(5) = 240$. Write an equation for the line tangent to the graph of $y = B(t)$ at $t = 5$.

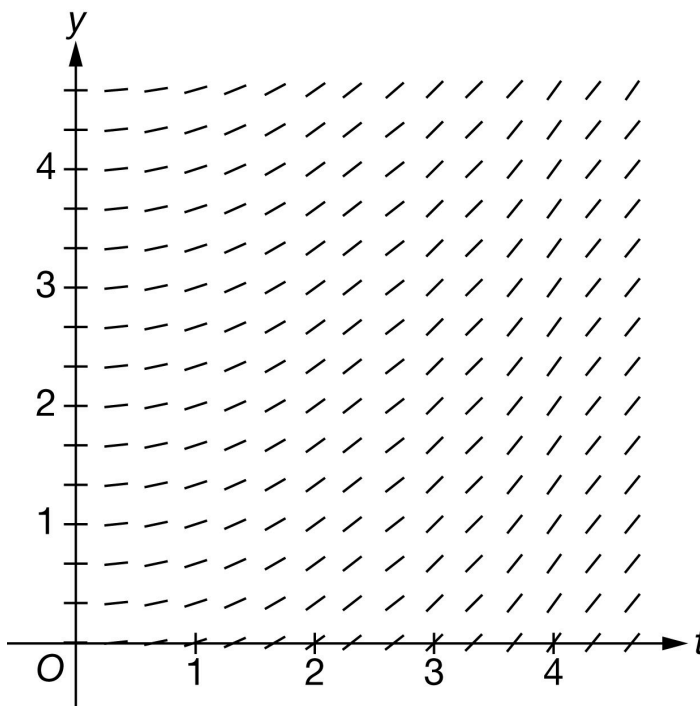
(d) Find the value of $\int_0^{4.5} S'(2t) \, dt$. Show the calculations that lead to your answer.

(e) Use a trapezoidal sum with the two subintervals indicated by the table to approximate $\int_0^9 S(t) \, dt$. Show the calculations that lead to your answer.

(f) Another student organization is also selling boxes of snacks. For $0 \leq t \leq 9$, the rate of change in the number of boxes of snacks remaining for the second student organization to sell is modeled by a decreasing, differentiable function M , where M is a function of t . Assume the weight of each box of snacks is 7 ounces. Write an expression in terms of M and S for the rate of change, in ounces per day, of the combined weight of the boxes of snacks remaining for the two organizations to sell at $t = 8$ days.

4-6-6

(g) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = 0.25y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

Answers without supporting work do not earn this point. We need to see both a difference and a quotient for the supporting work.

Response does not need to be simplified, but any simplification must be done correctly.

If a decimal is presented, it must be correct to three decimal places. We do not read anything past the third decimal place.

The second point is earned for the correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols.

The response will not be penalized for any verbal attempt to discuss the answer after the correct presentation of the approximation.

The minimum response needed to earn both points is $\frac{-16+70}{9}$ boxes per day per day

4-6-6



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$S'(5) \approx \frac{S(9) - S(0)}{9 - 0} = \frac{-16 - (-70)}{9} = 6 \text{ boxes per day per day}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This part is intended to be independent from part (a), so we do not want or need to see any reference to the value obtained in (a). If that value is referenced, conclusion must be consistent.

Three components of a correct answer:

- Indicating a rate of a rate (look for the word rate to be used correctly twice)
- Including the context of boxes
- Referring to the specific point in time

Units will not be considered for this point. They do not need to be present, and if they are, they do not need to be correct.

Any incorrect statements make the response ineligible for the point. For example:

- $S'(5)$ is the average (or an approximation of, or an estimate of) rate of change of . . .
- $S'(5)$ is the acceleration (or rate of change of velocity) of . . .

“Increasing” or “decreasing” may be used in place of “changing” as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Beware of phrases such as “decreasing at a negative rate.”

- Example: $S'(5)$ is the rate of increase of the rate of change of the number of boxes at 5 days. (earns the point)
- Example: $S'(5)$ means the number of boxes per day is increasing by 5 boxes/day/day. (does not earn the point)

Acceptable references to time include (but are not limited to):

- On day 5

4-6-6

- At $t = 5$ (with correct units of time, without correct units of time, or even with incorrect units of time)
- After $t = 5$ (with correct units of time, without correct units of time, or even with incorrect units of time)

If the response clearly indicates an interval of time (e.g. “from day 0 to day 5”) the response will not earn the point.

Examples that earn the point include:

- $S'(5)$ represents the rate at which the rate of change of the number of boxes is changing on day 5.
- $S'(5)$ is the rate of change in the rate of change of the number of boxes at 5 days.
- $S'(5)$ represents the rate at which the rate of change of the number of boxes is increasing at $t = 5$.
- “ $S'(5) = (\text{answer from part (a)})$ ” means the rate of change in the number of boxes per day is increasing at a rate of (answer from part (a)) at $t = 5$.
- $S'(5)$ represents the rate at which the rate of change of the number of boxes is decreasing on the fifth day.

Examples that do not earn the point include:

- $S'(5)$ represents the rate at which the number of boxes is changing on day 5.
- $S'(5)$ is the rate of change in the number of boxes per day at $t = 5$ days
- The rate of change in the number of boxes per day is increasing at $t = 5$.
- The rate of change in the rate of change of the number of boxes per day is increasing at $t = 5$.
- $S'(5) = -6$ means that the rate of change in the number of boxes is decreasing at a rate of -6 at $t = 5$ (double negative).



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$S'(5)$ is the rate at which the rate of change in the number of boxes per day is changing, in boxes per day per day, on day 5.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-6-6

The point is earned with a correct tangent line equation in any form. For example:

$$y = -30(t - 5) + 240$$

$$y - 240 = -30(t - 5)$$

$$y = -30t + 390$$

The student may write x in place of t and still earn the point.

No work is required to earn the point.

We will accept $B(t)$ in lieu of y in the equation of the tangent line.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = B'(5)(t - 5) + B(5)$$

$$y = -30(t - 5) + 240$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution.

Two things are required as evidence of correct substitution:

- Coefficient of $\frac{1}{2}$ correctly applied to an integral or an antiderivative
- Evidence that the correct limits of 0 to 9 will be applied

Examples that will earn the first point (these are just a few):

$$\cdot \frac{1}{2} \int_0^9 S'(u) \, du$$

$$\cdot \frac{1}{2} S(u) \Big|_0^9 \text{ or } \frac{1}{2} S(2t) \Big|_0^{4.5}$$

4-6-6

Responses that include misstatements such as using t -values for limits with u as the variable of integration in an intermediate step which is later corrected may still earn this point. Intermediate errors in bounds that are subsequently corrected will disqualify the student from earning the third point. If the bounds are never appropriately handled, the response cannot earn the first point. Scratch work that is not connected to the solution will not be read.

Examples:

$$\int_0^{4.5} S'(2t) dt = \frac{1}{2} \int_0^{4.5} S'(u) du = \frac{1}{2} S(2t) \Big|_0^{4.5} \text{ earns the first point, but will not earn the third point.}$$

$$\int_0^{4.5} S'(2t) dt = \frac{1}{2} \int_0^{4.5} S'(u) du = \frac{1}{2} [S(4.5) - S(0)] \text{ does not earn the first point.}$$

The second point is for the Fundamental Theorem of Calculus, demonstrating the knowledge that

$$\int_a^b S'(u) du = S(u) \Big|_a^b = S(b) - S(a)$$

This point can be earned with either $kS(2t) \Big|_a^b$, $kS(t) \Big|_a^b$, or $k(S(b) - S(a))$ where k is a nonzero constant [not necessarily correct], $a = 0$, and $b = 4.5$ or $b = 9$.

The response can earn this point without earning the first point.

$$\frac{1}{2} [S(9) - S(0)] \text{ earns the first two points.}$$

The third point is for the correct numerical value. Our answer only.

Values from the table must be used (i.e., $\frac{1}{2} [S(9) - S(0)]$ is not sufficient to earn the third point).

If the response includes any linkage error, (i.e., the use of equal signs to equate two expressions that are not equal), the third point will not be earned. If an indefinite integral is linked with a definite integral the response will not be penalized.

$$\int_0^{4.5} S'(2t) dt = \frac{1}{2} \int S'(u) du = \frac{1}{2} S(2t) \Big|_0^{4.5} = \dots \text{ is eligible for the third point.}$$

Eligibility for the third point: Must have earned the first two points.

The minimum response needed to earn all three points is: $\frac{1}{2}(-16 + 70)$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

☐ Substitution

4-6-6

- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 2t$. Then $du = 2 dt$.

When $t = 0$, $u = 0$.

When $t = 4.5$, $u = 9$.

$$\begin{aligned}\int_0^{4.5} S'(2t) dt &= \frac{1}{2} \int_0^9 S'(u) du \\ &= \frac{1}{2}(S(9) - S(0)) \\ &= \frac{1}{2}(-16 - (-70)) = \frac{1}{2}(54) = 27\end{aligned}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the form of a trapezoidal sum

Point is earned for the setup of the trapezoidal sum, with one numerical error allowed. Numerical values do not need to be pulled from the table for this point, but Δt values must be presented. Responses that contain one incorrect numerical value in the trapezoidal sum setup are still eligible to earn the first point, but will not earn the second.

Response may present a left Riemann sum, a right Riemann sum, then average the two. Again, one incorrect numerical value will not disqualify the response from earning the first point.

Response may compute trapezoids as areas of rectangles and triangles.

Trapezoidal areas may be given in a graph. In order to earn this point, the response must clearly show the computations leading to the values presented in the graph, and must show that these areas are added to obtain the final approximation value.

A response using subtraction instead of addition will not earn the point (e.g.

$\frac{1}{2}[5(S(0) - S(5)) + 4(S(5) - S(9))]$ earns no points).

Response utilizing “Trapezoidal Rule” will not earn this point, unless this represents only one error in the numerical values.

$\frac{1}{2}(5)[S(0) + 2S(5) + S(9)]$ is equivalent to $\frac{1}{2}[5(S(0) + S(5)) + 5(S(5) + S(9))]$, has only one incorrect numerical value (the second interval width is 5 instead of 4), so can earn the point.

4-6-6

$\cdot \frac{1}{2}(4.5) [S(0) + 2S(5) + S(9)]$ has two incorrect numerical values (both interval widths are incorrect) so will not earn the point.

The second point is for the correct numerical result.

Our answer only. Response does not need to be simplified, but if it is, it must be simplified correctly.

The answer must be accompanied with supporting work. Supporting work may appear in a graph, with clearly labeled areas. It is possible to earn the second point without earning the first if the supporting work is not sufficient to earn the first point.

Pictures without clear area computations and a clear summation of quantities can earn at most the second point.

Special case: A positive constant multiple of a complete and correctly expressed trapezoidal sum (with the correct values pulled from the table) will earn the first point but not the second:

$\cdot k \left[\left(\frac{-70-30}{2} \right) \cdot 5 + \left(\frac{-30-16}{2} \right) \cdot 4 \right]$ for $k > 0$ earns just the first point.

The minimum response needed to earn both points is: $\left(\frac{-100}{2} \right) \cdot 5 + \left(\frac{-46}{2} \right) \cdot 4$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of trapezoidal sum
- ☐ Answer

Solution:

$$\begin{aligned} \int_0^9 S(t) dt &\approx \left(\frac{S(0)+S(5)}{2} \right) (5-0) + \left(\frac{S(5)+S(9)}{2} \right) (9-5) \\ &= \left(\frac{-70-30}{2} \right) \cdot 5 + \left(\frac{-30-16}{2} \right) \cdot 4 = -342 \end{aligned}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for a correct constant multiple or correct sum.

Response must contain the constant 7 as a factor of the expression, and/or the correct sum $M(8) + S(8)$.

If the sum is incorrect, then the constant factor of 7 must be applied either to a sum that involves only $M(t)$ and $S(t)$, $M'(t)$ and $S'(t)$, or integrals of $M(t)$ and $S(t)$, or to one of the correct terms of the sum $M(8)$ or

4-6-6

$S(8)$, with no additional incorrect terms.

Examples that earn the point include:

· $7[M(t) + S(t)]$

· $7 \int [M(t) + S(t)] dt$

· $7[M' + S']$

· $7S(8)$

Examples that do not earn the point include:

· $7[M(t) + S'(t)]$

· $7 \int M(t) dt$

· $7M' + S'$

· $7S(8) + t$

Special case: $7M(8) + S(8)$ is an error in parentheses, so will earn the first point but is not eligible to earn the second point.

If the constant multiple is missing or incorrect, the sum must be correct. $k[M(8) + S(8)]$ for any nonzero constant k will earn the first point.

The second point is for the correct expression.

$7M(8) + 7S(8)$ (or an equivalent expression) earns both points.

Inclusion of units (correct or incorrect) will not affect the score.

A completely correct expression with one additional constant term added/subtracted would earn a just the first point. For example, $7M(8) + 7S(8) + 7(2)$ earns only the first point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Constant multiple or sum

☐ Expression

Solution:

4-6-6

The rate of change, in ounces per day, of the combined weight of the boxes of snacks remaining for the two organizations to sell at $t = 8$ days is $7M(8) + 7S(8)$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for a correct statement identifying specific inconsistencies between the given differential equation and the given slope field. There are many such correct statements, such as:

- The response references a specific point on the y -axis (such as $(0, 4)$), and indicates that the given slope field shows a slope of zero, yet the differential equation would yield a positive slope.
- The response references a pattern of slopes, and explains why that pattern does not fit our differential equation (e.g. “the slope field shows slopes that change as t changes, but our differential equation depends only on y , so slopes should remain constant for a given value of y as t changes.”)

Identifying a mismatch between the differential equation and the slope field at one point is sufficient to earn the point.

Response is not penalized for referencing “ x ” instead of “ t ”.

If the response contains any incorrect statement, the response will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point, nor will responses that present information beyond what we are given. For example:

- Responses that specify that the slope field shows a particular nonzero slope at a particular point will not earn the point, as this cannot actually be determined visually (i.e., the response should not indicate that at $(2, 2)$ the slope shown on the slope field is 1)
- Responses that state a specific slope value on the slope field for a general value of x or y that are not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

A response may involve a solution to the differential equation, with an argument based on the resulting function. If done correctly, and presented with a correct argument, this will earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

4-6-6

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

4-6-6

70. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	8	24	35
$P(t)$ (number of postcards remaining)	316	260	100	0

A principal is delivering postcards to teachers from students on a school trip. The principal begins with 316 postcards and delivers them over 35 minutes. For time t minutes, $0 \leq t \leq 35$, the total number of postcards remaining to deliver is modeled by a decreasing, twice-differentiable function P . Values of $P(t)$ at various times t are shown in the table above.

(a) Using correct units, approximate $P'(8)$ using the average rate of change of P over the interval $0 \leq t \leq 24$.

(b) Explain the meaning of $P'(8)$ in the context of the problem.

(c) It is known that $P'(24) = -11$. Write an equation for the line tangent to the graph of $y = P(t)$ at $t = 24$.

(d) Use a right Riemann sum with the three subintervals indicated by the table to approximate

$\int_0^{35} P(t) \, dt$. Show the computations that lead to your answer.

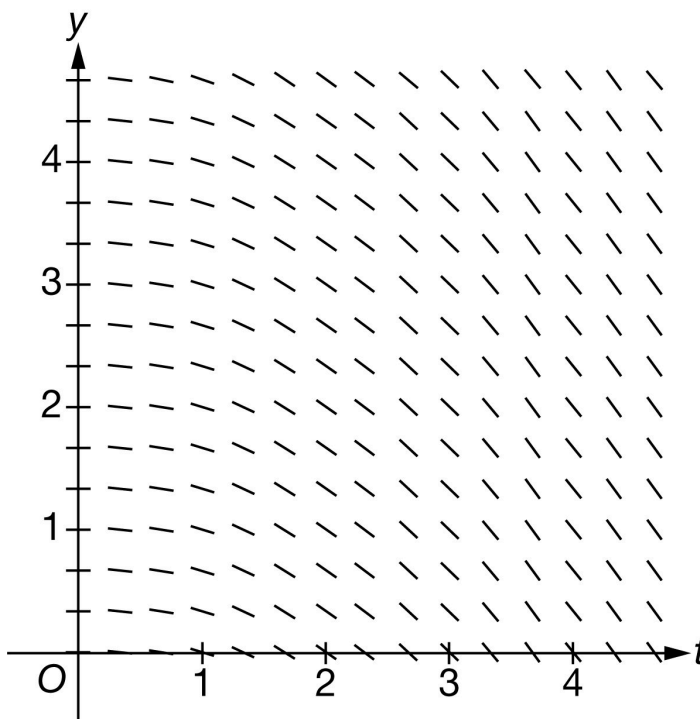
(e) Is the approximation from part (d) an overestimate or an underestimate for $\int_0^{35} P(t) \, dt$? Give a reason for your answer.

(f) Find the value of $\int_0^8 P'(3t) \, dt$. Show the computations that lead to your answer.

(g) For $0 \leq t \leq 35$, the total number of postcards remaining can also be modeled by the differentiable function R , where R is a function of t . Assume the weight of each postcard is 4 grams. Write an expression for the rate of change, in grams per minute, of the total weight of the postcards remaining at $t = 13$ minutes.

4-6-6

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = -0.4y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work. The supporting work must be at least a difference and a quotient. Correct values must be taken from the table to earn the point. The difference quotient does not need to be simplified.

Examples that earn the first point include:

$$\cdot \frac{100-316}{24-0}$$

$$\cdot \frac{P(24)-P(0)}{24} = \frac{-216}{24}$$

$$\cdot \frac{P(24)-P(0)}{24} = -9$$

An example that does not earn the first point is $\frac{P(24)-P(0)}{24}$.

Minimal response that earns the first point: $\frac{100-316}{24-0}$ or $\frac{100-316}{24}$

Neither $\frac{-216}{24}$ nor -9 , with no supporting work, will earn this point

The second point is for correct units

4-6-6

The correct units must be attached to a numerical value: either correct or incorrect.

The units may be written using words or a combination of words and symbols. For example:

- Postcards per minute
- Cards per min
- Postcards/min

The response will not be penalized for any verbal attempt to discuss the answer after the correct answer has been presented.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$P'(8) \approx \frac{P(24) - P(0)}{24 - 0} = \frac{100 - 316}{24} = -9 \text{ postcards per minute}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

- Indicating a rate of change
- Including the context of postcards
- Referring to the specific point in time, $t = 8$

The response is not penalized for incorrect units in this part. (Whether correct or incorrect, units have no effect on the score in this part.)

Responses may use “changing” or “decreasing” as a modifier as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Example: “the number of postcards is decreasing at 9 postcards per minute.” is okay, but “the number of postcards is decreasing at a rate of -9 postcards per day...” would not earn the point.

Acceptable reference to a specific point in time includes (but is not limited to):

4-6-6

- On minute 8 and after 8 minutes
- At $t = 8$
- At $t = 8$ hours or any other incorrect unit of time
- If the response clearly indicates an interval of time (e.g., “from minute 8 on,” “during the first 8 minutes”, or “for $t = 0$ to $t = 8$ ”), the point is not earned.

Examples that earn the point include:

- $P'(8)$ is the rate of change of the number of postcards per minute at time $t = 8$
- The rate of change of number of postcards is -9 postcards per minute at minute 8
- $P'(8)$ is the rate of change in the number of postcards at time $t = 8$ in postcards per hour
- -9 is the rate of decrease in the number of postcards per minute at minute 8

Examples that do not earn the point include:

- $P'(8)$ is the average rate of change of the total number of postcards /minute at minute 8
- $P'(8)$ is the number of postcards per minute at minute 8
- The rate of change is -9 postcards per minute (Missing “at $t = 8$ ”)
- $P'(8)$ is the slope of the tangent line at $t = 8$
- Rate of change of P at time $t = 8$
- Postcards per minute decreases on minute 8



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$P'(8)$ is the rate at which the number of postcards remaining is changing, in postcards per minute, at time $t = 8$ minutes.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-6-6

This point is earned with a correct tangent line equation in any form. The use of x in place of t is acceptable.

Some examples include:

$$\cdot y - 100 = -11(t - 24), y - 100 = -11(x - 24)$$

$$\cdot y = -11(t - 24) + 100, y = -11(x - 24) + 100$$

$$\cdot y = -11t + 364, y = -11x + 364$$

The key calculus concept being assessed in this part of the problem is that the value of $P'(24)$ is the slope of the tangent line. In that spirit, we are accepting $P'(t) = \dots$ in lieu of $y = \dots$, but we are not accepting $P'(24) = \dots$

Note that $P(t) = -11(t - 24) + 100$ earns the point, but $P(24) = -11(t - 24) + 100$ does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = P'(24)(t - 24) + P(24)$$

$$y = -11(t - 24) + 100$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a right Riemann sum. The setup must be a sum of three products.

· A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example in the table below).

· Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns. Implicit responses are eligible for both points.

A response presenting a left Riemann sum (instead of right), will not earn any points for part (d).

The second point is for the correct numerical result.

· In order to earn the second point, there must be supporting calculations.

4-6-6

- The numerical values for $P(8)$, $P(24)$, and $P(35)$ must be pulled from the table
- Any simplification shown must be correct.

Use the following as an illustration how to score both points. (In most cases, you see scores for which the expression on the left side represents the entirety of relevant work shown for the response.)

- A response of $260 \cdot (8 - 0) + 100 \cdot (24 - 8) + 0 \cdot (35 - 24)$ earns both points.
- A response of $8 \cdot P(8) + 16 \cdot P(24) + 11 \cdot P(35)$ earns the first point. Eligible for second point if more work is on the page.
- A response of $8 \cdot P(8) + 16 \cdot P(24) + 11 \cdot P(35) = 3680$ earns both points.
- A response of $8 \cdot 260 + 16 \cdot 100 + 11 \cdot 0$ earns both points.
- A response of $8 \cdot 260 + 12 \cdot 100 + 11 \cdot 0$ contains one numerical error. Earns the first point. Not eligible for the second point.
- A response of $8 \cdot 260 + 16 \cdot 100 + 11 \cdot 0 = 3860$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.
- A response of $2080 + 1600 + 0 = 3680$ or $2080 + 1600 = 3680$ does not earn the first point because we cannot see the sum of 3 products. However, this is the correct sum, so we see an answer with supporting work. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points.

$$8 \cdot 260 + 16 \cdot 100 + 11 \cdot 0$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Right Riemann sum
- ☐ Answer

Solution:

$$\begin{aligned} \int_0^{35} P(t) \, dt &\approx P(8)(8 - 0) + P(24)(24 - 8) + P(35)(35 - 24) \\ &= 260 \cdot 8 + 100 \cdot 16 + 0 \cdot 11 = 3680 \end{aligned}$$

4-6-6

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an underestimate because P is decreasing.

The response does not need an explicit reference to the approximation in part (d), nor does there need to be an answer given in part (d). The following statements are sufficient for earning this point:

- Underestimate because P is decreasing
- Underestimate because the function is decreasing

Note: At this point, there is only one function being discussed in this problem, so we will assume “the function” is P .

When a response includes additional information about concavity, we interpret that as a lack of understanding about the question, so this will not earn the point. When the response references a graph or the given table rather than the function P , the point is not earned.

Examples that earn the point:

- The right Riemann sum underestimates because the function is decreasing.
- The sum is an underestimate because $P(t)$ is decreasing.
- The sum is an underestimate because $P'(t) < 0$.

Examples that do not earn the point:

- The sum is an underestimate because $P'(t)$ is decreasing. ($P'(t)$ decreasing is not known)
- The right Riemann sum underestimates because the graph is decreasing.
- The right Riemann sum underestimates because the function is decreasing and concave up.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because P is a decreasing function, and this is a right Riemann sum, the answer from part (d) is an underestimate.

4-6-6

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are 3 components of substitution that we need to see:

- Uses $u = 3t$, which may be done implicitly
- Uses the correct constant of $\frac{1}{3}$
- Uses the correct limits for the substitution (0 and 24). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.
- Either of the expressions below will earn the substitution point without any additional supporting work:

$$\frac{1}{3} P(3t) \Big|_0^8 \text{ or } \frac{1}{3} P(u) \Big|_0^{24}$$

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point

- Limits for t attached to an integral with u as the variable of integration (or vice versa),
- Improper equals signs, such as an indefinite integral equal to a definite integral
- Reversed limits of integration, for example: $\int_{24}^0 P'(u) du$

The second point is for the Fundamental Theorem of Calculus. This point can be earned with any of the following, where k is a nonzero constant and $b = 8$ or $b = 24$. No explanation or additional supporting work is necessary.

- $kP(3t) \Big|_0^b$
- $k(P(b) - P(0))$

This point can be earned without earning the substitution point.

The third point is for the correct numerical value, which is -72 .

Values from the table must be used. $\frac{1}{3}[P(24) - P(0)]$ is not enough to earn the answer point.

The expression $\frac{1}{3}(100 - 316)$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they need not be correct in order to earn this point.

4-6-6

The response cannot earn this point without also earning the FTC point.

Minimalist response: The expression $\frac{1}{3}(-216)$ represents the minimum work that must be seen in a response in order to earn all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 3t$. Then $du = 3 dt$.

When $t = 0$, $u = 0$.

When $t = 8$, $u = 24$.

$$\begin{aligned}\int_0^8 P'(3t) dt &= \frac{1}{3} \int_0^{24} P'(u) du \\ &= \frac{1}{3}(P(24) - P(0)) \\ &= \frac{1}{3}(100 - 316) = -\frac{216}{3} = -72\end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct expression at $t = 13$. No supporting work is required.

The correct expression, $4R'(13)$ or $4P'(13)$, earns this point.

No justification or explanations are required.

Any incorrect statement such as, $P'(13) = 4R'(13)$, does not earn this point.



0	1
---	---

4-6-6

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in grams per minute, of the total weight of the postcards remaining at $t = 13$ minutes is $4R'(13)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between two things: the differential equation, $\frac{dy}{dt} = -0.4y$, and the given slope field.

The response does not need to explicitly reference both the differential equation and the given slope field., but it must explicitly reference one and implicitly reference the other.

Acceptable responses include

- Identifying an inconsistency between the differential equation and the slope field at one point, e.g., at the point $(4, 0)$, the slope of the segment should be 0, but it is negative.
- Referencing a pattern of slopes, and explaining why that pattern does not fit the given differential equation.

A response that lacks even an implicit reference to the one of two things being compared will not earn the point. For example, merely stating either $\frac{dy}{dt}$ is negative, or that the slope field is negative discretely does not earn the point.

- “The values for y are less than or equal to zero so $\frac{dy}{dt} \leq 0$ ” lacks any reference to the slope field and will not earn the point.
- “All the slopes in the slope field are negative” lacks any reference to the differential equation and will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point. Some examples are:

- “The field is decreasing”
- Responses that state a specific slope value on the slope field for a general value of x or y that is not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

A response may correctly solve or attempt to solve the differential equation and then provide a correct argument without reference to their solution. Such a response would earn the point.

4-6-6

A response might actually solve the differential equation and provide a correct argument on the resulting function.



0

1

The student response accurately includes the criteria below.

☐

Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

4-6-6

71. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	9	18	27	36
$P(t)$ (number of postcards remaining)	320	256	100	40	0

A principal is delivering postcards to teachers from students on a school trip. The principal begins with 320 postcards and delivers them over 36 minutes. For time t minutes, $0 \leq t \leq 36$, the total number of postcards remaining to deliver is modeled by a twice-differentiable function P . Values of $P(t)$ at various times t are shown in the table above.

(a) Use a midpoint Riemann sum with the two subintervals indicated by the table to approximate

$$\int_0^{36} P(t) \, dt.$$

Show the computations that lead to your answer.

(b) Find the value of $\int_0^9 P'(4t) \, dt$. Show the computations that lead to your answer.

(c) Using correct units, approximate $P'(18)$ using the average rate of change of P over the interval $9 \leq t \leq 27$.

(d) Explain the meaning of $P'(18)$ in the context of the problem.

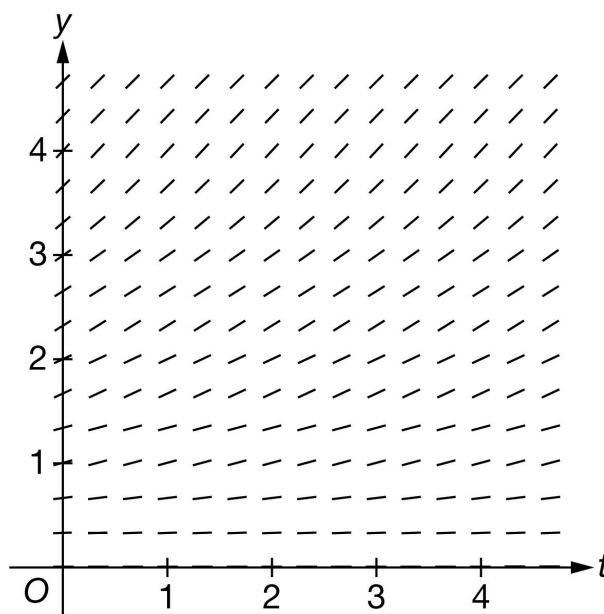
(e) It is known that $P'(9) = -12$. Write an equation for the line tangent to the graph of $y = P(t)$ at $t = 9$.

(f) It is known that $P''(t) < 0$ for $9 \leq t \leq 18$. If $P(15)$ is approximated using the equation from part (e), would the approximation be an overestimate or underestimate for the actual value of $P(15)$? Give a reason for your answer.

(g) For $0 \leq t \leq 36$, the total number of postcards remaining can be modeled by the differentiable function Z , where Z is a function of t . Assume the weight of each postcard is 4.5 grams. Write an expression for the rate of change, in grams per minute, of the total weight of the postcards remaining at $t = 19$ minutes.

4-6-6

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = -0.3y$.

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Form of midpoint sum): Philosophy: The response should provide clear evidence adding the area of two rectangles whose heights are found using the midpoint of the interval.

Acceptable evidence:

$$\cdot 18 \cdot (P(9) + P(27)) \text{ or } 18 \cdot P(9) + 18 \cdot P(27)$$

$$\cdot (18 - 0) \cdot 256 + (36 - 18) \cdot 40$$

$$\cdot 18 \cdot (256 + 40) \text{ or } 18 \cdot 256 + 18 \cdot 40$$

To earn this point the response can have at most one mistake in the expression or equation. So, expressions of the form $\frac{18}{b} \cdot (P(9) + P(27))$, for any nonzero value of b would earn the point. Note: The value $\frac{18}{b}$ might be distributed across the sum.

Expressions of the form $\frac{a}{b} \cdot (P(9) + P(27))$ for $a \neq 18$ will not earn this point.

Once the first point is earned, it cannot be lost.

Second Point (Answer)

To earn the point, the answer must be correct. It does not need to be simplified, but if it is, the simplification must be correct.

The answer point cannot be earned unless the first (midpoint sum) point was earned.

4-6-6

Minimal complete answers that earn both points:

$$\cdot 18 \cdot (256 + 40)$$

$$\cdot 18 \cdot 256 + 18 \cdot 40$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint sum
- ☐ Answer

Solution:

$$\int_0^{36} P(t) dt \approx P(9)(18 - 0) + P(27)(36 - 18)$$

$$= 256 \cdot 18 + 40 \cdot 18 = 5328$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Substitution) Philosophy: This point is earned when the response has all three components of the substitution done correctly.

Uses $u = 4t$, perhaps implicitly (any variable could be used for u)

Uses the correct constant of $\frac{1}{4}$

Uses the correct limits for their substitution.

· The limits might not be seen until the final evaluation, and at that point, response may have returned to the variable t .

· Limits for t attached to an integral involving u or vice versa will not earn this substitution point but could still earn the answer point, if the numerical answer is correct.

Second point: (Fundamental Theorem of Calculus) Philosophy: This point is earned when the FTC has been applied correctly to either $\int P'(4t) dt$ or $\int P'(u) du$.

The point is earned for any of the following:

4-6-6

· The equation $\int P'(4t) dt = P(4t)$ or $\int P'(u) du = P(u)$

· $\frac{1}{4}P(4t)\big|_0^9$ or $\frac{1}{4}P(u)\big|_0^{36}$

· $P(36) - P(0)$, $P(36)$, or $P(0)$

This point can be earned if the substitution point is not.

A response that applies the FTC to $P'(t)$ does not earn this point.

Note: $\int_a^b P'(u) du = P(b) - P(a)$ does not earn the point. The FTC point is for the response communicating that the antiderivative of P' is P .

Third Point (Answer)

Must be the correct numerical answer.

Must have earned the second point for the FTC.

Might have not earned the first point for substitution because of an incorrect use of limits, but will still earn the third point for the correct answer.

$P(36)$ and $P(0)$ must be evaluated; the expression $\frac{1}{4}(P(36) - P(0))$ by itself does not earn the third point.

The answer may be unsimplified, but if simplified it must be done correctly.

Scoring common responses:

The minimal work required to earn all three points is $\frac{1}{4}(0 - 320)$ or $-\frac{320}{4}$.

The following responses would earn the first two points and be eligible for the answer point if the response continues:

· $\frac{1}{4}P(4t)\big|_0^9$

· $\frac{1}{4}(P(36) - P(0))$

Responses of only the following would earn only the second point:

· $P(36) - P(0)$

· $P(9) - P(0)$

· $P(36)$

4-6-6

$$\cdot \int_0^9 P'(u) \, du = P(9)$$

$$\cdot P(4t) \Big|_0^9 \text{ or } P(u) \Big|_0^{36}$$

The following responses would earn no points:

· -80 in the absence of one of the first two points.

$$\cdot \int_0^9 P'(4t) \, dt = P(9)$$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 4t$. Then $du = 4 \, dt$.

When $t = 0$, $u = 0$.

When $t = 9$, $u = 36$.

$$\begin{aligned} \int_0^9 P'(4t) \, dt &= \frac{1}{4} \int_0^{36} P'(u) \, du \\ &= \frac{1}{4}(P(36) - P(0)) \\ &= \frac{1}{4}(0 - 320) = -80 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Approximation)

The response must be a difference and a quotient.

The answer does not need to be simplified, but if it is simplified it must be done correctly.

4-6-6

The minimal work that must be shown to earn this point is either $\frac{40-256}{18}$ or $\frac{-216}{27-9}$.

Second point (Units)

The units must be attached to a value (either correct or incorrect).

The response will not be penalized for any verbal attempt to discuss the answer after the correct approximation with units is presented.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$P'(18) \approx \frac{P(27)-P(9)}{27-9} = \frac{40-256}{18} = -12 \text{ postcards per minute}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The interpretation must include three components:

- indicate the rate of change
- the specific context of the number of postcards remaining
- the specific point in time, $t = 18$.

Responses do not need to include correct units or any units.

The word “decreasing” may be used in place of the word “changing”.

The response is not earned for:

- “the rate of postcards delivered in minute 18 ” as this does not indicate a change in the rate of delivery.
- “decreasing at a rate of -12 postcards per minute”

“the rate of change of P ” without specifying that P is the number of postcards delivered, does not earn the point.

Acceptable ways to provide the specific point in time:

4-6-6

- In minute 18 or after 18 minutes.
- At $t = 18$ (missing the unit of time)
- After 18 days (incorrect unit of time)

A response that clearly indicates an interval of time (e.g. “during the first 18 minutes” or “from minute 18 on”) does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$P'(18)$ is the rate at which the number of postcards remaining is changing, in postcards per minute, at time $t = 18$ minutes

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The use of $P(t)$ in place of y earns the point.

The use of x in place of t earns the point.

Tangent lines do not need to be simplified, but any simplification must be done correctly.

Any correct form of the tangent line earns the point. For example:

o $y = 256 - 12(t - 9)$

o $y - 256 = 12(t - 9)$

o $y = -12t + 108 + 256$

o $y = -12t + 364$



0	1
---	---

The student response accurately includes the criteria below.

4-6-6

☐ Tangent line equation
Solution:

$$y = P'(9)(t - 9) + P(9)$$

$$y = -12(t - 9) + 256$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point the response must include $P''(t) < 0$ and overestimate.

“ $P(t)$ is concave down” can be used instead of $P''(t) < 0$.

Information about $P'(t)$ by itself without reference to $P''(t) < 0$ is not sufficient and does not earn this point.

“Overestimate because it is concave down” is not sufficient by itself because of the use of “it”. The response must clearly, directly or indirectly, indicate to what “it” is referring.

If a response makes the correct conclusion and states $P''(t) < 0$ and then goes on and makes an incorrect statement or a statement that cannot be verified will not earn this point.

For example: “ $P'(t)$ is concave down, overestimate” does not earn this point.



0	1
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The student response accurately includes the criteria below.

☐ Answer with reason
Solution:

Because $P''(t) < 0$, the tangent line approximation would produce an overestimate.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression of only $4.5 \cdot Z'(19)$ earns this point.

Only expression $4.5 \cdot P'(19)$ also earns this point.

No justifications or explanations are required.

4-6-6

Any incorrect statement such as, $Z'(19) = 4.5 \cdot P'(19)$, does not earn this point.



0

1

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in grams per minute, of the total weight of the postcards remaining at $t = 19$ minutes is $4.5Z'(19)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Response clearly references what part of the differential equation is different from the slope field and the response needs only to clearly and explicitly reference one of them.

The response does not need to explicitly reference both the differential equation and the given slope field.

The distinction can be made with explicit reference to one of the two things and an implicit reference to the other

For example:

· “ $\frac{dy}{dt}$ is negative, not positive”. We will infer that negative is referring to the slopes shown in the slope field.

· “This would be correct for a positive derivative, but $\frac{dy}{dt}$ is negative.” We will read “This” as “the slope field” in this sentence.

· “All of the slopes in the slope field are positive, not negative.” We will infer that the values of $\frac{dy}{dt}$ at points in the first quadrant are negative.

A response that lacks even an implicit reference to either the differential equation or the slope field will not earn the point. For example:

· “ $\frac{dy}{dt}$ is negative” lacks any reference to the slope field and so will not earn the point.

· “All the slopes in the graph are positive” lacks any reference to the differential equation, and so will not earn the point.

· Vague answers such as “The slope field has positive points” or “The field is increasing” will not earn the point.

4-6-6



0

1

The student response accurately includes the criteria below.

☐

Explanation

Solution:

When $y > 0$, $\frac{dy}{dt} < 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are positive.

4-6-6

72. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	8	24	35
$P(t)$ (number of postcards remaining per minute)	-10	-8	-5	-4

A student is delivering postcards to teachers from students on a school trip. For time t minutes, $0 \leq t \leq 35$, the rate at which the total number of postcards remaining to deliver is changing, in number of postcards remaining per minute, is modeled by an increasing, twice-differentiable function P . Values of $P(t)$ at various times t are shown in the table above.

(a) Use a left Riemann sum with the three subintervals indicated by the table to approximate

$$\int_0^{35} P(t) \, dt. \text{ Show the calculations that lead to your answer.}$$

(b) Is your answer from part (a) an overestimate or an underestimate for $\int_0^{35} P(t) \, dt$? Give a reason for your answer.

(c) Using correct units, approximate $P'(24)$ using the average rate of change of P over the interval $8 \leq t \leq 35$.

(d) Explain the meaning of $P'(24)$ in the context of the problem.

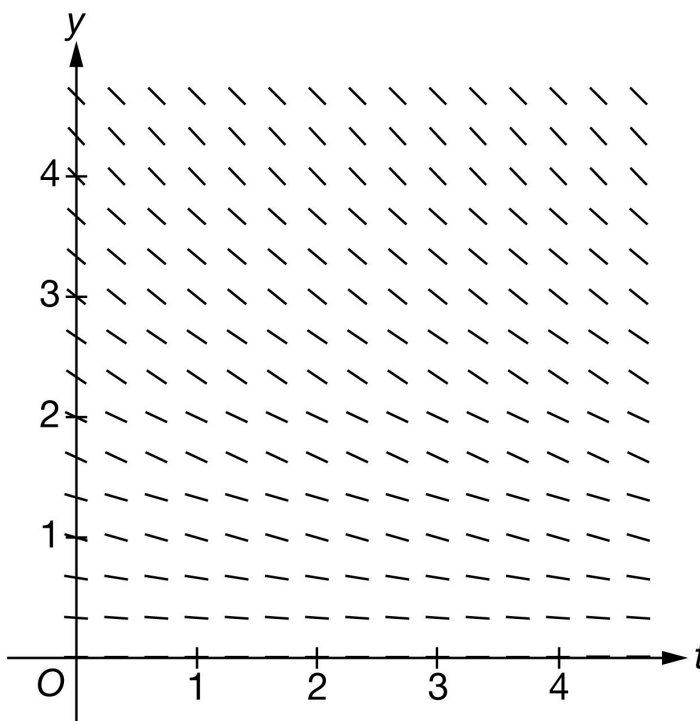
(e) It is known that $P'(8) = 0.2$. Write an equation for the line tangent to the graph of $y = P(t)$ at $t = 8$.

(f) Find the value of $\int_0^{24} P'\left(\frac{t}{3}\right) \, dt$. Show the calculations that lead to your answer.

(g) Another student brings a bag full of postcards to deliver as well. For $0 \leq t \leq 35$, the total number of postcards delivered to teachers by this student can be modeled by the increasing, differentiable function R , where R is a function of t . Assume the weight of each postcard is 4 grams. Write an expression for the rate of change, in grams per minute, of the total weight of postcards delivered by the second student at $t = 11$ minutes.

4-6-6

(h) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = R'(t)$ for the function R defined in part (g).



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a left Riemann sum. The setup must be a sum of three products.

A response may still earn this point if there is only 1 numerical error in the expression for the Riemann sum with values pulled from the table. (See example in the table below).

Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns, like so.

8	(-10)	-80
16	(-8)	-128
11	(-5)	+(-55)
		-263

or

(8 - 0)(-10)	= -80
(24 - 8)(-8)	= -128
(35 - 24)(-5)	= -55
	-263

Implicit responses such as those above are eligible for both points.

A response presenting a right Riemann sum (instead of left), will not earn any points for part (a).

The second point is for the correct numerical result.

4-6-6

In order to earn the second point, there must be supporting calculations.

The numerical values for $P(0)$, $P(8)$, and $P(24)$ must be pulled from the table

Any simplification shown must be correct.

Use the following as an illustration how to score of both points. (In most cases, you see scores for which the expression on the left side represents the entirety of relevant work shown for the response.)

- A response of $(-10) \cdot (8 - 0) + (-8) \cdot (24 - 8) + (-5) \cdot (35 - 24)$ earns both points.
- A response of $8 \cdot P(0) + 16 \cdot P(8) + 11 \cdot P(24)$ earns the first point. Eligible for second point if more work is on the page.
- A response of $8 \cdot P(0) + 16 \cdot P(8) + 11 \cdot P(24) = -263$ earns both points.
- A response of $8 \cdot (-10) + 16 \cdot (-8) + 11 \cdot (-5)$ earns both points.
- A response of $8 \cdot (-10) + 18 \cdot (-8) + 11 \cdot (-5)$ contains one numerical error. Earns the first point. Not eligible for the second point.
- A response of $8 \cdot (-10) + 16 \cdot (-8) + 11 \cdot (-5) = -200$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.
- A response of $(-80) + (-128) + (-55)$ or $(-80) + (-128) + (-55) = -263$ does not earn the first point because we cannot see the sum of three products. However, this is the correct sum, so we are not seeing an unsupported answer. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points. $8 \cdot (-10) + 16 \cdot (-8) + 11 \cdot (-5)$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Left Riemann sum
- ☐ Answer

Solution:

$$\int_0^{35} P(t) dt \approx P(0)(8 - 0) + P(8)(24 - 8) + P(24)(35 - 24)$$

$$= (-10) \cdot 8 + (-8) \cdot 16 + (-5) \cdot 11 = -263$$

4-6-6

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an underestimate because P is increasing.

The response does not need an explicit reference to part (a) or to a left Riemann sum.

A response that appeals to concavity as a reason will NOT earn this point.

The following examples earn the point:

- Underestimate because P is increasing
- Underestimate because $P'(t) > 0$
- Underestimate because the graph of P is increasing (we know they are talking about the given function P)
- Underestimate because the function is increasing. (There is only one function in the problem at this point.)

The following examples do not earn the point:

- Underestimate because P is concave down/up
- Underestimate because the graph/curve is increasing. Since no graph/curve is provided, we do not know what graph/curve they are talking about.
- Underestimate because the table [or table values] is/are increasing. The behavior of the values shown in the table is not the same as the behavior of the function P . The table does not tell us what is happening to P between the given points.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because P is an increasing function, and this is a left Riemann sum, this is an underestimate.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

4-6-6

The first point is earned for the correct approximation shown with supporting work.

We need to see both a difference and a quotient. They don't have to be together.

The expression does not need to be simplified, but values for $P(35)$ and $P(8)$ must be pulled from the table.

Decimal approximation is not expected, but if presented, then the decimal value must be correct to three decimal places, either rounded or truncated. We do not read anything past the third decimal place.

- If the first three places match 0.148 then the point is earned
- It does not matter whether any subsequent decimal places match the correct value or not. (For example, the value 0.1487 will earn the point because the first three decimal places match the correct value.)
- Responses with fewer than three decimal places (0.14, 0.15, 0.1, etc.) will not earn the point.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

The second point is for correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols. For example:

- Postcards per minute per minute or Postcards/min/min
- Postcards per min² or Postcards/min²
- Postcards per minute squared

We do not read work shown in part (d) for this point. The units must appear in part (c).

The letter “ P ” is not a standard abbreviation for “Postcards.” It will not be interpreted as such.

The response will not be penalized for any attempt to verbally describe their answer after a correct answer has been presented.

Minimalist response: The following expression and units represent the minimum work that must be seen in a response in order to earn both points. $\frac{-4+8}{27}$ postcards/min² or $\frac{-4-(-8)}{27}$ postcards/min²



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

4-6-6

Solution:

$$P'(24) \approx \frac{P(35) - P(8)}{35 - 8} = \frac{-4 - (-8)}{27} = \frac{4}{27} \text{ postcard per minute per minute}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

- Indicating a rate of a rate
- Including the context of postcard delivery (postcards remaining)
- Explicitly referring to the specified point in time $t = 24$

Incorrect or missing units will not be penalized as long as there is a clear connection to the context of the problem.

The words “increasing” and “decreasing” are interpreted as modifiers describing one of the specified rates.

If the numerical value from part (c) is used in this interpretation, then we expect that the choice of “increasing” or “decreasing” is consistent with that value.

The following examples use $\frac{4}{27}$ as the imported value from part (c). If a different [incorrect] value is imported from part (c), then these statements should be consistent with that value. (Remember that we are reading $=$ and $\approx$ as the same thing.)

- “... increasing at a rate of $\frac{4}{27}$ ” is OK [Units not required in part (d)] (the point is not earned with this phrase alone)
- “... decreasing at a rate of $\frac{4}{27}$ ” is OK [Units not required in part (d)] (the point is not earned with this phrase alone)

Note: the context is complicated. $P(t)$ is describing the rate of change of the remaining undelivered postcards. The values in the table indicate that the number of undelivered postcards is decreasing more slowly as time passes. Common sense might tell us that this equates to the idea that the rate at which postcards are delivered is decreasing. Explaining the meaning using “postcards delivered” rather than “postcards remaining” will not prevent the response from earning this point.

Rate of a rate

To earn this point, the response must explicitly discuss the rate of change of a rate of change. For example:

- The rate of change in the rate at which postcards are changing
- The rate of change in postcards is changing/increasing/decreasing at a rate of ...

4-6-6

The rate at which the rate of change of postcards is changing.

The words “increasing,” “decreasing,” or “changing” are not interpreted as rates.

Units are not interpreted as rates. (The phrase “postcards per minute” is not enough to indicate a rate.)

Slope of tangent line, velocity, and acceleration are not appropriate interpretations for this context.

“Rate of change of P ” is not enough without describing what P represents in this context.

Context

A key component of the context for this explanation is “postcards.”

Any misstatement about postcards delivered vs. postcards remaining to be delivered will not prevent the response from earning this point.

Explicit reference to $t = 24$

Acceptable references to a specific point in time include (but are not limited to):

- At minute 24 (where minute could be replaced with any other unit of time)
- At $t = 24$ (with or without units of time)
- After 24 minutes (where minutes could be replaced with any other unit of time).

Note: we are choosing to assume that “after” is being used in lieu of “at” unless we have additional evidence to the contrary.

If the response is explicitly indicating an interval of time (e.g., “from minute 24 on” or “during the first 24 minutes”), this point will not be earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$P'(24)$ is the rate at which the rate of change in the number of postcards remaining is changing, in postcards per minute per minute, at minute 24.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-6-6

This point is earned with a correct tangent line equation in any form. Any of these equations can be written as is without any explanation or additional supporting work. If rewriting of equations is done (transforming to slope-intercept form, perhaps), it must be done correctly.

- $y = 0.2(t - 8) - 8$

- $y + 8 = 0.2(t - 8)$

- $y = 0.2t - 9.6$

The key calculus concept being assessed in this part of the problem is that the value of $P'(8)$ is the slope of the tangent line. In that spirit:

- The response will not be penalized for using x instead of t .
- We are accepting $P(t)$ in lieu of y in all forms of the tangent line equation.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = P'(8)(t - 8) + P(8)$$

$$y = 0.2(t - 8) - 8$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are 3 components of substitution that we need to see:

- Uses $u = \frac{t}{3}$, which may be done implicitly
- Uses the correct constant of 3
- Uses the correct limits for the substitution (0 and 8). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.

Either of the expressions $3P\left(\frac{t}{3}\right)\bigg|_0^{24}$ or $3P(u)\bigg|_0^8$ will earn the substitution point without any additional supporting work.

Special Handling: Notation Errors

4-6-6

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point.

- Limits for t attached to an integral with u as the variable of integration (or vice versa),
- Improper equals signs, such as an indefinite integral equal to a definite integral
- Reversed limits of integration, for example: $\int_8^0 P'(u) \, du$

The second point is for the Fundamental Theorem of Calculus.

This point can be earned with any of the following, where k is a nonzero constant and $b = 8$ or $b = 24$. No explanation or additional supporting work is necessary.

- $kP\left(\frac{t}{3}\right)\Big|_0^b$
- $k(P(b) - P(0))$

This point can be earned without earning the substitution point.

The third point is for the correct numerical value, which is 6.

Values from the table must be used. $3(P(8) - P(0))$ is not enough to earn the answer point.

The expression $3(-8 - (-10))$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they do not need to be correct to earn this point.

The response cannot earn this point without also earning first two points.

Minimalist response: The expression $3(-8 - (-10))$ represents the minimum work that must be seen in a response in order to earn all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

4-6-6

Let $u = \frac{t}{3}$. Then $du = \frac{dt}{3}$.

When $t = 0$, $u = 0$.

When $t = 24$, $u = 8$.

$$\int_0^{24} P' \left(\frac{t}{3} \right) dt = 3 \int_0^8 P' (u) du$$

$$= 3 (P (8) - P (0))$$

$$= 3 (-8 - (-10)) = 6$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression at $t = 11$ earns the point. It may be written as $4 \cdot R' (11)$ or $4 \cdot \frac{dR}{dt} \Big|_{t=11}$

The expression $4 \cdot R' (t)$ is not sufficient because it does not use $t = 11$.

The response must use R (not M or G or any other function name) in order to earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in grams per minute, of the total weight of postcards delivered by the second student at $t = 11$ minutes is $4R' (11)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between the differential equation

$$\frac{dy}{dt} = R' (t) \text{ and the given slope field.}$$

In order to earn the point, the response must state that the values of the differential equation are positive while the slopes in the given slope field are negative or 0 (non-positive).

4-6-6

Some ways to say that the values of differential equation are positive:

- R is increasing
- $R'(t) > 0$
- R has positive slopes

When discussing the slopes in the slope field, the response must explicitly refer to the signs of the slopes.

The following examples could earn the point:

- The slope field has nonpositive slopes.
- The slopes shown are negative or 0.
- The slope field contains negative slopes.
- “The graph” or “The field” could be used in lieu of “The slope field.”

The following examples will not earn the point:

- The slope field is decreasing/negative.
- The slope field looks like it represents a decreasing function.
- The slope field represents a quadratic [or exponential] function.
- “It” (rather than “the slope field”)

Statements like “ R is an increasing function, so the signs of the slopes in the slope field should be positive” can earn this point because “should be” is indicating contrast with what is known.

The response will not be penalized for saying that the slopes in the slope field are “always” negative or “all” negative.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

R is an increasing function, so $\frac{dy}{dt} = R'(t) \geq 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are negative.

4-6-6

73. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	8	24
$P(t)$ (postcards per minute)	-6	-10	-12

A student is delivering postcards to teachers from students on a school trip. For time t minutes, $0 \leq t \leq 24$, the rate of change in the number of postcards remaining for the student to deliver is modeled by a twice-differentiable function P , in postcards remaining per minute. Values of $P(t)$ at various times t are shown in the table above.

(a) Using correct units, approximate $P'(8)$ using the average rate of change of P over the interval $0 \leq t \leq 24$.

(b) Explain the meaning of $P'(8)$ in the context of the problem.

(c) The function $N(t)$ models the number of postcards remaining for the student to deliver at time t minutes, where $N'(t) = P(t)$. It is known that $N(8) = 185$. Write an equation for the line tangent to the graph of $y = N(t)$ at $t = 8$.

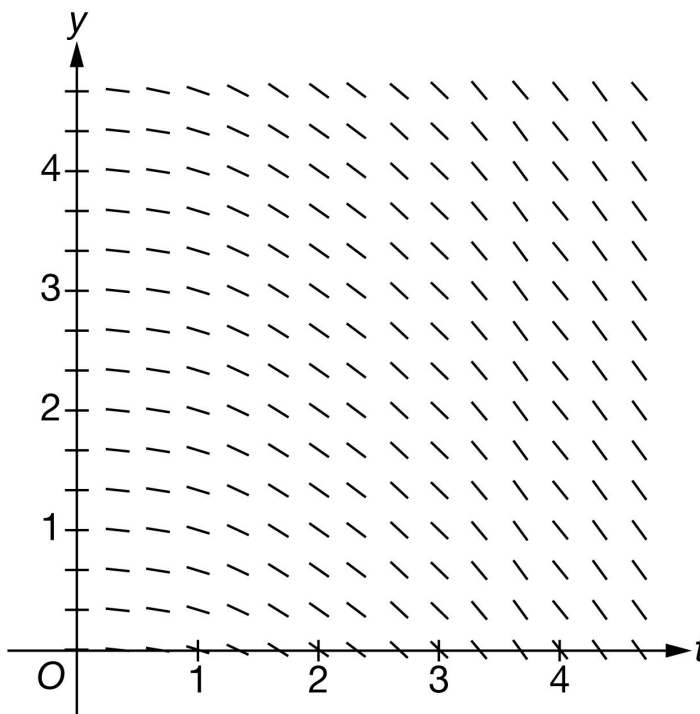
(d) Find the value of $\int_0^8 P'(3t) \, dt$. Show the calculations that lead to your answer.

(e) Use a trapezoidal sum with the two subintervals indicated by the table to approximate $\int_0^{24} P(t) \, dt$. Show the calculations that lead to your answer.

(f) Another student is also delivering postcards. For $0 \leq t \leq 24$, the rate of change in the number of postcards remaining for the second student to deliver, in postcards per minute, can be modeled by the differentiable function R . Assume the weight of each postcard is 4 grams. Write an expression in terms of P and R for the rate of change, in grams per minute, of the combined weight of the postcards remaining for both students at $t = 13$ minutes.

4-6-6

(g) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = -0.25y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

Answers without supporting work do not earn this point. Supporting work must be at least a difference and a quotient.

Response does not need to be simplified, but any simplification must be done correctly.

If a decimal is presented, it must be correct to three decimal places. We do not read anything past the third decimal place.

The second point is earned for the correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols.

The response will not be penalized for any verbal attempt to discuss the answer after the correct presentation of the approximation.

The minimum response needed to earn both points is: $\frac{-12+6}{24}$ postcards per minute per minute

4-6-6



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$P'(8) \approx \frac{P(24) - P(0)}{24 - 0} = \frac{-12 - (-6)}{24} = \frac{-6}{24}$$

$$= -\frac{1}{4} \text{ postcard per minute per minute}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This part is intended to be independent from part (a), so we do not want or need to see any reference to the value obtained in (a). If that value is referenced, conclusion must be consistent.

Three components of a correct answer:

- Indicating a rate of a rate (look for the word rate to be used correctly twice)
- Including the context of postcards
- Referring to the specific point in time

Units will not be considered for this point. They do not need to be present, and if they are, they do not need to be correct.

Any incorrect statements make the response ineligible for the point. For example:

- $P'(8)$ is the average (or an approximation, or an estimate of) rate of change of
- $P'(8)$ is the acceleration (or rate of change of velocity) of

“Increasing” or “decreasing” may be used in place of “changing” as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Beware of phrases such as “decreasing at a negative rate.”

- Example: $P'(8)$ is the rate of decrease of the rate of change of the number of postcards at 8 minutes. (earns the point)

4-6-6

· Example: $P'(8)$ means the number of postcards per minute is decreasing by $\frac{1}{4}$ postcards/minute/minute. (does not earn the point)

Acceptable references to time include (but are not limited to):

- On (or at) minute 8
- At $t = 8$ (with correct units of time, without correct units of time, or even with incorrect units of time)
- After $t = 8$ (with correct units of time, without correct units of time, or even with incorrect units of time)

If the response clearly indicates an interval of time (e.g. “from minute 0 to minute 8”) the response will not earn the point.

Examples that earn the point include:

- $P'(8)$ represents the rate at which the rate of change of the number of postcards is changing on minute 8.
- $P'(8)$ is the rate of change in the rate of change of the number of postcards at 8 minutes.
- $P'(8)$ represents the rate at which the rate of change of the number of postcards is decreasing at $t = 8$.
- $P'(8)$ equals (answer from part (a)) means the rate of change in the number of postcards per minute is increasing at a rate of (answer from part (a)) at $t = 8$.
- $P'(8)$ represents the rate at which the rate of change of the number of postcards is increasing on the eighth minute.

Examples that do not earn the point include:

- $P'(8)$ represents the rate at which the number of postcards is changing on minute 8.
- $P'(8)$ is the rate of change in the number of postcards per minute at $t = 8$ minutes
- The rate of change in the number of postcards per minute is decreasing at $t = 8$.
- The rate of change in the rate of change of the number of postcards per day is decreasing at $t = 8$.
- $P'(8) = -\frac{1}{4}$ means that the rate of change in the number of postcards is decreasing at a rate of $-\frac{1}{4} t = 8$. (double negative)



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

4-6-6

Solution:

$P'(8)$ is the rate at which the rate of change in the number of postcards remaining is changing, in postcards per minute per minute, at minute 8.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned with a correct tangent line equation in any form. Equivalent forms include (but are not limited to):

- $y = -10(t - 8) + 185$

- $y - 185 = -10(t - 8)$

- $y = -10t + 265$

The student may write x in place of t and still earn the point.

No work is required to earn the point.

We will accept $N(t)$ in lieu of y in the equation of the tangent line.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = N'(8)(t - 8) + N(8)$$

$$y = -10(t - 8) + 185$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution.

Two things are required as evidence of correct substitution:

- Coefficient of $\frac{1}{3}$ correctly applied to an integral or an antiderivative

4-6-6

- Evidence that the correct limits of 0 to 24 will be applied

Examples that will earn the first point:

$$\cdot \frac{1}{3} \int_0^{124} P'(u) du$$

$$\cdot \frac{1}{3} P(u) \Big|_0^{24} \text{ or } \frac{1}{3} P(3t) \Big|_0^8$$

Responses that include misstatements such as using t -values for limits with u as the variable of integration in an intermediate step which is later corrected may still earn this point. Intermediate errors in bounds that are subsequently corrected will disqualify the student from earning the third point. If the bounds are never appropriately handled, the response cannot earn the first point. Scratch work that is not connected to the solution will not be read.

Examples:

$$\cdot \int_0^8 P'(3t) dt = \frac{1}{3} \int_0^8 P'(u) du = \frac{1}{3} P(3t) \Big|_0^8 \text{ earns the first point, but will not earn the third point.}$$

$$\cdot \int_0^8 P'(3t) dt = \frac{1}{3} \int_0^8 P'(u) du = \frac{1}{3} [P(8) - P(0)] \text{ does not earn the first point.}$$

The second point is for the Fundamental Theorem of Calculus, demonstrating the knowledge that

$$\int_a^b P'(u) du = P(u) \Big|_a^b = P(b) - P(a)$$

This point can be earned with either $kP(3t) \Big|_a^b$, $kP(t) \Big|_a^b$, or $k(P(b) - P(a))$ where k is a nonzero constant [not necessarily correct], $a = 0$, and $b = 8$ or $b = 24$.

The response can earn this point without earning the first point.

$$\frac{1}{3} [P(24) - P(0)] \text{ earns the first two points.}$$

The third point is for the correct numerical value. Our answer only.

Values from the table must be used (i.e., $\frac{1}{3} [P(24) - P(0)]$ is not sufficient to earn the third point).

If the response includes any linkage error, (i.e., the use of equal signs to equate two expressions that are not equal), the third point will not be earned. If an indefinite integral is linked with a definite integral the response will not be penalized.

$$\cdot \int_0^8 P'(3t) dt = \frac{1}{3} \int P'(u) du = \frac{1}{3} P(3t) \Big|_0^8 = \dots \text{ is eligible for the third point.}$$

Eligibility: Must have earned the first two points.

The minimum response needed to earn all three points is: $\frac{1}{3}(-12 + 6)$.

4-6-6



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 3t$. Then $du = 3 dt$.

When $t = 0$, $u = 0$.

When $t = 8$, $u = 24$.

$$\begin{aligned}
 \int_0^8 P'(3t) dt &= \frac{1}{3} \int_0^{24} P'(u) du \\
 &= \frac{1}{3}(P(24) - P(0)) \\
 &= \frac{1}{3}(-12 - (-6)) = -\frac{6}{3} = -2
 \end{aligned}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the form of a trapezoidal sum

Point is earned for the setup of the trapezoidal sum. Numerical values do not need to be pulled from the table for this point, but Δt values must be presented. Responses that contain one incorrect numerical value in the trapezoidal sum setup are still eligible to earn the first point, but will not earn the second.

Response may present a left Riemann sum and a right Riemann sum, then average the two. Again, one incorrect numerical value will not disqualify the response from earning this first point.

Response may compute trapezoids as areas of rectangles and triangles.

Trapezoidal areas may be given in a graph. In order to earn this point, the response must clearly show the computations leading to the values presented in the graph, and must show that these areas are added to obtain the final approximation value.

4-6-6

A response using subtraction instead of addition will not earn the point (e.g., $\frac{1}{2}[8(P(0) - P(8)) + 16(P(8) - M(24))]$ earns no points).

Response utilizing “Trapezoidal Rule” will not earn this point, unless this represents only one error in the numerical values.

$\cdot \frac{1}{2}(8)[P(0) + 2P(8) + P(24)]$ is equivalent to $\frac{1}{2}[8(P(0) + P(8)) + 8(P(8) + P(24))]$, thus has only one incorrect numerical value (the second interval width is 8 instead of 16), so can earn the point.

$\cdot \frac{1}{2}(12)[P(0) + 2P(8) + P(24)]$ has two incorrect numerical values (both interval widths are incorrect) so will not earn the point.

The second point is for the correct numerical result.

Our answer only. Response does not need to be simplified, but if it is, it must be simplified correctly.

The answer must be accompanied with supporting work. Supporting work may appear in a graph, with clearly labeled areas. It is possible to earn the second point without earning the first if the supporting work is not sufficient to earn the first point.

Pictures without clear area computations and a clear summation (+) of quantities can earn at most the second point.

Special case: A positive constant multiple of a complete and correctly expressed trapezoidal sum (with the correct values pulled from the table) will only the first point:

$\cdot k \left[\left(\frac{-6-10}{2} \right) \cdot 8 + \left(\frac{-10-12}{2} \right) \cdot 16 \right]$ for $k > 0$ earns just the first point.

The minimum response needed to earn both points: $\left(\frac{-16}{2} \right) \cdot 8 + \left(\frac{-22}{2} \right) \cdot 16$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of trapezoidal sum
- ☐ Answer

Solution:

$$\begin{aligned} \int_0^{24} P(t) dt &\approx \left(\frac{P(0)+P(8)}{2} \right) (8-0) + \left(\frac{P(8)+P(24)}{2} \right) (24-8) \\ &= \left(\frac{-6-10}{2} \right) \cdot 8 + \left(\frac{-10-12}{2} \right) \cdot 16 = -240 \end{aligned}$$

4-6-6

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for a correct constant multiple or correct sum.

Response must contain the constant 4 as a factor of the expression, and/or the correct sum $P(13) + R(13)$.

If the sum is incorrect, then the constant factor of 4 must be applied either to a sum that involves only $P(t)$ and $R(t)$, $P'(t)$ and $R'(t)$, or integrals of $P(t)$ and $R(t)$, or to one of the correct terms of the sum $P(13)$ or $R(13)$, with no additional incorrect terms.

Examples that earn the point include:

$$\cdot 4[P(t) + R(t)]$$

$$\cdot 4 \int [P(t) + R(t)] dt$$

$$\cdot 4[P' + R']$$

$$\cdot 4P(13)$$

Examples that do not earn the point include:

$$\cdot 4[P(t) + R'(t)]$$

$$\cdot 4 \int P(t) dt$$

$$\cdot 4P' + R'$$

$$\cdot 4R(13) + t$$

Special case: $4P(13) + R(13)$ is an error in parentheses, so will earn the first point but is not eligible to earn the second point.

If the constant multiple is missing or incorrect, the sum must be correct. $k[P(13) + R(13)]$ for any nonzero constant k will earn the first point.

The second point is for the correct expression.

$4P(13) + 4R(13)$ (or an equivalent expression) earns both points.

Inclusion of units (correct or incorrect) will not affect the score.

A completely correct expression with one additional constant term added/subtracted would earn only the first point. For example, $4P(13) + 4R(13) + 4R(2)$ earns the first point only.

4-6-6



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Constant multiple or sum
- ☐ Expression

Solution:

The rate of change, in grams per minute, of the combined weight of the postcards remaining for both students at $t = 13$ minutes is $4P(13) + 4R(13)$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for a correct statement identifying specific inconsistencies between the given differential equation and the given slope field. There are many such correct statements, such as:

- The response references a specific point on the y -axis (such as $(0, 4)$), and indicates that the given slope field shows a slope of zero, yet the differential equation would yield a positive slope.
- The response references a pattern of slopes, and explains why that pattern does not fit our differential equation (e.g. “the slope field shows slopes that change as t changes, but our differential equation depends only on y , so slopes should remain constant for a given value of y as t changes.”)

Identifying a mismatch between the differential equation and the slope field at one point is sufficient to earn the point.

Response is not penalized for referencing “ x ” instead of “ t ”.

If the response contains any incorrect statement, the response will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point, nor will responses that present information beyond what we are given. For example:

- Responses that specify that the slope field shows a particular nonzero slope at a particular point will not earn the point, as this cannot actually be determined visually (i.e., the response should not indicate that at $(2, 2)$ the slope shown on the slope field is -1)
- Responses that state a specific slope value on the slope field for a general value of x or y that are not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

4-6-6

A response may involve a solution to the differential equation, with an argument based on the resulting function. If done correctly, and presented with a correct argument, this will earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

4-6-6

74. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	8	16	24	32
$D(t)$ (meters)	0	1500	2800	3900	5000

A student on the school track and field team is running laps around a track. For time t minutes, $0 \leq t \leq 32$, the total distance run, in meters, is modeled by a twice-differentiable function D . Values of $D(t)$ at various times t are shown in the table above.

(a) Use a midpoint Riemann sum with the two subintervals indicated by the table to approximate

$\int_0^{32} D(t) \, dt$. Show the computations that lead to your answer.

(b) Find the value of $\int_0^{16} D'(2t) \, dt$. Show the computations that lead to your answer.

(c) Using correct units, approximate $D'(16)$ using the average rate of change of D over the interval $8 \leq t \leq 24$.

(d) Explain the meaning of $D'(16)$ in the context of the problem.

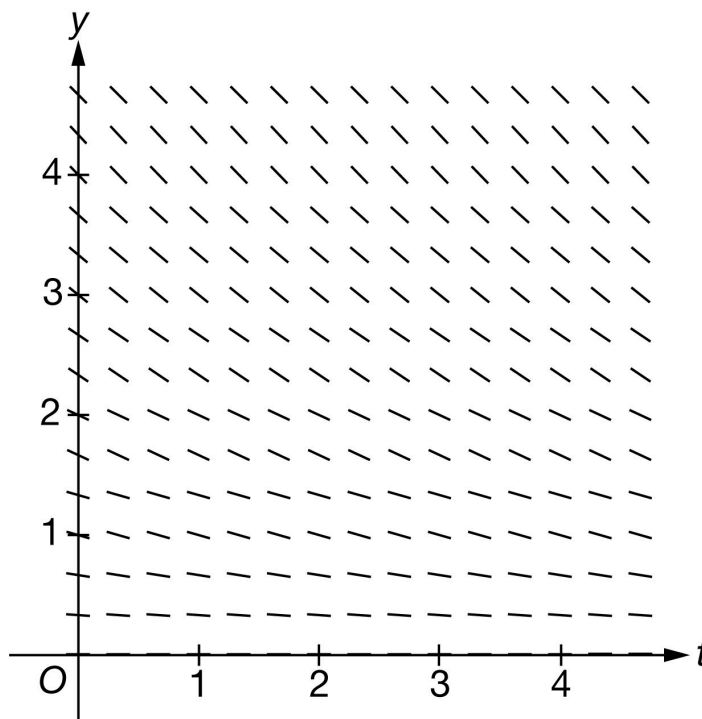
(e) It is known that $D'(24) = 140$. Write an equation for the line tangent to the graph of $y = D(t)$ at $t = 24$.

(f) It is known that $D''(t) < 0$ for $24 \leq t \leq 31$. If $D(30)$ is approximated using the equation from part (e), would the approximation be an overestimate or underestimate for the actual value of $D(30)$? Give a reason for your answer.

(g) A second student is raising funds by running laps for 60 minutes. For $0 \leq t \leq 60$, the total number of laps run by this student is modeled by $L(t)$, a differentiable function of t . For each lap run, the student raises 4 dollars. Write an expression for the rate of change, in dollars per minute, of the total money raised at time $t = 45$ minutes.

4-6-6

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = 0.25y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Form of midpoint sum): Philosophy: The response should provide clear evidence adding the area of two rectangles whose heights are found using the midpoint of the interval.

Acceptable evidence:

$$\cdot 16 \cdot (D(8) + D(24)) \text{ or } 16 \cdot D(8) + 16 \cdot D(24)$$

$$\cdot (16 - 0) \cdot 1500 + (32 - 16) \cdot 3900$$

$$\cdot 16 \cdot (1500 + 3900) \text{ or } 16 \cdot 1500 + 16 \cdot 3900$$

To earn the point the response can have at most one mistake in the expression or equation. So, expressions of the form $\frac{16}{b} \cdot (D(8) + D(24))$, for any nonzero value of b would earn the point. Note: The value $\frac{16}{b}$ might be distributed across the sum.

Expressions of the form $\frac{a}{b} \cdot (D(8) + D(24))$, for $a \neq 16$ will not earn this point.

Once the first point is earned, it cannot be lost.

Second Point (Answer)

4-6-6

To earn the point, the answer must be correct. It does not need to be simplified, but if it is, the simplification must be correct.

The answer point cannot be earned unless the first (midpoint sum) point was earned.

Minimal complete answers that earn both points:

$$\cdot 16 \cdot (1500 + 3900)$$

$$\cdot 16 \cdot 1500 + 16 \cdot 3900$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint sum
- ☐ Answer

Solution:

$$\begin{aligned} \int_0^{32} D(t) dt &\approx D(8)(16 - 0) + D(24)(32 - 16) \\ &= 1500 \cdot 16 + 3900 \cdot 16 = 86,400 \end{aligned}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Substitution) Philosophy: This point is earned when the response has all 3 components of the substitution done correctly.

Uses $u = 2t$, perhaps implicitly (any variable could be used for u)

Uses the correct constant of $\frac{1}{2}$

Uses the correct limits for their substitution.

· The limits might not be seen until the final evaluation, and at that point, response may have returned to the variable t .

· Limits for t attached to an integral involving u or vice versa will not earn this substitution point but could still earn the answer point, if the numerical answer is correct.

4-6-6

Second point: (Fundamental Theorem of Calculus) Philosophy: This point is earned when the FTC has been applied correctly to either $\int D'(2t) dt$ or $\int D'(u) du$.

The point is earned for any of the following:

- The equation $\int D'(2t) dt = D(2t)$ or $\int D'(u) du = D(u)$
- $\frac{1}{2}D(2t)\big|_0^{16}$ or $\frac{1}{2}D(u)\big|_0^{32}$
- $D(32) - D(0)$, $D(32)$, or $D(0)$

This point can be earned if the substitution point is not.

Note: $\int_a^b D'(u) du = D(b) - D(a)$ does not earn the point. The FTC point is for the response communicating that the antiderivative of D' is D .

Third Point (Answer)

Must be the correct numerical answer.

Must have earned the second point for the FTC.

Might have not earned the first point for substitution because of an incorrect use of limits, but will still earn the third point for the correct answer.

$D(32)$ and $D(0)$ must be evaluated; the expression $\frac{1}{2}(D(32) - D(0))$ by itself does not earn the third point.

The answer may be unsimplified, but if simplified it must be done correctly.

Scoring common responses:

The minimal work required to earn all three points is $\frac{1}{2}(5000 - 0)$ or $\frac{1}{2}(5000)$.

The following responses would earn the first two points and be eligible for the answer point if the response continues:

- $\frac{1}{2}D(2t)\big|_0^{16}$
- $\frac{1}{2}(D(32) - D(0))$

Responses of only the following would earn only the second point:

- $D(32) - D(0)$
- $D(16) - D(0)$

4-6-6

$$\cdot D(32)$$

$$\cdot \int_0^{16} D'(u) \, du = D(16)$$

$$\cdot D(2t)|_0^{16} \text{ or } D(u)|_0^{32}$$

The following responses would earn no points:

· 2500 in the absence of one of the first two points.

$$\cdot \int_0^{16} D'(2t) \, dt = D(16)$$



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 2t$. Then $du = 2 \, dt$.

When $t = 0$, $u = 0$.

When $t = 16$, $u = 32$.

$$\int_0^{16} D'(2t) \, dt = \frac{1}{2} \int_0^{32} D'(u) \, du$$

$$= \frac{1}{2}(D(32) - D(0))$$

$$= \frac{1}{2}(5000 - 0) = 2500$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Approximation)

The response must be a difference and a quotient.

4-6-6

The answer does not need to be simplified, but if it is simplified it must be done correctly.

The minimal work that must be shown to earn this point is either $\frac{3900-1500}{16}$ or $\frac{2400}{24-8}$.

Second point (Units)

The units must be attached to a value (either correct or incorrect).



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Approximation

☐ Units

Solution:

$$D'(16) \approx \frac{D(24) - D(8)}{24 - 8} = \frac{3900 - 1500}{16}$$

= 150 meters per minute

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The interpretation must include three components

- indicate the rate of change, the velocity, or the speed
- the specific context of the total distance run
- the specific point in time, $t = 16$.

Responses do not need to include correct units or any units.

The phrase “the student is running” may be used in place of “the total distance run is changing”.

The phrase “the student is running” is not enough. The response must attach “running” to the word “rate”, as in “the student is running at a rate of”.

The response is not earned for “the rate of total distance run” as this does not indicate a change in the rate of distance.

“the rate of change of D ” without specifying that D is the total distance run, does not earn the point.

4-6-6

Acceptable ways to provide the specific point in time:

- At minute 16 or after 16 minutes.
- At $t = 16$ (missing the unit of time)
- At 16 seconds (incorrect unit of time)

A response that clearly indicates an interval of time (e.g. “during the first 16 minutes” or “from minute 16 on”) does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$D'(16)$ is the rate at which the total distance run is changing, in meters per minute, at time $t = 16$ minutes.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The use of $D(t)$ in place of y earns the point.

The use of x in place of t earns the point.

Tangent lines do not need to be simplified, but any simplification must be done correctly.

Any correct form of the tangent line earns the point. For example:

- $y = 3900 + 140(t - 24)$
- $y - 3900 = 140(t - 24)$
- $y = 140t - 3360 + 3900$
- $y = 140t + 540$



0	1
---	---

The student response accurately includes the criteria below.

4-6-6

☐ Tangent line equation
Solution:

$$y = D'(24)(t - 24) + D(24)$$

$$y = 140(t - 24) + 3900$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point the response must include $D''(t) < 0$ and overestimate.

“ $D(t)$ is concave down” can be used instead of $D''(t) < 0$.

Information about $D'(t)$ by itself without reference to $D''(t) < 0$ is not sufficient and does not earn this point.

“Overestimate because it is concave down” is not sufficient by itself because of the use of “it”. The response must clearly, directly or indirectly, indicate to what “it” is referring.

If a response makes the correct conclusion and states $D''(t) < 0$ and then goes on and makes an incorrect statement or a statement that cannot be verified will not earn this point.

For example: “ $D'(t)$ is concave down, overestimate” does not earn this point.



0	1
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The student response accurately includes the criteria below.

☐ Answer with reason
Solution:

Because $D''(t) < 0$, the tangent line approximation would produce an overestimate.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression of only $4 \cdot L'(45)$ earns this point.

No justifications or explanations are required.

The expression $4 \cdot D'(45)$ does not earn this point.

4-6-6

Any incorrect statement such as, $L'(45) = 4 \cdot D'(45)$, does not earn this point.



0

1

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in dollars per minute, of the total money raised at time $t = 45$ minutes is $4L'(45)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Response clearly references what part of the differential equation is different from the slope field and the response needs only to clearly and explicitly reference one of them.

The response does not need to explicitly reference both the differential equation and the given slope field.

The distinction can be made with explicit reference to one of the two things and an implicit reference to the other

For example:

· “ $\frac{dy}{dt}$ is positive, not negative”. We will infer that negative is referring to the slopes shown in the slope field.

· “This would be correct for a negative derivative, but $\frac{dy}{dt}$ is positive.” We will read “This” as “the slope field” in this sentence.

· “All of the slopes in the slope field are negative, not positive.” We will infer that the values of $\frac{dy}{dt}$ at points in the first quadrant are positive.

A response that lacks even an implicit reference to either the differential equation or the slope field will not earn the point. For example:

· “ $\frac{dy}{dt}$ is positive” lacks any reference to the slope field and so will not earn the point.

· “All the slopes in the graph are negative” lacks any reference to the differential equation, and so will not earn the point.

· Vague answers such as “The slope field has negative points” or “The field is decreasing” will not earn the point.

4-6-6



0

1

The student response accurately includes the criteria below.

Solution:

When $y > 0$, $\frac{dy}{dt} > 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are negative.

4-6-6

75. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	5	20
$R(t)$ (meters per minute)	40	200	160

A student on the school track and field team is running on the track. For time t minutes, $0 \leq t \leq 20$, the rate at which the student runs, in meters per minute, is modeled by a twice-differentiable function, $R(t)$. Values of $R(t)$ at various times t are shown in the table above.

(a) Using correct units, approximate $R'(5)$ using the average rate of change of R over the interval $0 \leq t \leq 20$.

(b) Explain the meaning of $R'(5)$ in the context of the problem.

(c) The function $D(t)$ models the number of meters run by the student at time t minutes, where $D'(t) = R(t)$. It is known that $D(5) = 800$. Write an equation for the line tangent to the graph of $y = D(t)$ at $t = 5$.

(d) Find the value of $\int_0^5 R'(4t) \, dt$. Show the calculations that lead to your answer.

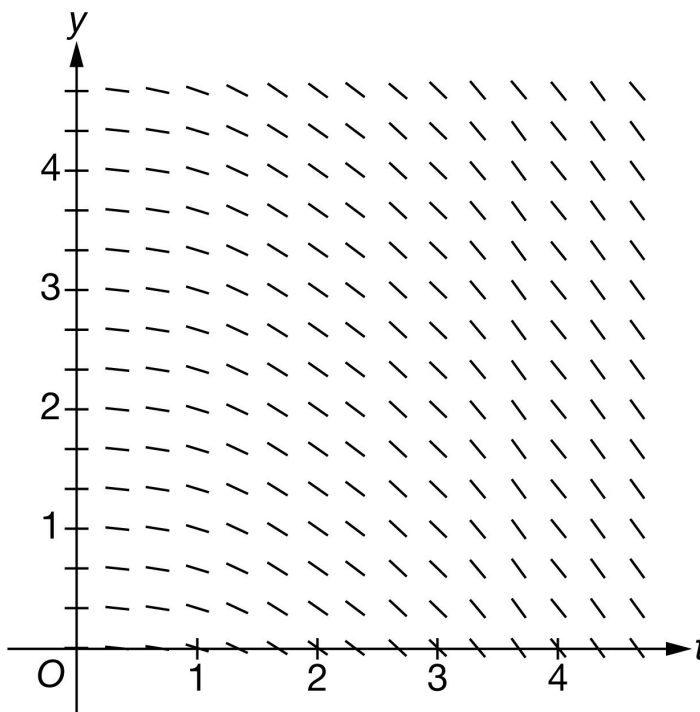
(e) Use a trapezoidal sum with the two subintervals indicated by the table to approximate $\int_0^{20} R(t) \, dt$.

Show the calculations that lead to your answer.

(f) The student and a friend team up to run on the school track for a fundraiser for 20 minutes. For $0 \leq t \leq 20$, the total number of laps run by the student is modeled by $M(t)$ and the total number of laps run by the friend is modeled by $L(t)$. Both M and L are differentiable functions of t . Each person raises 5 dollars for each lap run. Write an expression for the rate of change, in dollars per minute, of the total money raised at time $t = 18$ minutes.

4-6-6

(g) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = -0.35y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

Answers without supporting work do not earn this point. Supporting work must be at least a difference and a quotient.

Response does not need to be simplified, but any simplification must be done correctly.

If a decimal is presented, it must be correct to three decimal places. We do not read anything past the third decimal place.

The second point is earned for the correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols.

The response will not be penalized for any verbal attempt to discuss the answer after the correct presentation of the approximation.

The minimum response needed to earn both points is: $\frac{160-40}{20}$ meters per minute per minute.

4-6-6



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$R'(5) \approx \frac{R(20) - R(0)}{20 - 0} = \frac{160 - 40}{20} = 6 \text{ meters per minute per minute}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This part is intended to be independent from part (a), so we do not want or need to see any reference to the value obtained in (a). If that value is referenced, conclusion must be consistent.

Three components of a correct answer:

- Indicating a rate of a rate (look for the word rate to be used correctly twice)
- Including the context of the student running
- Referring to the specific point in time

Because the context is one of motion, a description of $R'(5)$ as the acceleration of the runner at time $t = 5$ earns the point.

Units will not be considered for this point. They do not need to be present, and if they are, they do not need to be correct.

Any incorrect statement makes the response ineligible for the point. For example:

$R'(5)$ is the average (or an approximation of, or an estimate of) rate of change of

“Increasing” or “decreasing” may be used in place of “changing” as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Beware of phrases such as “decreasing at a negative rate.”

- Example: $R'(5)$ is the rate of increase of the rate at which the student runs at 5 minutes. (earns the point)
- Example: $R'(5)$ means the rate at which the student runs is increasing by 6 meters / min / min. at time $t = 5$ (does not earn the point)

4-6-6

Acceptable references to time include (but are not limited to):

- At the fifth minute
- At $t = 5$ (with correct units of time, without correct units of time, or even with incorrect units of time)
- After $t = 5$ (with correct units of time, without correct units of time, or even with incorrect units of time)

If the response clearly indicates an interval of time (e.g. “during the first 5 minutes”) the response will not earn the point.

Examples that earn the point include:

- $R'(5)$ represents the rate at which the rate at which the student runs is changing on day 5.
- $R'(5)$ is the rate of change in the velocity of the runner at 5 days.
- $R'(5)$ represents the rate at which the velocity of the runner is increasing at $t = 5$.
- $R'(5)$ equals (answer from part (a)) means the velocity of the runner is increasing at a rate of (answer from part (a)) at $t = 5$.

Examples that do not earn the point include:

- $R'(5)$ represents the rate at which the meters per minute is changing on day 5.
- $R'(5)$ is the rate of change in the meters run at $t = 5$ days.
- The rate at which the velocity changes is increasing at $t = 5$.
- The rate of change in the rate of change of the runner is increasing at $t = 5$.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$R'(5)$ is the rate of change in the rate at which the student runs, in meters per minute per minute, at time $t = 5$ minutes.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-6-6

The point is earned with a correct tangent line equation in any form. Equivalent forms include (but are not limited to):

o $y = 200(t - 5) + 800$

o $y - 800 = 200(t - 5)$

o $y = 200t - 200$

The student may write x in place of t and still earn the point.

No work is required to earn the point.

We will accept $D(t)$ in lieu of y in the equation of the tangent line.



0	1
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The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = D'(5)(t - 5) + D(5)$$

$$y = 200(t - 5) + 800$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution.

Two things are required to award this point:

- Coefficient of $\frac{1}{4}$ correctly applied to an integral or an antiderivative
- Evidence that the correct limits of 0 to 20 will be or have been applied

Examples that will earn the first point (these are just a few):

$$\cdot \frac{1}{4} \int_0^{20} R'(u) \, du$$

$$\cdot \frac{1}{4} R(u) \Big|_0^{20} \text{ or } \frac{1}{4} R(4t) \Big|_0^5$$

4-6-6

Responses that include misstatements such as using t -values for limits with u as the variable of integration in an intermediate step which is later corrected may still earn this point. Intermediate errors in bounds that are subsequently corrected will disqualify the student from earning the third point. If the bounds are never appropriately handled, the response cannot earn the first point. Scratch work that is not connected to the solution will not be read.

Examples:

$$\int_0^5 R'(4t) dt = \frac{1}{4} \int_0^5 R'(u) du = \frac{1}{4} R(4t) \Big|_0^5 \text{ earns the first point, but will not earn the third point.}$$

$$\int_0^5 R'(4t) dt = \frac{1}{4} \int_0^5 R'(u) du = \frac{1}{4} [R(5) - R(0)] \text{ does not earn the first point.}$$

The second point is for the Fundamental Theorem of Calculus, demonstrating the knowledge that

$$\int_a^b R'(u) du = R(u) \Big|_a^b = R(b) - R(a)$$

This point can be earned with either $kR(4t) \Big|_a^b$, $kR(t) \Big|_a^b$, or $k(R(b) - R(a))$ where k is a nonzero constant [not necessarily correct], $a = 0$, and $b = 5$ or $b = 20$.

A response can earn this point without earning the first point.

$$\frac{1}{4} [R(20) - M(0)] \text{ earns the first two points.}$$

The third point is for the correct numerical value. Our answer only.

Values from the table must be used (i.e., $\frac{1}{4} [R(20) - M(0)]$ is not sufficient to earn the third point).

If the response includes any linkage error, (i.e., the use of equal signs to equate two expressions that are not equal), the third point will not be earned. If an indefinite integral is linked with a definite integral the response will not be penalized.

$$\int_0^5 R'(4t) dt = \frac{1}{4} \int R'(u) du = \frac{1}{4} R(4t) \Big|_0^5 = \dots \text{ is eligible for the third point.}$$

Eligibility: Must have earned the first two points.

The minimum response needed to earn all three points is: $\frac{1}{4}(160 - 40)$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

Substitution

4-6-6

☐☐

Fundamental Theorem of Calculus

☐

Answer

Solution:

Let $u = 4t$. Then $du = 4 dt$.

When $t = 0$, $u = 0$.

When $t = 5$, $u = 20$.

$$\begin{aligned}\int_0^5 R'(4t) dt &= \frac{1}{4} \int_0^{20} R'(u) du \\ &= \frac{1}{4}(R(20) - R(0)) \\ &= \frac{1}{4}(160 - 40) = 30\end{aligned}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The **first point** is for the form of a trapezoidal sum

Point is earned for the setup of the trapezoidal sum. Numerical values do not need to be pulled from the table for this point, but Δt values must be presented. Responses that contain one incorrect numerical value in the trapezoidal sum setup are still eligible to earn the first point, but will not earn the second.

Response may present a left Riemann sum and a right Riemann sum, then average the two. Again, one incorrect numerical value will not disqualify the response from earning this first point.

Response may compute trapezoids as areas of rectangles and triangles.

Trapezoidal areas may be given in a graph. In order to earn this point, the response must clearly show the computations leading to the values presented in the graph, and must show that these areas are added to obtain the final approximation value.

A response using subtraction instead of addition will not earn the point

(e.g. $\frac{1}{2}[5(R(0) - R(5)) + 15(R(5) - R(20))]$ earns no points).

Response utilizing “Trapezoidal Rule” will not earn this point, unless this represents only one incorrect numerical value.

4-6-6

$\cdot \frac{1}{2}(5) [R(0) + 2R(5) + R(20)]$ is equivalent to $\frac{1}{2}[5(R(0) + R(5)) + 5(M(5) + R(20))]$, thus has only one incorrect numerical value (the second interval width is 5 instead of 15), so can earn the point.

$\cdot \frac{1}{2}(10) [R(0) + 2R(5) + R(20)]$ has two incorrect numerical values (both interval widths are incorrect) so will not earn the point.

The second point is for the correct numerical result.

Our answer only. Response does not need to be simplified, but if it is, it must be simplified correctly.

The answer must be accompanied with supporting work. Supporting work may appear in a graph, with clearly labeled areas. It is possible to earn the second point without earning the first point if the supporting work is not sufficient to earn the first point.

Pictures without clear area computations and a clear summation (+) of quantities can earn at most the second point.

Special case: A positive constant multiple of a complete and correctly expressed trapezoidal sum (with the correct values pulled from the table) will earn the first but not the second point:

$\cdot k \left[\left(\frac{40+200}{2} \right) \cdot 5 + \left(\frac{200+160}{2} \cdot 15 \right) \right]$ for $k > 0$ earns only the first point.

The minimum response needed to earn both points: $\left(\frac{240}{2} \right) \cdot 5 + \left(\frac{360}{2} \right) \cdot 15$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of trapezoidal sum
- ☐ Answer

Solution:

$$\int_0^{20} R(t) dt \approx \left(\frac{R(0)+R(5)}{2} \right) (5-0) + \left(\frac{R(5)+R(20)}{2} \right) (20-5)$$

$$= \left(\frac{40+200}{2} \right) \cdot 5 + \left(\frac{200+160}{2} \right) \cdot 15 = 3300$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for a correct constant multiple or correct sum.

4-6-6

Response must contain the constant 5 as a factor of the expression, and/or the correct sum $L'(18) + M'(18)$.

If the sum is incorrect, then the constant factor of 5 must be applied either to a sum that involves only $L(t)$ and $M(t)$, $L'(t)$ and $M'(t)$, or integrals of $L(t)$ and $M(t)$, or to one of the correct terms of the sum $L'(18)$ or $M'(18)$, with no additional incorrect terms.

Examples that earn the point include:

- $5[M'(t) + L'(t)]$
- $5 \int [M(t) + L(t)] dt$
- $5[M + L]$
- $5M'(18)$

Examples that do not earn the point include:

- $5[M(t) + L'(t)]$
- $5 \int M(t) dt$
- $5M' + L$
- $5L'(18) + t$

Special case: $5L'(18) + M'(18)$ is an error in parentheses, so will earn the first point but is not eligible to earn the second point.

If the constant multiple is missing or incorrect, the sum must be correct. $k[L'(18) + M'(18)]$ for any nonzero constant k will earn the first point.

The second point is for the correct expression.

$5L'(18) + 5M'(18)$ (or an equivalent expression) earns both points.

Inclusion of units (correct or incorrect) will not affect the score.

A completely correct expression with one additional constant term added/subtracted would earn a just the first point. For example, $5L'(18) + 5M'(18) + 5(2)$ earns only the first point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

Constant multiple or sum

4-6-6

☐☐

Expression

Solution:

The rate of change, in dollars per minute, of the total money raised at time $t = 18$ minutes is $5M'(18) + 5L'(18)$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for a correct statement identifying specific inconsistencies between the given differential equation and the given slope field. There are many such correct statements, such as:

- The response references a specific point on the y -axis (such as $(0, 4)$), and indicates that the given slope field shows a slope of zero, yet the differential equation would yield a positive slope.
- The response references a pattern of slopes, and explains why that pattern does not fit our differential equation (e.g. “the slope field shows slopes that change as t changes, but our differential equation depends only on y , so slopes should remain constant for a given value of y as t changes.”)

Identifying a mismatch between the differential equation and the slope field at one point is sufficient to earn the point.

Response is not penalized for referencing “ x ” instead of “ t ”.

If the response contains any incorrect statement, the response will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point, nor will responses that present information beyond what we are given. For example:

- Responses that specify that the slope field shows a particular nonzero slope at a particular point will not earn the point, as this cannot actually be determined visually (i.e., the response should not indicate that at $(2, 2)$ the slope shown on the slope field is -1)
- Responses that state a specific slope value on the slope field for a general value of x or y that are not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

A response may involve a solution to the differential equation, with an argument based on the resulting function. If done correctly, and presented with a correct argument, this will earn the point.



0

1

4-6-6

The student response accurately includes the criteria below.

☐ Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

4-6-6

76. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	5	20	30
$D(t)$ (meters)	0	1000	4500	6500

A student on the school track and field team is running laps around a track. For time t minutes, $0 \leq t \leq 30$, the total distance run, in meters, is modeled by an increasing, twice-differentiable function D . Values of $D(t)$ at various times t are shown in the table above.

(a) Using correct units, approximate $D'(20)$ using the average rate of change of D over the interval $5 \leq t \leq 30$.

(b) Explain the meaning of $D'(20)$ in the context of the problem.

(c) It is known that $D'(5) = 200$. Write an equation for the line tangent to the graph of $y = D(t)$ at $t = 5$.

(d) Use a left Riemann sum with the three subintervals indicated by the table to approximate

$\int_0^{30} D(t) \, dt$. Show the computations that lead to your answer.

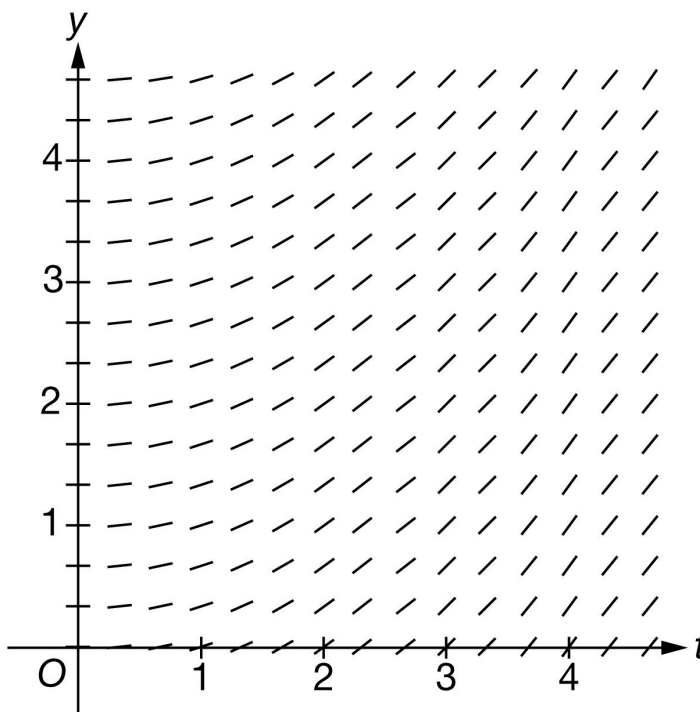
(e) Is the approximation from part (d) an overestimate or an underestimate for $\int_0^{30} D(t) \, dt$? Give a reason for your answer.

(f) Find the value of $\int_0^5 D'(4t) \, dt$. Show the computations that lead to your answer.

(g) The student is using their running practice as a fundraiser. For $0 \leq t \leq 30$, the total number of laps run can be modeled by the differentiable function L , where L is a function of t . For every lap run, the student raises 5 dollars. Write an expression for the rate of change, in dollars per minute, of the total money raised at $t = 22$ minutes.

4-6-6

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = 0.5y$.

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work. The supporting work must be at least a difference and a quotient. Correct values must be taken from the table to earn the point. The difference quotient does not need to be simplified.

Examples that earn the first point include:

$$\cdot \frac{6500 - 1000}{30 - 5}$$

$$\cdot \frac{D(30) - D(5)}{25} = \frac{5500}{25}$$

$$\cdot \frac{D(30) - D(5)}{25} = 220$$

An example that does not earn the first point is $\frac{D(30) - D(5)}{30 - 5}$.

Minimal response that earns the point: $\frac{5500 - 1000}{30 - 5}$ or $\frac{6500 - 1000}{25}$

Neither $\frac{5500}{25}$ nor 220, with no supporting work, will earn this point

The second point is for correct units

4-6-6

The correct units must be attached to a numerical value: either correct or incorrect.

The units may be written using words or a combination of words and symbols. For example:

- Meters per minute
- Meters/minute
- m/min

The response will not be penalized for any verbal attempt to discuss the answer after the correct answer has been presented.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$D'(20) \approx \frac{D(30) - D(5)}{30 - 5} = \frac{6500 - 1000}{25} = 220 \text{ meters per minute}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

- Indicating a rate of change
- Including the context of student running
- Referring to the specific point in time $t = 20$

The response is not penalized for incorrect units in this part. (Whether correct or incorrect, units have no effect on the score in this part.)

Responses may use “changing” or “increasing” as a modifier as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Example: “The number of meters is increasing at a rate of 220 meters per minute” is okay.

Acceptable reference to a specific point in time includes (but is not limited to):

4-6-6

- On minute 20 and after 20 minutes
- At $t = 20$
- At $t = 20$ hours or any incorrect unit of time

If the response clearly indicates an interval of time (e.g., “from minute 20 on,” “during the first 20 minutes,” or “for $t = 0$ to $t = 20$ ”), the point is not earned.

Examples that earn the point include:

- $D'(20)$ is the rate of change of the distance of meters per minute at time $t = 20$
- The rate of change (or speed/velocity/ how fast) is 220 meters per minute at 20 minutes
- $D'(20)$ is the rate of change in the distance ran at time $t = 20$ in meters per *second*

Examples that do not earn the point include:

- 220 is the rate of increase in the distance of meters per minute at 20 minutes
- $D'(20)$ is the average rate of change of the total meters ran per minute at minute 20
- $D'(20)$ is the number of meters per minute at minute 20
- The rate of change is 220 meters per minute (Missing “at $t = 20$ ”)
- $D'(20)$ is the slope of the tangent line at $t = 20$
- Rate of change of D at time $t = 20$
- Increase in number of meters at $t = 20$



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$D'(20)$ is the rate at which the total distance run is changing, in meters per minute, at time $t = 20$ minutes.

– OR –

$D'(20)$ is the speed at which the student is running, in meters per minute, at time $t = 20$ minutes.

4-6-6

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. The use of x in place of t is acceptable.

Some examples include

- $y = 200(t - 5) + 1000, y = 200(x - 5) + 1000$

- $y - 1000 = 200(t - 5),$

- $y - 1000 = 200(x - 5)$

- $y = 200t, y = 200x$

The key calculus concept being assessed in this part of the problem is that the value of $D'(5)$ is the slope of the tangent line. In that spirit, we are accepting $D'(t) = \dots$ in lieu of $y = \dots$, but we are not accepting $D'(5) = \dots$

Note that $D(t) = 200(t - 5) + 1000$ earns the point, but $D(5) = 200(t - 5) + 1000$ does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = D'(5)(t - 5) + D(5)$$

$$y = 200(t - 5) + 1000$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a left Riemann sum. The setup must be a sum of three products.

- A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example in the table below).

- Responses with the numerical error described above are not eligible for the second point.

4-6-6

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns. Implicit responses are eligible for both points.

A response presenting a right Riemann sum (instead of left), will not earn any points for part (a).

The second point is for the correct numerical result.

- In order to earn the second point, there must be supporting calculations.
- The numerical values for $D(0)$, $D(5)$, and $D(20)$ must be pulled from the table
- Any simplification shown must be correct.

Use the following as an illustration how to score both points. (In most cases, you see scores for which the expression on the left side represents the entirety of relevant work shown for the response.)

- A response of $0 \cdot (5 - 0) + 1000 \cdot (20 - 5) + 4500 \cdot (30 - 20)$ earns both points.
- A response of $5 \cdot D(0) + 15 \cdot D(5) + 10 \cdot D(20)$ earns the first point. Eligible for second point if more work is on the page.
- A response of $5 \cdot D(0) + 15 \cdot D(5) + 10 \cdot D(20) = 6000$ earns both points.
- A response of $5 \cdot 0 + 15 \cdot 1000 + 10 \cdot 4500$ earns both points.
- A response of $5 \cdot 0 + 10 \cdot 1000 + 10 \cdot 4500$ contains one numerical error. Earns the first point. Not eligible for the second point.
- A response of $5 \cdot 0 + 15 \cdot 1000 + 10 \cdot 4500 = 50,000$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.
- A response of $0 + 15,000 + 45,000$ or $0 + 15,000 + 45,000 = 60,000$ does not earn the first point because we cannot see the sum of 3 products. However, this is the correct sum, so we see an answer with supporting work. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points.

$$5 \cdot 0 + 15 \cdot 1000 + 10 \cdot 4500$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Left Riemann sum

4-6-6

☐ Answer**Solution:**

$$\int_0^{30} D(t) dt \approx D(0)(5-0) + D(5)(20-5) + D(20)(30-20)$$

$$= 0 \cdot 5 + 1000 \cdot 15 + 4500 \cdot 10 = 60,000$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an underestimate because D is increasing.

The response does not need an explicit reference in part (d). The following statements are sufficient for earning this point:

- Underestimate because D is increasing
- Underestimate because the function is increasing

Note: At this point, there is only one function being discussed in this problem, so we will assume “the function” is D .

When a response includes additional information about concavity, we interpret that as a lack of understanding about the question, so this will not earn the point. When the response references a graph or the given table rather than the function D , the point is not earned.

Examples that earn the point:

- The left Riemann sum underestimates because the function is increasing.
- The sum is an underestimate because $D(t)$ is increasing.
- The sum is an underestimate $D'(t) > 0$.

Examples that do not earn the point:

- The sum is an underestimate because $D'(t)$ is increasing. ($D'(t)$ increasing is not known)
- The left Riemann sum underestimates because the graph is increasing.
- The left Riemann sum underestimates because the function is increasing and concave up.



0	1
---	---

4-6-6

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because D is an increasing function, and this is a left Riemann sum, the answer from part (d) is an underestimate.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are 3 components of substitution that we need to see:

- Uses $u = 4t$, which may be done implicitly
- Uses the correct constant of $\frac{1}{4}$
- Uses the correct limits for the substitution (0 and 20). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.
- Either of the expressions below will earn the substitution point without any additional supporting work:

$$\frac{1}{4} D(4t) \Big|_0^5 \text{ or } \frac{1}{4} D(u) \Big|_0^{20}$$

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point

- Limits for t attached to an integral with u as the variable of integration (or vice versa),
- Improper equals signs, such as an indefinite integral equal to a definite integral
- Reversed limits of integration, for example: $\int_{20}^0 D'(u) du$

The second point is for the Fundamental Theorem of Calculus. This point can be earned with any of the following, where k is a nonzero constant and $b = 5$ or $b = 20$. No explanation or additional supporting work is necessary.

- $kD(4t) \Big|_0^b$
- $k(D(b) - D(0))$

The response cannot earn this point without also earning the FTC point.

The third point is for the correct numerical value, which is 1125.

4-6-6

Values from the table must be used. $\frac{1}{4}[D(20) - D(0)]$ is not enough to earn the answer point.

The expression $\frac{1}{4}(4500 - 0)$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they need not be correct in order to earn this point.

The response cannot earn this point without also earning first two points.

Minimalist response: The expression $\frac{1}{4}(4500)$ represents the minimum work that must be seen in a response in order to earn all three points.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 4t$. Then $du = 4 dt$.

When $t = 0$, $u = 0$.

When $t = 5$, $u = 20$.

$$\begin{aligned}\int_0^5 D'(4t) dt &= \frac{1}{4} \int_0^{20} D'(u) du \\ &= \frac{1}{4}(D(20) - D(0)) \\ &= \frac{1}{4}(4500 - 0) = 1125\end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct expression at $t = 22$. No supporting work is required.

The correct expression of $5 \cdot L'(22)$ earns this point.

No justifications or explanations are required.

4-6-6

Any incorrect statement such as, $D'(22) = 5 \cdot L'(22)$, does not earn this point.



0

1

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in dollars per minute, of the total money raised at $t = 22$ minutes is $5L'(22)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between two things: the differential equation, $\frac{dy}{dt} = 0.5y$, and the given slope field.

The response does not need to explicitly reference both the differential equation and the given slope field, but it must explicitly reference one and implicitly reference the other.

Acceptable responses include

- Identifying an inconsistency between the differential equation and the slope field at one point, e.g., at the point $(4, 0)$, the slope of the segment should be 0, but it is positive.
- Referencing a pattern of slopes, and explaining why that pattern does not fit the given differential equation.

A response that lacks even an implicit reference to the one of two things being compared will not earn the point.

For example, merely stating either $\frac{dy}{dt}$ is positive, or that the slope field is positive does not earn the point.

- “The values for are greater than or equal to zero so $\frac{dy}{dt} \geq 0$ ” lacks any reference to the slope field and will not earn the point.
- “All the slopes in the slope field are positive” lacks any reference to the differential equation and will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point. Some examples are:

- “The field is increasing”
- Responses that state a specific slope value on the slope field for a general value of x or y that is not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an

4-6-6

incorrect statement)

A response may correctly solve or attempt to solve the differential equation and then provide a correct argument without reference to their solution. Such a response would earn the point.

A response might actually solve the differential equation and provide a correct argument on the resulting function.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

4-6-6

77. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	5	20	30
$D(t)$ (meters per minute)	0	100	120	150

A student on the school track and field team is running laps around a track. For time t minutes, $0 \leq t \leq 30$, the rate at which the student runs, in meters per minute, is modeled by an increasing, twice-differentiable function D . Values of $D(t)$ at various times t are shown in the table above.

(a) Use a right Riemann sum with the three subintervals indicated by the table to approximate

$\int_0^{30} D(t) \, dt$. Show the calculations that lead to your answer.

(b) Is your answer from part (a) an overestimate or an underestimate for $\int_0^{30} D(t) \, dt$? Give a reason for your answer.

(c) Using correct units, approximate $D'(5)$ using the average rate of change of D over the interval $0 \leq t \leq 20$.

(d) Explain the meaning of $D'(5)$ in the context of the problem.

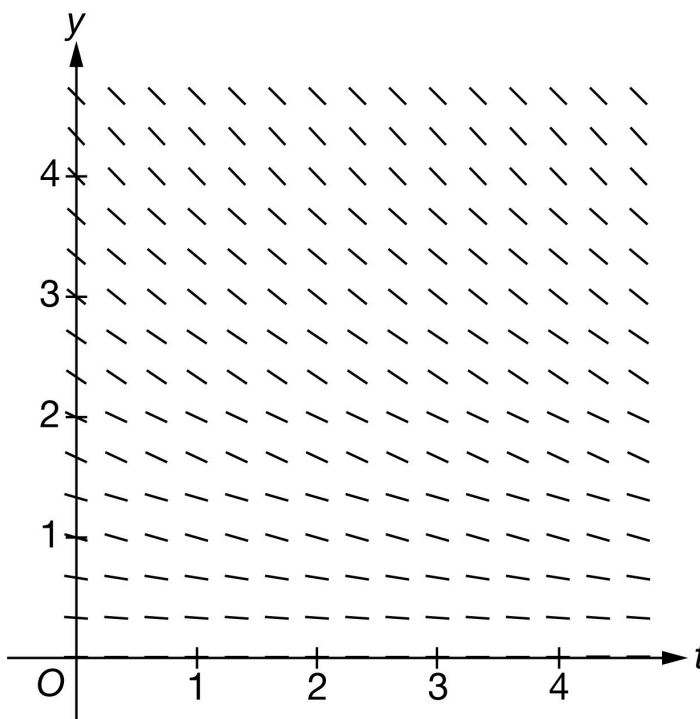
(e) It is known that $D'(20) = 2.5$. Write an equation for the line tangent to the graph of $y = D(t)$ at $t = 20$.

(f) Find the value of $\int_0^{20} D'\left(\frac{t}{4}\right) \, dt$. Show the calculations that lead to your answer.

(g) A friend on the track and field team is using her running practice as a fundraiser. For $0 \leq t \leq 30$, the total number of laps run by the friend can be modeled by the increasing, differentiable function L , where L is a function of t . For every lap run, the friend raises 4 dollars. Write an expression for the rate of change of the money, in dollars per minute, raised by the friend at $t = 19$ minutes.

4-6-6

(h) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = L'(t)$ for the function L defined in part (g).

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a right Riemann sum. The setup must be a sum of three products.

A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example in the table below).

Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns, like so.

100	5	500
120	15	1800
150	10	+1500
		3800

or

$(5 - 0)(100)$	$= 500$
$(20 - 5)(120)$	$= 1800$
$(30 - 20)(150)$	$= 1500$
	3800

Implicit responses such as those above are eligible for both points.

4-6-6

A response presenting a left Riemann sum (instead of right), will not earn any points for part (a).

The second point is for the correct numerical result.

In order to earn the second point, there must be supporting calculations.

The numerical values for $D(5)$, $D(20)$, and $D(30)$ must be pulled from the table.

Any simplification shown must be correct.

Use the following as an illustration of how to score both points. (In most cases, you see scores for which the expression represents the entirety of relevant work shown for the response.)

- A response of $100 \cdot (5 - 0) + 120 \cdot (20 - 5) + 150 \cdot (30 - 20)$ earns both points.
- A response of $5 \cdot D(5) + 15 \cdot D(20) + 10 \cdot D(30)$ earns the first point. Eligible for second point if more work is on the page.
- A response of $5 \cdot D(5) + 15 \cdot D(20) + 10 \cdot D(30) = 3800$ earns both points.
- A response of $5 \cdot 100 + 15 \cdot 120 + 10 \cdot 150$ earns both points.
- A response of $5 \cdot 100 + 25 \cdot 120 + 10 \cdot 150$ contains one numerical error. Earns the first point. Not eligible for the second point.
- A response of $5 \cdot 100 + 15 \cdot 120 + 10 \cdot 150 = 3300$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.
- A response of $500 + 1800 + 1500$ or $500 + 1800 + 1500 = 3800$ does not earn the first point because we cannot see the sum of three products. However, this is the correct sum, so we are not seeing an unsupported answer. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points. $5 \cdot 100 + 15 \cdot 120 + 10 \cdot 150$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Right Riemann sum
- ☐ Answer

Solution:

4-6-6

$$\int_0^{30} D(t) dt \approx D(5)(5-0) + D(20)(20-5) + D(30)(30-20)$$

$$= 100 \cdot 5 + 120 \cdot 15 + 150 \cdot 10 = 3800$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an overestimate because D is increasing.

The response does not need an explicit reference to part (a) or to a right Riemann sum.

A response that appeals to concavity as a reason will NOT earn this point.

The following table contains some examples of what WILL or will NOT earn the point.

The following examples earn the point:

- Overestimate because D is increasing
- Overestimate because $D'(t) > 0$
- Overestimate because the graph of D is increasing (we know they are talking about the given function D)
- Overestimate because the function is increasing. (There is only one function in the problem at this point.)

The following examples do not earn the point:

- Overestimate because D is concave down/up
- Overestimate because the graph/curve is increasing. Since no graph/curve is provided, we do not know what graph/curve they are talking about.
- Overestimate because the table [or table values] is/are increasing. The behavior of the values shown in the table is not the same as the behavior of the function D . The table does not tell us what is happening to D between the given points.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because D is an increasing function, and this is a right Riemann sum, this is an overestimate.

4-6-6

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

We need to see both a difference and a quotient. They don't have to be together.

The expression does not need to be simplified, but values for $D(20)$ and $D(0)$ must be pulled from the table.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

The second point is for correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols. For example:

- Meters per minute per minute or Meters/min/min
- Meters per min^2 or Meters/ min^2
- Meters per minute squared

We do not read work shown in part (d) for this point. The units must appear in part (c).

Use caution when interpreting abbreviations: m should be meters and min should be minutes, but responses may vary. For example,

- if a response includes units “Meters/m/m ” we read “ m ” as minutes; or
- if a response includes units “m/min/min ” we read “ m ” as meters; or
- if a response includes units “m/m/m ” we cannot interpret the units and the response will not earn the second point.

The response will not be penalized for any attempt to verbally describe their answer after a correct answer has been presented.

Minimalist response: The following expression and units represent the minimum work that must be seen in a response in order to earn both points.

$$\frac{120-0}{20} \text{ meters/min}^2 \text{ or } \frac{120}{20-0} \text{ meters/min}^2$$

In the minimalist response, we still need to see subtraction (a difference), so subtracting 0 must be explicit either in the numerator or in the denominator.

4-6-6



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$D'(5) \approx \frac{D(20) - D(0)}{20 - 0} = \frac{120 - 0}{20} = 6 \text{ meters per minute per minute}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

- Three components of a correct answer:
- Indicating a rate of a rate
- Including the context of running
- Explicitly referring to the specified point in time ($t = 5$)

Incorrect or missing units will not be penalized as long as there is a clear connection to the context of the problem.

The words “increasing” and “decreasing” are interpreted as modifiers describing one of the specified rates.

If the numerical value from part (c) is used in this interpretation, then we expect that the choice of “increasing” or “decreasing” is consistent with that value.

The following examples use 6 as the imported value from part (c). If a different [incorrect] value is imported from part (c), then these statements should be consistent with that value. (Remember that we are reading $=$ and $\approx$ as the same thing.)

- “... increasing at a rate of 6 ” is OK [Units not required in part (d)] (the point is not earned with this phrase alone)
- “... decreasing at a rate of 6 ” is not OK and the point is not earned.

Rate of a rate

To earn this point, the response must explicitly discuss the rate of change of a rate of change in the context of motion [running]. For example:

4-6-6

- The rate of change in the rate at which students are running
- The rate of change in speed is changing (or increasing) at a rate of . . .

Since the runners are unlikely to be running backwards, we interpret speed as being the same as velocity.

- The rate at which the velocity of the runner is changing
- Acceleration of the runner (this is correct for the context)

The words “increasing,” “decreasing,” or “changing” are not interpreted as rates

Units are not interpreted as rates. (The phrase “meters per minute” is not enough to indicate a rate.)

Slope of tangent line is not an appropriate interpretation for this context.

“Rate of change of D ” is not enough without describing what D represents in this context.

Context

The context is running, which could be indicated by words such as (but not limited to): meters, minutes, laps, speed, velocity, runner

Explicit reference to $t = 5$

Acceptable references to a specific point in time include (but are not limited to):

- At minute 5 (where minute could be replaced with any other unit of time)
- At $t = 5$ (with or without units of time)
- After 5 minutes (where minutes could be replaced with any other unit of time).

Note: we are choosing to assume that “after” is being used in lieu of “at” unless we have additional evidence to the contrary.

If the response is explicitly indicating an interval of time (e.g., “from minute 5 on” or “during the first 5 minutes”), this point will not be earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

4-6-6

$D'(5)$ is the rate at which the rate at which the student runs is changing, in meters per minute per minute, at minute 5.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. Any of these equations can be written as is without any explanation or additional supporting work. If rewriting of equations is done (transforming to slope-intercept form, perhaps), it must be done correctly.

· $y = 2.5(t - 20) + 120$

· $y - 120 = 2.5(t - 20)$

· $y = 2.5t + 70$

The key calculus concept being assessed in this part of the problem is that the value of $D'(20)$ is the slope of the tangent line. In that spirit:

- The response will not be penalized for using x instead of t .
- We are accepting $D(t)$ in lieu of y in all forms of the tangent line equation.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = D'(20)(t - 20) + D(20)$$

$$y = 2.5(t - 20) + 120$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are three components of substitution that we need to see:

- Uses $u = \frac{t}{4}$, which may be done implicitly
- Uses the correct constant of 4

4-6-6

· Uses the correct limits for the substitution (0 and 5). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.

Either of the expressions $4D\left(\frac{t}{4}\right)\bigg|_0^{20}$ or $4D(u)\bigg|_0^5$ will earn the substitution point without any additional supporting work.

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point.

- Limits for t attached to an integral with u as the variable of integration (or vice versa),
- Improper equals signs, such as an indefinite integral equal to a definite integral
- Reversed limits of integration, for example: $\int_5^0 D'(u) du$

The second point is for the Fundamental Theorem of Calculus.

This point can be earned with any of the following, where k is a nonzero constant and $b = 5$ or $b = 20$. No explanation or additional supporting work is necessary.

- $kD\left(\frac{t}{4}\right)\bigg|_0^b$
- $k(D(b) - D(0))$ or $kD(b)$ (because $D(0) = 0$)

This point can be earned without earning the substitution point.

The third point is for the correct numerical value, which is 400.

Values from the table must be used. $4(D(5) - D(0))$ is not enough to earn the answer point.

The expressions $4(100 - 0)$ or $4(100)$ are sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they do not need to be correct to earn this point.

The response cannot earn this point without also earning the first two points.

Minimalist response: The expression $4(100)$ represents the minimum work that must be seen in a response in order to earn all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

4-6-6

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = \frac{t}{4}$. Then $du = \frac{dt}{4}$.

When $t = 0$, $u = 0$.

When $t = 20$, $u = 5$.

$$\begin{aligned}\int_0^{20} D' \left(\frac{t}{4} \right) dt &= 4 \int_0^5 D' (u) du \\ &= 4 (D (5) - D (0)) \\ &= 4 (100 - 0) = 400\end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression at $t = 19$ earns the point. It may be written as $4 \cdot L' (19)$ or $4 \cdot \frac{dL}{dt} \Big|_{t=19}$.

The expression $4 \cdot L' (t)$ is not sufficient because it does not use $t = 19$.

The response must use L (not M or D or any other function name) in order to earn the point.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Expression

Solution:

The rate of change of the money, in dollars per minute, raised by the friend at $t = 19$ minutes is $4L' (19)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-6-6

This point is earned for a correct statement identifying the distinction between the differential equation

$$\frac{dy}{dt} = L'(t) \text{ and the given slope field.}$$

In order to earn the point, the response must:

State that the values of the differential equation are positive while the slopes in the given slope field are negative or 0.

Some ways to say that the values of differential equation are positive:

- L is increasing
- $L'(t) > 0$
- L has positive slopes

When discussing the slopes in the slope field, the response must explicitly refer to the signs of the slopes.

The following examples could earn the point:

- The slope field has negative (non-positive) slopes.
- The slopes shown are negative or 0.
- “The graph” or “The field” could be used in lieu of “The slope field.”

The following examples will not earn the point:

- The slope field is decreasing/negative.
- The slope field looks like it represents a decreasing function.
- The slope field represents a quadratic [or exponential] function.
- “It” (rather than “the slope field”)

Statements like “ L is an increasing function, so the signs of the slopes in the slope field should be positive” can earn this point because “should be” is indicating contrast with what is known.

The response will not be penalized for saying that the slopes in the slope field are “always” negative or “all” negative.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

4-6-6**Solution:**

L is an increasing function, so $\frac{dy}{dt} = L'(t) \geq 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are negative.

4-6-6

78. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

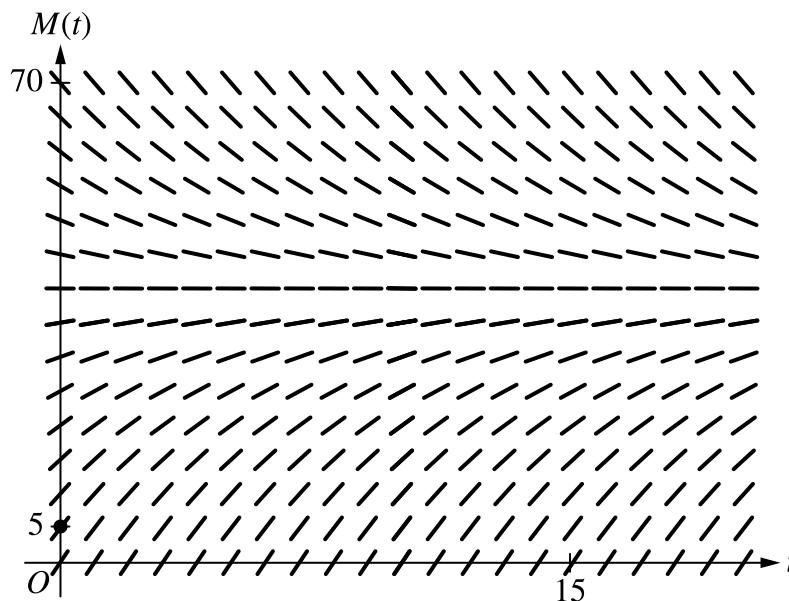
Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

A bottle of milk is taken out of a refrigerator and placed in a pan of hot water to be warmed. The increasing function M models the temperature of the milk at time t , where $M(t)$ is measured in degrees Celsius ($^{\circ}\text{C}$) and t is the number of minutes since the bottle was placed in the pan. M satisfies the differential equation $\frac{dM}{dt} = \frac{1}{4}(40 - M)$. At time $t = 0$, the temperature of the milk is 5°C . It can be shown that $M(t) < 40$ for all values of t .

- (a) A slope field for the differential equation $\frac{dM}{dt} = \frac{1}{4}(40 - M)$ is shown. Sketch the solution curve through the point $(0, 5)$.



- (b) Use the line tangent to the graph of M at $t = 0$ to approximate $M(2)$, the temperature of the milk at time $t = 2$ minutes.

- (c) Write an expression for $\frac{d^2M}{dt^2}$ in terms of M . Use $\frac{d^2M}{dt^2}$ to determine whether the approximation from part (b) is an underestimate or an overestimate for the actual value of $M(2)$. Give a reason for your answer.

4-6-6

(d) Use separation of variables to find an expression for $M(t)$, the particular solution to the differential equation $\frac{dM}{dt} = \frac{1}{4}(40 - M)$ with initial condition $M(0) = 5$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The solution curve must pass through the point $(0, 5)$, extend reasonably close to the left and right edges of the rectangle, and have no obvious conflicts with the given slope lines.

Only portions of the solution curve within the given slope field are considered.

The solution curve must lie entirely below the horizontal line segments at $M = 40$.



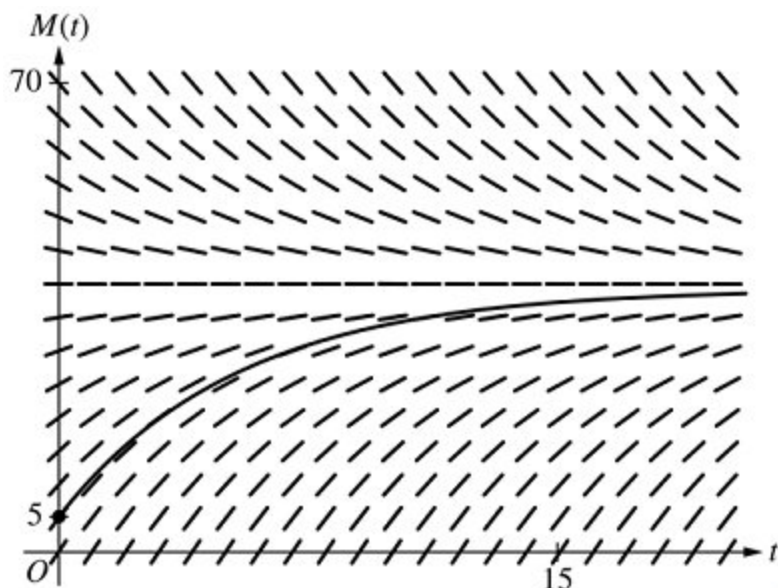
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1

The student response accurately includes the criteria below.

☐

Solution curve

Solution:**Part B**

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The value of the slope may appear in a tangent line equation or approximation.

4-6-6

A response of $5 + \frac{35}{4} \cdot 2$ is the minimal response to earn both points.

A response of $\frac{1}{4}(40 - 5)$ earns the first point. If there are any subsequent errors in simplification, the response does not earn the second point.

In order to earn the second point the response must present an approximation found by using a tangent line that:

- passes through the point $(0, 5)$ and
- has slope $\frac{35}{4}$ or a nonzero slope that is declared to be the value of $\frac{dM}{dt}$.

An unsupported approximation does not earn the second point.

The approximation need not be simplified, but the response does not earn the second point if the approximation is simplified incorrectly.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $\frac{dM}{dt} \Big|_{t=0}$
- ☐ Approximation

Solution:

$$\frac{dM}{dt} \Big|_{t=0} = \frac{1}{4}(40 - 5) = \frac{35}{4}$$

The tangent line equation is $y = 5 + \frac{35}{4}(t - 0)$.

$$M(2) \approx 5 + \frac{35}{4} \cdot 2 = 22.5$$

The temperature of the milk at time $t = 2$ minutes is approximately 22.5° Celsius.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for either $\frac{d^2M}{dt^2} = -\frac{1}{4}\left(\frac{1}{4}(40 - M)\right)$ or $\frac{d^2M}{dt^2} = -\frac{1}{16}(40 - M)$ (or equivalent). A response that presents any subsequent simplification error does not earn the second point.

4-6-6

A response that presents an expression for $\frac{d^2M}{dt^2}$ in terms of $\frac{dM}{dt}$ but fails to continue to an expression in terms of (i.e., $\frac{d^2M}{dt^2} = -\frac{1}{4}\frac{dM}{dt}$) does not earn the first point but is eligible for the second point.

If the response presents an expression for $\frac{d^2M}{dt^2}$ that is incorrect, the response is eligible for the second point only if the expression is a nonconstant linear function that is negative for $5 < M < 40$.

- Special case: A response that presents $\frac{d^2M}{dt^2} = \frac{1}{16}(40 - M)$ does not earn the first point but is eligible to earn the second point for a consistent answer and reason.

To earn the second point a response must include $\frac{d^2M}{dt^2} < 0$, or $\frac{dM}{dt}$ is decreasing, or the graph of M is concave down, as well as the conclusion that the approximation is an overestimate.

A response that presents an argument based on $\frac{d^2M}{dt^2}$ or concavity at a single point does not earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $\frac{d^2M}{dt^2}$
- ☐ Overestimate with reason

Solution:

$$\frac{d^2M}{dt^2} = -\frac{1}{4}\frac{dM}{dt} = -\frac{1}{4}\left(\frac{1}{4}(40 - M)\right) = -\frac{1}{16}(40 - M)$$

Because $M(t) < 40$, $\frac{d^2M}{dt^2} < 0$, so the graph of M is concave down. Therefore, the tangent line approximation of $M(2)$ is an overestimate.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with no separation of variables earns 0 out of 4 points.

A response that presents an antiderivative of $-\ln(40 - M)$ without absolute value symbols is eligible for all 4 points.

A response with no constant of integration can earn at most the first 2 points.

4-6-6

A response is eligible for the third point only if it has earned the first 2 points.

- Special Case: A response that presents $+\ln(40 - M) = \frac{t}{4} + C$ (or equivalent) does not earn the second point, is eligible for the third point, but not eligible for the fourth.

An eligible response earns the third point by correctly including the constant of integration in an equation and substituting 0 for t and 5 for M .

A response is eligible for the fourth point only if it has earned the first 3 points.

A response earns the fourth point only for an answer of $M = 40 - 35e^{-\frac{t}{4}}$ or equivalent.



0	1	2	3	4
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The student response accurately includes **all four** of the criteria below.

- ☐ Separates variables
- ☐ Finds antiderivatives
- ☐ Constant of integration and uses initial condition
- ☐ Solves for M

Solution:

$$\frac{dM}{40-M} = \frac{1}{4}dt$$

$$\int \frac{dM}{40-M} = \int \frac{1}{4}dt$$

$$-\ln|40-M| = \frac{1}{4}t + C$$

$$-\ln|40-5| = 0 + C \Rightarrow C = -\ln 35$$

$$M(t) < 40 \Rightarrow 40 - M > 0 \Rightarrow |40 - M| = 40 - M$$

$$-\ln(40-M) = \frac{1}{4}t - \ln 35$$

$$\ln(40-M) = -\frac{1}{4}t + \ln 35$$

$$40 - M = 35e^{-\frac{t}{4}}$$

$$M = 40 - 35e^{-\frac{t}{4}}$$

4-6-6

79. Let f be the function given by $f(x) = \ln x$. What is the approximation for $\ln\left(\frac{3}{2}\right)$ found by using the line tangent to the graph of f at $x = 1$?
- (A) $\frac{1}{3}$
- (B) $\frac{1}{2}$ ✓
- (C) $\frac{2}{3}$
- (D) 1

Answer B

Correct. An equation of the line tangent to the graph of f at $x = a$ is $y = f(a) + f'(a)(x - a)$. In this question, $a = 1$. The value of y when $x = \frac{3}{2}$ would be an approximation to $\ln\left(\frac{3}{2}\right) = f\left(\frac{3}{2}\right)$. Since $f'(x) = \frac{1}{x}$, $f'(1) = 1$.

$$f\left(\frac{3}{2}\right) \approx f(1) + 1 \cdot \left(\frac{3}{2} - 1\right) = 0 + \frac{1}{2} = \frac{1}{2}$$

80. Let g be a differentiable function such that $g(3) = 2$ and $g'(3) = -\frac{3}{4}$. The graph of g is concave down on the interval $(2, 4)$. Which of the following is true about the approximation for $g(2.6)$ found using the line tangent to the graph of g at $x = 3$?
- (A) $g(2.6) \approx 1.7$ and this approximation is an overestimate of the value of $g(2.6)$.
- (B) $g(2.6) \approx 1.7$ and this approximation is an underestimate of the value of $g(2.6)$.
- (C) $g(2.6) \approx 2.3$ and this approximation is an overestimate of the value of $g(2.6)$. ✓
- (D) $g(2.6) \approx 2.3$ and this approximation is an underestimate of the value of $g(2.6)$.

Answer C

Correct. An equation for the tangent line at $x = a$ is $y = g(a) + g'(a)(x - a)$. Here $a = 3$ and $g'(a) = -\frac{3}{4}$. The value of y when $x = 2.6$ would be the approximation to $g(2.6)$, as follows.

$$g(2.6) \approx g(3) - \frac{3}{4}(2.6 - 3) = 2 - \frac{3}{4}(-0.4) = 2 + 0.3 = 2.3$$

Because the graph of g is concave down on the interval $(2, 4)$, the line tangent to the graph of g at $x = 3$ will overestimate the value of $g(2.6)$.

4-6-6

81.  The derivative of the function A is given by $A'(t) = 2 + 9e^{0.4 \sin t}$, and $A(1.2) = 7.5$. If the linear approximation to $A(t)$ at $t = 1.2$ is used to estimate $A(t)$, at what value of t does the linear approximation estimate that $A(t) = 15$?

(A) 0.498

(B) 1.166

(C) 1.698

(D) 2.400

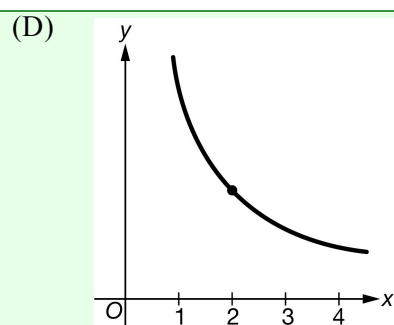
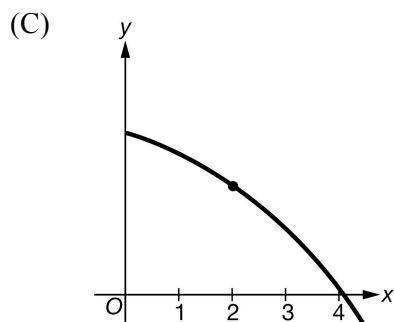
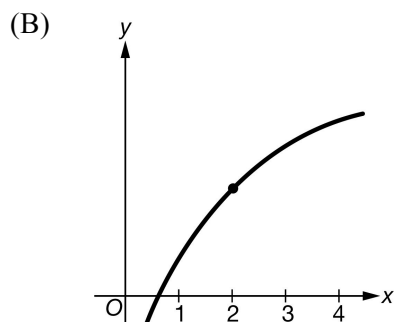
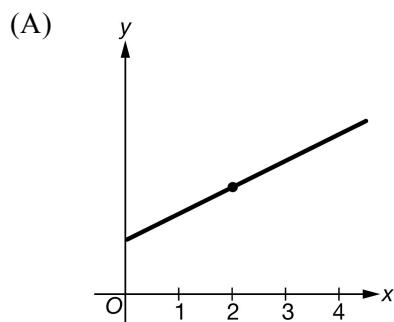
**Answer C**

Correct. The linear approximation to $A(t)$ at $t = 1.2$ will be $y = A(1.2) + A'(1.2)(t - 1.2)$. Since $A'(1.2) = 2 + 9e^{0.4 \sin 1.2} = 15.06635$, the linear approximation is $y = 7.5 + 15.06635(t - 1.2)$. The linear approximation will estimate that $A(t) = 15$ when

$$y = 7.5 + 15.06635(t - 1.2) = 15 \Rightarrow t = \frac{15 - 7.5}{15.06635} + 1.2 = 1.698.$$

4-6-6

82. The locally linear approximation of the differentiable function f at $x = 2$ is used to approximate the value of $f(2.3)$. The approximation at $x = 2.3$ is an underestimate of the corresponding function value at $x = 2.3$. Which of the following could be the graph of f ?

**Answer D**

Correct. The tangent line is the graph of the locally linear approximation of the function f near the point of tangency at $x = 2$. Since this graph is concave up on the interval $(1, 3)$, the tangent line would lie under the graph and the tangent line value at $x = 2.3$ would be less than $f(2.3)$. Therefore, this could be the graph of f . The tangent line

4-6-6

approximation is an overestimate when the line is above the curve at the point. The tangent line approximation is an underestimate when the line is below the curve at the point.

83. The line tangent to the graph of the twice-differentiable function f at the point $x = 5$ is used to approximate the value of $f(5.25)$. Which of the following statements guarantees that the tangent line approximation at $x = 5.25$ is an overestimate of $f(5.25)$?
- (A) The function f is decreasing on the interval $5 \leq x \leq 5.25$.
 - (B) The function f is increasing on the interval $5 \leq x \leq 5.25$.
 - (C) The graph of the function f is concave down on the interval $5 \leq x \leq 5.25$. ✓
 - (D) The graph of the function f is concave up on the interval $5 \leq x \leq 5.25$.

Answer C

Correct. Because the graph is concave down on the interval $5 \leq x \leq 5.25$, the line tangent to the graph at $x = 5$ will lie above the graph of f over the interval $5 \leq x \leq 5.25$. Therefore the point at $x = 5.25$ on the tangent line will be above the point at $x = 5.25$ on the graph of f , and so the tangent line value at $x = 5.25$ will be greater than $f(5.25)$. The tangent line approximation is an overestimate when the line is above the curve at the point. The tangent line approximation is an underestimate when the line is below the curve at the point.

4-6-6

84.  Let f be a function such that at each point (x, y) on the graph of f , the slope is given by $\frac{dy}{dx} = \frac{1}{2}x - \frac{1}{4}y^2$. The graph of f passes through the point $(1, -2)$ and is concave up on the interval $1 < x < 1.5$. Let k be the approximation for $f(1.3)$ found by using the locally linear approximation of f at $x = 1$. Which of the following statements about k is true?
- (A) $k = -2.65$ and is an underestimate for $f(1.3)$.
- (B) $k = -2.65$ and is an overestimate for $f(1.3)$.
- (C) $k = -2.15$ and is an underestimate for $f(1.3)$. ✓
- (D) $k = -2.15$ and is an overestimate for $f(1.3)$.

Answer C

Correct. The tangent line is the graph of the locally linear approximation of f at $x = 1$. The slope of the tangent line at $x = 1$ is $f'(1) = \left. \frac{dy}{dx} \right|_{(1, -2)} = \frac{1}{2}(1) - \frac{1}{4}(-2)^2 = \frac{1}{2} - 1 = -\frac{1}{2}$. An equation of the tangent line at the point $(1, -2)$ is therefore $y = -2 - \frac{1}{2}(x - 1)$. The value of y when $x = 1.3$ would be an approximation to $f(1.3)$, as follows.

$$f(1.3) \approx -2 - \frac{1}{2}(1.3 - 1) = -2 - 0.15 = -2.15$$

Because the graph of f is concave up on the interval $1 < x < 1.5$, the locally linear approximation of f at $x = 1$ will underestimate the value of $f(1.3)$.

85. Let f be a differentiable function such that $f(4) = 7$ and $f'(4) = \frac{1}{5}$. The graph of f is concave up on the interval $(3, 5)$. Which of the following is true about the approximation for $f(3.5)$ found using the line tangent to the graph of f at $x = 4$?
- (A) $f(3.5) \approx 6.9$ and this approximation is an overestimate of the value of $f(3.5)$.
- (B) $f(3.5) \approx 6.9$ and this approximation is an underestimate of the value of $f(3.5)$. ✓
- (C) $f(3.5) \approx 7.1$ and this approximation is an overestimate of the value of $f(3.5)$.
- (D) $f(3.5) \approx 7.1$ and this approximation is an underestimate of the value of $f(3.5)$.

Answer B

Correct. An equation for the tangent line at $x = a$ is $y = f(a) + f'(a)(x - a)$. Here $a = 4$ and $f'(a) = \frac{1}{5}$. The value of y when $x = 3.5$ would be the approximation to $f(3.5)$, as follows.

$$f(3.5) \approx f(4) + \frac{1}{5}(3.5 - 4) = 7 + \frac{1}{5}(-0.5) = 7 - 0.1 = 6.9$$

Because the graph of f is concave up on the interval $(3, 5)$, the line tangent to the graph of f at $x = 4$ will underestimate the value of $f(3.5)$.

4-6-6

86.

x	2.8	3.0	3.2	3.4
$g'(x)$	1.05	-1.2	-0.8	1.3

Selected values of the derivative of the function g are given in the table above. It is known that $g(3) = 17$. What is the approximation for $g(3.2)$ found using the line tangent to the graph of g at $x = 3$?

(A) 16.76



(B) 16.80

(C) 16.84

(D) 17.40

Answer A

Correct. An equation of the tangent line at $x = a$ is $y = g(a) + g'(a)(x - a)$. Here $a = 3$ and $g'(a) = -1.2$. The value of y when $x = 3.2$ would be an approximation to $g(3.2)$, as follows.
 $g(3.2) \approx g(3) - 1.2(3.2 - 3) = 17 - 1.2(0.2) = 17 - 0.24 = 16.76$

87.  The derivative of the function B is given by $B'(t) = 8e^{0.2 \cos t}$, and $B(2.2) = 4.5$. If the linear approximation to $B(t)$ at $t = 2.2$ is used to estimate $B(t)$, at what value of t does the linear approximation estimate that $B(t) = 9$?

(A) 0.633

(B) 0.941

(C) 2.833



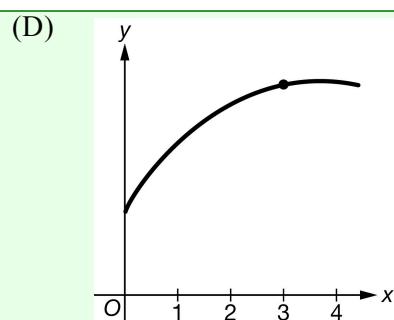
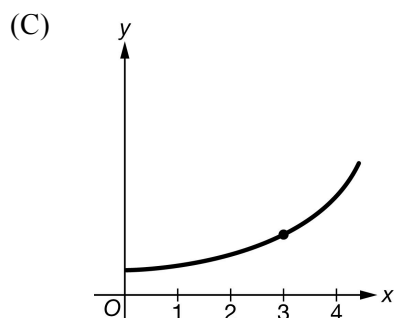
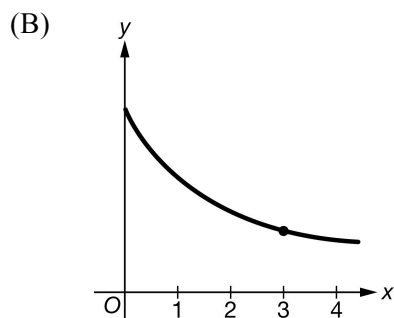
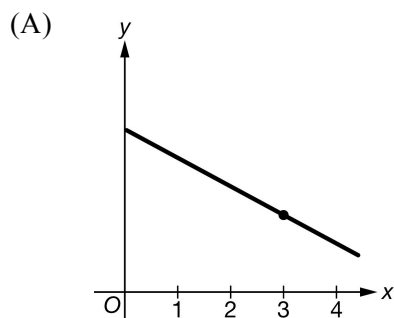
(D) 4.400

Answer C

Correct. The linear approximation to $B(t)$ at $t = 2.2$ will be
 $y = B(2.2) + B'(2.2)(t - 2.2)$. Since $B'(2.2) = 8e^{0.2 \cos 2.2} = 7.1117$, the linear approximation is
 $y = 4.5 + 7.1117(t - 2.2)$. The linear approximation will estimate that $B(t) = 9$ when
 $y = 4.5 + 7.1117(t - 2.2) = 9 \Rightarrow t = 2.833$.

4-6-6

88. The locally linear approximation of the differentiable function f at $x = 3$ is used to approximate the value of $f(3.2)$. The approximation at $x = 3.2$ is an overestimate of the corresponding function value at $x = 3.2$. Which of the following could be the graph of f ?

**Answer D**

Correct. The tangent line is the graph of the locally linear approximation of the function f near the point of tangency at $x = 3$. Since this graph is concave down on the interval $(2, 4)$, the tangent line would lie above the graph and the tangent line value at $x = 3.2$ would be greater than $f(3.2)$. Therefore, this could be the graph of f . The tangent line

4-6-6

approximation is an overestimate when the line is above the curve at the point. The tangent line approximation is an underestimate when the line is below the curve at the point.

89. The line tangent to the graph of the twice-differentiable function f at the point $x = 3$ is used to approximate the value of $f(3.25)$. Which of the following statements guarantees that the tangent line approximation at $x = 3.25$ is an underestimate of $f(3.25)$?
- (A) The function f is decreasing on the interval $3 \leq x \leq 3.25$.
 - (B) The function f is increasing on the interval $3 \leq x \leq 3.25$.
 - (C) The graph of the function f is concave down on the interval $3 \leq x \leq 3.25$.
 - (D) The graph of the function f is concave up on the interval $3 \leq x \leq 3.25$. ✓

Answer D

Correct. Because the graph is concave up the interval $3 \leq x \leq 3.25$, the line tangent to the graph at $x = 3$ will lie below the graph of f over the interval $3 \leq x \leq 3.25$. Therefore the point at $x = 3.25$ on the tangent line will be below the point at $x = 3.25$ on the graph of f , and so the tangent line value at $x = 3.25$ will be less than $f(3.25)$. The tangent line approximation is an overestimate when the line is above the curve at the point. The tangent line approximation is an underestimate when the line is below the curve at the point.

90.

x	3.8	4.0	4.2	4.4
$g'(x)$	-0.8	2.2	1.8	-1.2

Selected values of the derivative of the function g are given in the table above. It is known that $g(4) = 12$. What is the approximation for $g(4.2)$ found using the line tangent to the graph of g at $x = 4$?

- (A) 12.44 ✓
- (B) 12.40
- (C) 12.36
- (D) 11.60

Answer A

Correct. An equation for the tangent line at $x = a$ is $y = g(a) + g'(a)(x - a)$. Here $a = 4$ and $g'(a) = 2.2$. The value of y when $x = 4.2$ would be an approximation to $g(4.2)$, as follows.

$$g(4.2) \approx g(4) + 2.2(4.2 - 4) = 12 + 2.2(0.2) = 12 + 0.44 = 12.44$$

4-6-6

91. Let f be a function such that at each point (x, y) on the graph of f , the slope is given by $\frac{dy}{dx} = y^2 - x$. The graph of f passes through the point $(1, 2)$ and is concave down on the interval $1 < x < 1.5$. Let k be the approximation for $f(1.2)$ found by using the locally linear approximation of f at $x = 1$. Which of the following statements about k is true?
- (A) $k = 5.6$ and is an overestimate for $f(1.2)$.
- (B) $k = 5.6$ and is an underestimate for $f(1.2)$.
- (C) $k = 2.6$ and is an overestimate for $f(1.2)$. ✓
- (D) $k = 2.6$ and is an underestimate for $f(1.2)$.

Answer C

Correct. The tangent line is the graph of the locally linear approximation of f at $x = 1$. The slope of the tangent line at $x = 1$ is $f'(1) = \left. \frac{dy}{dx} \right|_{(1,2)} = 2^2 - 1 = 3$. An equation of the tangent line at the point $(1, 2)$ is therefore

$y = 2 + 3(x - 1)$. The value of y when $x = 1.2$ would be an approximation to $f(1.2)$, as follows.

$$f(1.2) \approx 2 + 3(1.2 - 1) = 2 + 0.6 = 2.6$$

Because the graph of f is concave down on the interval $1 < x < 1.5$, the locally linear approximation of f at $x = 1$ will overestimate the value of $f(1.2)$.

92. Let f be a differentiable function with $f(2) = 3$ and $f'(2) = \frac{1}{2}$. Using the line tangent to the graph of f at $x = 2$ as a local linear approximation for f , what is the estimate for $f(1.8)$?
- (A) 2.5
- (B) 2.8
- (C) 2.9 ✓
- (D) 3.1

Answer C

Correct. An equation of the line tangent to the graph of f at $x = a$ is $y = f(a) + f'(a)(x - a)$. Here $a = 2$ and $f'(a) = \frac{1}{2}$. The value of y when $x = 1.8$ would be an approximation to $f(1.8)$, as follows.

$$f(1.8) \approx f(2) + \frac{1}{2}(1.8 - 2) = 3 + \frac{1}{2}(-0.2) = 2.9$$

4-6-6

93. At time t , $0 < t < 2$, the velocity of a particle moving along the x -axis is given by $v(t) = t \sin(t^3)$. Let $t = b$ be the time at which the particle changes direction from moving left to moving right.



What is the total distance traveled by the particle during the time interval $0 < t < b$?

- (A) 0.212
(B) 0.612
(C) 1.011
(D) 1.208



Answer C

Correct. The graph of the velocity over the interval $(0, 2)$ shows that the velocity changes from positive to negative, then back to positive. The time at which the particle changes direction from moving left to moving right, therefore, is the second zero of $v(t)$, where the velocity changes from negative to positive. This zero is at $t = b = 1.84527$. Store this value in the calculator, and use the stored value for the limit of integration in order to ensure accuracy. The total distance traveled by the particle during the time interval $0 < t < b$ is

$$\int_0^b |v(t)| dt = \int_0^{1.84527} |t \sin(t^3)| dt = 1.011.$$

94. The line tangent to the graph of the polynomial function f at $x = 1$ is used to approximate the value of $f(1.2)$. Which of the following conditions guarantees that the approximation for $f(1.2)$ is an underestimate?
- (A) $f'(1)$ is negative.
(B) $f''(1)$ is positive.
(C) $f''(x)$ is negative for $1 < x < 1.2$.
(D) $f''(x)$ is positive for $1 < x < 1.2$.



Answer D

Correct. For a tangent line approximation, the function's behavior near the point of tangency may determine whether a tangent line value is an underestimate or an overestimate of the corresponding function value. The tangent line value at $x = 1.2$ will be an underestimate if the graph of f is concave up on the interval $1 < x < 1.2$. The condition that $f''(x)$ is positive for $1 < x < 1.2$ will guarantee that the graph is concave up on that interval.

4-6-6

95. Let f be a function with $f(3) = -7$ and $f'(3) = 5$. What is the approximation of $f(2.8)$ obtained by using the line tangent to the graph of f at $x = 3$?

(A) -8 (B) -6 (C) 4 (D) 6 **Answer A**

Correct. The tangent line to the graph of f at $x = 3$ is $y = f(3) + f'(3)(x - 3) = -7 + 5(x - 3)$. The value of y when $x = 2.8$ would be an approximation to $f(2.8)$.

$$f(2.8) \approx -7 + 5(-0.2) = -8$$

96. Let f be the function defined by $f(x) = \sqrt{x+4}$. What is the approximation of $\sqrt{4.2}$ found by using the line tangent to the graph of f at $x = 0$?

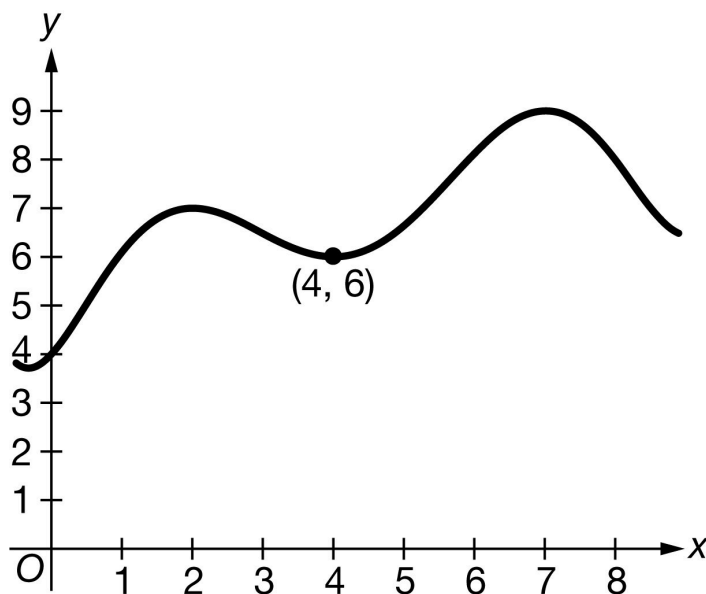
(A) 2.05 (B) 2.20 (C) 2.25 (D) 2.40 **Answer A**

Correct. An equation for the line tangent to the graph of f at $x = a$ is $y = f(a) + f'(a)(x - a)$. In this question, $a = 0$. The value of y when $x = 0.2$ would approximate $\sqrt{4.2} = f(0.2)$. Since $f'(x) = \frac{1}{2\sqrt{x+4}}$, $f'(0) = \frac{1}{4}$.

$$f(0.2) \approx f(0) + f'(0)(0.2) = 2 + \frac{1}{4}(0.2) = 2 + 0.05 = 2.05$$

4-6-6

97.

Graph of f'

The graph of f' , the derivative of the function f , is shown above. If $f(4) = -1$, what is the approximation for $f(4.5)$ using the line tangent to the graph of f at $x = 4$?

- (A) -4
- (B) -1
- (C) 2
- (D) 6

**Answer C**

Correct. An equation of the tangent line at $x = a$ is $y = f(a) + f'(a)(x - a)$. Here $a = 4$, so $f(a) = -1$ and $f'(a) = 6$. The value of y when $x = 4.5$ would be an approximation to $f(4.5)$, as follows.

$$f(4.5) \approx f(4) + 6(4.5 - 4) = -1 + 3 = 2$$

4-6-6

98.

x	2	3	5	8	13
$f(x)$	1	4	-2	3	6

Let f be a function that is twice differentiable for all real numbers. The table above gives values of f for selected points in the closed interval $2 \leq x \leq 13$.

(a) Estimate $f'(4)$. Show the work that leads to your answer.

(b) Evaluate $\int_2^{13} (3 - 5f'(x)) dx$. Show the work that leads to your answer.

(c) Use a left Riemann sum with subintervals indicated by the data in the table to approximate

$\int_2^{13} f(x) dx$. Show the work that leads to your answer.

(d) Suppose $f(5) = 3$ and $f''(x) < 0$ for all x in the closed interval $5 \leq x \leq 8$. Use the line tangent to the graph of f at $x = 5$ to show that $f(7) \leq 4$. Use the secant line for the graph of f on $5 \leq x \leq 8$ to show that $f(7) \geq 4/3$.

Part A

1 point is earned for the answer

$$f'(4) \approx \frac{f(5) - f(3)}{5 - 3} = -3$$



0	1
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The student response earns all of the following points:

1 point is earned for the answer

$$f'(4) \approx \frac{f(5) - f(3)}{5 - 3} = -3$$

Part B

1 point is earned if uses Fundamental Theorem of Calculus

1 point is earned for the answer

4-6-6

$$\begin{aligned}\int_2^{13} (3 - 5f'(x)) dx &= \int_2^{13} 3dx - 5 \int_2^{13} f'(x) dx \\ &= 3(13 - 2) - 5(f(13) - f(2)) = 8\end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned if uses Fundamental Theorem of Calculus

1 point is earned for the answer

$$\begin{aligned}\int_2^{13} (3 - 5f'(x)) dx &= \int_2^{13} 3dx - 5 \int_2^{13} f'(x) dx \\ &= 3(13 - 2) - 5(f(13) - f(2)) = 8\end{aligned}$$

Part C

1 point is earned for left Riemann sum

1 point is earned for the answer

$$\int_2^{13} f(x) dx \approx f(2)(3 - 2) + f(3)(5 - 3) + f(5)(8 - 5) + f(8)(13 - 8) = 18$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for left Riemann sum

1 point is earned for the answer

$$\int_2^{13} f(x) dx \approx f(2)(3 - 2) + f(3)(5 - 3) + f(5)(8 - 5) + f(8)(13 - 8) = 18$$

4-6-6

Part D

1 point is earned for the tangent line

1 point is earned for showing $f(7) \leq 4$

1 point is earned for secant line

1 point is earned if shows $f(7) \geq 4/3$

An equation for the tangent line is $y = -2 + 3(x - 5)$.

Since $f''(x) < 0$ for all x in the interval $5 \leq x \leq 8$, the line tangent to the graph of $y = f(x)$ at $x = 5$ lies above the graph for all x in the interval $5 < x \leq 8$.

Therefore, $f(7) \leq -2 + 3 \cdot 2 = 4$.

An equation for the secant line is $y = -2 + \frac{5}{3}(x - 5)$.

Since $f''(x) < 0$ for all x in the interval $5 \leq x \leq 8$, the secant line connecting $(5, f(5))$ and $(8, f(8))$ lies below the graph of $y = f(x)$ for all x in the interval $5 < x < 8$.

Therefore, $f(7) \geq -2 + \frac{5}{3} \cdot 2 = \frac{4}{3}$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the tangent line

1 point is earned for showing $f(7) \leq 4$

1 point is earned for secant line

1 point is earned if shows $f(7) \geq 4/3$

An equation for the tangent line is $y = -2 + 3(x - 5)$.

Since $f''(x) < 0$ for all x in the interval $5 \leq x \leq 8$, the line tangent to the graph of $y = f(x)$ at $x = 5$ lies above the graph for all x in the interval $5 < x \leq 8$.

4-6-6

Therefore, $f(7) \leq -2 + 3 \cdot 2 = 4$.

An equation for the secant line is $y = -2 + \frac{5}{3}(x - 5)$.

Since $f''(x) < 0$ for all x in the interval $5 \leq x \leq 8$, the secant line connecting $(5, f(5))$ and $(8, f(8))$ lies below the graph of $y = f(x)$ for all x in the interval $5 < x < 8$.

Therefore, $f(7) \geq -2 + \frac{5}{3} \cdot 2 = \frac{4}{3}$.

4-6-6

99.

t (days)	0	10	22	30
$W'(t)$ (GL per day)	0.6	0.7	1.0	0.5

The twice-differentiable function W models the volume of water in a reservoir at time t , where $W(t)$ is measured in gigaleters (GL) and t is measured in days. The table above gives values of $W'(t)$ sampled at various times during the time interval $0 \leq t \leq 30$ days. At time $t = 30$, the reservoir contains 125 gigaleters of water.

(a) Use the tangent line approximation to W at time $t = 30$ to predict the volume of water $W(t)$, in gigaleters, in the reservoir at time $t = 32$. Show the computations that lead to your answer.

(b) Use a left Riemann sum, with the three subintervals indicated by the data in the table, to approximate $\int_0^{30} W'(t) dt$. Use this approximation to estimate the volume of water $W(t)$, in gigaleters, in the reservoir at time $t = 0$. Show the computations that lead to your answer.

(c) Explain why there must be at least one time t , other than $t = 10$, such that $W'(t) = 0.7$ GL/day.

(d) The equation $A = 0.3W^{2/3}$ gives the relationship between the area A , in square kilometers, of the surface of the reservoir, and the volume of water $W(t)$, in gigaleters, in the reservoir. Find the instantaneous rate of change of A , in square kilometers per day, with respect to t when $t = 30$ days.

Part A

The response can earn up to 1 point:

1 point: For the correct answer

An equation of the tangent line is $y = 0.5(t - 30) + 125$.

$$W(32) \approx 0.5(32 - 30) + 125 = 126$$



0	1
---	---

The response earns all of the following point:

1 point: For the correct answer

An equation of the tangent line is $y = 0.5(t - 30) + 125$.

$$W(32) \approx 0.5(32 - 30) + 125 = 126$$

4-6-6

Part B

The response can earn up to 3 points:

1 point: For the left Riemann sum

1 point: For the approximation

1 point: For the correct answer

$$\int_0^{30} W'(t) dt \approx (10)(0.6) + (12)(0.7) + (8)(1.0) = 22.4$$

$$W(0) = W(30) - \int_0^{30} W'(t) dt = 125 - 22.4 = 102.6$$



0	1	2	3
---	---	---	---

The response earns all of the following points:

1 point: For the left Riemann sum

1 point: For the approximation

1 point: For the correct answer

$$\int_0^{30} W'(t) dt \approx (10)(0.6) + (12)(0.7) + (8)(1.0) = 22.4$$

$$W(0) = W(30) - \int_0^{30} W'(t) dt = 125 - 22.4 = 102.6$$

Part C

The response can earn up to 2 points:

2 points: For the explanation

W' is differentiable $\Rightarrow W'$ is continuous.

$$W'(30) = 0.5 < 0.7 < 1.0 = W'(22)$$

By the Intermediate Value Theorem, there must be at least one time t , $22 \leq t \leq 30$, such that $W'(t) = 0.7$.

4-6-6



0	1	2
---	---	---

The response earns all of the following points:

2 points: For the explanation

W' is differentiable $\Rightarrow W'$ is continuous.

$$W'(30) = 0.5 < 0.7 < 1.0 = W'(22)$$

By the Intermediate Value Theorem, there must be at least one time t , $22 \leq t \leq 30$, such that $W'(t) = 0.7$.

Part D

The response can earn up to 3 points:

2 points: For stating $\frac{dA}{dt}$

1 point: For the correct answer

$$\frac{dA}{dt} = (0.3)^{\frac{2}{3}} W^{-1/3} \cdot \frac{dW}{dt} = \frac{0.2}{\sqrt[3]{W}} \cdot \frac{dW}{dt}$$

$$\left. \frac{dA}{dt} \right|_{t=30} = \frac{0.2}{\sqrt[3]{125}} \cdot 0.5 = 0.02$$



0	1	2	3
---	---	---	---

The response earns all of the following points:

2 points: For stating $\frac{dA}{dt}$

1 point: For the correct answer

$$\frac{dA}{dt} = (0.3)^{\frac{2}{3}} W^{-1/3} \cdot \frac{dW}{dt} = \frac{0.2}{\sqrt[3]{W}} \cdot \frac{dW}{dt}$$

$$\left. \frac{dA}{dt} \right|_{t=30} = \frac{0.2}{\sqrt[3]{125}} \cdot 0.5 = 0.02$$

4-6-6

100. Let f be the function defined by $f(x) = (1 + \tan x)^{3/2}$ for $-\pi/4 < x < \pi/2$.

(a) Write an equation for the line tangent to the graph of f at the point where $x = 0$.

(b) Using the equation found in part (a), approximate $f(0.02)$.

(c) Let f^{-1} denote the inverse function of f . Write an expression that gives $f^{-1}(x)$ for all x in the domain of f^{-1} .

Part A

1 point is earned for $f(0) = 1$

2 points are earned for $f'(x)$

1 point is earned for $f'(0)$

1 point is earned for equation of line. Must use viable $f(0)$ and $f'(0)$

Note: max 2/5 for answer only

$$f(0) = 1$$

$$f'(x) = \frac{3}{2}(1 + \tan(x))^{\frac{1}{2}} (\sec^2(x))$$

$$f'(0) = \frac{3}{2}$$

$$y - 1 = \frac{3}{2}x \text{ or } y = \frac{3}{2}x + 1$$



0	1	2	3	4	5
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The student response earns all of the following points:

1 point is earned for $f(0) = 1$

2 points are earned for $f'(x)$

1 point is earned for $f'(0)$

1 point is earned for equation of line. Must use viable $f(0)$ and $f'(0)$

Note: max 2/5 for answer only

$$f(0) = 1$$

4-6-6

$$f'(x) = \frac{3}{2}(1 + \tan(x))^{\frac{1}{2}} (\sec^2(x))$$

$$f'(0) = \frac{3}{2}$$

$$y - 1 = \frac{3}{2}x \quad \text{or} \quad y = \frac{3}{2}x + 1$$

Part B

1 point is earned for evaluating their equation from part (a) at $x = 0.02$

$$f(0.02) \approx \frac{3}{2}(0.02) + 1$$



0	1
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The student response earns all of the following points:

1 point is earned for evaluating their equation from part (a) at $x = 0.02$

$$f(0.02) \approx \frac{3}{2}(0.02) + 1$$

Part C

1 point is earned for reversing variables anywhere in process

1 point is earned for solving for $\tan y$

1 point is earned for solving for y

$$x = (1 + \tan y)^{3/2}$$

$$x^{2/3} = 1 + \tan y$$

$$\tan y = x^{2/3} - 1$$

$$y = \arctan(x^{2/3} - 1)$$

OR

$$f^{-1}(x) = \arctan(x^{2/3} - 1)$$

4-6-6



0	1	2	3
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The student response earns all of the following points:

1 point is earned for reversing variables anywhere in process

1 point is earned for solving for $\tan y$

1 point is earned for solving for y

$$x = (1 + \tan y)^{3/2}$$

$$x^{2/3} = 1 + \tan y$$

$$\tan y = x^{2/3} - 1$$

$y = \arctan(x^{2/3} - 1)$ OR $f^{-1}(x) = \arctan(x^{2/3} - 1)$
